DATA SHEET

(DOC No. HX8252-A-DS)

HX8252-A

2402CH TFT LCD Source Driver with MIPI TCON

Version 06 October, 2019

Himax Technologies, Inc.
http://www.himax.com.tw

Revision History

October, 2019

Version	Date	Description of changes
01	2017/01/13	New setup.
02	2017/02/02	Page 76~77 1. Modify THF min. value. Page 97 2. Modify CHIPID value. Page 103 3. Add page 6 0xC2,0xC6 registers.
03	2017/02/11	Page 22~27 1. Modify power on and power off sequence.
04	2017/02/17	Page 22, 24, 26 1. Modify power on. Page 149 2. Modify MIPI data-clock timing specification.
05	2017/02/20	All pages 1. Remove “preliminary” wording from the data sheet. Page 17~21 2. Add VSN to VGL schottky diode. Page 22, 24, 26 3. Modify power on sequence. Page 81~122 4. Modify default value. Page 147 5. Modify current consumption.
06	2018/07/09	Page 17~21 1. Add VCL to VSN Capacitor.

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1. General Description

HX8252-A is 2402 channel outputs source driver with MIPI TCON. The TCON generates the horizontal and vertical control timing to source driver and GOA.

The HX8252-A two-chip cascade solution supports the resolution of 1600RGBx(640~2560), 1536RGBx(640~2560), 1440RGBx(640~2560), 1280RGBx(640~2048), 1200RGBx(640~2048), 1080RGBx(640~1920) and full 8-bits output which are more superior in color depth. Since the output circuit of this source driver incorporates an operational amplifier with low power dissipation, and performs wide voltage supply range and small output deviation. Therefore, a high quality display with less crosstalk can be achieved.

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2. Features

- MIPI interface support one port with 2/4 lane
- Support R/L type MIPI interface
- Two chip cascade solution for high resolution TFT LCD driver
- 2402 channels output, build in source driver and TCON
- Display resolution option:
  - Two chip solution for
  -1600RGBx(640~2560)
  -1536RGBx(640~2560)
  -1440RGBx(640~2560)
  -1280RGBx(640~2048)
  -1200RGBx(640~2048)
  -1080RGBx(640~1920)
- Support gamma curve adjustment by analog gamma and the RGB separated positive/negative 10-bit digital gamma
- 256 gray scale driving output compliant to 8bit display data
- Support GOA gate control signal
- Support Inversion :1/2/4 dot inversion, column inversion, 1+2 dot inversion
- Support stripe and zigzag panel
- Support CABC function
- Support Color enhancement function
- Support VCOM is programmable adjustment by OTP
- Build-in PFM Booster controller to drive DC/DC converter circuit -- VSP & VSN
- Build-in Charge Pump controller to drive DC/DC converter circuit – VGH, VGL

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3. Block Diagram

3.1 Function block diagram

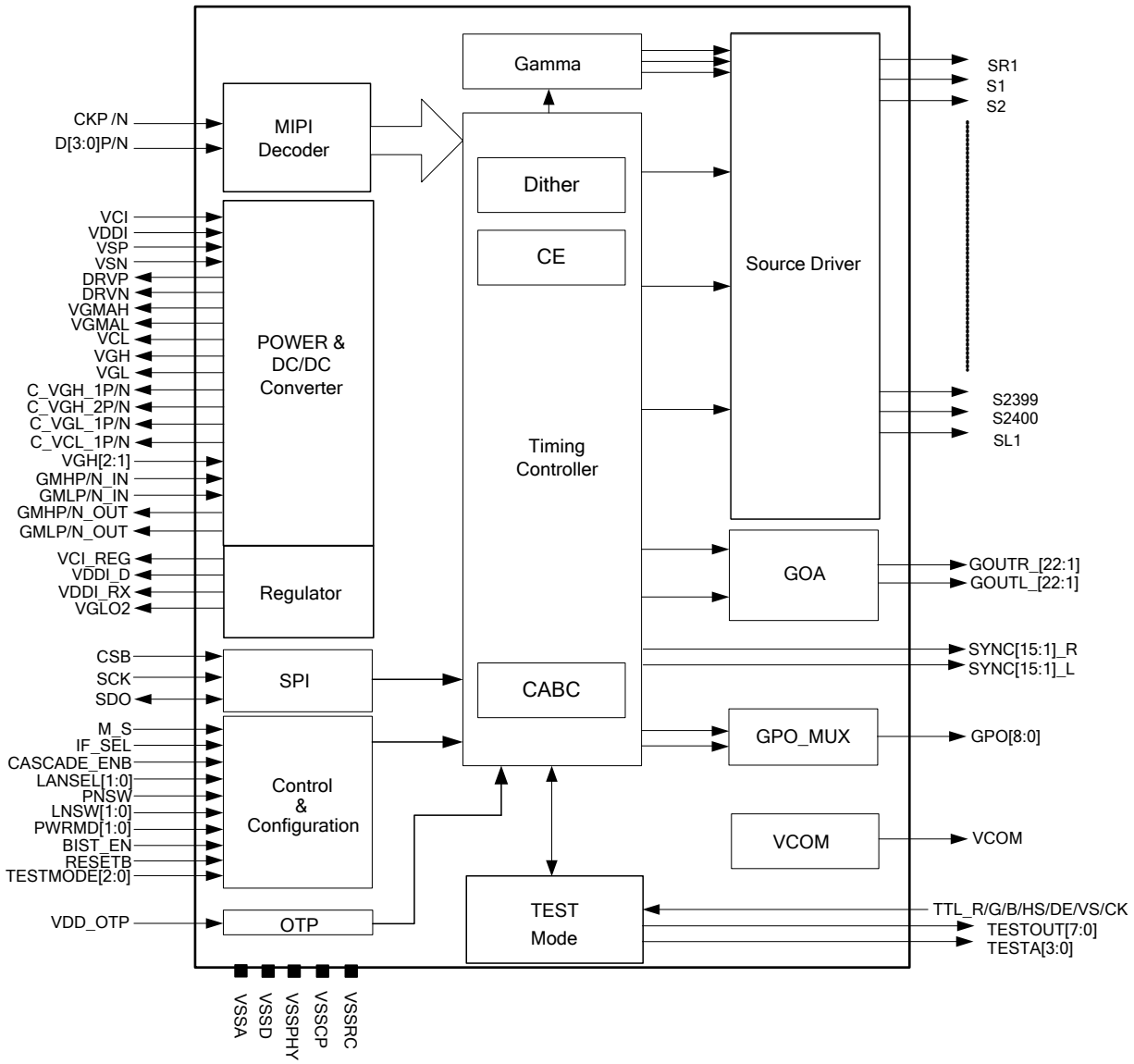


Figure 3.1: Function block diagram

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4. Pin Description

Pin types: **I**=Input, **O**=Output, **P**=Power, **G**=Ground, **N**=No Connection.

4.1 Pin description

<table>
<tr><th>Pin name</th><th>Type</th><th>Description</th></tr>
<tr><td>RESETB</td><td>I</td><td>Global reset. (Default pull high)</td></tr>
<tr><td>IF_SEL</td><td>I</td><td>Interface selection. (Default pull high)
IF_SEL=1: MIPI mode.
IF_SEL=0: Reserved.</td></tr>
<tr><td>CASCADE_ENB</td><td>I</td><td>Chip cascade selection. (Default pull high)
CASCADE_ENB=1: Single.
CASCADE_ENB=0: Cascade.</td></tr>
<tr><td>M_S</td><td>I</td><td>Define chip is master or slave. (Default pull high)
M_S=1: Master.
M_S=0: Slave.</td></tr>
<tr><td>LNSW[1:0]</td><td>I</td><td>MIPI lane swap function pin. (Default LNSW[1:0]=2’b11)
<table>
<tr><th colspan="2">Pad</th><th>D2P</th><th>D2N</th><th>D1P</th><th>D1N</th><th>CLKP</th><th>CLKN</th><th>D0P</th><th>D0N</th><th>D3P</th><th>D3N</th></tr>
<tr><th>LNSW1</th><th>LNSW0</th><th colspan="10">MIPI lanes mapping table</th></tr>
<tr><td>0</td><td>0</td><td>D3P</td><td>D3N</td><td>D2P</td><td>D2N</td><td>CLKP</td><td>CLKN</td><td>D1P</td><td>D1N</td><td>D0P</td><td>D0N</td></tr>
<tr><td>0</td><td>1</td><td>D3P</td><td>D3N</td><td>D0P</td><td>D0N</td><td>CLKP</td><td>CLKN</td><td>D1P</td><td>D1N</td><td>D2P</td><td>D2N</td></tr>
<tr><td>1</td><td>0</td><td>D0P</td><td>D0N</td><td>D1P</td><td>D1N</td><td>CLKP</td><td>CLKN</td><td>D2P</td><td>D2N</td><td>D3P</td><td>D3N</td></tr>
<tr><td>1</td><td>1</td><td>D2P</td><td>D2N</td><td>D1P</td><td>D1N</td><td>CLKP</td><td>CLKN</td><td>D0P</td><td>D0N</td><td>D3P</td><td>D3N</td></tr>
</table></td></tr>
<tr><td>PNSW</td><td>I</td><td>Pin change polarity of MIPI pin. (Default pull high)
PNSW=1: P/N normal.
PNSW=0: P/N swap.</td></tr>
<tr><td>PWRMD[1:0]</td><td>I</td><td>Power mode contorl pin. (Default PWRMD[1:0]=2’b11)
<table>
<tr><th>PWRMD1</th><th>PWRMD0</th><th>Driving mode (Note)</th></tr>
<tr><td>0</td><td>0</td><td>Support HX5186-C power mode.</td></tr>
<tr><td>0</td><td>1</td><td>Suppot PFM circuit power mode.</td></tr>
<tr><td>1</td><td>0</td><td>External VSP, VSN, VGH, VGL power mode.</td></tr>
<tr><td>1</td><td>1</td><td>External VSP, VSN power mode.</td></tr>
</table>
(When RB3h[5:4]=00b at Page.0 , PWRMD_XOR function)</td></tr>
<tr><td>LANSEL[1:0]</td><td>I</td><td>MIPI Lane number control pin. (Default LNSW[1:0]=2’b11)
<table>
<tr><th>LANSEL1</th><th>LANSEL0</th><th>MIPI lane</th></tr>
<tr><td>0</td><td>0</td><td>Reserved</td></tr>
<tr><td>0</td><td>1</td><td>2 Lane</td></tr>
<tr><td>1</td><td>0</td><td>Reserved</td></tr>
<tr><td>1</td><td>1</td><td>4 Lane</td></tr>
</table></td></tr>
<tr><td>BIST_EN</td><td>I</td><td>BIST pattern selection.(Default pull low)
BIST_EN=0: Self test mode is disabled.
BIST_EN=1: Self test mode is enabled.</td></tr>
<tr><td>GPO[8:0]</td><td>O</td><td>General purpose output.</td></tr>
</table>

Table 4.1: Global pin

Pin name	Type	Description
SYNC[15:1]_R	I/O	Synchronized signals between master and slave. These pins at right side of chip.
SYNC[15:1]_L	I/O	Synchronized signals between master and slave. These pins at left side of chip.

Table 4.2: Synchronize pin

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Pin name	Type	Description
CKP/CKN	I	MIPI clock input pin.
D0P/D0N D1P/D1N D2P/D2N D3P/D3N	I	MIPI data input pin.

Table 4.3: MIPI interface pin

Pin name	Type	Description
GOUTR_[22:1]	O	GOA control signal at right side.**(VGL~VGH)**
GOUTL_[22:1]	O	GOA control signal at left side. **(VGL~VGH)**

Note: (1) IO cell voltage is between VGH and VGL.

Table 4.4: GOA control pin

Pin name	Type	Description
S1 ~ S2400	O	Source output pin.
SL1,SR1	O	Source output pin. It is available when zigzag function enable.

Note: (1) IO cell voltage is between VSP and VSN.

Table 4.5: Source output pin

Pin name	Type	Description
VCI	P	Power input 2.7V ~ 6.5V.
VSP	P	Positive power input for source driver and power circuits **(4.0V ~ 6.5V)**.
VSN	P	Negative power input for source driver and power circuits **(-4.0V ~ -6.5V)**.
VDDI	P	Power input 1.7V ~ 3.6V for I/O and regulator output VDDI_D,VDDI_RX.
VDDI_D	P	Regulator output 1.5V for internal TCON and Logic.
VDDI_RX	P	Regulator output 1.5V for MIPI RX.
VDD_OTP	P	Power input for OTP programming **(8.5V)**.Leave this pin open when not programming OTP.
DRVP	O	CLK for VSP PFM and HX5186-C.
DRVN	O	CLK for VSN PFM and HX5186-C.
VGLO2	P	VGLO Regulator output 2 from VGL for GOUTR/L[13:15].If VGLO2 disable,please floating the VGLO2 and the level of GOUTR/L.[13:15] are VGL.
VGMAH	P	Regulator output for positive gamma high voltage.
VGMAL	P	Regulator output for negative gamma high voltage.
VCI_REG	P	Regulator output for internal reference.
VCL	P	VCL charge pump output.
C_VCL_1P C_VCL_1N	O	VCL flying cap. pin.
VGH	P	VGH charge pump output.
VGH1 VGH2	I	For GOA VGH voltage used, please connect to VGH.
C_VGH_1P C_VGH_1N C_VGH_2P C_VGH_2N	O	VGH flying cap. pin.
VGL	P	VGL charge pump output.
C_VGL_1P C_VGL_1N	O	VGL flying cap. pin.
VCOM	P	VCOM Regulator output.
GMHP_IN GMLP_IN GMHN_IN GMLN_IN	I	Gamma voltage link when use cascade function **(CASCADE_ENB="0")** Connect GMHP_OUT **(Master)** and GMHP_IN **(Slave)** together. Connect GMLP_OUT **(Master)** and GMLP_IN **(Slave)** together. Connect GMHN_OUT **(Master)** and GMHN_IN **(Slave)** together.

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Pin name	Type	Description
GMHP_OUT GMLP_OUT GMHN_OUT GMLN_OUT	O	Connect GMLN_OUT **(Master)** and GMLN_IN **(Slave)** together.
VSSD	G	Digital circuit ground **(0V)**.
VSSA	G	Analog circuit ground **(0V)**.
VSSPHY	G	Analog circuit ground **(0V)**.
VSSCP	G	Ground for charge pump circuit **(0V)**.
VSSRC	G	Ground for gamma reference voltage. **(0V)**

Table 4.6: Power and GND pin

Pin name	Type	Description
TESTOUT[7:0]	O	Test pin. Float these pin for normal operation.
TESTMODE[2:0]	I	Test pin. Float these pin for normal operation. **(Default pull high)**
TESTA[3:0]	O	Test pin. Float these pin for normal operation.
TTL_R[7:0] TTL_G[7:0] TTL_B[7:0] TTL_DE TTL_CK TTL_HS TTL_VS	I	Test pin. Float these pin for normal operation. **(Default pull high)**
CSB	I	Test pin. SPI chip enable signal. **(Default pull high)** Float these pin for normal operation.
SDO	I/O	Test pin. SPI data signal. **(Default pull high)** Float these pin for normal operation.
SCK	I	Test pin. SPI clock signal. **(Default pull high)** Float these pin for normal operation.
DUMMY DUMMY_R1 DUMMY_R2	N	No connection.

Table 4.7: Others pin

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4.2 Value of wiring resistance to each pin

The pin wiring resistance values affect power or signal integrity and the display quality. So be sure to design using values that do not exceed those recommendations as below.

Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
Power & Ground	VSSD	< 3	-
	VDDI	< 5	-
	VDDI_D	< 5	-
	VDDI_RX	< 5	-
	VCI	< 5	-
	VSSA	< 3	-
	VSP	< 5	-
	VSN	< 5	-
	VSSCP	< 5	-
	VSSRC	< 5	-
	VDD_OTP	< 10	-
	VSSPHY	< 5	-

Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
PFM & Charge Pump & Regulator	VCOM	< 5	-
	VCI_REG	< 5	-
	VGLO2	< 5	-
	VGL	< 10	-
	VGH/VGH1/VGH2	< 10	-
	VGMAH	< 10	-
	VGMAL	< 10	-
	VCL	< 5	-
	C_VCL_1P,C_VCL_1N	< 5	-
	C_VGH_1P,C_VGH_1N C_VGH_2P,C_VGH_2N	< 5	-
	DRVN,DRVP	< 5	-
	C_VGL_1P,C_VGL_1N	< 3	-

Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
GOA	GOUTR_1~GOUTR_22	< 100	-
	GOUTL_1~GOUTL_22	< 100	-

Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
Cascade	SYNC[15:1]_R	RC < 40ns	RC < 40ns
	SYNC[15:1]_L	RC < 40ns	RC < 40ns
	GMHP_IN / GMHP_OUT	< 10	-
	GMLP_IN / GMLP_OUT	< 10	-
	GMHN_IN / GMHN_OUT	< 10	-
	GMLN_IN / GMLN_OUT	< 10	-

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Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
MIPI Interface	CKP, CKN	< 10	< 0.9
	D0P, D0N		
	D1P, D1N		
	D2P, D2N		
	D3P, D3N		

Pin type	Pin name	Resistance value (Ω)	Capacitance value (pF)
Global	GPO	< 30	-
	PWRMD, RESETB, LNSW, IF_SEL, BIST_EN, PNSW, CASCADE_ENB, M_S	< 100	-
Other	CSB,SCL,SDO	< 100	-

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5. Power Application

5.1 Power architecture

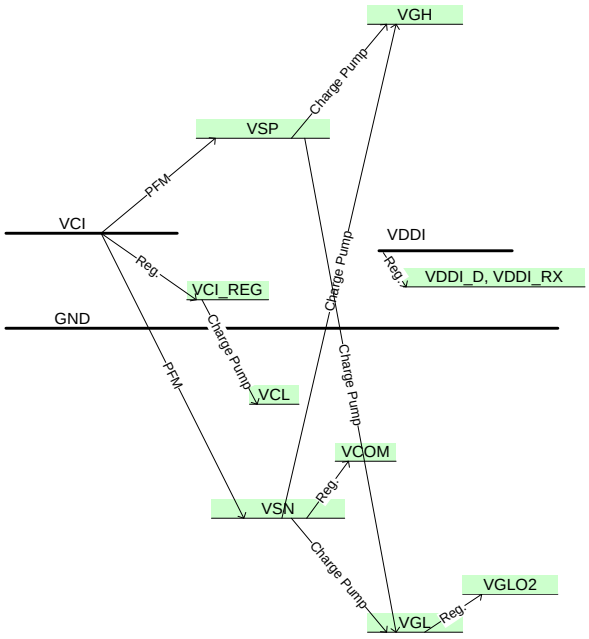
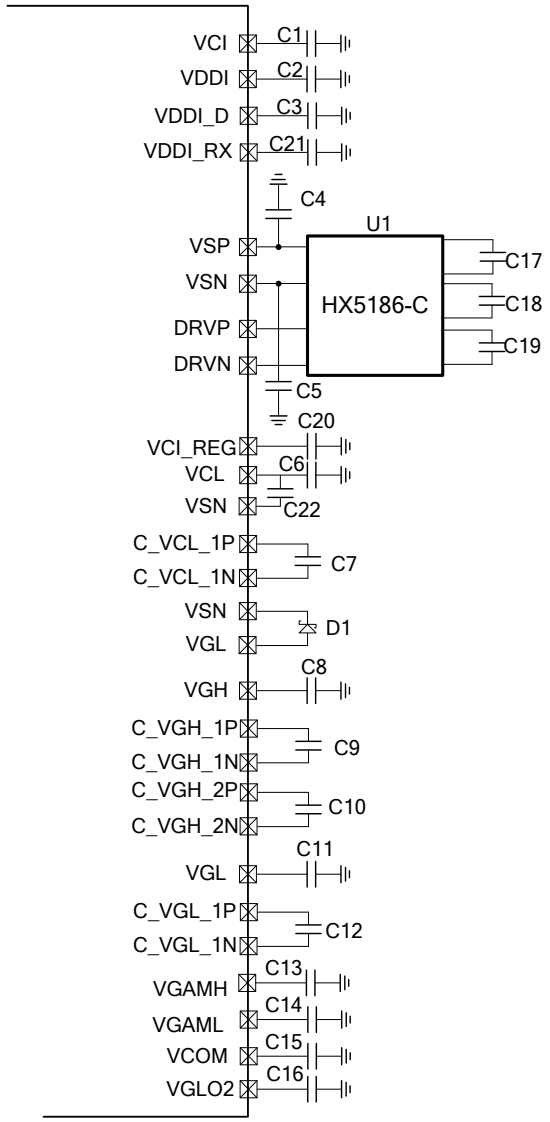


Figure 5.1: Power architecture

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5.2 PWRMD[1:0]=00b

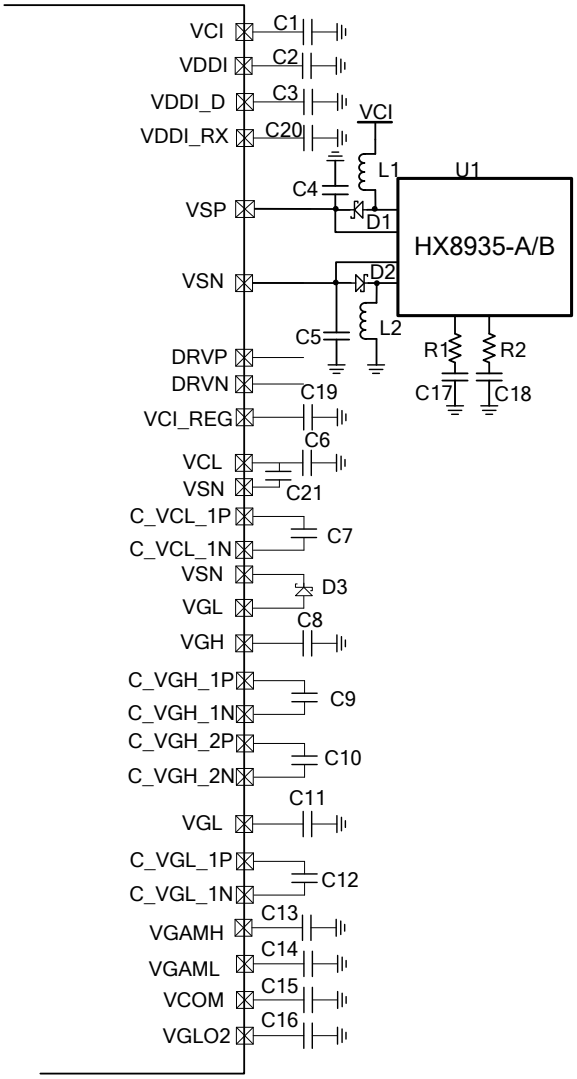
- **HX5186-C example circuit**



Component	Value
C1	2.2μF/10V
C2	2.2μF/6V
C3	2.2μF/6V
C4	2.2μF/10V
C5	2.2μF/10V
C6	1μF/6V
C7	1μF/6V
C8	1μF/25V
C9	1μF/16V
C10	1μF/16V
C11	1μF/25V
C12	1μF/16V
C13	1μF/10V
C14	1μF/10V
C15	1μF/6V
C16	1μF/25V
C17	Please refer HX5186-C datasheet
C18	Please refer HX5186-C datasheet
C19	Please refer HX5186-C datasheet
C20	1μF/6V
C21	2.2μF/6V
C22	1μF/10V
U1	HX5186-C
D1	Schottky diode

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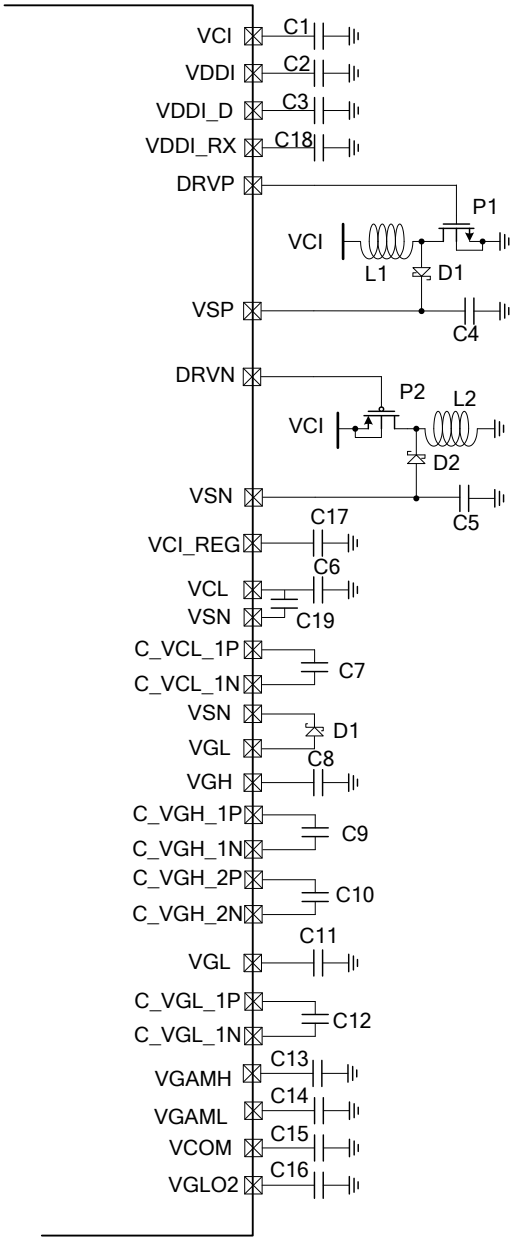
- **HX8935-A/B example circuit**



Component	Value
C1	2.2μF/10V
C2	2.2μF/6V
C3	2.2μF/6V
C4	2.2μF/10V
C5	2.2μF/10V
C6	1μF/6V
C7	1μF/6V
C8	1μF/25V
C9	1μF/16V
C10	1μF/16V
C11	1μF/25V
C12	1μF/16V
C13	1μF/10V
C14	1μF/10V
C15	1μF/6V
C16	1μF/25V
C17,C18	1μF/6.3V
C19	1μF/6V
C20	2.2μF/6V
C21	1μF/10V
R1,R2	100K
L1,L2	4.7μH
D1,D2	VF<0.4V /20mA, VR>30V recommand diode :RB521S-30
U1	HX8935-A/B
D3	Schottky diode

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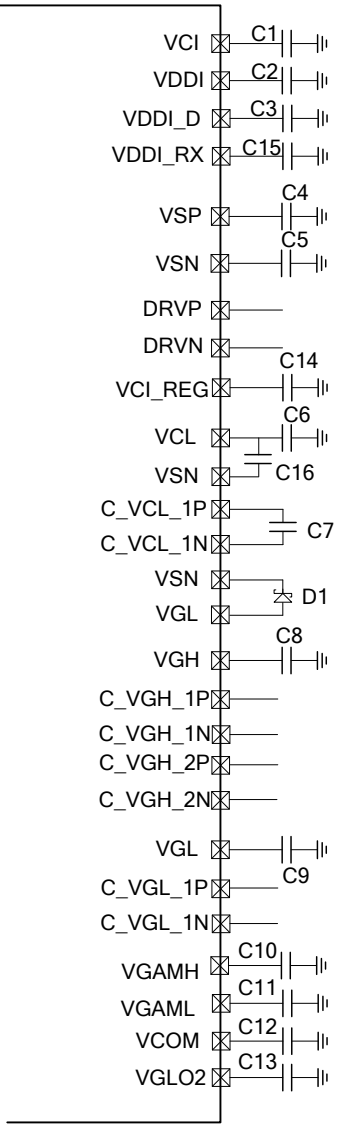
5.3 PWRMD[1:0]=01b



Component	Value
C1	2.2μF/10V
C2	2.2μF/6V
C3	2.2μF/6V
C4	2.2μF/10V
C5	2.2μF/10V
C6	1μF/6V
C7	1μF/6V
C8	1μF/25V
C9	1μF/16V
C10	1μF/16V
C11	1μF/25V
C12	1μF/16V
C13	1μF/10V
C14	1μF/10V
C15	1μF/6V
C16	1μF/25V
C17	1μF/6V
C18	2.2μF/6V
C19	1μF/10V
D1,D2	VF<0.4V /20mA, VR>30V recommand diode :RB521S-30
P1	NMOS
P2	PMOS
L1	10μH
L2	10μH
D1	Schottky diode

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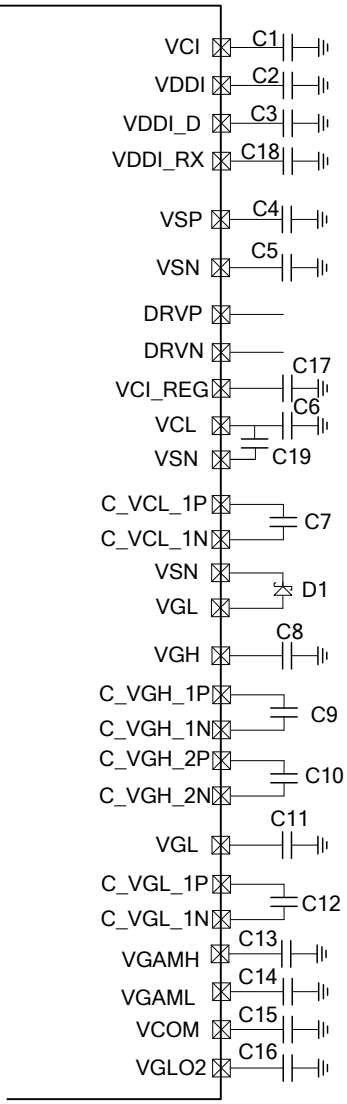
5.4 PWRMD[1:0]=10b



Component	Value
C1	2.2µF/10V
C2	2.2µF/6V
C3	2.2µF/6V
C4	2.2µF/10V
C5	2.2µF/10V
C6	1µF/6V
C7	1µF/6V
C8	1µF/25V
C9	1µF/25V
C10	1µF/10V
C11	1µF/10V
C12	1µF/6V
C13	1µF/25V
C14	1µF/6V
C15	2.2µF/6V
C16	1µF/6V
D1	Schottky diode

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5.5 PWRMD[1:0]=11b



Component	Value
C1	2.2μF/10V
C2	2.2μF/6V
C3	2.2μF/6V
C4	2.2μF/10V
C5	2.2μF/10V
C6	1μF/6V
C7	1μF/6V
C8	1μF/25V
C9	1μF/16V
C10	1μF/16V
C11	1μF/25V
C12	1μF/16V
C13	1μF/10V
C14	1μF/10V
C15	1μF/6V
C16	1μF/25V
C17	1μF/6V
C18	2.2μF/6V
C19	1μF/6V
D1	Schottky diode

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5.6 Power on/off sequence

5.6.1 Power on sequence PWRMD[1:0]=00b or 01b

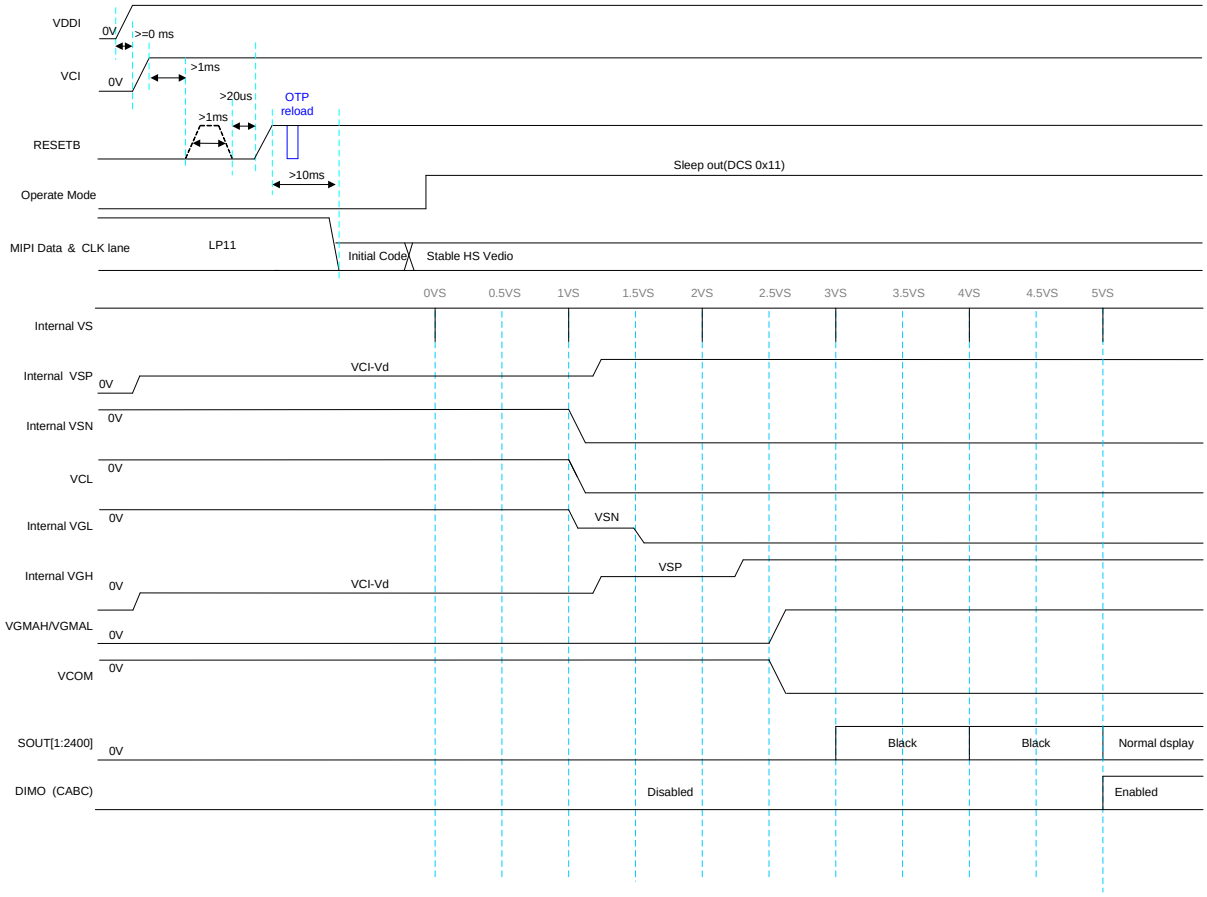


Figure 5.2: Power on sequence with PWRMD[1:0]=00b or 01b

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5.6.2 Power off sequence PWRMD[1:0]=00b or 01b

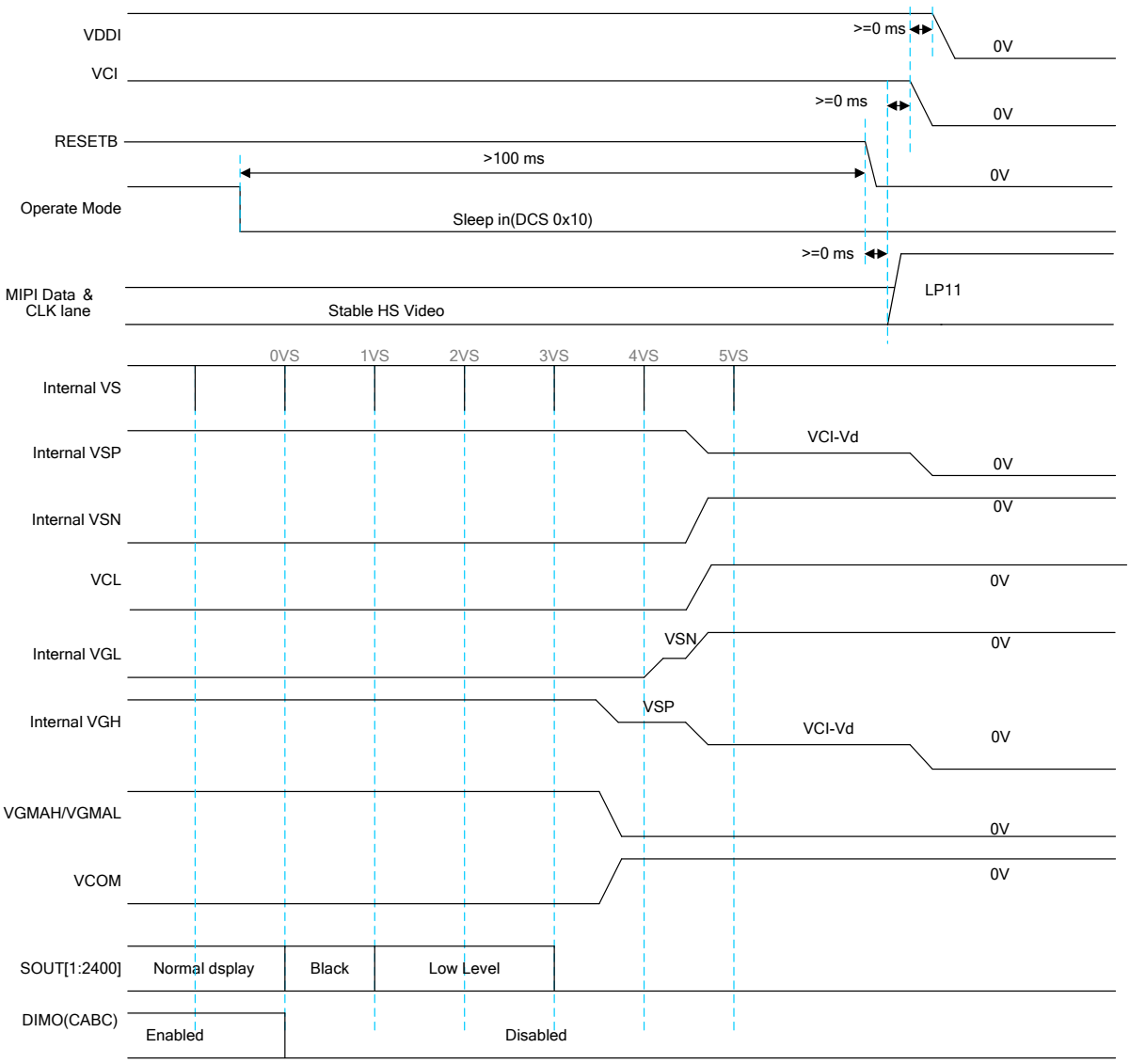


Figure 5.3: Power sequence in power off or standby mode with PWRMD[1:0]=00b or 01b

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5.6.3 Power on sequence PWRMD[1:0]=10b

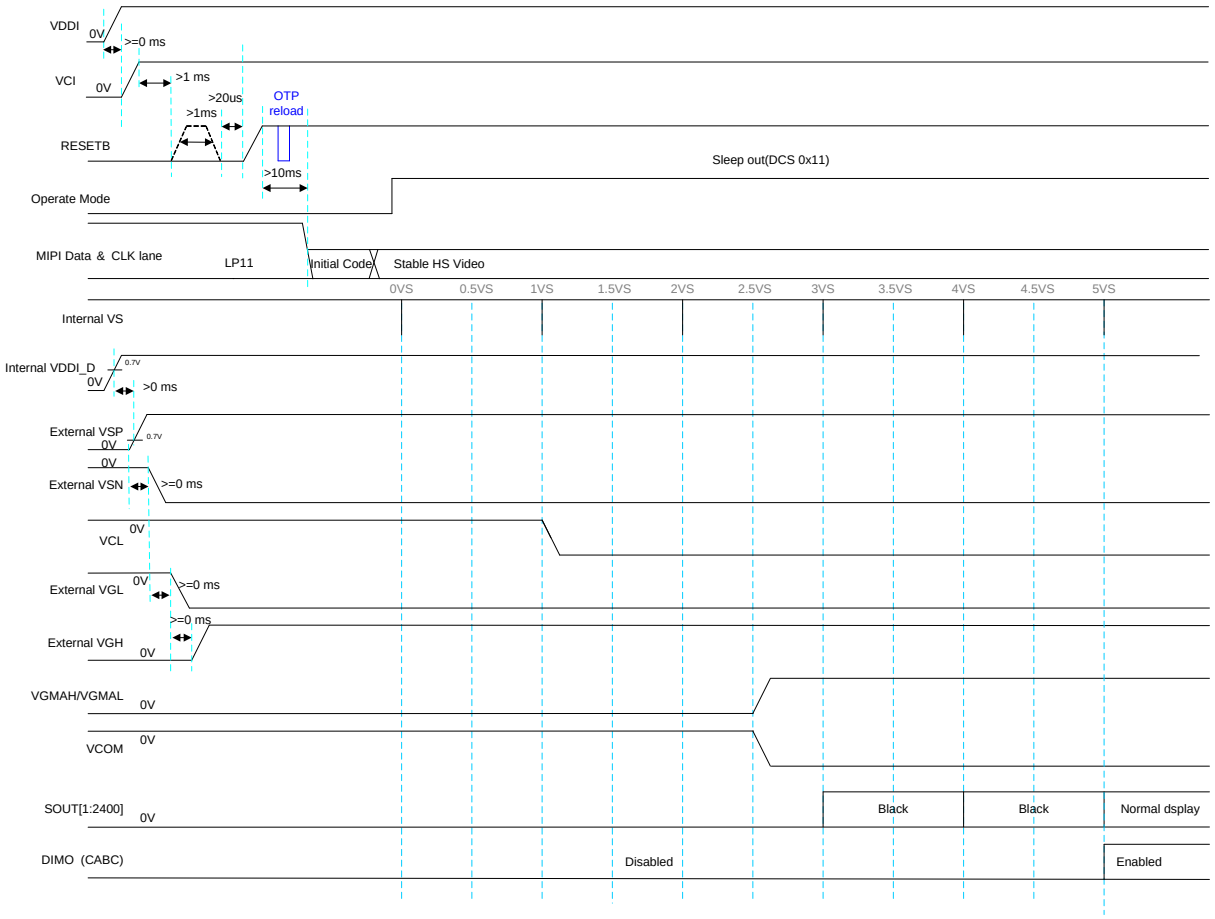


Figure 5.4: Power on sequence with PWRMD[1:0]=10b

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5.6.4 Power off sequence PWRMD[1:0]=10b

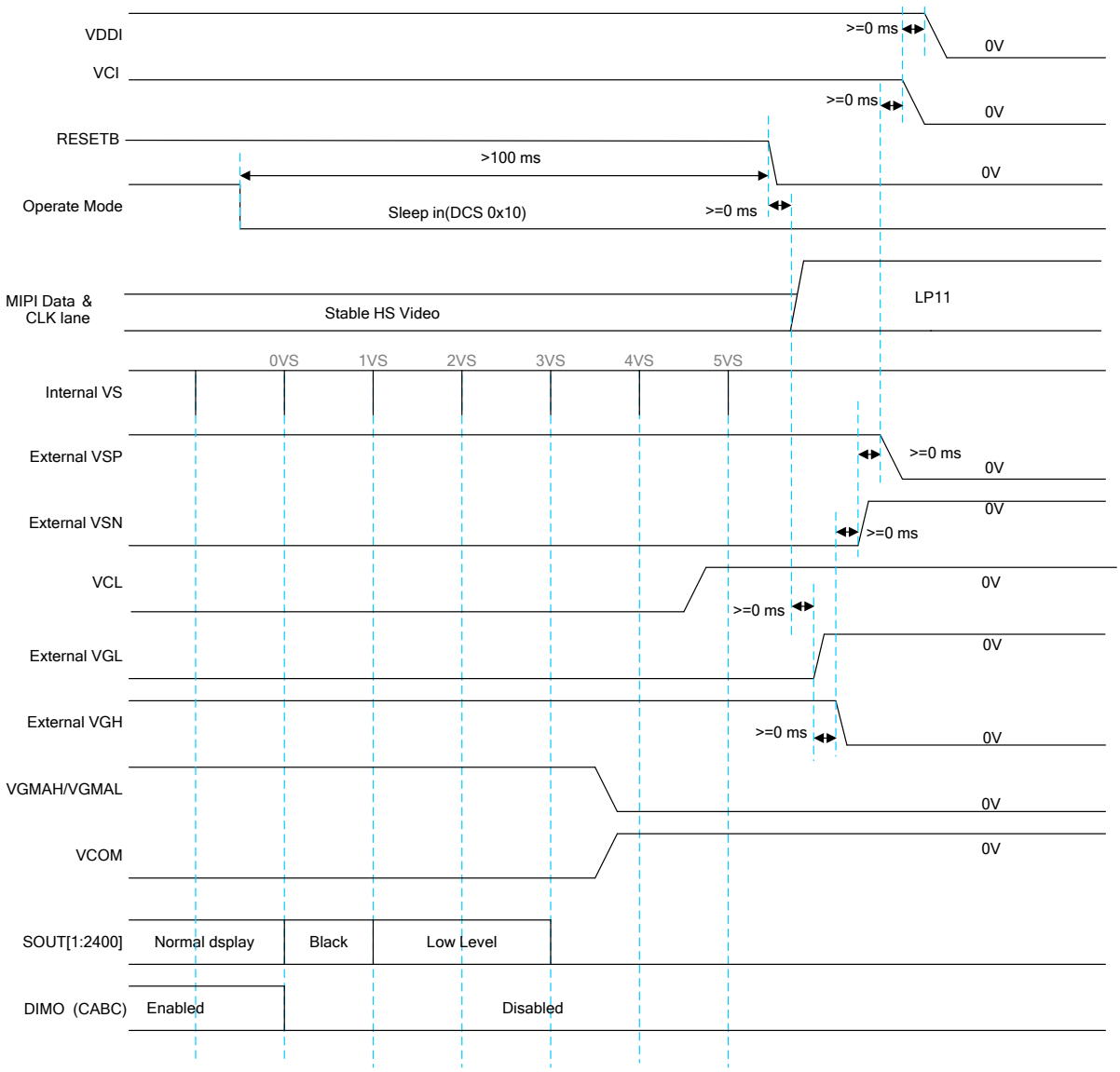


Figure 5.5: Power sequence in power off or standby mode with PWRMD[1:0]=10b

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5.6.5 Power on sequence PWRMD[1:0]=11b

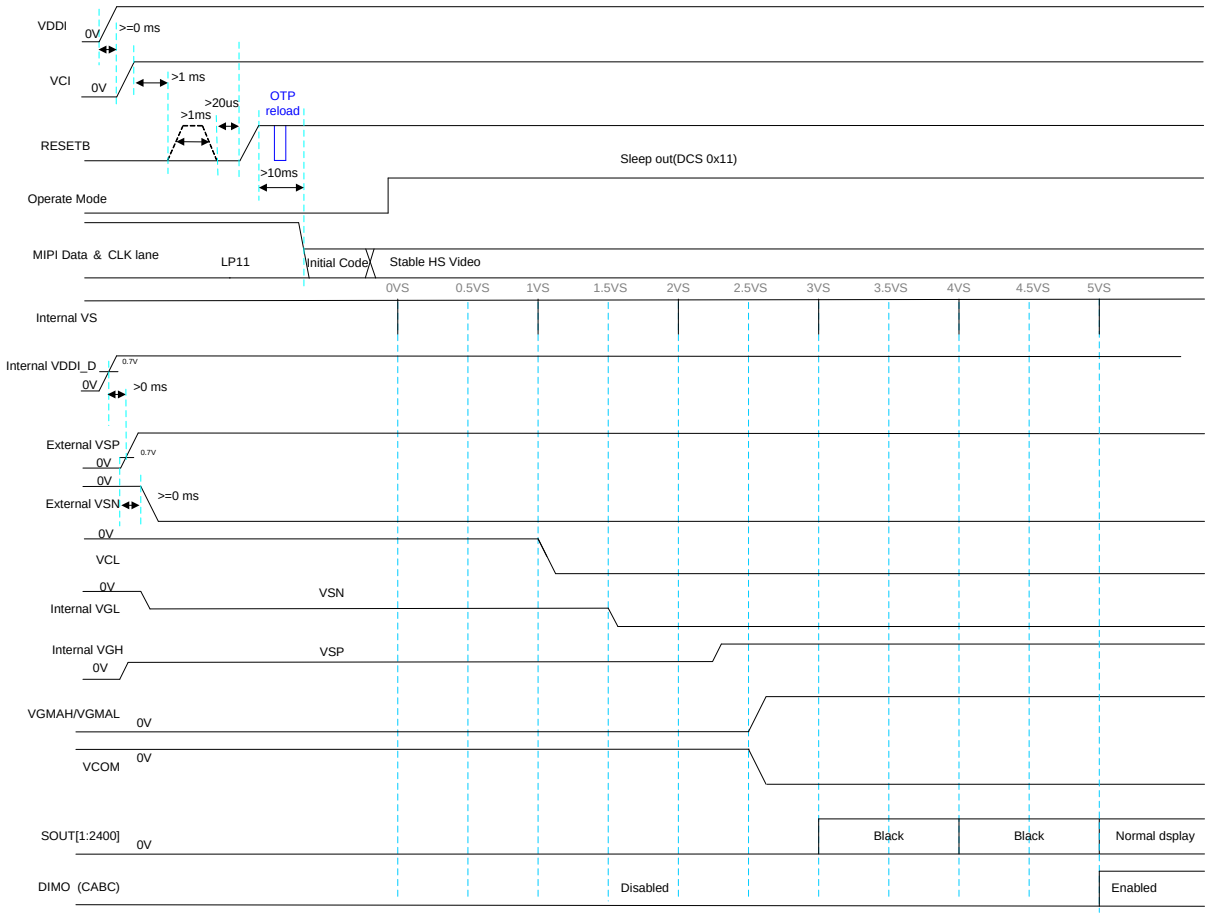


Figure 5.6: Power on sequence with PWRMD[1:0]=11b

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5.6.6 Power off sequence PWRMD[1:0]=11b

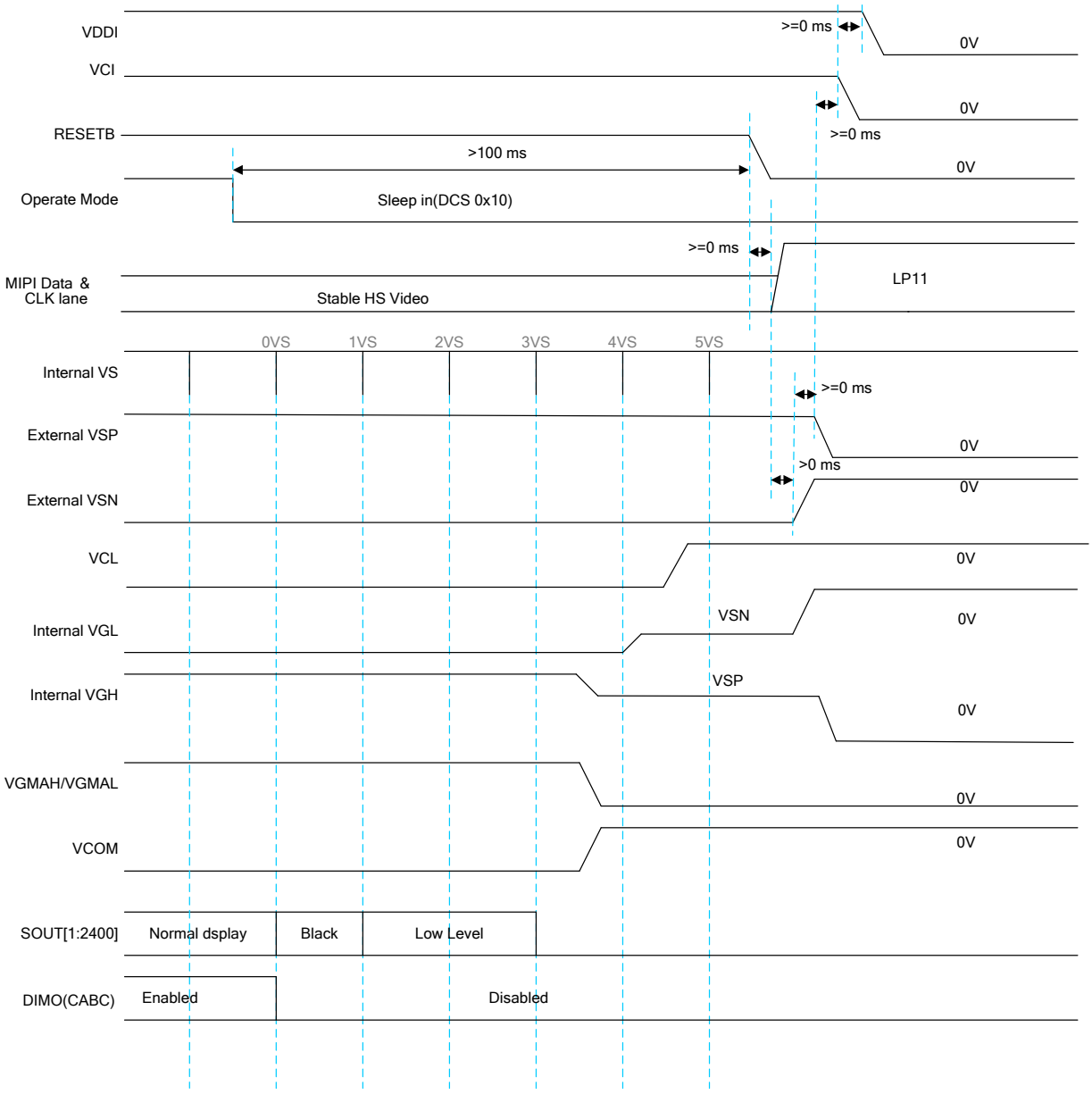


Figure 5.7: Power sequence in power off or standby mode with PWRMD[1:0]=11b

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6. Panel Application

The HX8252-A supports cascade mode. The resolution of cascade mode is 1600RGBx2560, 1536RGBx2560 1440RGBx2560, 1280RGBx2048, 1200RGBx2048, 1080RGBx1920, and 1440RGBx2160.

The TCON also can generate gate controller timing. These signals only support for GOA **(Gate driver on Array)**.

6.1 Display resolution

6.1.1 Display resolution configuration

Resolution selection can set by register RES[2:0], address is located at 0xB3 of page0.

RES[2:0]	Resolution
000	1600RGB x 2560 **(Default)**
001	1536RGB x 2560
010	1440RGB x 2560
011	1280RGB x 2048
100	1200RGB x 2048
101	1080RGB x 1920
110	1440RGB x 2160

Table 6.1: Display resolution setting

6.1.2 Source output channel valid range

The relationship between resolution and enable source channel is shows below. The channel of available output depends on the horizontal resolution, and non-available channel always locate at centre channel numbers. The source output pin must be floating if not used.

Horizontal resolution	Strip panel (Master/Slave IC)			Zigzag panel (Master/Slave IC)		
	Enable channel	Disable channel	Enable channel	Enable channel	Disable channel	Enable channel
1600RGB	1 - 1200	x	1201 - 2400	1 **(SR1)** - 1200	x	1201 – 2400 **(SL1)**
1536RGB	1 - 1152	1153 - 1248	1249 - 2400	1 **(SR1)** - 1152	1153 - 1248	1249 - 2400 **(SL1)**
1440RGB	1 - 1080	1081 - 1320	1321 - 2400	1 **(SR1)** - 1080	1081 - 1320	1321 - 2400 **(SL1)**
1280RGB	1 - 960	961 - 1440	1441 - 2400	1 **(SR1)** - 960	961 - 1440	1441 - 2400 **(SL1)**
1200RGB	1 - 900	901 - 1500	1501 - 2400	1 **(SR1)** - 900	901 - 1500	1501 - 2400 **(SL1)**
1080RGB	1 - 810	811 - 1590	1591 - 2400	1 **(SR1)** - 810	811 - 1590	1591 - 2400 **(SL1)**

Note: (1) Blue mark is typical application.
(2) For zigzag panel type, HX8252-A will enable SR1 and SL1 channel.
For strip panel type, HX8252-A will disable SR1 and SL1 channel.

Table 6.2: Source output enable/disable channel

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6.2 Cascade connection

6.2.1 Cascade mode of synchronized signals

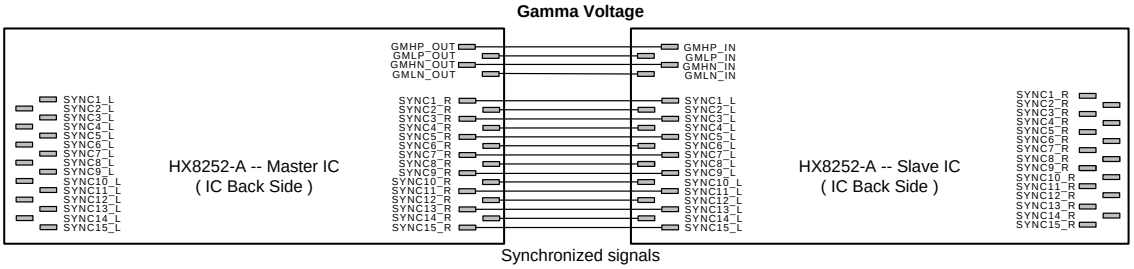


Figure 6.1: Cascade mode of synchronized signals

6.2.2 Synchronized signals of two chips cascade

Name	Type	Description of master	Type	Description of slave
SYNC1_R	I/O	Internal cascade signal.		-
SYNC2_R	I/O			
SYNC3_R	I/O			
SYNC4_R	I/O			
SYNC5_R	I/O			
SYNC6_R	I/O			
SYNC7_R	I/O			
SYNC8_R	I/O			
SYNC9_R	I/O			
SYNC10_R	I/O			
SYNC11_R	I/O			
SYNC12_R	I/O			
SYNC13_R	I/O			
SYNC14_R	I/O			
SYNC15_R	I/O			
Name	**Type**	**Description of master**	**Type**	**Description of slave**
SYNC1_L		-	I/O	Internal cascade signal.
SYNC2_L			I/O	
SYNC3_L			I/O	
SYNC4_L			I/O	
SYNC5_L			I/O	
SYNC6_L			I/O	
SYNC7_L			I/O	
SYNC8_L			I/O	
SYNC9_L			I/O	
SYNC10_L			I/O	
SYNC11_L			I/O	
SYNC12_L			I/O	
SYNC13_L			I/O	
SYNC14_L			I/O	
SYNC15_L			I/O	

Table 6.3: Synchronized signals description with two chip cascade

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6.3 GOA connection

The HX8252-A can support GOA **(Gate driver on array)** function.

GOA output pin define can set by register. A multiplexer is built in GOA function that selects one of several GOA signals. GOA function showed as below.The detail GOA output signal setting please refer application note.

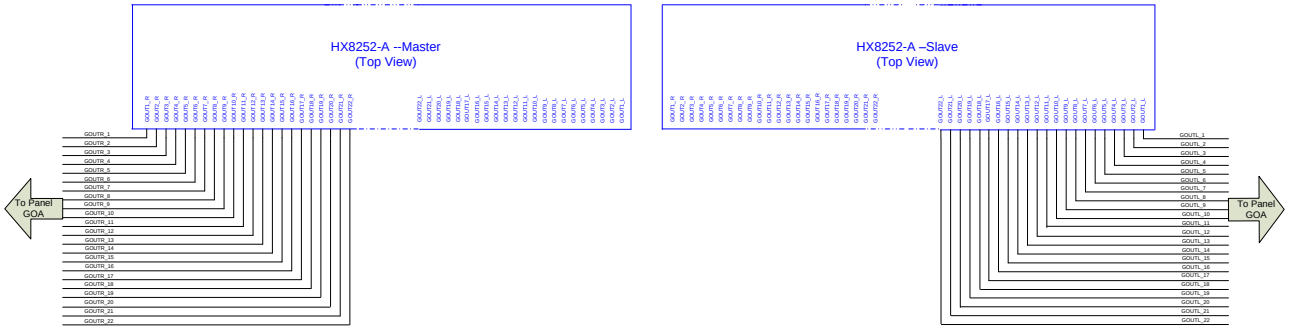


Figure 6.2: GOA wire example

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6.4 Panel structure

HX8252-A can support 2 types of driving method – stripe and zigzag. User could control Register: ZIGZAG_SEL and ZTYPE_SEL [1:0], address is located at 0xB2 and 0xB3 of page0, select panel type as following Figure:

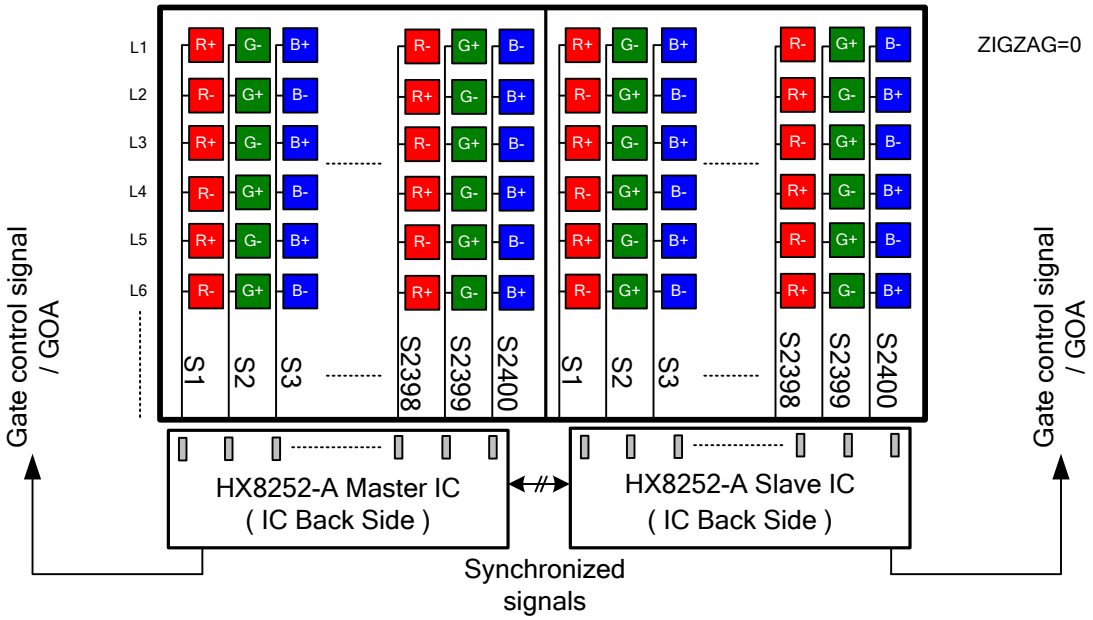


Figure 6.3: Stripe driving method

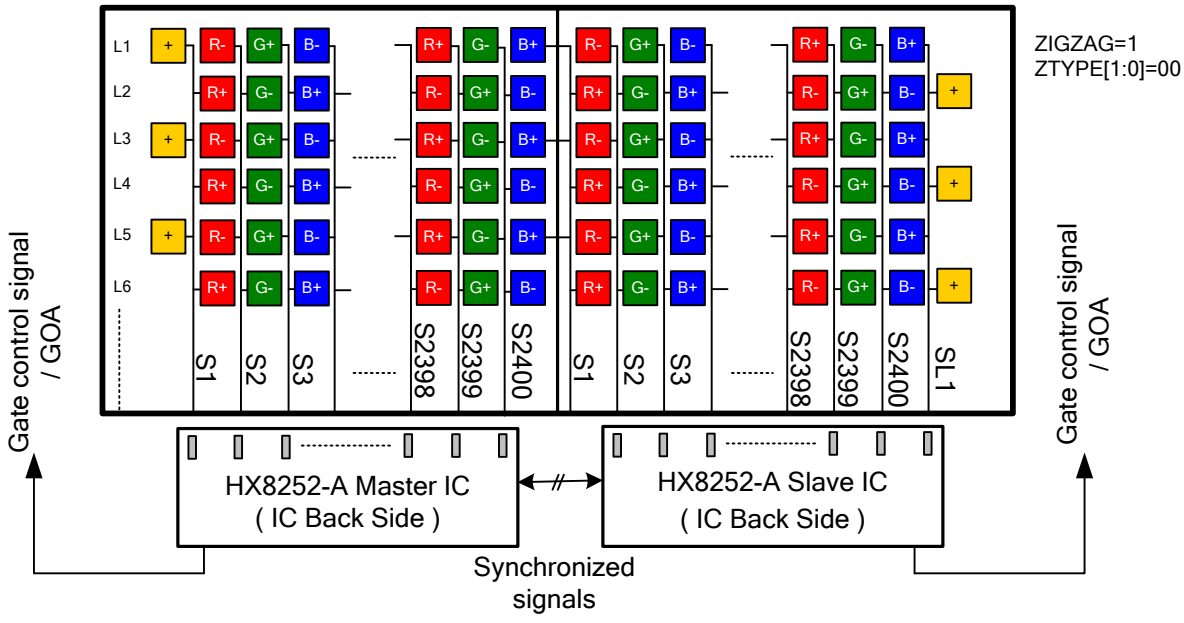


Figure 6.4: Zigzag type0 driving method

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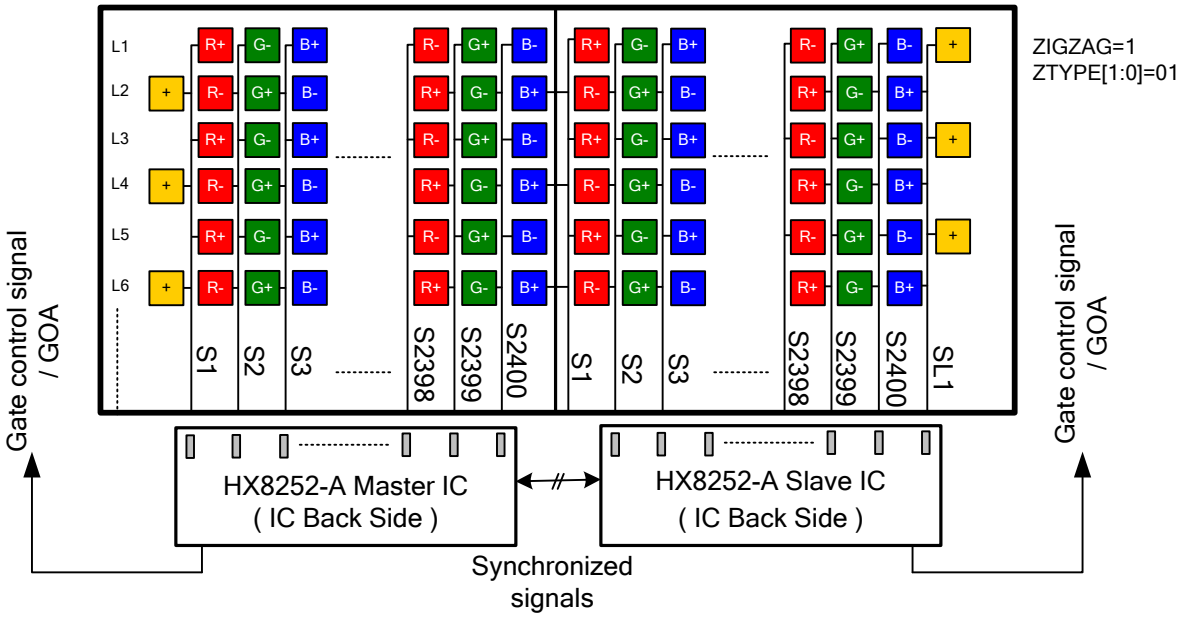


Figure 6.5: Zigzag type1 driving method

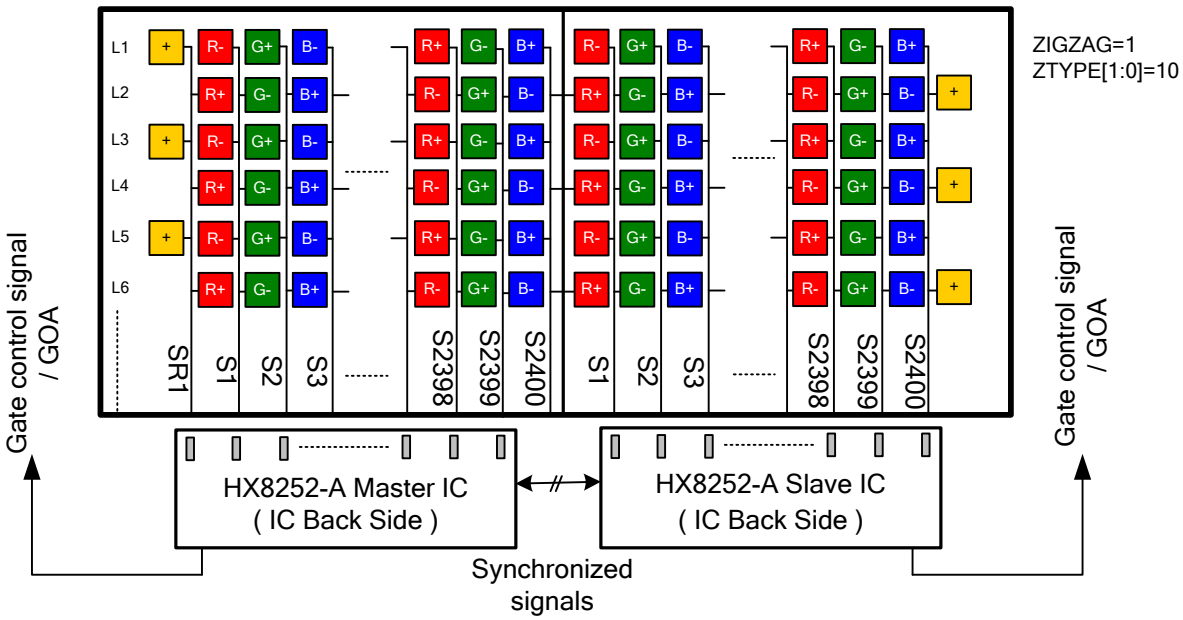


Figure 6.6: Zigzag type2 driving method

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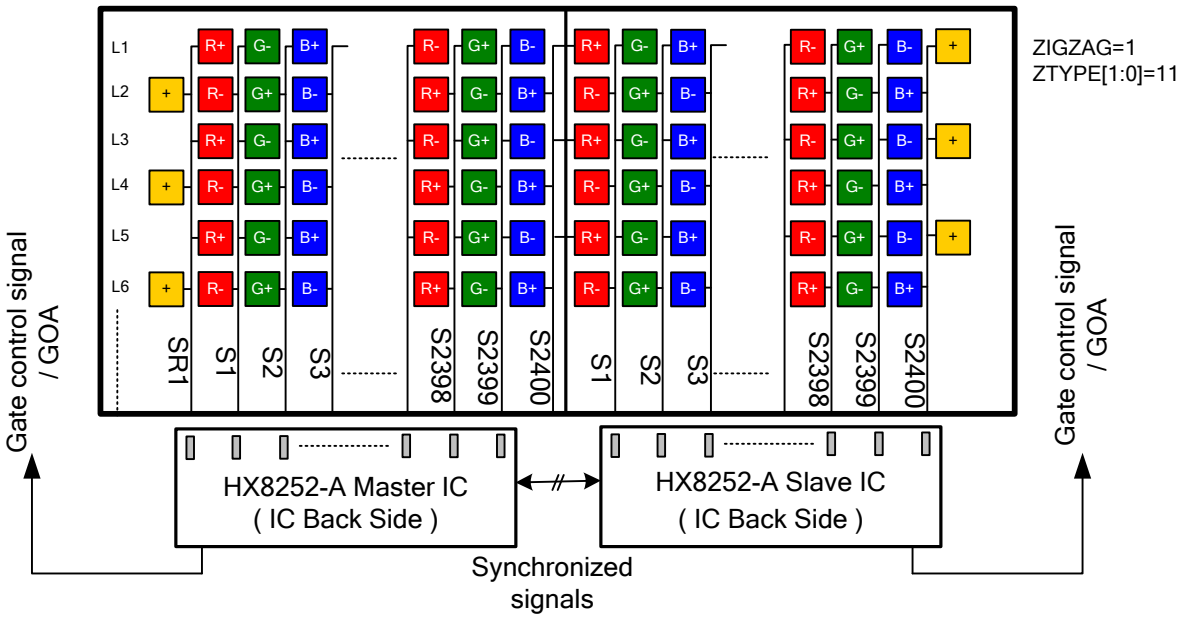


Figure 6.7: Zigzag type3 driving method

6.5 MIPI interface and pin mapping

HX8252-A support MIPI interface, and user could configure data lane arrangement by hardware pin PNSW and LNSW[1:0].

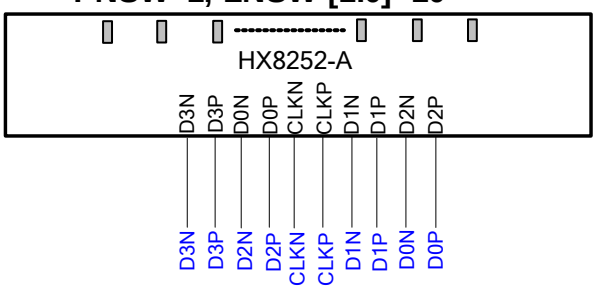
- PNSW is for swap differential pair polarity
- LNSW[1:0] are for swap MIPI data pair

Following table showed the MIPI lane swap pin mapping:

Configuration / Pad name			D2P	D2N	D1P	D1N	CLKP	CLKN	D0P	D0N	D3P	D3N
PNSW	LNSW1	LNSW0	MIPI lanes mapping table									
1	0	0	D3P	D3N	D2P	D2N	CLKP	CLKN	D1P	D1N	D0P	D0N
1	0	1	D3P	D3N	D0P	D0N	CLKP	CLKN	D1P	D1N	D2P	D2N
1	1	0	D0P	D0N	D1P	D1N	CLKP	CLKN	D2P	D2N	D3P	D3N
1	1	1	D2P	D2N	D1P	D1N	CLKP	CLKN	D0P	D0N	D3P	D3N
0	0	0	D3N	D3P	D2N	D2P	CLKN	CLKP	D1N	D1P	D0N	D0P
0	0	1	D3N	D3P	D0N	D0P	CLKN	CLKP	D1N	D1P	D2N	D2P
0	1	0	D0N	D0P	D1N	D1P	CLKN	CLKP	D2N	D2P	D3N	D3P
0	1	1	D2N	D2P	D1N	D1P	CLKN	CLKP	D0N	D0P	D3N	D3P

Setting example:

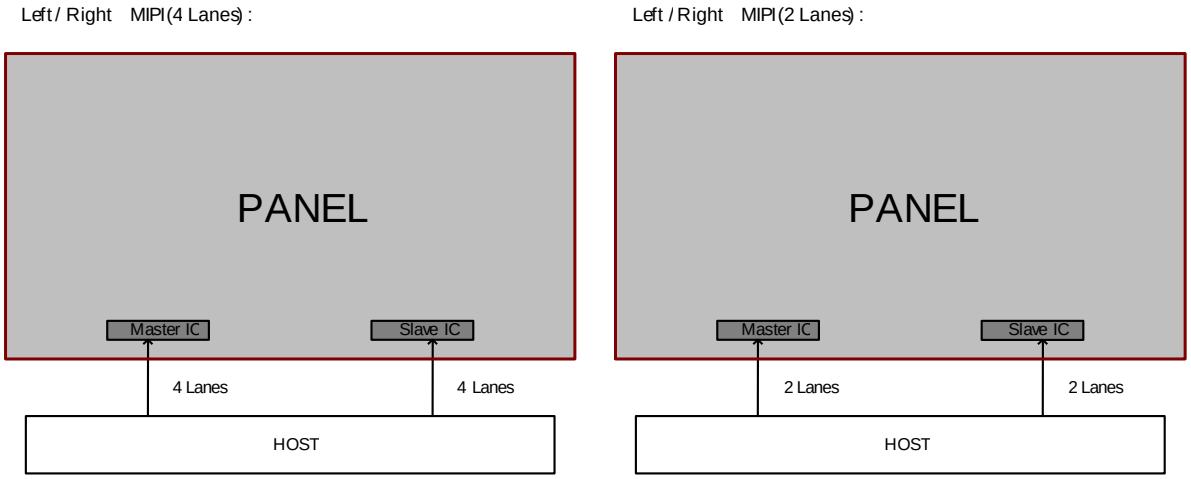
PNSW=1, LNSW [1:0]=10



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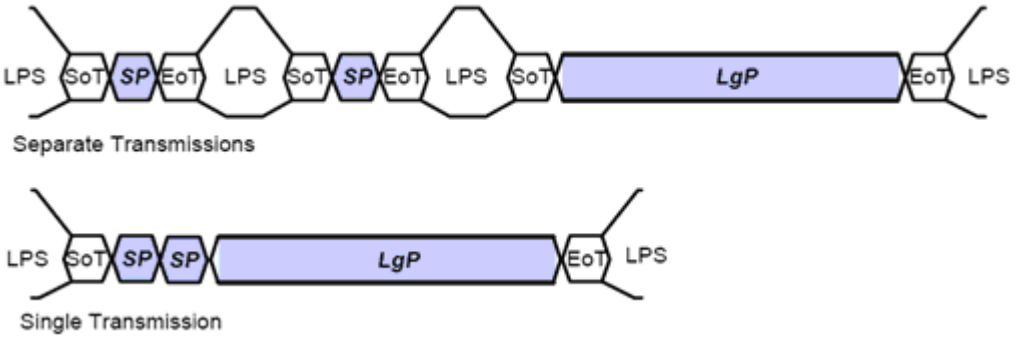
7. MIPI Interface

HX8252-A supports Left/Right type MIPI interface with 2+2 and 4+4 Lanes.



7.1 DSI protocol

The protocol layer appends packet-protocol information and headers. The receiver side of a DSI Link performs the converse of the transmitter side, decomposing the packet into parallel data, signal events and commands. The DSI protocol permits multiple packets which is useful for events such as peripheral initialization, where many registers may be loaded separate write commands at system startup. Figure 7.3 illustrates multiple HS Transmission packets.



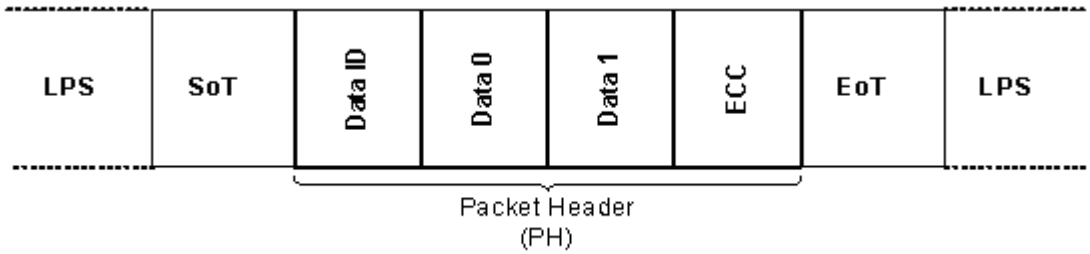
Note: (1) a. LPS: Low power state
b. SoT: Start of Transmission
c. SP: Short Packet
d. LgP: Long Packet
e. EoT: End of Transmission

Figure 7.1: Multiple packets transmission

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The packet includes two types which are Long packet and Short packet. The first byte of the packet, the Data Identifier **(DI)**, includes information specifying the type of the packet. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

Short packets are four bytes in length including the ECC. Short packet is used for most Command Mode commands and associated parameters. Where Short packets format include an 8-bit Data ID followed by two command or data bytes and an 8-bit ECC. Figure 7.4 shows the structure of the Short packet.



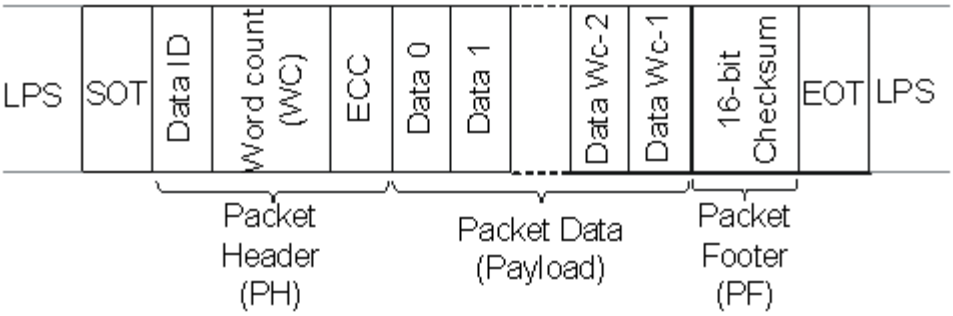
Note: (1) a. DI **(Data ID)**: Contain Virtual Channel Identifier and Data Type.
b. ECC **(Error Correction Code)**: The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Short Packet.

Figure 7.2: Structure of the short packet

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Long packets specify the payload length using a two-byte Word Count field and then the payload maybe from 0 to 65,535 bytes in length. Thus Long packets permit transmission of large blocks of pixel or other data.. Figure 7.5 shows the structure of the Long packet. Long Packet Header composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. An application-specific Data Payload has Word Count * bytes following the Packet Header. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.

Where 65,541 bytes = 4 bytes PH + (2^{16}-1) bytes Payload + 2 bytes PF



Note: (1) a. DI **(Data ID)**: Contain Virtual Channel Identifier and Data Type.
b. WC **(Word Count)**: The receiver uses WC to determine the packet end.
c. ECC **(Error Correction Code)**: The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.
d. PF **(Packet Footer)**: Mean 16-bit Checksum.

Figure 7.3: Structure of the long packet

According to packet form, basic elements include DI and ECC. Table 7.1 shows format of Data ID.

DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
VC **(Virtual Channel)**		DT **(Data Type)**					

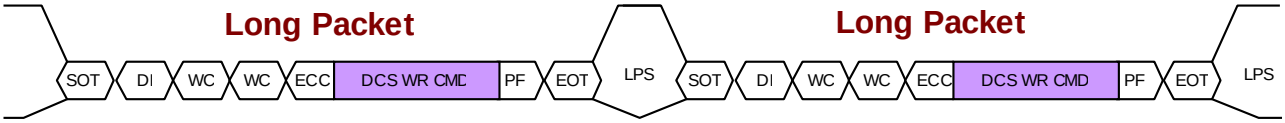
Note: (1) a. DI[7:6]: These two bits identify the data as directed to one of four virtual channels.
b. DI[5:0]: These six bits specify the Data Type, which specifies the size, format and, in some cases, the interpretation of the packet contents.

Figure 7.4: The format of data ID

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Due to Data Type **(DT)** mean format of transmission type, following figure Short- / Long-packet transmission command sequence.

Long packet writes Command / Parameters / Pixel Data



Short packet writes Command / Parameters

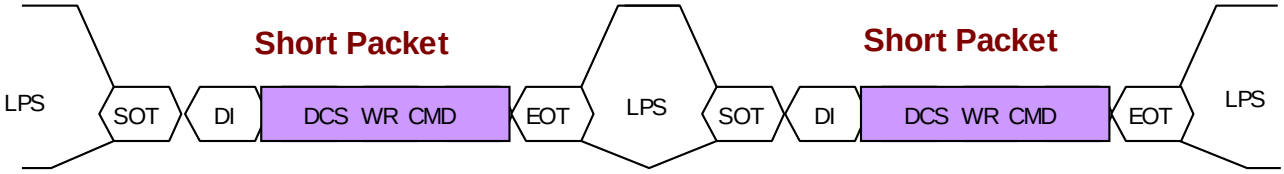
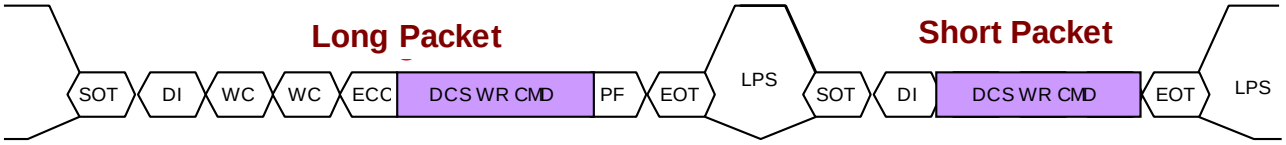
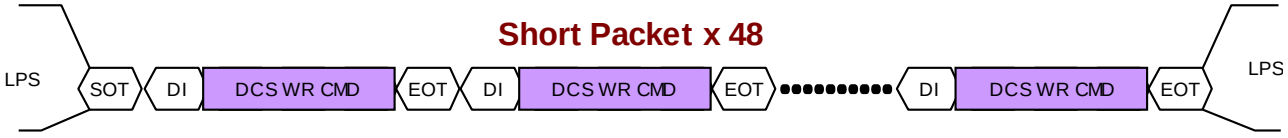


Figure 7.5: Show short- / long-packet transmission command sequence

Long packet and short packet to access Command



Short packets with brust write
Don't send more 48 commands in one HS.



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7.1.1 Processor to peripheral (Forward direction) packets data types

The set of transaction types sent from the host processor to a peripheral, such as a display module, are shown in Table 7.2 Data Types for Processor-sourced Packets.

Data type, hex	Data type, binary	Description packet	Size
01h	00 0001	Sync event, V Sync start.	Short
11h	01 0001	Sync event, V Sync end.	Short
21h	10 0001	Sync event, H Sync start.	Short
31h	11 0001	Sync event, H Sync end.	Short
08h	00 1000	End of transmission packet **(EoTp)**.	Short
02h	00 0010	Color Mode **(CM)** off command.	Short
12h	01 0010	Color Mode **(CM)** on command.	Short
22h	10 0010	Shut down peripheral command.	Short
32h	11 0010	Turn on peripheral command.	Short
03h	00 0011	Generic short WRITE, no parameter.	Short
13h	01 0011	Generic short WRITE, 1 parameter.	Short
23h	10 0011	Generic short WRITE, 2 parameter.	Short
04h	00 0100	Generic READ, no parameter.	Short
14h	01 0100	Generic READ, 1 parameter.	Short
24h	10 0100	Generic READ, 2 parameters.	Short
05h	00 0101	DCS short WRITE, no parameter.	Short
15h	01 0101	DCS short WRITE, 1 parameter.	Short
06h	00 0110	DCS READ, no parameters.	Short
37h	11 0111	Set maximum return packet size.	Short
09h	00 1001	Null packet, no data.	Long
19h	01 1001	Blanking packet, no data.	Long
29h	10 1001	Generic long write.	Long
39h	11 1001	DCS long write/write LUT command packet.	Long
0Eh	00 1110	Packed pixel stream, 16-bit RGB, 5-6-5 format.	Long
1Eh	01 1110	Packed pixel stream, 18-bit RGB, 6-6-6 format.	Long
2Eh	10 1110	Loosely packed pixel stream, 18-bit RGB, 6-6-6 format.	Long
3Eh	11 1110	Packed pixel stream, 24-bit RGB, 8-8-8 format.	Long
X0h and XFh, unspecified	xx 0000 xx 1111	DO NOT USE. All unspecified codes are reserved.	-

Table 7.1: Data types for processor-sourced packets

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Under tables list all detail function of all data types.

Sync event (H start, H end, V start, V end), data type=xx 0001 (x1h)		
Data type, hex	**Function description**	**Number of bytes**
01h	V Sync start, start of VSA pulse.	4 bytes **(DI + 00h + 00h + ECC)**
11h	V Sync end, end of VSA pulse.	
21h	H Sync start, start of HSA pulse.	
31h	H Sync end, end of HSA pulse.	

Note: (1) V Sync start and V Sync end event represents the start and end of the VSA, respectively. Similarly H Sync start and H Sync end event represents the start and end of the HSA, respectively.

End of transmission packet (EoTp)		
Data type, hex	**Function description**	**Number of bytes**
08h	End of transmission packet **(EoTp)**.	4 bytes **(DI + 00h + 00h + ECC)**

Color mode status (Color mode on, color mode off)		
Data type, hex	**Function description**	**Number of bytes**
02h	Color mode on that switches a video mode display module to a low-color mode for power saving.	4 bytes **(DI + 00h + 00h + ECC)**
12h	Color mode off that switches a video mode display module from low-color display to normal display.	

Display status (Shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
22h	Shutdown peripheral command that turns off the display in a video mode display for power saving.	4 bytes **(DI + 00h + 00h + ECC)**
32h	Turn on peripheral command that turns on the display in video mode display for normal display.	

Note: (1) When use shutdown command; interface shall remain powered in order to receive the turn-on, or wake-up, command.

Generic short WRITE packet with 0,1,2 parameter		
Data type, hex	**Function description**	**Number of bytes**
03h	Generic short WRITE, no parameter.	**(DI + 00h + 00h + ECC)**
13h	Generic short WRITE, 1 parameter.	**(DI + P1 + 00h + ECC)**
23h	Generic short WRITE, 2 parameters. **(P1=Addr, P2=Data)**	**(DI + P1 + P2 + ECC)**

Note: (1) P1=Parameter1, P2=Parameter2.

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Generic READ request with 0,1,2 parameter		
Data type, hex	**Function description**	**Number of bytes**
04h	Generic READ, no parameter.	**(DI + 00h + 00h + ECC)**
14h	Generic READ, 1 parameter. **(P1=Addr)** only read 1 Data.	**(DI + P1 + 00h + ECC)**
24h	Generic READ, 2 parameters. **(P1=Addr, P2=Burst read length)**	**(DI + P1 + P2 + ECC)**

Note: (1) P1=parameter1, P2=parameter2.

DCS show WRITE command with 0,1 parameter		
Data type, hex	**Function description**	**Number of bytes**
05h	DCS short WRITE, no parameter.	**(DI + DCS + 00h + ECC)**
15h	DCS short WRITE, 1 parameter. **(P1=DCS's data)**	**(DI + DCS + P1 + ECC)**

Note: (1) P1=Parameter1, DCS=DCS command.

DCS command setting		
Data type, hex	**Function description**	**Number of bytes**
06h	DCS Read command, the returned data may be of Short or Long packet format.	4 bytes **(DI + DCS CMD.+00h + ECC)**
39h	DCS Long Write/ Write _ LUT Command is used to send larger blocks of data to a display module that implements the Display Command Set.	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA(WC-1) + PF)**

Note: (1) For write part, If DCS Short Write command is followed by BTA, the peripheral shall respond with ACK when no error was detected in the transmission **(Host → Slave)**. Unless an error was detected, the peripheral shall respond with Acknowledge with Error Report.
(2) When use DCS Read Command, the Set Max Return Packet Size command will limit the size of returning packets.
(3) The peripheral shall respond to DCS Read Command Request in one of the following ways:
- If an error was detected and corrected in Packet Header field by the peripheral, it shall send *Acknowledge with Error Report*. So the peripheral shall transmit the requested READ data packet with suitable ECC in the same transmission.
- If no error was detected by the peripheral, it shall send the requested READ packet **(Short or Long)** with appropriate ECC and Checksum, if either or both features are enabled.

(4) One byte ≦ Length of payload DATA ≦ 2^{16}-1

Generic long write		
Data type, hex	**Function description**	**Number of bytes**
29h	Generic long write packek is used to transmit arbitrary blocks of data from a host processor to peripheral in a Long packet. Support burst write : Parameter_1 → MCS Addr **(p1>B0)** Parameter_2 → Address's data Parameter_3 → Address+1 's data Parameter_4 → Address+2 's data : Parameter_N → Address+N-1 's data	Up to 65535 bytes **(DI + WC + ECC + Payload DATA(WC) + PF)**

Note: (1) For write part, If Short Write command is followed by BTA, the peripheral shall respond with ACK when no error was detected in the transmission **(Host → Slave)**. Unless an error was detected, the peripheral shall respond with Acknowledge with Error Report.
(2) When use Read Command, the Set Max Return Packet Size command will limit the size of returning packets.
(3) The peripheral shall respond to Read Command Request in one of the following ways:
- If an error was detected and corrected in Packet Header field by the peripheral, it shall send *Acknowledge with Error Report*. So the peripheral shall transmit the requested READ data packet with suitable ECC in the same transmission.
- If no error was detected by the peripheral, it shall send the requested READ packet **(Short or Long)** with appropriate ECC and Checksum, if either or both features are enabled.

(4) One byte <= Length of payload DATA <= 2^{16}-1

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Return packet size setting		
Data type, hex	**Function description**	**Number of bytes**
37h	Set Maximum Return Packet Size that specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.	4 bytes **(DI + Maximum Return Packet Size + ECC)**

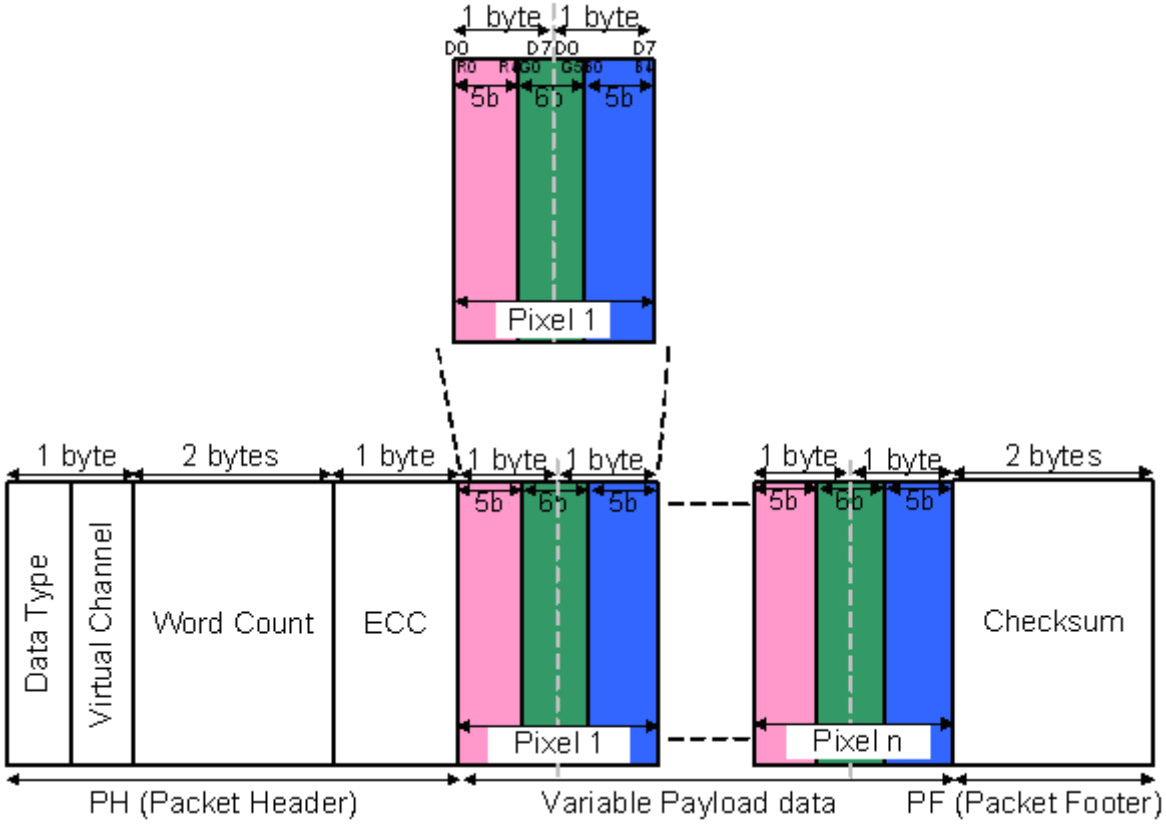
Note: (1) The two-byte value is transmitted with LS byte first. And during a power-on or Reset sequence, the Maximum Return Packet Size shall be set by the peripheral to a default value of one.

Variable data packet		
Data type, hex	**Function description**	**Number of bytes**
09h	Null Packet is a mechanism for keeping the serial Data Lane**(s)** in High-Speed mode while sending dummy data.	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**
19h	Blanking packet is used to convey blanking timing information in a Long packet.	

Note: (1) When Null Packet, the Payload Data belong "null" Data, actual data values sent are irrelevant because the peripheral does not capture or store the data.
(2) When Blanking packet, the packet represents a period between active scan lines of a Video Mode display.

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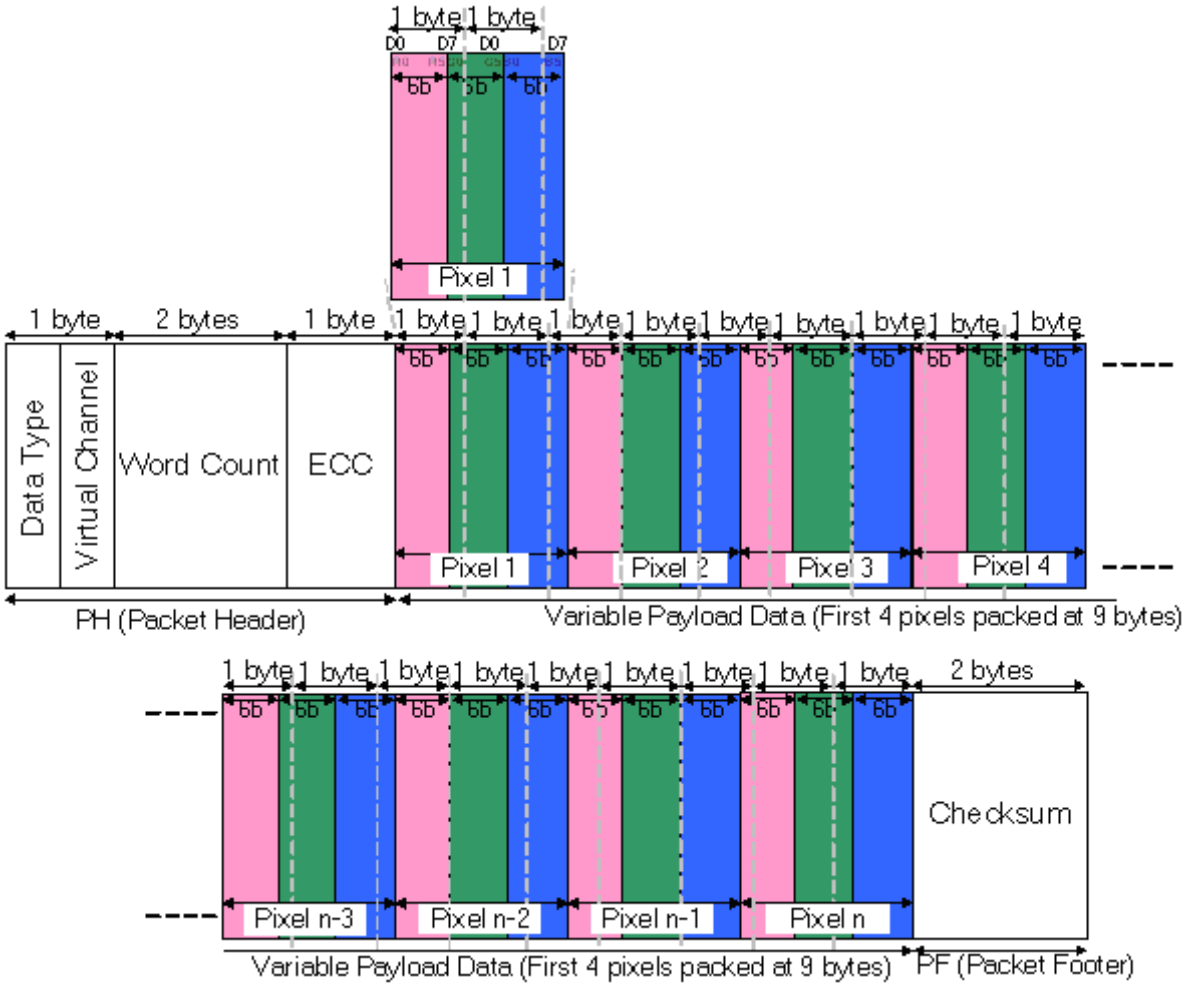
Data stream format – 16-bit format		
Data type, hex	**Function description**	**Number of bytes**
0Eh	Packed Pixel Stream 16-Bit Format is used to transmit image data formatted as 16-bit pixels to a Video Mode display module. Pixel format is "**(5 bits)** red, **(6 bits)** green and **(5 bits)** blue".	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: (1) Within a color component, the "LSB is sent first, the MSB last ".

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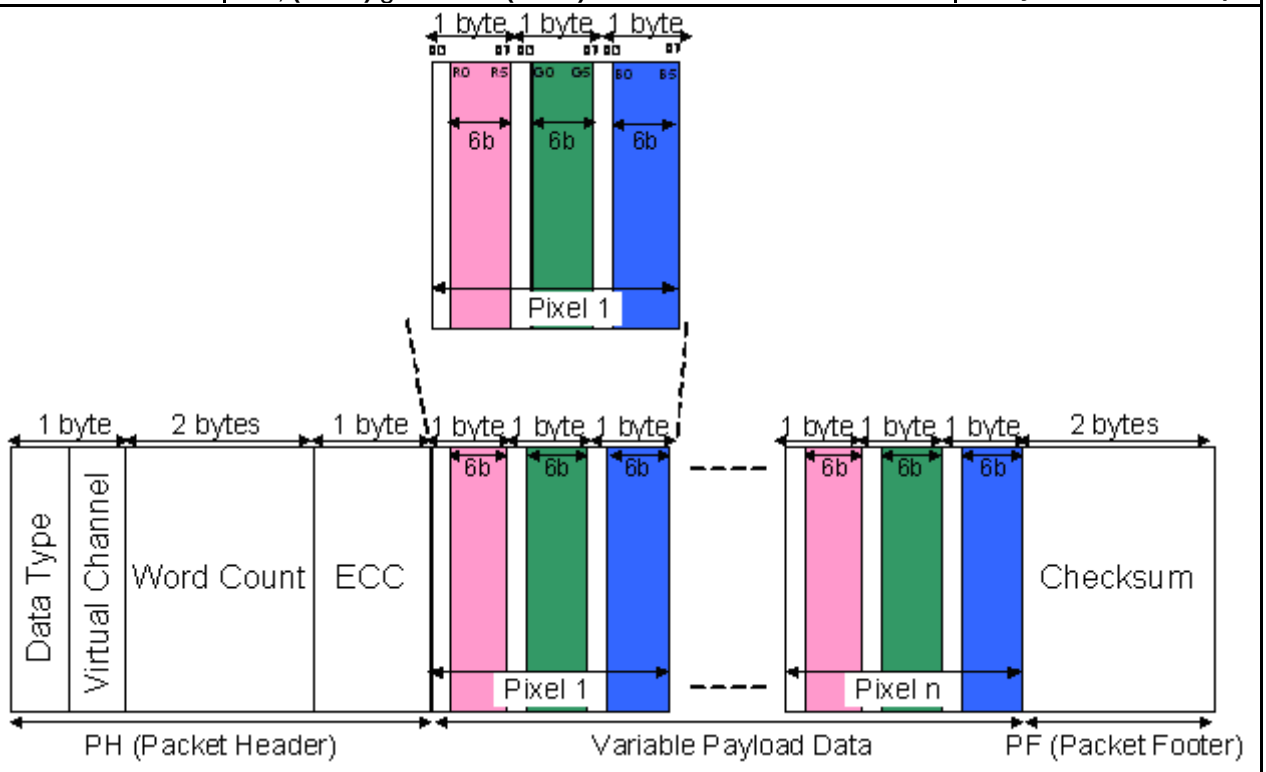
Data stream format – 18-bit format		
Data type, hex	**Function description**	**Number of bytes**
1Eh	Packed Pixel Stream 18-Bit Format is used to transmit image data formatted as 18-bit pixels to a Video Mode display module. Pixel format is "**(6 bits)** red, **(6 bits)** green and **(6 bits)** blue".	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: (1) Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up w ith byte boundaries every four pixels **(Nine bytes)**. Preferably, display modules employing this format have a horizontal extent **(Width in pixels)** evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line w idth that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a "clean start" for the next line.

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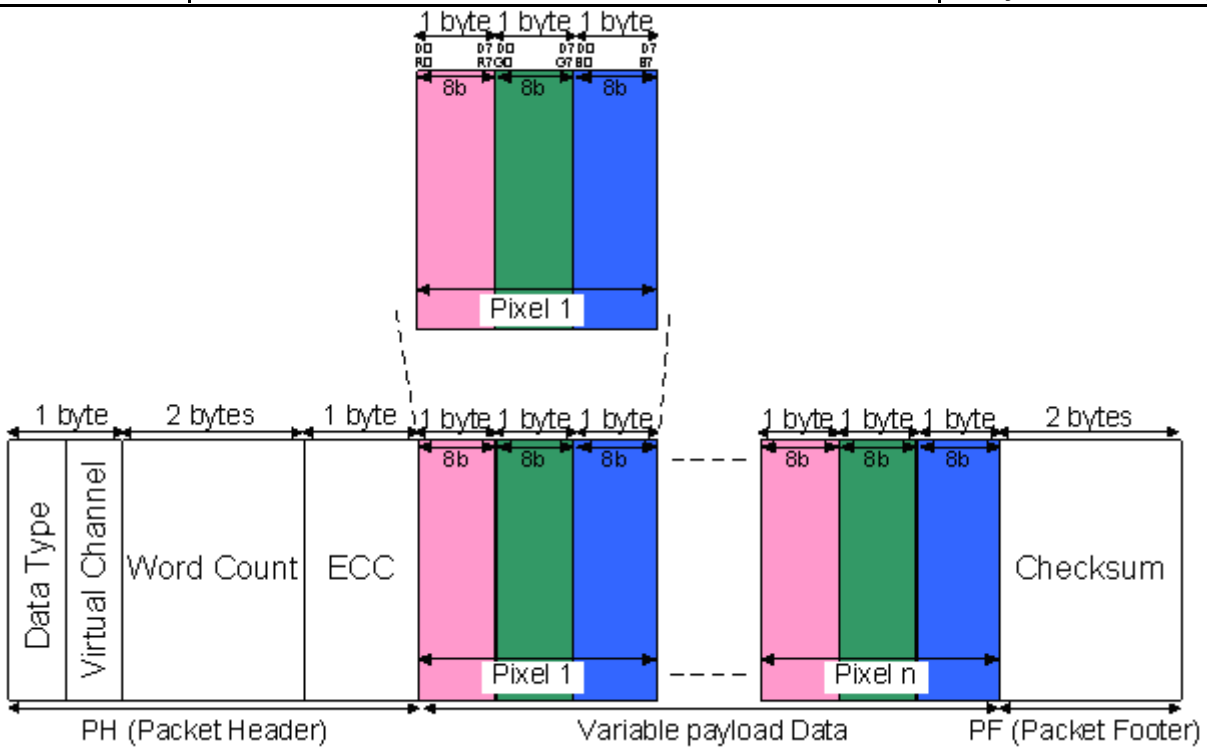
Data stream format – 18-bit format		
Data type, hex	**Function description**	**Number of bytes**
2Eh	Packed Pixel Stream 18-Bit Format, each R, G, or B color component is one byte form, but the valid pixel bits occupy bits [7:2] and bits [1:0] of are ignored. Pixel format is "**(6 bits)** red, **(6 bits)** green and **(6 bits)** blue".	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: (1) Within a color component, the LSB is sent first, the MSB last. With this format, pixel boundaries line up with byte boundaries every three bytes.

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Data stream format – 24-bit format		
Data type, Hex	**Function Description**	**Number of bytes**
3Eh	Packed Pixel Stream 24-Bit Format is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. Pixel format is **(8 bits)** red, **(8 bits)** green and **(8 bits)** blue.	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: (1) Within a color component, the LSB is sent first, the MSB last. With this format, pixel boundaries line up with byte boundaries every three bytes.

7.1.2 Peripheral to processor (Reverse direction) packet data type

Table 7.3 present the complete set of peripheral-to-processor Data Types

Data type, hex	Data type, binary	Description packet	Size
0x02	00 0010	Acknowledge and error report.	Short
0x08	00 1000	End of transmission packet **(EoTp)**.	Short
0x11	01 0001	Generic short read response, 1 byte returned.	Short
0x12	01 0010	Generic short read response, 2 byte returned.	Short
0x1A	01 1010	Generic long read response.	Long
0x1C	01 1100	DCS long read response.	Long
0x21	10 0001	DCS short read response, 1 byte returned.	Short
0x22	10 0010	DCS short read response, 2 byte returned.	Short

Table 7.2: Data types for peripheral-sourced packets

Acknowledge types		
Data type, hex	**Function description**	**Number of bytes**
02h	Get Acknowledge with Error report when Error occurs from processor transmission.	4 bytes **(DI + bit0 ~ bit15 + ECC)**

Note: (1) When processor transmits complete Payload, following signal by BTA, peripheral must respond to processor. With error➔Acknowledge with error report **(Short packet)**, without error➔ request READ data or Acknowledge **(Trigger message)**.

Bit	Description
0	SoT error.
1	SoT Sync error.
2	EoT Sync error.
3	Escape mode entry command error.
4	Low-power transmit Sync error.
5	Peripheral timeout error.
6	False control error.
7	Contention detected.
8	ECC error, single-bit **(Detected and corrected)**.
9	ECC error, multi-bit **(Detected, not corrected)**.
10	Checksum error **(Long packet only)**.
11	DSI data type not recognized.
12	DSI VC ID invalid.
13	Invalid transmission length.
14	Reserved.
15	DSI protocol violation.

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Generic short read response (1 byte returned)		
Data type, hex	**Function description**	**Number of bytes**
11h	This is the short-packet to generic read request. **(1 byte returned)**.	4 bytes **(DI + R1 + 00h + ECC)**

Note: (1) R1=Returned byte 1.

Generic short read response (1 byte returned)		
Data type, hex	**Function description**	**Number of bytes**
12h	This is the short-packet to generic read request. **(1 byte returned)**.	4 bytes **(DI + R1 + R2 + ECC)**

Note: (1) R1=Returned byte 1, R2=Returned byte 2.

Generic long read response		
Data type, hex	**Function description**	**Number of bytes**
1Ah	This is the long-packet response to generic long read request.	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**

Note: (1) If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h.
If the command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the Acknowledge with Error Report packet shall be sent.

DCS read response		
Data type, hex	**Function description**	**Number of bytes**
1Ch	This is the long-packet response to DCS long read request.	Up to 65535 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**

Note: (1) If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h.
If the DCS command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the Acknowledge with Error Report packet shall be sent.

DCS short read response (1 byte returned)		
Data type, hex	**Function description**	**Number of bytes**
21h	This is the short-packet to DCS Read Request. **(1 byte returned)**.	4 bytes **(DI + R1 + 00h + ECC)**

Note: (1) R1=Returned byte 1.

DCS short read response (2 byte returned)		
Data type, hex	**Function description**	**Number of bytes**
22h	This is the short-packet to DCS read request. **(2 byte returned)**.	4 bytes **(DI + R1 + R2 + ECC)**

Note: (1) R1=Returned byte 1, R2=Returned byte 2.

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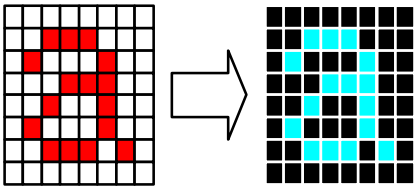
7.2 Display command set (DCS)

Command	Hex code	Description	Number of parameters
Enter invert mode	21h	Displayed image colors are inverted.	0
Enter sleep mode	10h	Power for the display panel is off.	0
Exit invert mode	20h	Displayed image colors are not inverted.	0
Exit sleep mode	11h	Power for the display panel is on.	0
Get address mode	0Bh	Get data order for transfers from the Host to the display module and from the frame memory to the display device.	1
Get display mode	0Dh	Get the current display mode from the peripheral.	1
Get pixel format	0Ch	Get the current pixel format.	1
Get power mode	0Ah	Get the current power mode.	1
Get signal mode	0Eh	Get display module signaling mode.	1
Nop	00h	No operation.	0
Read DDB continue	A8h	Continue reading the DDB from the last read location.	Variable
Read DDB start	A1h	Read the DDB from the provided location.	Variable
Set address mode	36h	Set the data order for transfers from the host to the display module and from the frame memory to the display device.	1
Set display off	28h	Blanks the display device.	0
Set display on	29h	Show the image on the display device.	0
Set pixel format	3Ah	Defines how many bits per pixel are used in the interface.	1
Set tear off	34h	Synchronization information is not sent from the display module to the host processor.	0
Set tear on	35h	Synchronization information is sent from the display module to the host processor at the start of VFP.	1
Set tear scan line	44h	Synchronization information is sent from the display module to the host processor when the display device refresh reaches the provided scan line.	2
Soft reset	01h	Software Reset.	0
Enter idle mode	39h	Reduced color depth is used on the display panel.	0
Exit idle mode	38h	Full color depth is used on the display panel.	0
Get diagnostic result	0Fh	Get Peripheral Self-Diagnostic Reslut.	1
RDNUMED	05h	RDNUMED **(Read number of the errors on DSI)**.	1

Table 7.3: DCS command list

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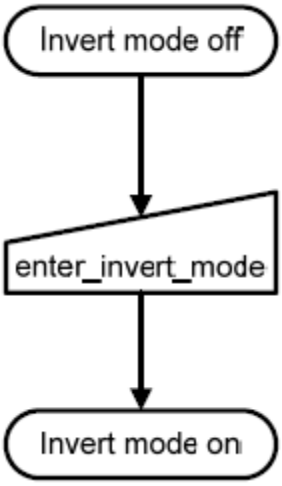
7.2.1 Enter_invert_mode (21h)

21H	Enter_invert_mode (Display inversion on)→INVON									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	0	0	0	0	1	21
Parameter	No parameter.									
Description	This command is used to enter into display inversion mode. This command makes no change of contents of frame memory. Every bit is inverted from the frame memory to the display. This command does not change any other status. (Example) Host display 									
Restriction	This command has no effect when module is already in inversion on mode.									

Register Availability	
Status	**Availability**
Sleep Out	Yes
Sleep In	Yes

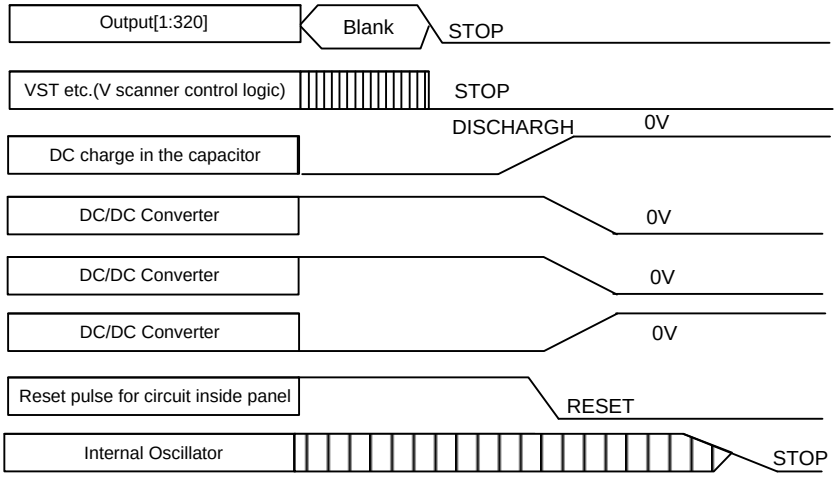
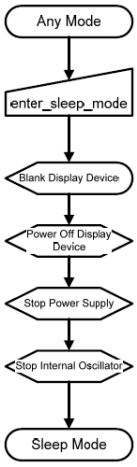
Default	
Status	**Default value**
Power On Sequence	Display Inversion Off
S/W Reset	Display Inversion Off
H/W Reset	Display Inversion Off

Flow Chart



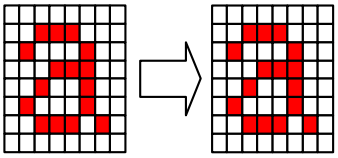
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7.2.2 Enter_sleep_mode (10h)

10H	Enter_sleep_mode(Sleep In)→SLPIN									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	1	0	0	0	0	10
Parameter	No parameter.									
Description	This command causes the LCD module to enter the minimum power consumption mode. In this mode the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped.  MCU interface and memory are still working and the memory keeps its contents.									
Restriction	This command has no effect when module is already in sleep in mode. Sleep In Mode can only be left by the Sleep Out Command **(11h)**. It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command **(When in Sleep In Mode)** before Sleep In command can be sent.									
Register Availability	**Status** / **Availability** Sleep Out / Yes Sleep In / Yes									
Default	**Status** / **Default value** Power On Sequence / Sleep In Mode S/W Reset / Sleep In Mode H/W Reset / Sleep In Mode									
Flow Chart	It takes 120msec to get into Sleep In mode after SLPIN command issued.  									

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7.2.3 Exit_invert_mode (20h)

20H	Exit_invert_mode (Display inversion off)→INVOFF									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	0	0	0	0	0	20
Parameter	No parameter.									
Description	This command is used to recover from display inversion mode. This command makes no change of contents of frame memory. This command does not change any other status. (Example) Host display 									
Restriction	This command has no effect when module is already in inversion off mode.									

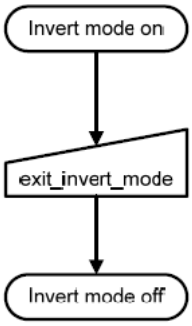
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

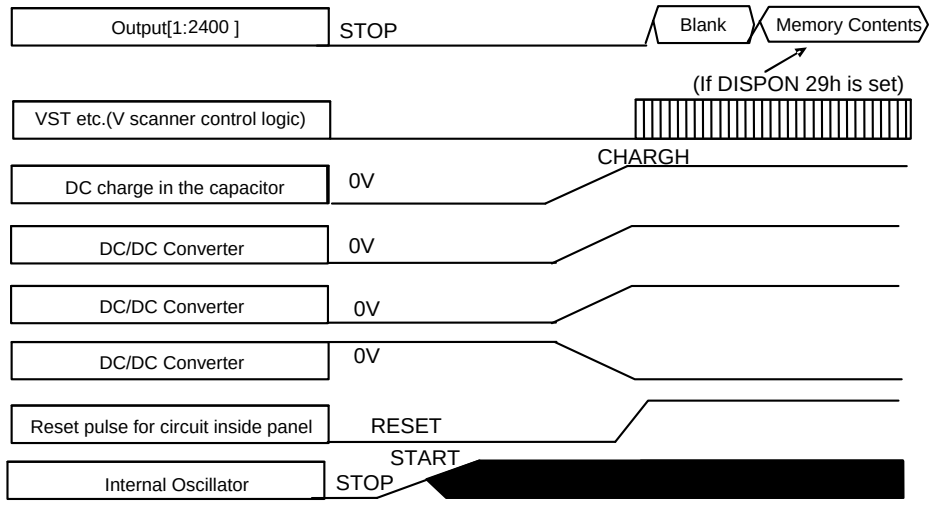
Status	Default value
Power On Sequence	Display Inversion Off
S/W Reset	Display Inversion Off
H/W Reset	Display Inversion Off

Flow Chart



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7.2.4 Exit_sleep_mode (11h)

11H	Exit_sleep_mode (Sleep Out)→SLPOUT									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	1	0	0	0	1	11
Parameter	No parameter.									
Description	This command turns off sleep mode. In this mode the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started. 									
Restriction	This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command **(10h)**. It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. The display module loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode. The display module is doing self-diagnostic functions during this 5msec. It will be necessary to wait 120msec after sending Sleep In command **(when in Sleep Out mode)** before Sleep Out command can be sent.									

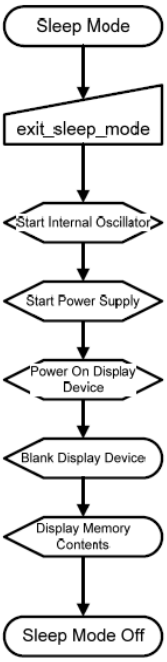
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	Sleep In Mode
S/W Reset	Sleep In Mode
H/W Reset	Sleep In Mode

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Flow Chart	It takes 120msec to become Sleep Out mode after SLPOUT command issued. 

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7.2.5 Get_address _mode (0Bh)

0BH	Get_address_mode (Read display MADCTL)→ RDDMADCTL									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	1	0	1	1	0B
1st parameter	D→H	D7	D6	0	0	D3	0	0	0	xx

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Not defined	-
D6	Not defined	-
D5	Not defined	Set to '0'
D4	Not defined	Set to '0'
D3	RGB/BGR order	-
D2	Not defined	Set to '0'
D1	Flip horizontal	Set to '0'
D0	Flip vertical	Set to '0'

Bit D3 – RGB/BGR Order
'0'=RGB **(When MADCTL B3='0')**.
'1'=BGR **(When MADCTL B3='1')**.
Bit D1 – Flip Horizontal
'0'=Display from Left to Right.
'1'=Display from Right to Left.
Bit D0 – Flip Vertical
'0'=Display from Top to Bottom.
'1'=Display from Bottom to Top.

Restrictions

-

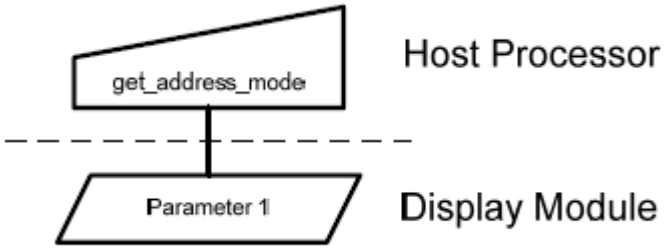
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

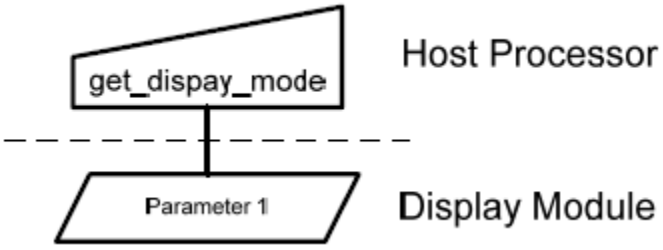
Status	Default value
Power On Sequence	00h
S/W Reset	No change
H/W Reset	00h

Flow Chart



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7.2.6 Get_display _mode (0Dh)

0DH	Get_display_mode (Read display image mode)→ RDDIM									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	1	1	0	1	0D
1st parameter	D→H	0	0	D5	0	0	0	0	0	xx
Description	This command indicates the current status of the display as described in the table below: **Bit D5 – Inversion On/Off** '0'=Inversion is Off. '1'=Inversion is On. **Bit D7,D6,D4,D3,D2,D1,D0- Not Defined** Set to '0'									
Restrictions	-									
Register Availability	**Status** / **Availability** Sleep Out / Yes Sleep In / Yes									
Default	**Status** / **Default value** Power On Sequence / 00h S/W Reset / No change H/W Reset / 00h									
Flow Chart										

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7.2.7 Get_pixel_format (0Ch)

0CH	Get_pixel_format (Read display COLMOD)→RDDCOLMOD									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	1	1	0	0	0C
1st parameter	D→H	0	D6	D5	D4	0	0	0	0	xx

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Reserved	Set to '0'
D6	DPI Interface Pixel format	-
D5		-
D4		-
D3	Reserved	Set to '0'
D2	DBI Interface Pixel format→Not defined	Set to '0'
D1		Set to '0'
D0		Set to '0'

Bits D6, D5, D4 – DPI Pixel Format Definition
Bits D2, D1, D0 – DBI Pixel Format Definition→Not defined.

Interface Color Format	D6	D5	D4
Not defined	0	0	0
Not defined	0	0	1
Not defined	0	1	0
Not defined	0	1	1
Not defined	1	0	0
16-bit/pixel	1	0	1
18-bit/pixel	1	1	0
24-bit/pixel	1	1	1

If a particular interface, either DSI or DPI, is not used then the corresponding bits in the parameter returned from the display module are undefined.

Restrictions: -

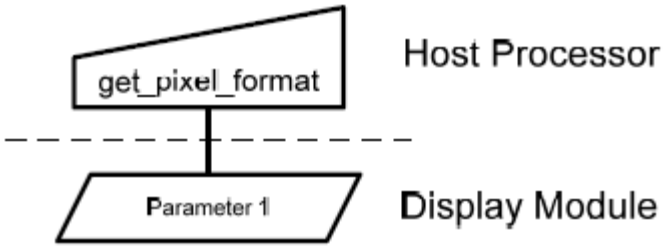
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	24-bit/pixel
S/W Reset	24-bit/pixel
H/W Reset	24-bit/pixel

Flow Chart



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7.2.8 Get_power_mode (0Ah)

0AH	Get_power_mode (Read display power mode)→RDDPM									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	1	0	1	0	0A
1st parameter	D→H	0	0	0	D4	0	D2	0	0	xx

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Not defined	Set to '0'
D6	Not defined	Set to '0'
D5	Not defined	Set to '0'
D4	Sleep In/Out	-
D3	Not defined	Set to '0'
D2	Display on/off	-
D1	Not defined	Set to '0'
D0	Not defined	Set to '0'

Bit D4 – Sleep In/Out
'0'=Sleep In Mode.
'1'=Sleep Out Mode.
Bit D2 – Display On/Off
'0'=Display is Off.
'1'=Display is On.
Bit D7,D6,D5,D3,D1,D0 – Not defined
Set to '0'.

Restrictions

-

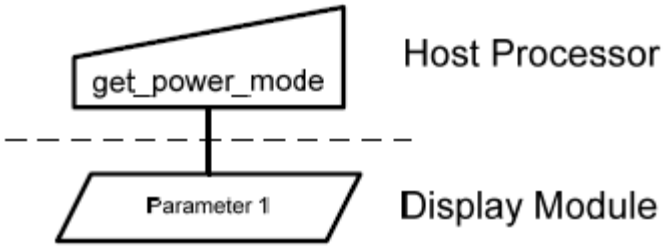
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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7.2.9 Get_signal_mode (0Eh)

0EH	Get_signal_mode (Read display signal mode)→RDDSM									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	1	1	1	0	0E
1st parameter	D→H	D7	D6	0	0	0	0	0	D0	xx
Description	This command indicates the current status of the display as described in the table below: **Bit D7 – Tearing Effect Line On/Off** '0'=Tearing Effect Line Off. '1'=Tearing Effect On. **Bit D6 – Tearing Effect Line Output Mode** See section set_tear_on **(35h)** for mode definitions. '0'=Mode 0. **(M=0)** '1'=Mode 1. **(M=1)** **Bit [D5:D1]** –reserved and set to '0'. **Bit D0 – Error report bit** '0'=No Error happened. '1'=Error happened. **(After BTA ,Bit D0 will set to '0')**									
Restrictions	-									

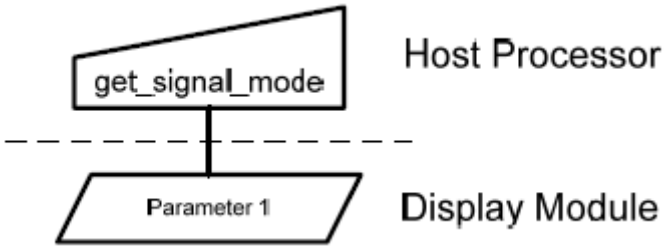
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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7.2.10 Nop: No operation (00h)

00H	NOP (No operation)									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	0	0	0	0	00
Parameter	NO PARAMETER									
Description	This command is an empty command; it does not have any effect on the display module. However it can be used to terminate Frame Memory Write as described in RAMWR **(Memory Write)** or RAMRD **(Memory Read)** command.									
Restriction	-									
Register Availability										

Status	Availability
Sleep Out	Yes
Sleep In	Yes

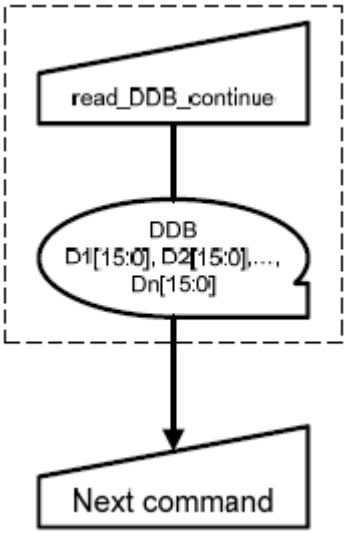
Default

Status	Default value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart -

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7.2.11 Read_DDB_continue (A8h)

<table>
<tr><th rowspan="2">A8H</th><th colspan="10">Read_DDB_continue</th></tr>
<tr><th>Direction</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>H→D</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>A8</td></tr>
<tr><td>1st parameter</td><td>D→H</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>xx</td></tr>
<tr><td>2nd parameter</td><td>D→H</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>xx</td></tr>
<tr><td>:</td><td>D→H</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>xx</td></tr>
<tr><td>Nth parameter</td><td>D→H</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="10">A read_DDB_start command should be executed at least once before a read_DDB_continue command to define the read location. Otherwise, data read with a read_DDB_continue command is undefined.</td></tr>
<tr><td>Restrictions</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default value</th></tr><tr><td>Power On Sequence</td><td>Read PA 1st~6th is the same as 00h , and the 7th read is FFh.</td></tr><tr><td>S/W Reset</td><td>Read PA 1st~6th is the same as 00h , and the 7th read is FFh.</td></tr><tr><td>H/W Reset</td><td>Read PA 1st~6th is the same as 00h , and the 7th read is FFh.</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"> 
</td></tr>
</table>

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7.2.12 Read_DDB_start (A1h)

A1H	Read_DDB_start									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	1	0	1	0	0	0	0	1	A1
1st parameter	D→H	x	x	x	x	x	x	x	X	xx
2nd parameter	D→H	x	x	x	x	x	x	x	x	xx
3rd parameter	D→H	x	x	x	x	x	x	x	x	xx
4th parameter	D→H	x	x	x	x	x	x	x	x	xx
:	D→H	x	x	x	x	x	x	x	x	xx
7th parameter	D→H	1	1	1	1	1	1	1	1	FF

Description

The format of returned data is as follows:
Parameter 1: LS **(Least significant)** byte of Supplier ID. Supplier ID is a unique value assigned to each peripheral supplier by the MIPI organization.
Parameter 2: MS **(Most significant)** byte of Supplier ID.
Parameter 3: LS **(Least significant)** byte of Supplier Elective Data. This is a byte of information that is determined by the supplier. It could include model number or revision information, for example.
Parameter 4: MS **(Most significant)** byte of Supplier Elective Data
Parameter 7: single-byte *Escape or Exit Code* **(EEC)**. The code is interpreted as follows:
- FFh - Exit code – There is no more data in the Descriptor Block
- 00h - Escape code – There is supplier-proprietary data in the Descriptor Block

(Does not conform to any MIPI standard)

Restrictions

-

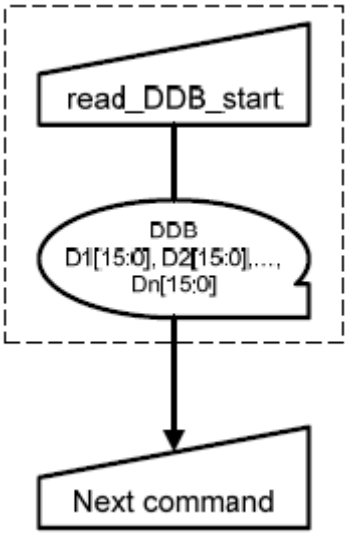
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	PA 1st~6th =00h, PA 7th =FFh
S/W Reset	PA 1st~6th =00h, PA 7th =FFh
H/W Reset	PA 1st~6th =00h, PA 7th =FFh

Flow Chart



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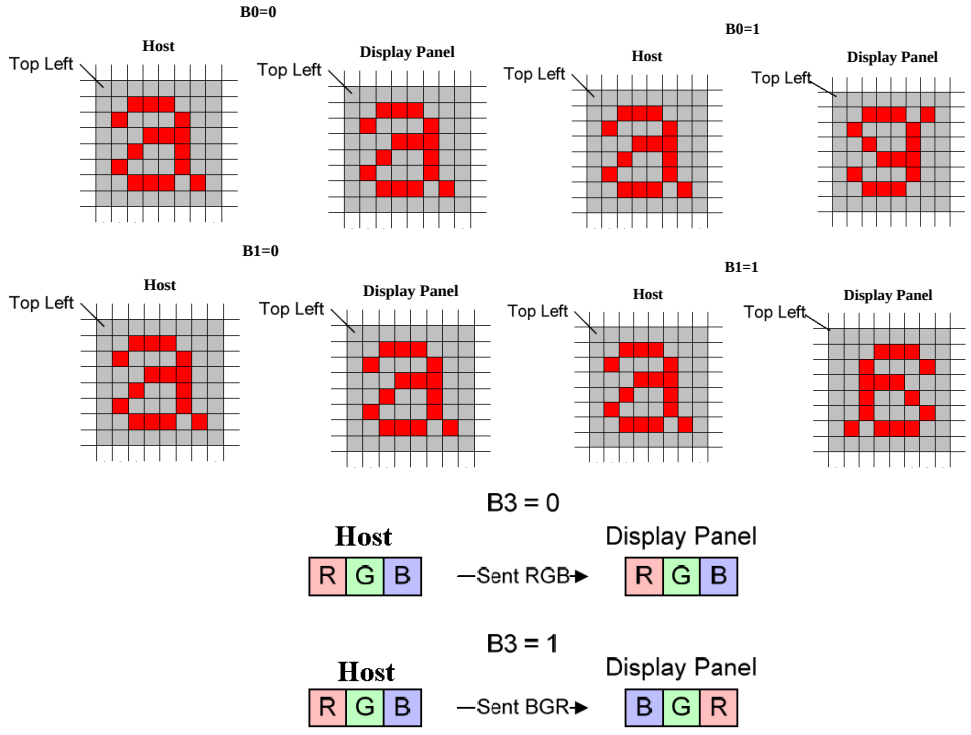
7.2.13 Set_address_mode (36h)

36H	Set_address_mode (Memory access control)→MADCTL									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	1	0	1	1	0	36
1st parameter	H→D	0	0	0	0	BGR	0	Flip V	Flip H	-

Description

This command defines read/write scanning direction of frame memory.
This command makes no change on the other driver status.
Bit Assignment

Bit	Name	Description
B7	-	-
B6	-	-
B5	Not defined	-
B4	Not defined	-
B3	RGB-BGR ORDER **(BGR)**	Color selector switch control. 0=RGB color filter panel. 1=BGR color filter panel.
B2	Not defined	-
B1	Flip Horizontal	Flip Horizontal. 0=Display from Left to Right. 1= Display from Right to Left.
B0	Flip Vertical	Flip Vertical 0=Display from Top to Bottom. 1=Display from Bottom to Top.



Restriction: -

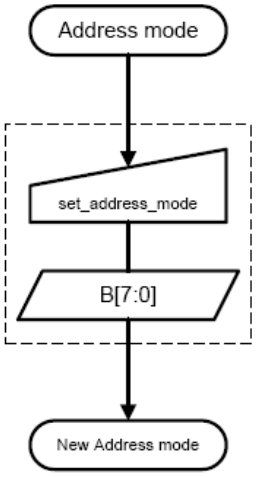
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

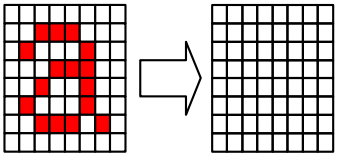
Status	Default value
Power On Sequence	00h
S/W Reset	No change
H/W Reset	00h

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Flow Chart	

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7.2.14 Set_display_off (28h)

28H	Set_display_off (Display off)→DISPOFF									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	0	1	0	0	0	28
Parameter	No parameter.									
Description	This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted. This command makes no change of contents of frame memory. This command does not change any other status. There will be no abnormal visible effect on the display. (Example) Host display 									
Restriction	This command has no effect when module is already in display off mode.									

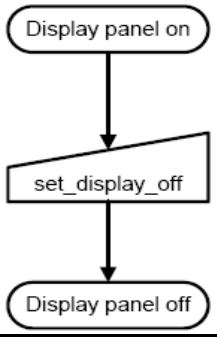
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

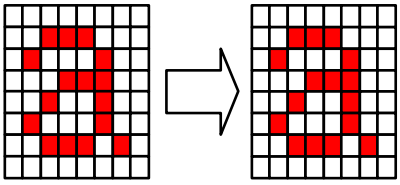
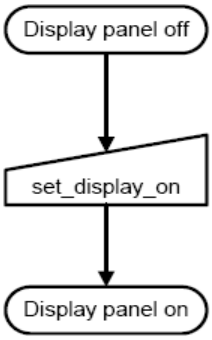
Status	Default value
Power On Sequence	Display Off
S/W Reset	Display Off
H/W Reset	Display Off

Flow Chart



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7.2.15 Set_display_on (29h)

<table>
<tr><td rowspan="2">29H</td><td colspan="10">Set_display_on (Display on)→DISPON</td></tr>
<tr><td>Direction</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>H→D</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>29</td></tr>
<tr><td>Parameter</td><td colspan="10">No parameter.</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to recover from DISPLAY OFF mode. Output from the Frame Memory is enabled.
This command makes no change of contents of frame memory.
This command does not change any other status.
(Example)
memory display
  </td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in display on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default value</th></tr><tr><td>Power On Sequence</td><td>Display Off</td></tr><tr><td>S/W Reset</td><td>Display Off</td></tr><tr><td>H/W Reset</td><td>Display Off</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"> 
</td></tr>
</table>

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7.2.16 Set_pixel_format (3Ah)

3AH	Set_pixel_format (Interface pixel format)→COLMOD									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	1	1	0	1	0	3A
1st parameter	H→D	0	D6	D5	D4	0	0	0	0	-

Description

This command is used to define the format of RGB picture data, which is to be transfer via the system and RGB interface. The formats are shown in the table:

Bit D6,D5,D4 - DPI pixel format definition

DPI interface:

Interface format	D6	D5	D4
Not defined	0	0	0
Not defined	0	0	1
Not defined	0	1	0
Not defined	0	1	1
Not defined	1	0	0
16-bit/pixel	1	0	1
18-bit/pixel	1	1	0
24-bit/pixel	1	1	1

Bit D7,D3,D2,D1,D0 – Not defined

Set to '0'

Restriction

There is no visible effect until the Frame Memory is written to.

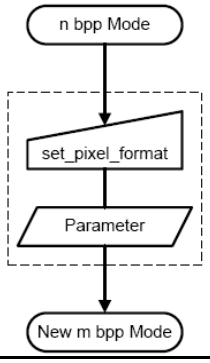
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

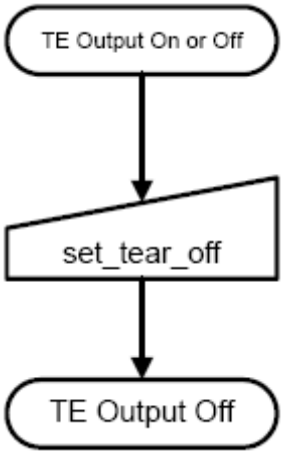
Status	Default value
Power On Sequence	24-bit/pixel
S/W Reset	No Change
H/W Reset	24-bit/pixel

Flow Chart



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7.2.17 Set_tear_off (34h)

<table>
<tr><td rowspan="2">34H</td><td colspan="10">Set_tear_off (Tearing effect line off)→TEOFF</td></tr>
<tr><td>Direction</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>H→D</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>34</td></tr>
<tr><td>Parameter</td><td colspan="10">No parameter.</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to turn OFF the Tearing Effect output signal from the TE signal line.</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when Tearing Effect output is already OFF.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default value</th></tr><tr><td>Power On Sequence</td><td>Tearing Effect Off</td></tr><tr><td>S/W Reset</td><td>Tearing Effect Off</td></tr><tr><td>H/W Reset</td><td>Tearing Effect Off</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"> 
</td></tr>
</table>

7.2.18 Set_tear_on (35h)

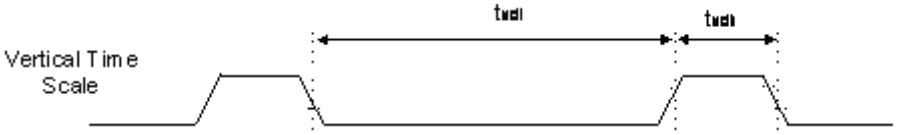
35H	Set_tear_off (Tearing effect line on)→TEON									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	1	0	1	0	1	35
1st parameter	H→D	-	-	-	-	-	-	-	M	-

Description

This command is used to turn ON the Tearing Effect output signal from the TE signal line. The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line. **(X=Don't Care)**.

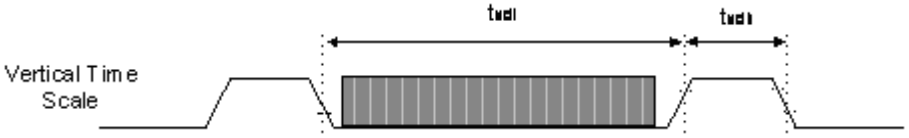
When **M=0 (Mode 0)**:

The Tearing Effect Output line consists of V-Blanking information only:



When **M=1 (Mode 1)** :

The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information:



(During Sleep In mode with tearing effect line on, tearing effect output pin will be active low.)

Restriction

This command has no effect when Tearing Effect output is already ON.

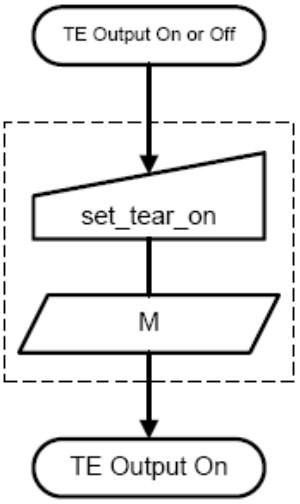
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

Status	Default value
Power On Sequence	Tearing Effect Off
S/W Reset	Tearing Effect Off
H/W Reset	Tearing Effect Off

Flow Chart



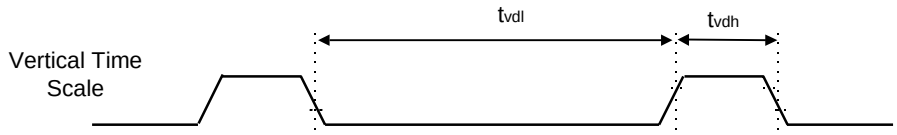
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7.2.19 Set_tear_scanline (44h)

44H	Set_tear_scanline (Tear effect scan lines)→TESL									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	1	0	0	0	1	0	0	44
1st parameter	H→D	N15	N14	N13	N12	N11	N10	N9	N8	00..FF
2nd parameter	H→D	N7	N6	N5	N4	N3	N2	N1	N0	00..FF

Description

This command is turns on the display module's Tearing Effect output signal on the TE signal Line when the display module reacfes line N.
The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line.
The Tearing Effect Output line consists of V-Blanking information only:



(That N=0 is equivalent to set_tear_on with M=0.
The Tearing Effect Output Line shall be active low when the display module is in Sleep mode.)

Restriction

The command has no effect when Tearing Effect output is already ON.

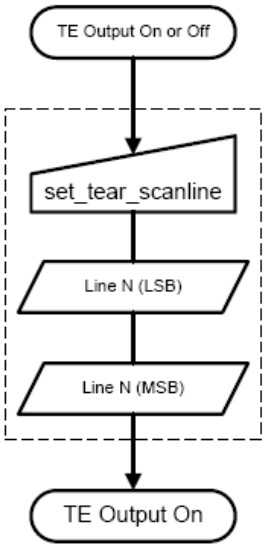
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

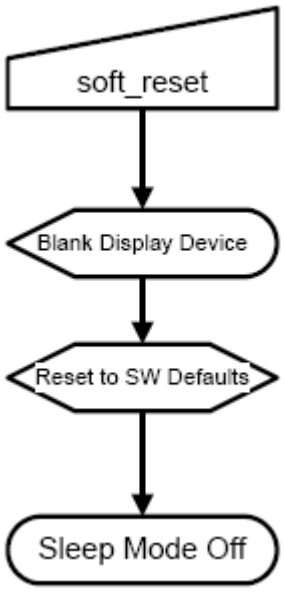
Status	Default value
Power On Sequence	N[15:0]=0000h
S/W Reset	N[15:0]=0000h
H/W Reset	N[15:0]=0000h

Flow Chart



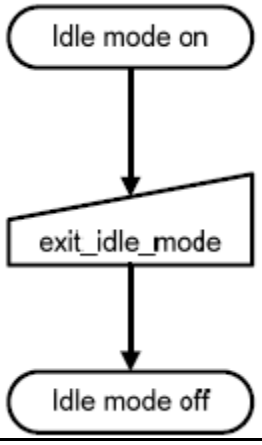
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7.2.20 Soft_reset (01h)

01H	Soft reset (Software reset)→SWRESET									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	0	0	0	0	0	1	01
Parameter	No parameter.									
Description	When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values. **(See default tables in each command description.)** The display is blank immediately. **(The frame memory contents are unaffected by this command.)**									
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display suppliers' factory default values to the registers during this 5m sec. If SW Reset is applied during Sleep Out mode, it will be necessary to wait 120m sec before sending Sleep Out command. SW Reset command cannot be sent during Sleep Out sequence.									
Register Availability	**Status** / **Availability** Sleep Out / Yes Sleep In / Yes									
Default	**Status** / **Default value** Power On Sequence / N/A S/W Reset / N/A H/W Reset / N/A									
Flow Chart										

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7.2.21 Exit idle mode (38h)

38H	Exit idle mode (Idle mode off)									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	1	1	0	0	0	38
Parameter	No parameter.									
Description	This command causes the display module to exit Idle mode. In the idle off mode, LCD can display maximum 262,144 colors.									
Restriction	This command has no effect when module is already in idle off mode.									
Register Availability	Status: Sleep Out — Availability: Yes Status: Sleep In — Availability: Yes									
Default	Status: Power On Sequence — Default value: Idle Mode Off Status: S/W Reset — Default value: Idle Mode Off Status: H/W Reset — Default value: Idle Mode Off									
Flow Chart										

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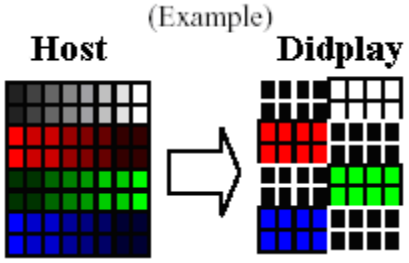
7.2.22 Enter idle mode (39h)

39H	Enter idle mode (Idle modeon)									
	Direction	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	H→D	0	0	1	1	1	0	0	1	39
Parameter	No parameter.									

Description

This command causes the display module to enter Idle Mode.
In the idle on mode, color expression is reduced. The primary and the secondary colors using MSB of each R, G and B in the Frame Memory, 8 color depth data is displayed.

(Example)



Memory contents vs. Display color			
	R5 R4 R3 R2 R1 R0	G5 G4 G3 G2 G1 G0	B5 B4 B3 B2 B1 B0
Black	0XXXXX	0XXXXX	0XXXXX
Blue	0XXXXX	0XXXXX	1XXXXX
Red	1XXXXX	0XXXXX	0XXXXX
Magenta	1XXXXX	0XXXXX	1XXXXX
Green	0XXXXX	1XXXXX	0XXXXX
Cyan	0XXXXX	1XXXXX	1XXXXX
Yellow	1XXXXX	1XXXXX	0XXXXX
White	1XXXXX	1XXXXX	1XXXXX

Restriction

This command has no effect when module is already in idle off mode.

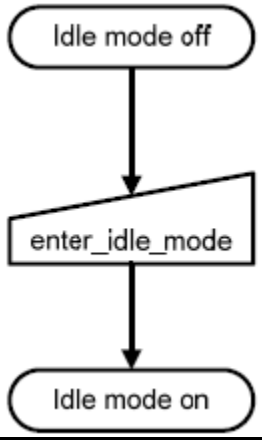
Register Availability

Status	Availability
Sleep Out	Yes
Sleep In	Yes

Default

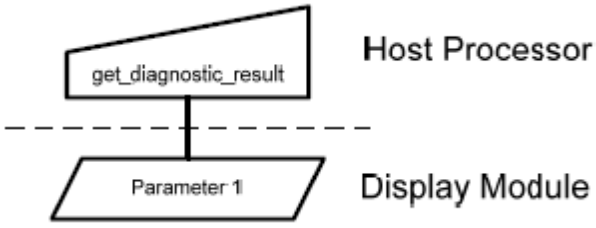
Status	Default value
Power On Sequence	Idle Mode Off
S/W Reset	Idle Mode Off
H/W Reset	Idle Mode Off

Flow Chart



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7.2.23 Get diagnostic result (0Fh)

<table>
<tr><td rowspan="2">0FH</td><td colspan="10">Get diagnostic result (Read display self-diagnostic result)→RDDSDR</td></tr>
<tr><td>Direction</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>H→D</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0F</td></tr>
<tr><td>1st parameter</td><td>H→D</td><td>D7</td><td>D6</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>-</td></tr>
<tr><td>Description</td><td colspan="10">The display module returns the self-diagnostic results following a Sleep Out command.
Bit D7 – Register Loading Detection
Bit D6 – Functionality Detection
Bits D[5:0] – Reserved, Set to ‘0’.</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"> 
</td></tr>
</table>

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7.2.24 RDNUMED (05h)

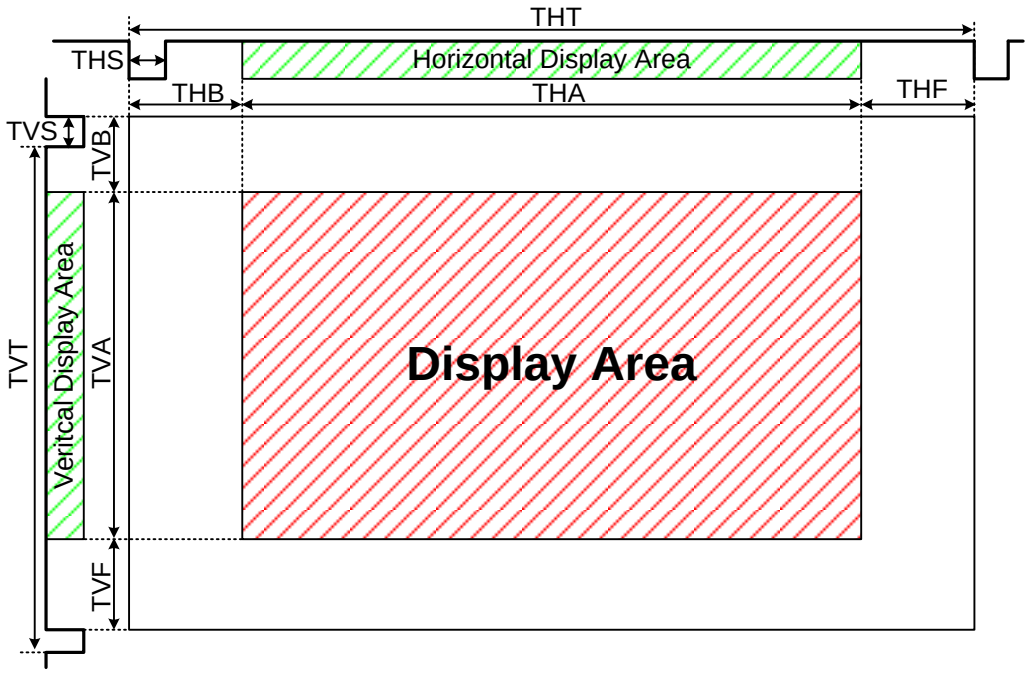
<table>
<tr><td rowspan="2">05H</td><td colspan="10">RDNUMED (Read number of the errors on DSI)</td></tr>
<tr><td>Direction</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>H→D</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>05</td></tr>
<tr><td>1st parameter</td><td>H→D</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>-</td></tr>
<tr><td>Description</td><td colspan="10">The parameter is telling a number of the errors on DSI.

D[6:0] bits are telling number of the errors.
D[7] is set to ‘1’ if there is overflow with D[6:0] bits.
D[7:0] bits are set to ‘00h’ (As well as get_signal_mode (0Eh)’s D0 is setting to ‘0’ at the same time).

Please also refer to get_signal_mode (0Eh).</td></tr>
<tr><td>Restriction</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10">-</td></tr>
</table>

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7.2.25 MIPI video input timing



HX8252-A supports Left/right type MIPI interface only. Below figures are R/L type input timing diagram when normal and reverse scan.

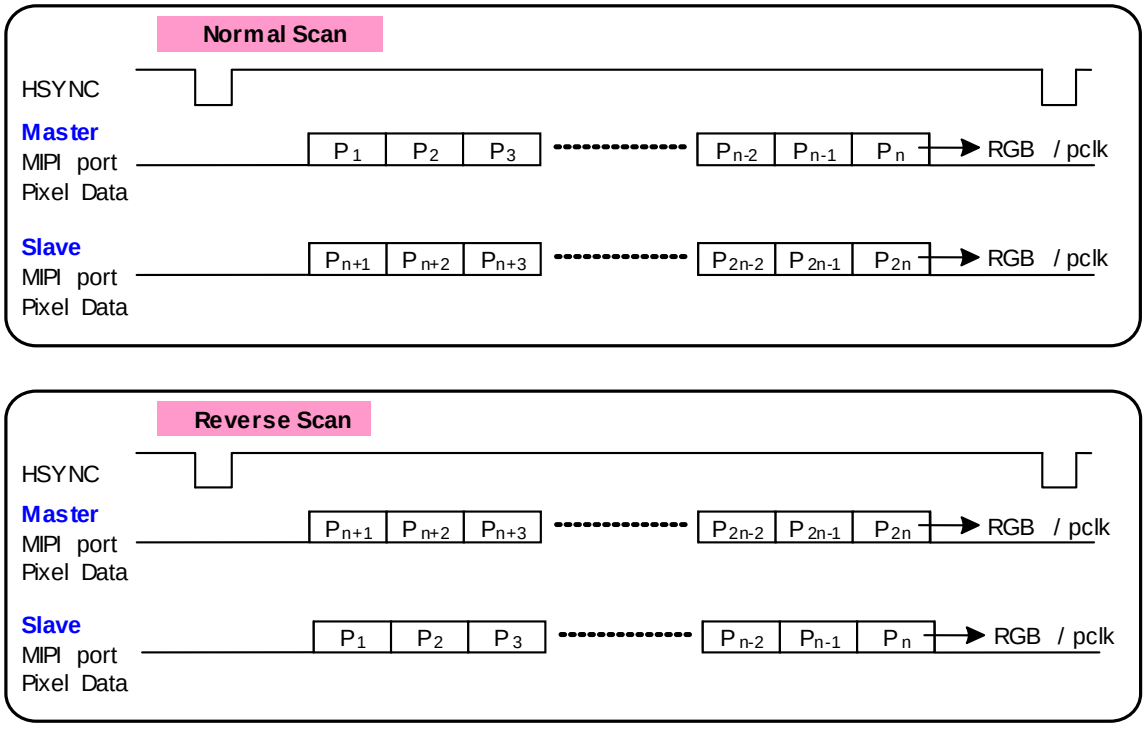


Figure 7.6: MIPI video input timing

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Parameter	Symbol	Resolution						Unit
		1600RGBx2560			1536RGBx2560			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
PCLK frequency	-	-	149	-		144		MHz
Horizontal total	THT	940	960	1600	908	928	1536	DCLK
Horizontal synchronization	THS	10	24	-	10	24	-	DCLK
Horizontal back porch	THB	70	80	-	70	80	-	DCLK
Horizontal address	THA	-	800	-	-	768	-	DCLK
Horizontal front porch	THF	50	80	-	50	80	-	DCLK
Vertical frequency	-	-	60	-		60		Hz
Vertical total	TVT	2584	2592	3000	2584	2592	3000	THT
Vertical synchronization	TVS	1	2	-	1	2	-	THT
Vertical back porch	TVB	8	12	-	8	12	-	THT
Vertical address	TVA	-	2560	-	-	2560		THT
Vertical front porch	TVF	16	20	-	16	20	-	THT

Parameter	Symbol	Resolution						Unit
		1440RGBx2560			1280RGBx2048			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
PCLK frequency	-	-	137	-	-	100	-	MHz
Horizontal total	THT	860	880	1440	780	800	1280	DCLK
Horizontal synchronization	THS	10	24	-	10	24	-	DCLK
Horizontal back porch	THB	70	80	-	70	80	-	DCLK
Horizontal address	THA	-	720	-	-	640	-	DCLK
Horizontal front porch	THF	50	80	-	50	80	-	DCLK
Vertical frequency	-	-	60	-	-	60	-	Hz
Vertical total	TVT	2584	2592	3000	2072	2080	3000	THT
Vertical synchronization	TVS	1	2	-	1	2	-	THT
Vertical back porch	TVB	8	12	-	8	12	-	THT
Vertical address	TVA	-	2560	-	-	2048	-	THT
Vertical front porch	TVF	16	20	-	16	20	-	THT

Parameter	Symbol	Resolution						Unit
		1200RGBx2048			1080RGBx1920			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
PCLK frequency	-	-	95	-	-	82	-	MHz
Horizontal total	THT	740	760	1200	680	700	1200	DCLK
Horizontal synchronization	THS	10	24	-	10	24	-	DCLK
Horizontal back porch	THB	70	80	-	70	80	-	DCLK
Horizontal address	THA	-	600	-	-	540	-	DCLK
Horizontal front porch	THF	50	80	-	50	80	-	DCLK
Vertical frequency	-	-	60	-	-	60	-	Hz
Vertical total	TVT	2072	2080	3000	1944	1952	3000	THT
Vertical synchronization	TVS	1	2	-	1	2	-	THT
Vertical back porch	TVB	8	12	-	8	12	-	THT
Vertical address	TVA	-	2048	-	-	1920	--	THT
Vertical front porch	TVF	16	20	-	16	20	-	THT

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Parameter	Symbol	Resolution			Unit
		1440RGBx2160			
		Min.	Typ.	Max.	
PCLK frequency	-		116		MHz
Horizontal total	THT	860	880	1440	DCLK
Horizontal synchronization	THS	10	24	-	DCLK
Horizontal back porch	THB	70	80	-	DCLK
Horizontal address	THA	-	720	-	DCLK
Horizontal front porch	THF	50	80	-	DCLK
Vertical frequency	-	-	60	-	Hz
Vertical total	TVT	2184	2192	3000	THT
Vertical synchronization	TVS	1	2	-	THT
Vertical back porch	TVB	8	12	-	THT
Vertical address	TVA	-	2160	-	THT
Vertical front porch	TVF	16	20	-	THT

Table 7.4: MIPI video input timing

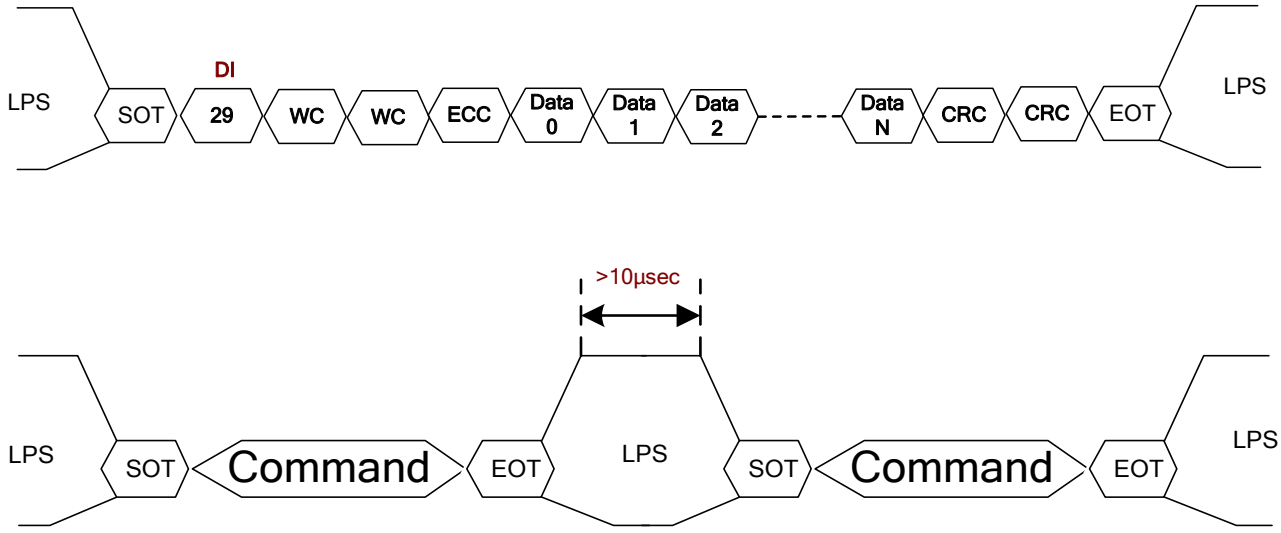
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8. Register Table

The HX8252-A supports set internal register by MIPI interface.

8.1 MIPI command mode control register

The HX8252-A supports the generic long write command to set internal register. User could use HS or LP mode to write internal register. Figure 8.1 showed the reference long package structure. There is a LP timing request between HS command to next HS command. This time needs more than 10µsec as figure 8.1. Detail MIPI commands describation showed in Section 8.3.



Note: (1) Data ID: Contain virtual channel identifier and data type.
(2) ECC **(Error Correction Code)**: The error correction code allows single-bit errors to be corrected and 2-bit errors to be detected in the short packet.

Figure 8.1: Support the DSI data long write

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8.2 Page change

The HX8252-A supports multi-page for register write.

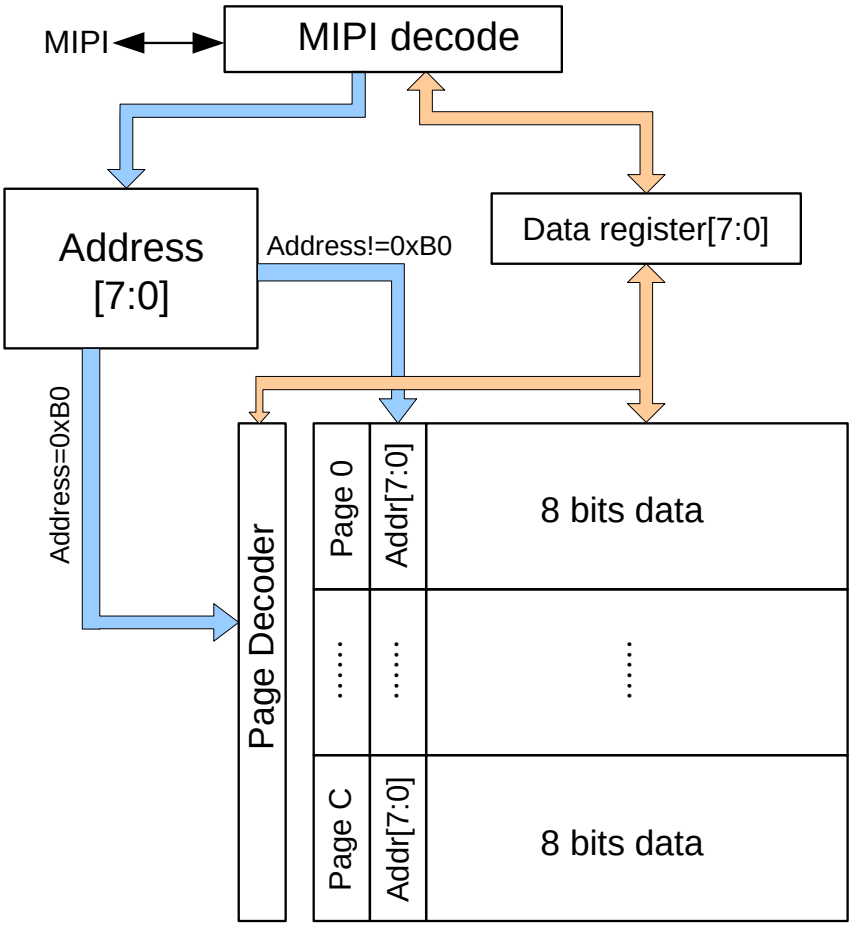


Figure 8.2: Page change for register

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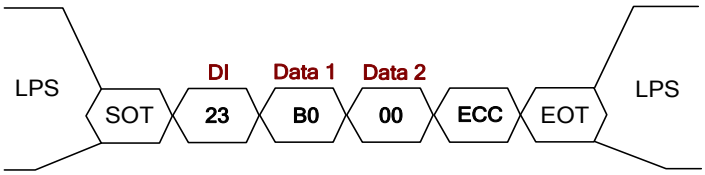
8.3 User define command list and description

8.3.1 User define command list table

User could use HS or LP command send data.

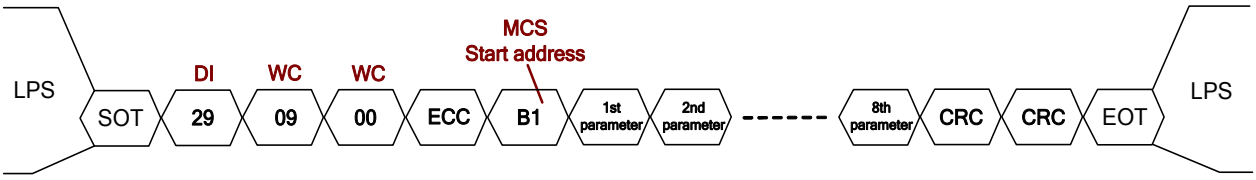
- Page 0

Step 1: Set to Page 0



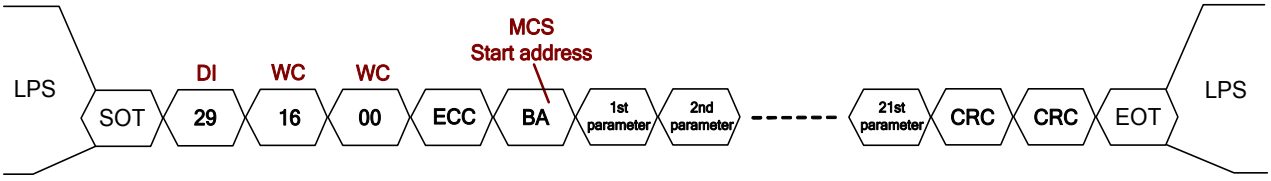
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(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	VENDER_ID								00h
	2	-	-	-	-	ZIGZAG_SEL	-	NBW_SEL	BIST_XOR	02h
	3	ZTYPE_SEL		PWRMD_XOR		VRES_FIX	RES			03h
	4	VRES[7:0]								F0h
	5	ZDATA								A5h
	6	VRES[8]	-	ZZ_2PL	RB_SWAP	CABC_CTRL		DITHER_EN	D_GAM_EN	00h
	7	-	-	-	-	-	-	BC_CTRL	CASCADE XOR	02h
	8	GOA_EN	VCOM_EN	PNSW	-	LNSW_XOR		LANSEL_XOR		C0h

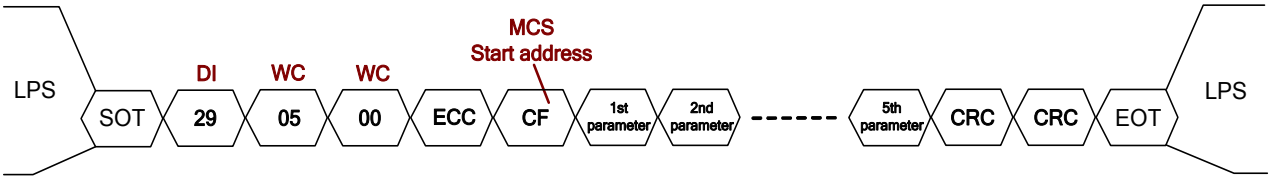
(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	1	0	1	0	Hex
BA	1	BLREV			BLREVONOFF	VDD_OTP_SW	INV_SEL			43h
	2	VCOMS								67h
	3	-	-	-	VSPS					0Ah
	4	VDD_OTPOS			VSNS					4Ah
	5	-	-	VGHS						23h
	6	-	-	VGLS						14h
	7	VDD_OTP_EN	VGLO_EN	VGLOS						14h
	8	VSPON_T				VSPOFF_T				66h
	9	VSNON_T				VSNOFF_T				66h
	10	VCL_cpctrl[2]	VGLXSP	VGHXSP		PFMFREN	tontoff_offset	dpfm_clk		61h
	11	Reserve								04h
	12	VCL_cpctrl[1:0]		VGL_cpctrl[2:0]			VGH_cpctrl[2:0]			00h
	13	cgpp_inv		cgpp_soft_en	cgpp_clksel			cgpp_cmd_sel		ADh
	14	Reserve								66h
	15	Reserve								66h
	16	Reserve								66h
	17	Reserve								66h
	18	-	-	VGMHP						1Ch
	19	-	-	VGMLP						04h
	20	-	-	VGMHN						1Ch
	21	-	-	VGMLN						04h

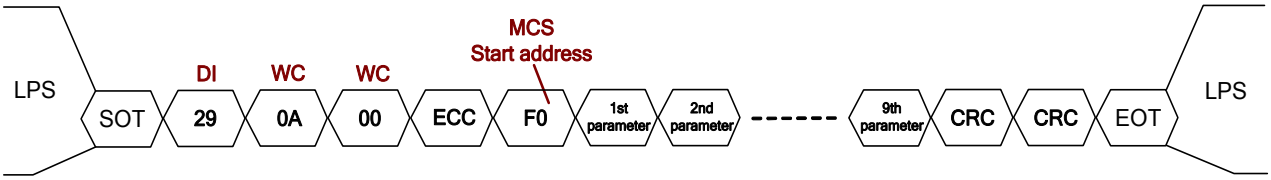
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(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	0	0	1	1	1	1	Hex
CF	1	-	POCSD_CTL			SD_EQW				00h
	2	-	ISSEL_N_CAS			-	ISSEL_P_CAS			00h
	3	-	ISSEL_N_NOTCAS			-	ISSEL_P_NOTCAS			44h
	4	-	-	-	-	T_VBP		T_VBN		0Ah

(Step 2): Set function control register

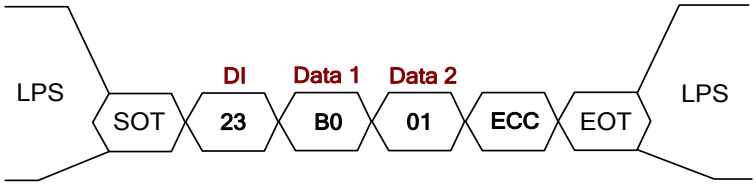


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	1	1	0	0	0	0	Hex
F0	1	GPO1_SEL				GPO0_SEL				92h
	2	GPO3_SEL				GPO2_SEL				23h
	3	GPO5_SEL				GPO4_SEL				00h
	4	GPO7_SEL				GPO6_SEL				00h
	5	-	-	-	-	GPO8_SEL				00h
	6	-	-	-	-	-	-	-	cta_en	00h
	7	cta_gain_r								FFh
	8	cta_gain_g								FFh
	9	cta_gain_b								FFh

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- Page 1

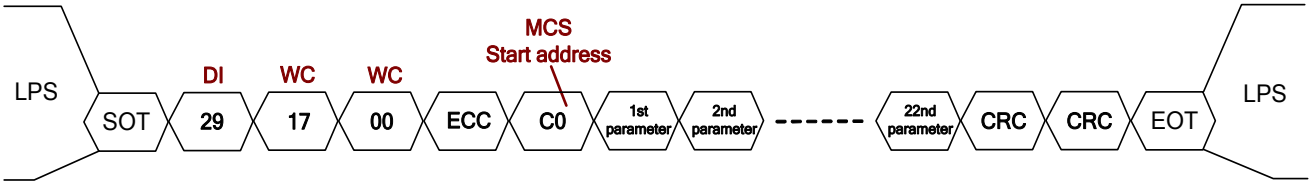
Step 1: Set to Page 1



(Step 2): Reference "10.3 OTP function"

Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	-	-	OTP_Group						00h
	2	OTP_pwd								5Ah
	3	-	-	-	-	-	OTP_Re_Load	OTP_RD	OTP_WR	00h
	4	-	-	-	-	-	-	OTP_ADDR[9:8]		00h
	5	OTP_ADDR[7:0]								00h
	6	OTP_PDOB **(READ only)**								00h

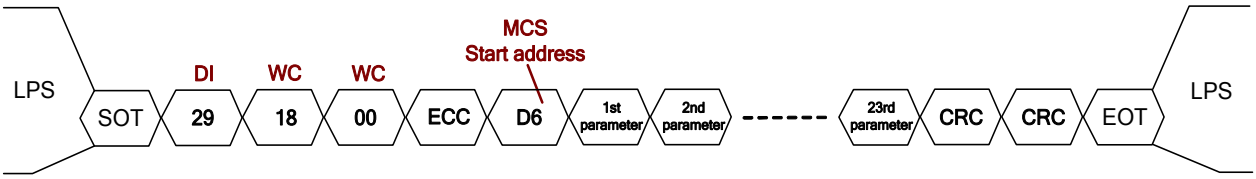
(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	0	0	0	0	0	0	Hex
C0	1	GOUTL1_STBYB		GOUTL_1_SEL						25h
	2	GOUTL2_STBYB		GOUTL_2_SEL						25h
	3	GOUTL3_STBYB		GOUTL_3_SEL						02h
	4	GOUTL4_STBYB		GOUTL_4_SEL						02h
	5	GOUTL5_STBYB		GOUTL_5_SEL						00h
	6	GOUTL6_STBYB		GOUTL_6_SEL						00h
	7	GOUTL7_STBYB		GOUTL_7_SEL						00h
	8	GOUTL8_STBYB		GOUTL_8_SEL						25h
	9	GOUTL9_STBYB		GOUTL_9_SEL						08h
	10	GOUTL10_STBYB		GOUTL_10_SEL						08h
	11	GOUTL11_STBYB		GOUTL_11_SEL						06h
	12	GOUTL12_STBYB		GOUTL_12_SEL						06h
	13	GOUTL13_STBYB		GOUTL_13_SEL						0Ch
	14	GOUTL14_STBYB		GOUTL_14_SEL						0Ch
	15	GOUTL15_STBYB		GOUTL_15_SEL						0Ah
	16	GOUTL16_STBYB		GOUTL_16_SEL						0Ah
	17	GOUTL17_STBYB		GOUTL_17_SEL						00h
	18	GOUTL18_STBYB		GOUTL_18_SEL						00h
	19	GOUTL19_STBYB		GOUTL_19_SEL						00h
	20	GOUTL20_STBYB		GOUTL_20_SEL						00h
	21	GOUTL21_STBYB		GOUTL_21_SEL						00h
	22	GOUTL22_STBYB		GOUTL_22_SEL						00h

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(Step 2): Set function control register

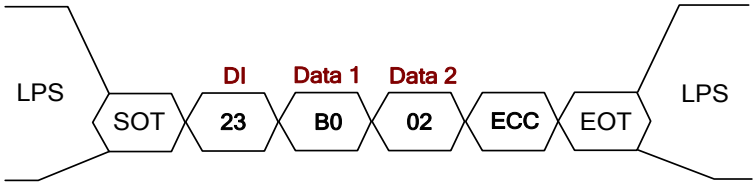


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	0	1	0	1	1	0	Hex
D6	1	GOUTR1_STBYB		GOUTR_1_SEL						25h
	2	GOUTR2_STBYB		GOUTR_2_SEL						25h
	3	GOUTR3_STBYB		GOUTR_3_SEL						01h
	4	GOUTR4_STBYB		GOUTR_4_SEL						01h
	5	GOUTR5_STBYB		GOUTR_5_SEL						00h
	6	GOUTR6_STBYB		GOUTR_6_SEL						00h
	7	GOUTR7_STBYB		GOUTR_7_SEL						00h
	8	GOUTR8_STBYB		GOUTR_8_SEL						25h
	9	GOUTR9_STBYB		GOUTR_9_SEL						07h
	10	GOUTR10_STBYB		GOUTR_10_SEL						07h
	11	GOUTR11_STBYB		GOUTR_11_SEL						05h
	12	GOUTR12_STBYB		GOUTR_12_SEL						05h
	13	GOUTR13_STBYB		GOUTR_13_SEL						0Bh
	14	GOUTR14_STBYB		GOUTR_14_SEL						0Bh
	15	GOUTR15_STBYB		GOUTR_15_SEL						09h
	16	GOUTR16_STBYB		GOUTR_16_SEL						09h
	17	GOUTR17_STBYB		GOUTR_17_SEL						00h
	18	GOUTR18_STBYB		GOUTR_18_SEL						00h
	19	GOUTR19_STBYB		GOUTR_19_SEL						00h
	20	GOUTR20_STBYB		GOUTR_20_SEL						00h
	21	GOUTR21_STBYB		GOUTR_21_SEL						00h
	22	GOUTR22_STBYB		GOUTR_22_SEL						00h
	23	-	-	GOA_HZ_EN_L	GOA_HZ_EN_R	-	-	-	-	C0h

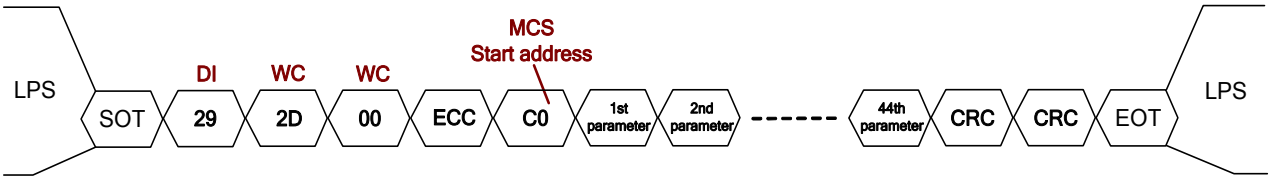
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- Page 2

Step 1: Set to Page 2



(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	0	0	0	0	0	0	Hex
C0	1	-	-	T_PVP0						00h
	2	-	-	T_PVP1						00h
	3	-	-	T_PVP2						0Fh
	4	-	-	T_PVP3						17h
	5	-	-	T_PVP4						29h
	6	-	-	T_PVP5						27h
	7	-	-	T_PVP6						28h
	8	-	-	T_PVP7						28h
	9	-	-	T_PVP8						26h
	10	-	-	T_PVP9						23h
	11	-	-	T_PVP10						28h
	12	-	-	T_PVP11						27h
	13	-	-	T_PVP12						28h
	14	-	-	T_PVP13						25h
	15	-	-	T_PVP14						2Fh
	16	-	-	T_PVP15						2Dh
	17	-	-	T_PVP16						32h
	18	-	-	T_PVP17						32h
	19	-	-	T_PVP18						34h
	20	-	-	T_PVP19						32h
	21	-	-	T_PVP20						3Ah
	22	-	-	T_PVP21						1Fh
	23	-	-	T_PVN0						00h
	24	-	-	T_PVN1						00h
	25	-	-	T_PVN2						0Fh
	26	-	-	T_PVN3						17h
	27	-	-	T_PVN4						29h
	28	-	-	T_PVN5						23h
	29	-	-	T_PVN6						2Bh
	30	-	-	T_PVN7						2Ch
	31	-	-	T_PVN8						2Ah
	32	-	-	T_PVN9						25h
	33	-	-	T_PVN10						28h
	34	-	-	T_PVN11						27h
	35	-	-	T_PVN12						34h
	36	-	-	T_PVN13						33h
	37	-	-	T_PVN14						3Dh
	38	-	-	T_PVN15						39h
	39	-	-	T_PVN16						3Eh

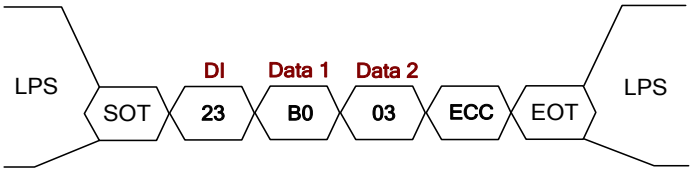
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Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	1	0	0	0	0	0	0	Hex
	40	-	-	T_PVN17						36h
	41	-	-	T_PVN18						34h
	42	-	-	T_PVN19						32h
	43	-	-	T_PVN20						3Ah
	44	-	-	T_PVN21						1Fh

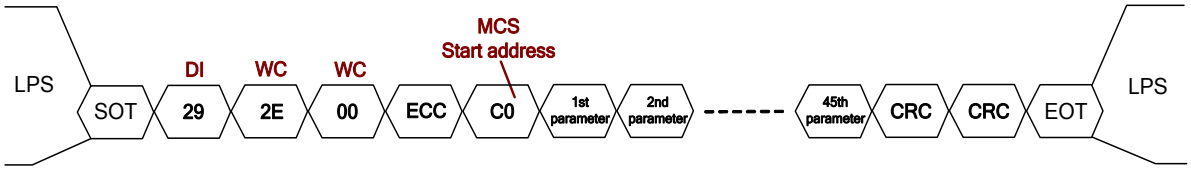
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- Page 3 **(For GOA function)**

Step 1: Set to Page 3



(Step 2): Set function control register

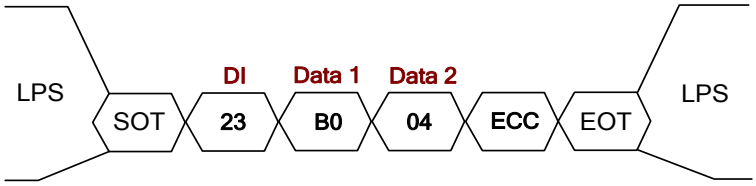


Note: (1) GOA function please reference application note.

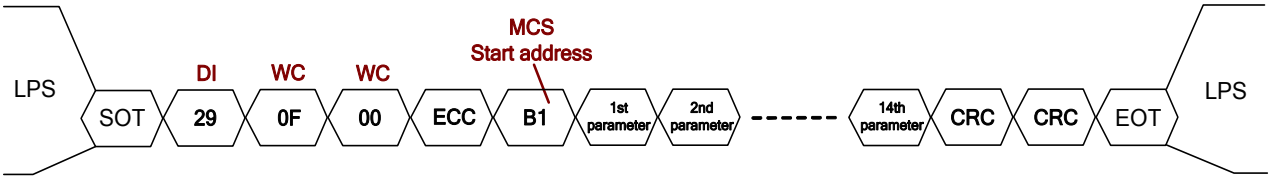
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- Page 4

Step 1: Set to Page 4



(Step 2): Set function control register

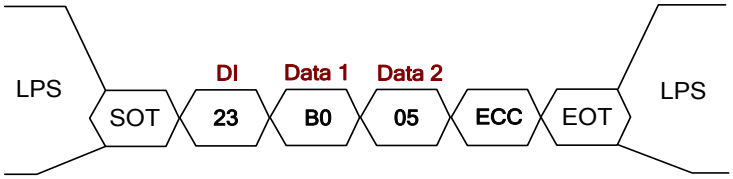


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	-	DBG_UD_EN	DIM_EN	BL_EN	PWM_POL	-	-	-	39h
	2	-	-	dim_step		-	dim_fm			21h
	3	Reserve								FFh
	4	CABC_MB								10h
	5	PWM_PRD								00h
	6	-	-	-	-	-	PWM_DIV			02h
	7	DBG0								00h
	8	DBG1								20h
	9	DBG2								40h
	10	DBG3								60h
	11	DBG4								80h
	12	DBG5								A0h
	13	DBG6								C0h
	14	DBG7								E0h

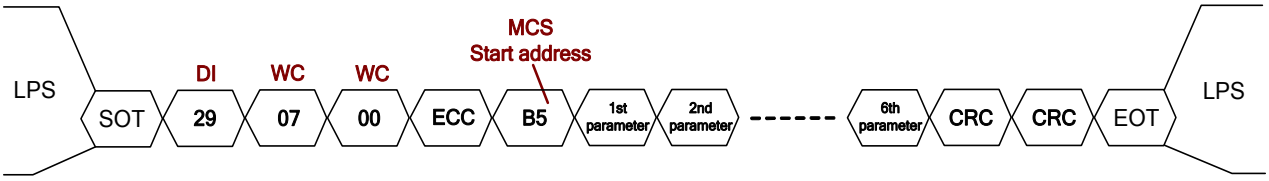
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- Page 5

Step 1: Set to Page 5



(Step 2): Set function control register

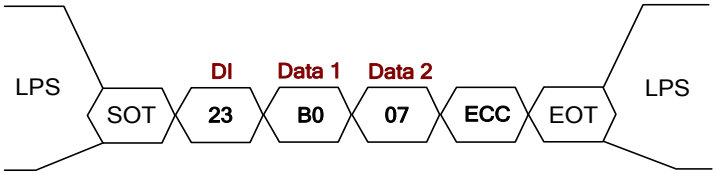


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	1	0	1	Hex
B5	1	-	RT3_M			-	RT2_M			55h
	2	-	RT1_M			-	RT0_M			55h
	3	-	-	-	-	-	RTC_M			05h
	4	-	RT3_S			-	RT2_S			55h
	5	-	RT1_S			-	RT0_S			55h
	6	-	-	-	-	-	RTC_S			05h

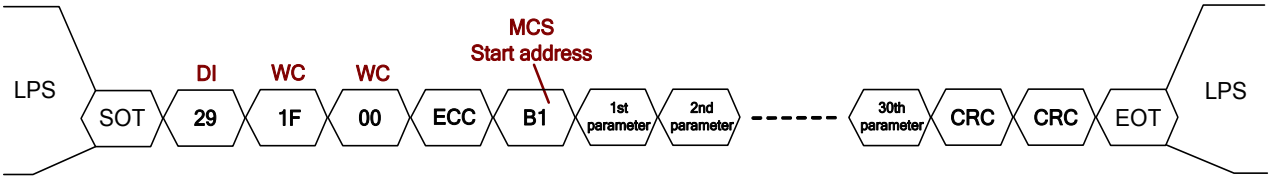
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- Page 7

Step 1: Set to Page 7



(Step 2): Set function control register

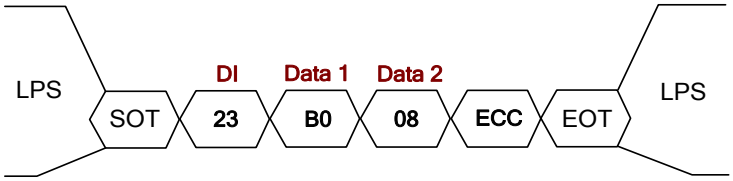


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	PGMA1R[7:0]								00h
	2	PGMA2R[7:0]								04h
	3	PGMA3R[7:0]								0Bh
	4	PGMA4R[7:0]								19h
	5	PGMA5R[7:0]								28h
	6	PGMA6R[7:0]								37h
	7	PGMA7R[7:0]								55h
	8	PGMA8R[7:0]								71h
	9	PGMA9R[7:0]								AEh
	10	PGMA10R[7:0]								F1h
	11	PGMA11R[7:0]								7Ah
	12	PGMA12R[7:0]								F9h
	13	PGMA13R[7:0]								FDh
	14	PGMA14R[7:0]								7Ah
	15	PGMA15R[7:0]								FBh
	16	PGMA16R[7:0]								3Bh
	17	PGMA17R[7:0]								7Eh
	18	PGMA18R[7:0]								9Fh
	19	PGMA19R[7:0]								C0h
	20	PGMA20R[7:0]								D1h
	21	PGMA21R[7:0]								E0h
	22	PGMA22R[7:0]								F1h
	23	PGMA23R[7:0]								F8h
	24	PGMA24R[7:0]								FCh
	25	PGMA1R[9:8]		PGMA2R[9:8]		PGMA3R[9:8]		PGMA4R[9:8]		00h
	26	PGMA5R[9:8]		PGMA6R[9:8]		PGMA7R[9:8]		PGMA8R[9:8]		00h
	27	PGMA9R[9:8]		PGMA10R[9:8]		PGMA11R[9:8]		PGMA12R[9:8]		05h
	28	PGMA13R[9:8]		PGMA14R[9:8]		PGMA15R[9:8]		PGMA16R[9:8]		6Bh
	29	PGMA17R[9:8]		PGMA18R[9:8]		PGMA19R[9:8]		PGMA20R[9:8]		FFh
	30	PGMA21R[9:8]		PGMA22R[9:8]		PGMA23R[9:8]		PGMA24R[9:8]		FFh

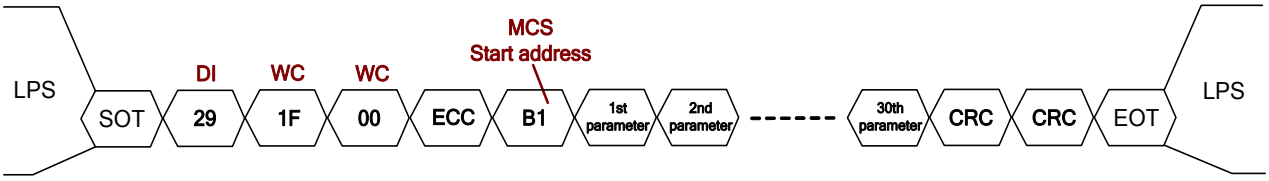
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- Page 8

Step 1: Set to Page 8



(Step 2): Set function control register

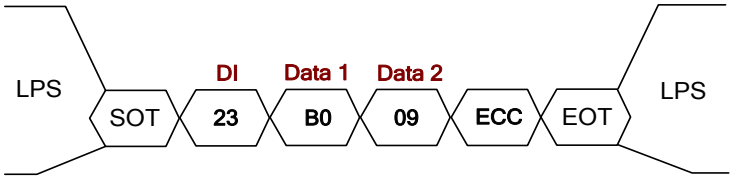


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	PGMA1G[7:0]								00h
	2	PGMA2G[7:0]								04h
	3	PGMA3G[7:0]								0Bh
	4	PGMA4G[7:0]								19h
	5	PGMA5G[7:0]								27h
	6	PGMA6G[7:0]								35h
	7	PGMA7G[7:0]								51h
	8	PGMA8G[7:0]								6Ah
	9	PGMA9G[7:0]								A4h
	10	PGMA10G[7:0]								E5h
	11	PGMA11G[7:0]								66h
	12	PGMA12G[7:0]								E0h
	13	PGMA13G[7:0]								E4h
	14	PGMA14G[7:0]								5Bh
	15	PGMA15G[7:0]								D7h
	16	PGMA16G[7:0]								14h
	17	PGMA17G[7:0]								52h
	18	PGMA18G[7:0]								71h
	19	PGMA19G[7:0]								90h
	20	PGMA20G[7:0]								A0h
	21	PGMA21G[7:0]								AFh
	22	PGMA22G[7:0]								C0h
	23	PGMA23G[7:0]								C7h
	24	PGMA24G[7:0]								CCh
	25	PGMA1G[9:8]		PGMA2G[9:8]		PGMA3G[9:8]		PGMA4G[9:8]		00h
	26	PGMA5G[9:8]		PGMA6G[9:8]		PGMA7G[9:8]		PGMA8G[9:8]		00h
	27	PGMA9G[9:8]		PGMA10G[9:8]		PGMA11G[9:8]		PGMA12G[9:8]		05h
	28	PGMA13G[9:8]		PGMA14G[9:8]		PGMA15G[9:8]		PGMA16G[9:8]		6Bh
	29	PGMA17G[9:8]		PGMA18G[9:8]		PGMA19G[9:8]		PGMA20G[9:8]		FFh
	30	PGMA21G[9:8]		PGMA22G[9:8]		PGMA23G[9:8]		PGMA24G[9:8]		FFh

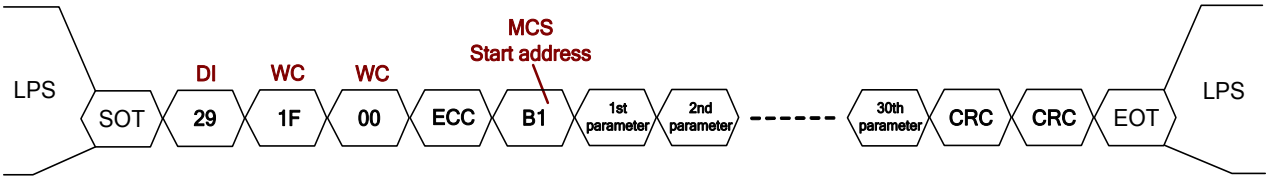
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Step 1: Set to Page 9



(Step 2): Set function control register

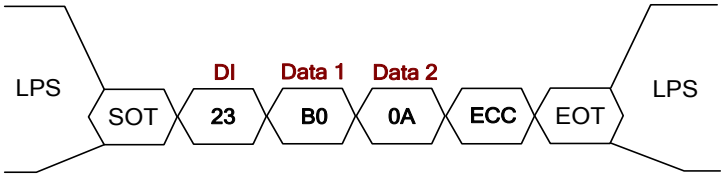


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	PGMA1B[7:0]								00h
	2	PGMA2B[7:0]								03h
	3	PGMA3B[7:0]								09h
	4	PGMA4B[7:0]								16h
	5	PGMA5B[7:0]								23h
	6	PGMA6B[7:0]								2Fh
	7	PGMA7B[7:0]								48h
	8	PGMA8B[7:0]								67h
	9	PGMA9B[7:0]								A6h
	10	PGMA10B[7:0]								E9h
	11	PGMA11B[7:0]								74h
	12	PGMA12B[7:0]								F0h
	13	PGMA13B[7:0]								F4h
	14	PGMA14B[7:0]								70h
	15	PGMA15B[7:0]								ECh
	16	PGMA16B[7:0]								29h
	17	PGMA17B[7:0]								67h
	18	PGMA18B[7:0]								84h
	19	PGMA19B[7:0]								A1h
	20	PGMA20B[7:0]								B1h
	21	PGMA21B[7:0]								C0h
	22	PGMA22B[7:0]								CCh
	23	PGMA23B[7:0]								D4h
	24	PGMA24B[7:0]								D8h
	25	PGMA1B[9:8]		PGMA2B[9:8]		PGMA3B[9:8]		PGMA4B[9:8]		00h
	26	PGMA5B[9:8]		PGMA6B[9:8]		PGMA7B[9:8]		PGMA8B[9:8]		00h
	27	PGMA9B[9:8]		PGMA10B[9:8]		PGMA11B[9:8]		PGMA12B[9:8]		05h
	28	PGMA13B[9:8]		PGMA14B[9:8]		PGMA15B[9:8]		PGMA16B[9:8]		6Bh
	29	PGMA17B[9:8]		PGMA18B[9:8]		PGMA19B[9:8]		PGMA20B[9:8]		FFh
	30	PGMA21B[9:8]		PGMA22B[9:8]		PGMA23B[9:8]		PGMA24B[9:8]		FFh

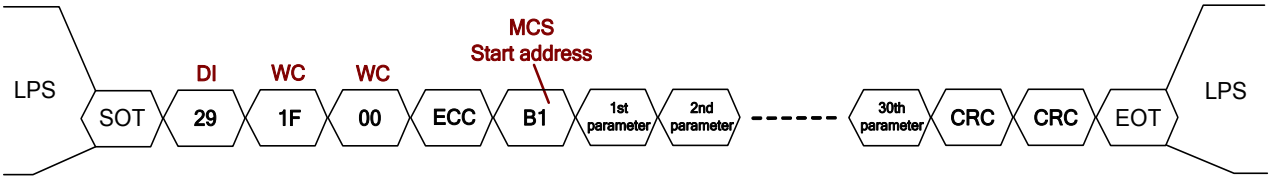
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- Page 10

Step 1: Set to Page 10



(Step 2): Set function control register

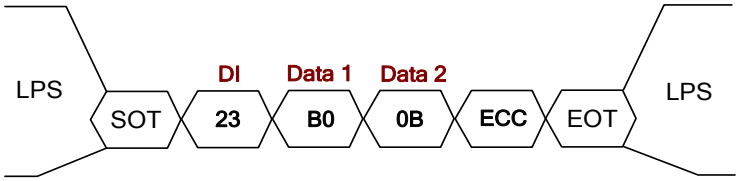


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	NGMA1R[7:0]								00h
	2	NGMA2R[7:0]								04h
	3	NGMA3R[7:0]								0Bh
	4	NGMA4R[7:0]								19h
	5	NGMA5R[7:0]								28h
	6	NGMA6R[7:0]								37h
	7	NGMA7R[7:0]								55h
	8	NGMA8R[7:0]								71h
	9	NGMA9R[7:0]								AEh
	10	NGMA10R[7:0]								F1h
	11	NGMA11R[7:0]								7Ah
	12	NGMA12R[7:0]								F9h
	13	NGMA13R[7:0]								FDh
	14	NGMA14R[7:0]								7Ah
	15	NGMA15R[7:0]								FBh
	16	NGMA16R[7:0]								3Bh
	17	NGMA17R[7:0]								7Eh
	18	NGMA18R[7:0]								9Fh
	19	NGMA19R[7:0]								C0h
	20	NGMA20R[7:0]								D1h
	21	NGMA21R[7:0]								E0h
	22	NGMA22R[7:0]								F1h
	23	NGMA23R[7:0]								F8h
	24	NGMA24R[7:0]								FCh
	25	NGMA1R[9:8]		NGMA2R[9:8]		NGMA3R[9:8]		NGMA4R[9:8]		00h
	26	NGMA5R[9:8]		NGMA6R[9:8]		NGMA7R[9:8]		NGMA8R[9:8]		00h
	27	NGMA9R[9:8]		NGMA10R[9:8]		NGMA11R[9:8]		NGMA12R[9:8]		05h
	28	NGMA13R[9:8]		NGMA14R[9:8]		NGMA15R[9:8]		NGMA16R[9:8]		6Bh
	29	NGMA17R[9:8]		NGMA18R[9:8]		NGMA19R[9:8]		NGMA20R[9:8]		FFh
	30	NGMA21R[9:8]		NGMA22R[9:8]		NGMA23R[9:8]		NGMA24R[9:8]		FFh

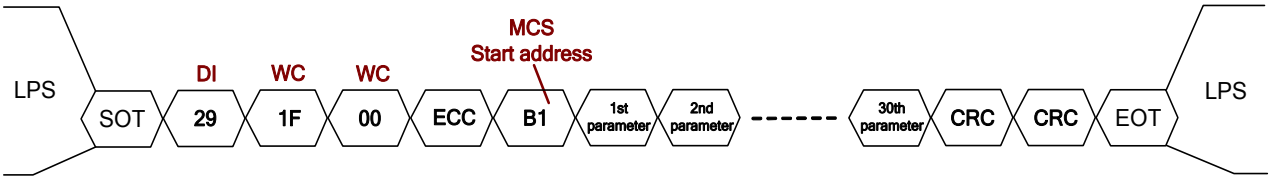
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- Page 11

Step 1: Set to Page 11



(Step 2): Set function control register

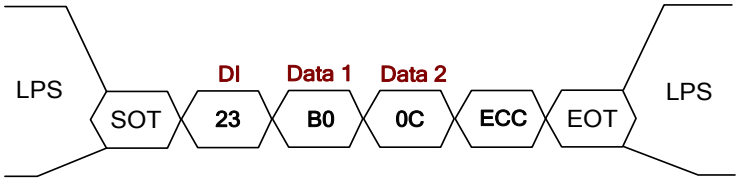


Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	NGMA1G[7:0]								00h
	2	NGMA2G[7:0]								04h
	3	NGMA3G[7:0]								0Bh
	4	NGMA4G[7:0]								19h
	5	NGMA5G[7:0]								27h
	6	NGMA6G[7:0]								35h
	7	NGMA7G[7:0]								51h
	8	NGMA8G[7:0]								6Ah
	9	NGMA9G[7:0]								A4h
	10	NGMA10G[7:0]								E5h
	11	NGMA11G[7:0]								66h
	12	NGMA12G[7:0]								E0h
	13	NGMA13G[7:0]								E4h
	14	NGMA14G[7:0]								5Bh
	15	NGMA15G[7:0]								D7h
	16	NGMA16G[7:0]								14h
	17	NGMA17G[7:0]								52h
	18	NGMA18G[7:0]								71h
	19	NGMA19G[7:0]								90h
	20	NGMA20G[7:0]								A0h
	21	NGMA21G[7:0]								AFh
	22	NGMA22G[7:0]								C0h
	23	NGMA23G[7:0]								C7h
	24	NGMA24G[7:0]								CCh
	25	NGMA1G[9:8]		NGMA2G[9:8]		NGMA3G[9:8]		NGMA4G[9:8]		00h
	26	NGMA5G[9:8]		NGMA6G[9:8]		NGMA7G[9:8]		NGMA8G[9:8]		00h
	27	NGMA9G[9:8]		NGMA10G[9:8]		NGMA11G[9:8]		NGMA12G[9:8]		05h
	28	NGMA13G[9:8]		NGMA14G[9:8]		NGMA15G[9:8]		NGMA16G[9:8]		6Bh
	29	NGMA17G[9:8]		NGMA18G[9:8]		NGMA19G[9:8]		NGMA20G[9:8]		FFh
	30	NGMA21G[9:8]		NGMA22G[9:8]		NGMA23G[9:8]		NGMA24G[9:8]		FFh

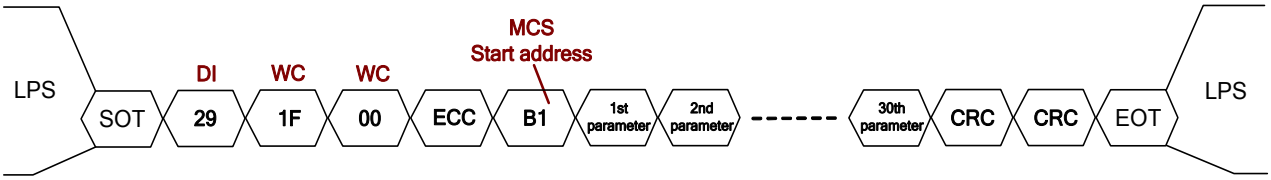
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Step 1: Set to Page 12



(Step 2): Set function control register



Hex	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default
		1	0	1	1	0	0	0	1	Hex
B1	1	NGMA1B[7:0]								00h
	2	NGMA2B[7:0]								03h
	3	NGMA3B[7:0]								09h
	4	NGMA4B[7:0]								16h
	5	NGMA5B[7:0]								23h
	6	NGMA6B[7:0]								2Fh
	7	NGMA7B[7:0]								48h
	8	NGMA8B[7:0]								67h
	9	NGMA9B[7:0]								A6h
	10	NGMA10B[7:0]								E9h
	11	NGMA11B[7:0]								74h
	12	NGMA12B[7:0]								F0h
	13	NGMA13B[7:0]								F4h
	14	NGMA14B[7:0]								70h
	15	NGMA15B[7:0]								ECh
	16	NGMA16B[7:0]								29h
	17	NGMA17B[7:0]								67h
	18	NGMA18B[7:0]								84h
	19	NGMA19B[7:0]								A1h
	20	NGMA20B[7:0]								B1h
	21	NGMA21B[7:0]								C0h
	22	NGMA22B[7:0]								CCh
	23	NGMA23B[7:0]								D4h
	24	NGMA24B[7:0]								D8h
	25	NGMA1B[9:8]		NGMA2B[9:8]		NGMA3B[9:8]		NGMA4B[9:8]		00h
	26	NGMA5B[9:8]		NGMA6B[9:8]		NGMA7B[9:8]		NGMA8B[9:8]		00h
	27	NGMA9B[9:8]		NGMA10B[9:8]		NGMA11B[9:8]		NGMA12B[9:8]		05h
	28	NGMA13B[9:8]		NGMA14B[9:8]		NGMA15B[9:8]		NGMA16B[9:8]		6Bh
	29	NGMA17B[9:8]		NGMA18B[9:8]		NGMA19B[9:8]		NGMA20B[9:8]		FFh
	30	NGMA21B[9:8]		NGMA22B[9:8]		NGMA23B[9:8]		NGMA24B[9:8]		FFh

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8.4 Register table list

Register page selection.

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB0	0Fh	R/W	[4:0]	Page_sel	Register page selection 0~9.	-

8.4.1 Register of page 0

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB1	00h	R/W	[7:0]	VENDER_ID	Driver ID and module ID.	44
0xB2	02h	R/W	[4]	LR_XOR	H-scan direction control.	2
		R/W	[3]	ZIGZAG_SEL	Panel type selection.	
		R/W	[1]	NBW_SEL	Panel normal w ihte/black selection.	
		R/W	[0]	BIST_XOR	BIST mode control.	
0xB3	03h	R/W	[7:6]	ZTYPE_SEL	Zigzag type selection.	
		R/W	[5:4]	PWRMD_XOR	Pow er mode selection.	
		R/W	[3]	VRES_FIX	Display vertical line decide by(1):RES[2:0] or (0) VRES.	
		R/W	[2:0]	RES	Resolution selection.	
0xB4	F0h	R/W	[7:0]	VRES[7:0]	Vertical resolution selection.	
0xB5	A5h	R/W	[7:0]	ZDATA[7:0]	ZigZag dummy data selection.	
0xB6	00h	R/W	[7]	VRES[8]	Vertical resolution selection.	
		R/W	[5]	ZZ_2PL	Zigzag 2H panel type selection.	
		R/W	[4]	RB_SWAP	RB data sw ap on/off control.	
		R/W	[3:2]	CABC_CTRL	CABC-Mode selection.	
		R/W	[1]	DITHER_EN	Dithering on/off control.	
		R/W	[0]	D_GAM_EN	Digital gamma on/off control.	
0xB7	02h	R/W	[1]	BC_CTRL	Back light control.	
		R/W	[0]	CASCADE_XOR	Cascade mode on/off control.	
0xB8	C0h	R/W	[7]	GOA_EN	GOA on/off control.	
		R/W	[6]	VCOM_EN	VCOM on/off control.	
		R/W	[5]	PNSW	MIPI P-N sw ap.	
		R/W	[3:2]	LNSW_XOR	MIPI lane sw ap.	
		R/W	[1:0]	LANSEL_XOR	MIPI lane number selection.	
0xBA	43h	R/W	[7:5]	BLREV	source output at V-blanking.	
		R/W	[4]	BLREVONOFF	source output at pow er on off.	
		R/W	[3]	VDD_OTP_SW	VDD_OTP pow er path sw itch selection.	
		R/W	[2:0]	INV_SEL	Data inverion select.	
0xBB	67h	R/W	[7:0]	VCOMS	VCOM voltage setting.	0
0xBC	0Ah	R/W	[4:0]	VSPS	VSP voltage setting.	3
0xBD	4Ah	R/W	[7:5]	VDD_OTPOS	VDD_OTP voltage setting.	
		R/W	[4:0]	VSNS	VSN voltage setting.	
0xBE	23h	R/W	[5:0]	VGHS	VGH voltage selection.	
0xBF	14h	R/W	[5:0]	VGLS	VGL voltage selection.	
0xC0	14h	R/W	[7]	VDD_OTP_EN	VDD_OTP on/off control.	
		R/W	[6]	VGLO_EN	VGLO2 on/off control.	3
		R/W	[5:0]	VGLOS	VGLO2 voltage setting.	
0xC1	66h	R/W	[7:4]	VSPON_T	Set PFM TON duty for VSP voltage.	
		R/W	[3:0]	VSPOFF_T	Set PFM TOFF duty for VSP voltage.	
0xC2	66h	R/W	[7:4]	VSNON_T	Set PFM TON duty for VSN voltage.	
		R/W	[3:0]	VSNOFF_T	Set PFM TOFF duty for VSN voltage.	
0xC3	61h	R/W	[7]	VCL_cpctrl[2]	VCL charge pump clock frequency [2].	
		R/W	[6]	VGLXSP	VGL boost function selection.	
		R/W	[5:4]	VGHXSP	VGH boost function selection.	
		R/W	[3]	PFMFREN	Enable PFM frequency randomizer of VSP and VSN.	
			[2]	tontoff_offset	PFM ton/toff offset.	
			[1:0]	dpfm_clk	DPFM clock selection.	
0xC5	00h	R/W	[7:6]	VCL_cpctrl[1:0]	VCL charge pump clock frequency [1:0].	
		R/W	[5:3]	VGL_cpctrl[2:0]	VGL charge pump clock frequency.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
		R/W	[2:0]	VGH_cpctrl[2:0]	VGH charge pump clock frequency.	
0xC6	ADh	R/W	[7:6]	cgpp_inv	Set DRVN/DRVP polarity.	
		R/W	[5]	cgpp_soft_en	Charge pump soft start.	
		R/W	[4:2]	cgpp_clksel	Charge pump clock selection.	
		R/W	[1:0]	cgpp_cmd_sel	Charge pump command selection.	
0xCB	1Ch	R/W	[5:0]	VGMHP	VGMHP voltage setting.	4
0xCC	04h	R/W	[5:0]	VGMLP	VGMLP voltage setting.	
0xCD	1Ch	R/W	[5:0]	VGMHN	VGMHN voltage setting.	
0xCE	04h	R/W	[5:0]	VGMLN	VGMLN voltage setting.	
0xCF	00h	R/W	[6:5]	POCSD_CTL	Source output offset cancel method selection.	5
		R/W	[4:0]	SD_EQW	Source EQ time setting.	
0xD0	00h	R/W	[6:4]	ISSEL_N_CAS	Source N bias current.	6
		R/W	[2:0]	ISSEL_P_CAS	Source P bias current.	
0xD1	44h	R/W	[6:4]	ISSEL_N_NOTCAS	Source N bias current when NOT-CASCADE mode.	
		R/W	[2:0]	ISSEL_P_NOTCAS	Source P bias current when NOT-CASCADE mode.	
0xD2	0Ah	R/W	[3:2]	T_VBP	Gamma P bias current selection.	
		R/W	[1:0]	T_VBN	Gamma N bias current selection.	
0xF0	92h	R/W	[7:4]	GPO1_SEL	GPO1 output signal selection.	7
		R/W	[3:0]	GPO0_SEL	GPO0 output signal selection.	
0xF1	23h	R/W	[7:4]	GPO3_SEL	GPO3 output signal selection.	
		R/W	[3:0]	GPO2_SEL	GPO2 output signal selection.	
0xF2	00h	R/W	[7:4]	GPO5_SEL	GPO5 output signal selection.	
		R/W	[3:0]	GPO4_SEL	GPO4 output signal selection.	
0xF3	00h	R/W	[7:4]	GPO7_SEL	GPO7 output signal selection.	
		R/W	[3:0]	GPO6_SEL	GPO6 output signal selection.	
0xF4	00h	R/W	[3:0]	GPO8_SEL	GPO8 output signal selection.	
0xF5	00h	R/W	[0]	cta_en	CTA function enable selection.	8
0xF6	FFh	R/W	[7:0]	cta_en_r	CTA R gain setting.	
0xF7	FFh	R/W	[7:0]	cta_en_g	CTA G gain setting.	
0xF8	FFh	R/W	[7:0]	cta_en_b	CTA B gain setting.	
0xFF	0Ch	R	[7:0]	CHIPID	IC Version.	-

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8.4.2 Register of page 1 (OTP & GOA MUX)

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB1	00h	R/W	[5:0]	OTP Group	OTP trimming group select. The group range is from group0 to group 44.	-
0xB2	5Ah	R/W	[7:0]	OTP pwd	Set to 0xA5 to enable 0xB3 command.	-
0xB3	00h	R/W	[2]	OTP_Re_Load	OTP auto re-load command.	-
		R/W	[1]	OTP_RD	OTP read control.	-
		R/W	[0]	OTP_WR	OTP write control.	-
0XB4	00h	R/W	[1:0]	OTP_ADDR[9:8]	Set OTP address[9:8].	-
0xB5	00h	R/W	[7:0]	OTP_ADDR[7:0]	Set OTP address[7:0].	-
0xB6	00h	R	[7:0]	OTP_PDOB	Read data from OTP function.	-
0xC0	25h	R/W	[7:6]	GOUTL_1_STBYB	Set signal voltage at standby mode.	9
		R/W	[5:0]	GOUTL_1_SEL	Mux goa signal to GINL1 ref GOUT SEL mapping table.	
0xC1	25h	R/W	[7:6]	GOUTL_2_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_2_SEL	Mux goa signal to GINL2 ref GOUT SEL mapping table.	
0xC2	02h	R/W	[7:6]	GOUTL_3_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_3_SEL	Mux goa signal to GINL3 ref GOUT SEL mapping table.	
0xC3	02h	R/W	[7:6]	GOUTL_4_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_4_SEL	Mux goa signal to GINL4 ref GOUT SEL mapping table.	
0xC4	00h	R/W	[7:6]	GOUTL_5_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_5_SEL	Mux goa signal to GINL5 ref GOUT SEL mapping table.	
0xC5	00h	R/W	[7:6]	GOUTL_6_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_6_SEL	Mux goa signal to GINL6 ref GOUT SEL mapping table.	
0xC6	00h	R/W	[7:6]	GOUTL_7_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_7_SEL	Mux goa signal to GINL7 ref GOUT SEL mapping table.	
0xC7	25h	R/W	[7:6]	GOUTL_8_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_8_SEL	Mux goa signal to GINL8 ref GOUT SEL mapping table.	
0xC8	08h	R/W	[7:6]	GOUTL_9_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_9_SEL	Mux goa signal to GINL9 ref GOUT SEL mapping table.	
0xC9	08h	R/W	[7:6]	GOUTL_10_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_10_SEL	Mux goa signal to GINL10 ref GOUT SEL mapping table.	
0xCA	06h	R/W	[7:6]	GOUTL_11_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_11_SEL	Mux goa signal to GINL11 ref GOUT SEL mapping table.	
0xCB	06h	R/W	[7:6]	GOUTL_12_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_12_SEL	Mux goa signal to GINL12 ref GOUT SEL mapping table.	
0xCC	0Ch	R/W	[7:6]	GOUTL_13_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_13_SEL	Mux goa signal to GINL13 ref GOUT SEL mapping table.	
0xCD	0Ch	R/W	[7:6]	GOUTL_14_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_14_SEL	Mux goa signal to GINL14 ref GOUT SEL mapping table.	
0xCE	0Ah	R/W	[7:6]	GOUTL_15_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_15_SEL	Mux goa signal to GINL15 ref GOUT SEL mapping table.	
0xCF	0Ah	R/W	[7:6]	GOUTL_16_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_16_SEL	Mux goa signal to GINL16 ref GOUT SEL mapping table.	
0xD0	00h	R/W	[7:6]	GOUTL_17_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_17_SEL	Mux goa signal to GINL17 ref GOUT SEL mapping table.	
0xD1	00h	R/W	[7:6]	GOUTL_18_STBYB	Set signal voltage at standby mode.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
			[5:0]	GOUTL_18_SEL	Mux goa signal to GINL18 ref GOUT SEL mapping table.	
0xD2	00h	R/W	[7:6]	GOUTL_19_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_19_SEL	Mux goa signal to GINL19 ref GOUT SEL mapping table.	
0xD3	00h	R/W	[7:6]	GOUTL_20_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_20_SEL	Mux goa signal to GINL20 ref GOUT SEL mapping table.	
0xD4	00h	R/W	[7:6]	GOUTL_21_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_21_SEL	Mux goa signal to GINL21 ref GOUT SEL mapping table.	
0xD5	00h	R/W	[7:6]	GOUTL_22_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTL_22_SEL	Mux goa signal to GINL22 ref GOUT SEL mapping table.	
0xD6	25h	R/W	[7:6]	GOUTR_1_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_1_SEL	Mux goa signal to GINR1 ref GOUT SEL mapping table.	
0xD7	25h	R/W	[7:6]	GOUTR_2_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_2_SEL	Mux goa signal to GINR2 ref GOUT SEL mapping table.	
0xD8	01h	R/W	[7:6]	GOUTL_3_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_3_SEL	Mux goa signal to GINR3 ref GOUT SEL mapping table.	
0xD9	01h	R/W	[7:6]	GOUTR_4_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_4_SEL	Mux goa signal to GINR4 ref GOUT SEL mapping table.	
0xDA	00h	R/W	[7:6]	GOUTR_5_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_5_SEL	Mux goa signal to GINR5 ref GOUT SEL mapping table.	
0xDB	00h	R/W	[7:6]	GOUTR_6_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_6_SEL	Mux goa signal to GINR6 ref GOUT SEL mapping table.	
0xDC	00h	R/W	[7:6]	GOUTR_7_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_7_SEL	Mux goa signal to GINR7 ref GOUT SEL mapping table.	
0xDD	25h	R/W	[7:6]	GOUTR_8_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_8_SEL	Mux goa signal to GINR8 ref GOUT SEL mapping table.	
0xDE	07h	R/W	[7:6]	GOUTR_9_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_9_SEL	Mux goa signal to GINR9 ref GOUT SEL mapping table.	
0xDF	07h	R/W	[7:6]	GOUTR_10_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_10_SEL	Mux goa signal to GINR10 ref GOUT SEL mapping table.	
0xE0	05h	R/W	[7:6]	GOUTR_11_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_11_SEL	Mux goa signal to GINR11 ref GOUT SEL mapping table.	
0xE1	05h	R/W	[7:6]	GOUTR_12_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_12_SEL	Mux goa signal to GINR12 ref GOUT SEL mapping table.	
0xE2	0Bh	R/W	[7:6]	GOUTR_13_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_13_SEL	Mux goa signal to GINR13 ref GOUT SEL mapping table.	
0xE3	0Bh	R/W	[7:6]	GOUTR_14_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_14_SEL	Mux goa signal to GINR14 ref GOUT SEL mapping table.	
0xE4	09h	R/W	[7:6]	GOUTR_15_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_15_SEL	Mux goa signal to GINR15 ref GOUT SEL mapping table.	
0xE5	09h	R/W	[7:6]	GOUTR_16_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_16_SEL	Mux goa signal to GINR16 ref GOUT SEL mapping table.	
0xE6	00h	R/W	[7:6]	GOUTR_17_STBYB	Set signal voltage at standby mode.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
			[5:0]	GOUTR_17_SEL	Mux goa signal to GINR17 ref GOUT SEL mapping table.	
0xE7	00h	R/W	[7:6]	GOUTR_18_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_18_SEL	Mux goa signal to GINR18 ref GOUT SEL mapping table.	
0xE8	00h	R/W	[7:6]	GOUTR_19_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_19_SEL	Mux goa signal to GINR19 ref GOUT SEL mapping table.	
0xE9	00h	R/W	[7:6]	GOUTR_20_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_20_SEL	Mux goa signal to GINR20 ref GOUT SEL mapping table.	
0xEA	00h	R/W	[7:6]	GOUTR_21_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_21_SEL	Mux goa signal to GINR21 ref GOUT SEL mapping table.	
0xEB	00h	R/W	[7:6]	GOUTR_22_STBYB	Set signal voltage at standby mode.	
			[5:0]	GOUTR_22_SEL	Mux goa signal to GINR22 ref GOUT SEL mapping table.	
0xEC	C0h	R/W	[5]	GOA_HZ_EN_L	GOUTL [22:1] “Hi-Z” enable selection.	
			[4]	GOA_HZ_EN_R	GOUTR [22:1] “Hi-Z” enable selection.	

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8.4.3 Register of page 2 (Analog gamma)

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xC0	00h	R/W	[4:0]	T_PVP0[4:0]	V0 positive gamma op's input voltage.	10
0xC1	00h	R/W	[4:0]	T_PVP1[4:0]	V1 positive gamma op's input voltage.	
0xC2	0Fh	R/W	[5:0]	T_PVP2[5:0]	V4 positive gamma op's input voltage.	
0xC3	17h	R/W	[5:0]	T_PVP3[5:0]	V8 positive gamma op's input voltage.	
0xC4	29h	R/W	[5:0]	T_PVP4[5:0]	V16 positive gamma op's input voltage.	
0xC5	27h	R/W	[5:0]	T_PVP5[5:0]	V28 positive gamma op's input voltage.	
0xC6	28h	R/W	[5:0]	T_PVP6[5:0]	V40 positive gamma op's input voltage.	
0xC7	28h	R/W	[5:0]	T_PVP7[5:0]	V56 positive gamma op's input voltage.	
0xC8	26h	R/W	[5:0]	T_PVP8[5:0]	V72 positive gamma op's input voltage.	
0xC9	23h	R/W	[5:0]	T_PVP9[5:0]	V96 positive gamma op's input voltage.	
0xCA	28h	R/W	[5:0]	T_PVP10[5:0]	V128 positive gamma op's input voltage.	
0xCB	27h	R/W	[5:0]	T_PVP11[5:0]	V152 positive gamma op's input voltage.	
0xCC	28h	R/W	[5:0]	T_PVP12[5:0]	V176 positive gamma op's input voltage.	
0xCD	25h	R/W	[5:0]	T_PVP13[5:0]	V192 positive gamma op's input voltage.	
0XCE	2Fh	R/W	[5:0]	T_PVP14[5:0]	V204 positive gamma op's input voltage.	
0xCF	2Dh	R/W	[5:0]	T_PVP15[5:0]	V216 positive gamma op's input voltage.	
0xD0	32h	R/W	[5:0]	T_PVP16[5:0]	V228 positive gamma op's input voltage.	
0XD1	32h	R/W	[5:0]	T_PVP17[5:0]	V240 positive gamma op's input voltage.	
0xD2	34h	R/W	[5:0]	T_PVP185:0]	V248 positive gamma op's input voltage.	
0xD3	32h	R/W	[5:0]	T_PVP19[5:0]	V252 positive gamma op's input voltage.	
0xD4	3Ah	R/W	[4:0]	T_PVP20[4:0]	V254 positive gamma op's input voltage.	
0xD5	1Fh	R/W	[4:0]	T_PVP21[4:0]	V255 positive gamma op's input voltage.	
0xD6	00h	R/W	[4:0]	T_PVN0[4:0]	V0 negative gamma op's input voltage.	
0xD7	00h	R/W	[4:0]	T_PVN1[4:0]	V1 negative gamma op's input voltage.	
0xD8	0Fh	R/W	[5:0]	T_PVN2[5:0]	V4 negative gamma op's input voltage.	
0xD9	17h	R/W	[5:0]	T_PVN3[5:0]	V8 negative gamma op's input voltage.	
0xDA	29h	R/W	[5:0]	T_PVN4[5:0]	V16 negative gamma op's input voltage.	
0xDB	23h	R/W	[5:0]	T_PVN5[5:0]	V28 negative gamma op's input voltage.	
0xDC	2Bh	R/W	[5:0]	T_PVN6[5:0]	V40 negative gamma op's input voltage.	
0xDD	2Ch	R/W	[5:0]	T_PVN7[5:0]	V56 negative gamma op's input voltage.	
0xDE	2Ah	R/W	[5:0]	T_PVN8[5:0]	V72 negative gamma op's input voltage.	
0xDF	25h	R/W	[5:0]	T_PVN9[5:0]	V96 negative gamma op's input voltage.	
0xE0	28h	R/W	[5:0]	T_PVN10[5:0]	V128 negative gamma op's input voltage.	
0xE1	27h	R/W	[5:0]	T_PVN11[5:0]	V152 negative gamma op's input voltage.	
0xE2	34h	R/W	[5:0]	T_PVN12[5:0]	V176 negative gamma op's input voltage.	
0xE3	33h	R/W	[5:0]	T_PVN13[5:0]	V192 negative gamma op's input voltage.	
0xE4	3Dh	R/W	[5:0]	T_PVN14[5:0]	V204 negative gamma op's input voltage.	
0xE5	39h	R/W	[5:0]	T_PVN15[5:0]	V216 negative gamma op's input voltage.	
0xE6	3Eh	R/W	[5:0]	T_PVN16[5:0]	V228 negative gamma op's input voltage.	
0xE7	36h	R/W	[5:0]	T_PVN17[5:0]	V240 negative gamma op's input voltage.	
0xE8	34h	R/W	[5:0]	T_PVN18[5:0]	V248 negative gamma op's input voltage.	
0xE9	32h	R/W	[5:0]	T_PVN19[5:0]	V252 negative gamma op's input voltage.	
0xEA	3Ah	R/W	[4:0]	T_PVN20[4:0]	V254 negative gamma op's input voltage.	
0xEB	1Fh	R/W	[4:0]	T_PVN21[4:0]	V255 negative gamma op's input voltage.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xC0	01h	R/W	[3]	CKV_BLANKON	Blanking all on enable.	
		R/W	[2:0]	GOA_PHASE[2:0]	GOA phase selection.	
0xC1	10h	R/W	[5:4]	GOA_Tn_FACTOR	GOA timing **(T0~T3)** adjustment factor selection.	
0xC2	00h	R/W	[7:0]	GOA_T0[7:0]	Odd STV falling position adjustment.	
0xC3	01h	R/W	[7:0]	GOA_T1[7:0]	Odd STV rising position adjustment.	
0xC4	00h	R/W	[7:0]	GOA_T2[7:0]	Odd CKV falling position adjustment.	
0xC5	23h	R/W	[7:0]	GOA_T3[7:0]	Odd CKV rising position adjustment.	
0xC8	09h	R/W	[4:0]	STV_LEAD[4:0]	GOA STV leads time setting.	
0xC9	07h	R/W	[4:0]	CKV_LEAD[4:0]	GOA CKV leads time setting.	
0xCA	41h	R/W	[7]	CKV_NONOVERLAP	CKV non-overlap control.	
		R/W	[5:0]	CKV_DUMMY[5:0]	GOA CKV dummy number in V-blanking.	
0xCB	34h	R/W	[7:0]	CKV_RISE_PREC[7:0]	CKV rising pre-charge width adjustment.	
0xCC	44h	R/W	[7:4]	STV_WIDTH[3:0]	GOA STV width setting.	
		R/W	[3:0]	CKV_WIDTH[3:0]	GOA CKV width setting.	
0xCD	07h	R/W	[7:0]	GOA_FLC[7:0]	GOA FLC toggle frame setting.	
0xCE	03h	R/W	[6:0]	GOA_FLCA_LEAD[6:0]	FLCA lead 1st DE toggle number.	
0xCF	10h	R/W	[7]	CLR4_LEAD	GOA CLR4 offset direction setting.	
		R/W	[6]	CLR3_LEAD	GOA CLR3 offset direction setting.	
		R/W	[5]	CLR2_LEAD	GOA CLR2 offset direction setting.	
0xD0	05h	R/W	[7:0]	CKV_BLANK_SHIFT[7:0]	Blank CKV shift from the last DE.	
0xD1	06h	R/W	[7:0]	CKV_BLANK_WIDTH[7:0]	Blank CKV width adjustment.	
0xD2	00h	R/W	[7:0]	CLR1_WD[7:0]	GOA CLR1 width adjustment.	
0xD3	00h	R/W	[7:0]	CLR234_WD[7:0]	GOA CLR2/3/4 width adjustment.	
0xD4	88h	R/W	[7]	CLR1_POL	CLR1 polarity setting.	
		R/W	[6:0]	CLR1_START[6:0]	CLR1 start position.	
0xD5	09h	R/W	[7]	CLR2_POL	CLR2 polarity setting.	
		R/W	[6:0]	CLR2_START[6:0]	CLR2 start position.	
0xD6	03h	R/W	[7]	CLR3_POL	CLR3 polarity setting.	
		R/W	[6:0]	CLR3_START[6:0]	CLR3 start position.	
0xD7	04h	R/W	[7]	CLR4_POL	CLR4 polarity setting.	
		R/W	[6:0]	CLR4_START[6:0]	CLR4 start position.	
0xD8	00h	R/W	[4:0]	CLR1_START_MSB[4:0]	MSB[11:7] of CLR1 start position.	
0xD9	00h	R/W	[4:0]	CLR2/3/4_START_MSB[4:0]	MSB[11:7] of CLR2/3/4 start position.	
0xDB	00h	R/W	[3:1]	FLC_NONOVLAP[1:0]	FLC nonoverlap setting.	
		R/W	[0]	CLR1_LEAD	GOA CLR1 offset direction setting.	
0xDD	01h	R/W	[7:0]	GOA_T2B[7:0]	Tune even CKV falling time.	
0xDE	23h	R/W	[7:0]	GOA_T3B[7:0]	Tune even CKV rising time.	
0xE6	00h	R/W	[7:0]	GOA_T0B[7:0]	Tune even STV falling time.	
0xE7	01h	R/W	[7:0]	GOA_T1B[7:0]	Tune even STV rising time.	
0xEA	00h	R/W	[2:0]	STV_PREC	STV pre-charge width adjustment.	13
0xEB	00h	R/W	[7:0]	CKV_FALL_PREC[7:0]	CKV falling pre-charge width adjustment.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB1	39h	R/W	[6]	DBG_UD_EN	DBG user define enable.	14
		R/W	[5]	DIM_EN	Back light dimming enable.	
		R/W	[4]	BL_EN	CABC back light control enable.	
		R/W	[3]	PWM_POL	Polarity of PWM control signal setting.	
0xB2	21h	R/W	[5:4]	dim_step	Dimming frame scale A.	
		R/W	[2:0]	dim_fm	Dimming frame scale B.	
0xB4	10h	R/W	[7:0]	CABC_MB	Set PWM minimum duty.	
0xB5	00h	R/W	[7:0]	PWM_PRD	PWM_FRQ_B for PWM period generation.	
0xB6	02h	R/W	[2:0]	PWM_DIV	PWM_FRQ_A for PWM period generation.	
0xB7	00h	R/W	[7:0]	DBG0	Back Light gain at frame average 0.	
0xB8	20h	R/W	[7:0]	DBG1	Back Light gain at frame average 32.	
0xB9	40h	R/W	[7:0]	DBG2	Back Light gain at frame average 64.	
0xBA	60h	R/W	[7:0]	DBG3	Back Light gain at frame average 96.	
0xBB	80h	R/W	[7:0]	DBG4	Back Light gain at frame average 128.	
0xBC	A0h	R/W	[7:0]	DBG5	Back Light gain at frame average 160.	
0xBD	C0h	R/W	[7:0]	DBG6	Back Light gain at frame average 192.	
0xBE	E0h	R/W	[7:0]	DBG7	Back Light gain at frame average 224.	

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MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB5	55h	R/W	[6:4]	RT3_M	Lane 3 termination resistance control for Master.	15
			[2:0]	RT2_M	Lane 2 termination resistance control for Master.	
0xB6	55h	R/W	[6:4]	RT1_M	Lane 1 termination resistance control for Master.	
			[2:0]	RT0_M	Lane 0 termination resistance control for Master.	
0xB7	05h	R/W	[2:0]	RTC_M	Clock Lane termination resistance control for Master.	

8.4.7 Register of page 6 (Engineer)

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB8	5Ah	R/W	[7:0]	ENGINEER_PWD	engineer_pwd=8'hA5 enable engineer mode.	-
0xC0	5Ah	R/W	[7:0]	Function_EN	Function_EN=8'hA5 enable in-house function.	-
0xC2	23h	R/W	[7:6]	SD_OPM	Source slew rate adjust.	23
0xC6	69h	R/W	[1:0]	IBSEL	Whole chip bias adjust.	24
0xD5	32h	R/W	[7:0]	GOE_WD	Source delay time adjustment **(CKV falling to Source off)**.	30

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8.4.8 Register of digital gamma

Page 7/A: Red positive/negative gamma
Page 8/B: Green positive/negative gamma
Page 9/C: Blue positive/negative gamma

MIPI Addr[7:0]	Default[7:0]	Read/ Write	D[7:0]	Bit name	Description	OTP group
0xB1	00h/00h/00h	R/W	[7:0]	DGMA1[7:0]	Digital gamma V0 reference.	
0xB2	04h/04h/03h	R/W	[7:0]	DGMA2[7:0]	Digital gamma V1 reference.	
0xB3	0Bh/0Bh/09h	R/W	[7:0]	DGMA3[7:0]	Digital gamma V3 reference.	
0xB4	19h/19h/16h	R/W	[7:0]	DGMA4[7:0]	Digital gamma V7 reference.	
0xB5	28h/27h/23h	R/W	[7:0]	DGMA5[7:0]	Digital gamma V11 reference.	
0xB6	37h/35h/2Fh	R/W	[7:0]	DGMA6[7:0]	Digital gamma V15 reference.	
0xB7	55h/51h/48h	R/W	[7:0]	DGMA7[7:0]	Digital gamma V23 reference.	
0xB8	71h/6Ah/67h	R/W	[7:0]	DGMA8[7:0]	Digital gamma V31 reference.	
0xB9	AEh/A4h/A6h	R/W	[7:0]	DGMA9[7:0]	Digital gamma V47 reference.	
0xBA	F1h/E5h/E9h	R/W	[7:0]	DGMA10[7:0]	Digital gamma V63 reference.	
0xBB	7Ah/66h/74h	R/W	[7:0]	DGMA11[7:0]	Digital gamma V95 reference.	
0xBC	F9h/E0h/F0h	R/W	[7:0]	DGMA12[7:0]	Digital gamma V127 reference.	
0xBD	FDh/E4h/F4h	R/W	[7:0]	DGMA13[7:0]	Digital gamma V128 reference.	
0xBE	7Ah/5Bh/70h	R/W	[7:0]	DGMA14[7:0]	Digital gamma V160 reference.	
0xBF	FBh/D7h/ECh	R/W	[7:0]	DGMA15[7:0]	Digital gamma V192 reference.	
0xC0	3Bh/14h/29h	R/W	[7:0]	DGMA16[7:0]	Digital gamma V208 reference.	
0xC1	7Eh/52h/67h	R/W	[7:0]	DGMA17[7:0]	Digital gamma V224 reference.	
0xC2	9Fh/71h/84h	R/W	[7:0]	DGMA18[7:0]	Digital gamma V232 reference.	
0xC3	C0h/90h/A1h	R/W	[7:0]	DGMA19[7:0]	Digital gamma V240 reference.	
0xC4	D1h/A0h/B1h	R/W	[7:0]	DGMA20[7:0]	Digital gamma V244 reference.	
0xC5	E0h/AFh/C0h	R/W	[7:0]	DGMA21[7:0]	Digital gamma V248 reference.	
0xC6	F1h/C0h/CCh	R/W	[7:0]	DGMA22[7:0]	Digital gamma V252 reference.	31**(R)**
0xC7	F8h/C7h/D4h	R/W	[7:0]	DGMA23[7:0]	Digital gamma V254 reference.	32**(G)**
0xC8	FCh/CCh/D8h	R/W	[7:0]	DGMA24[7:0]	Digital gamma V255 reference.	33**(B)**
0xC9	00h/00h/00h	R/W	[7:6]	DGMA1[9:8]	Digital gamma V0 reference.	34**(R)**
		R/W	[5:4]	DGMA2[9:8]	Digital gamma V1 reference.	35**(G)**
		R/W	[3:2]	DGMA3[9:8]	Digital gamma V3 reference.	36**(B)**
		R/W	[1:0]	DGMA4[9:8]	Digital gamma V7 reference.	
0xCA	00h/00h/00h	R/W	[7:6]	DGMA5[9:8]	Digital gamma V11 reference.	
		R/W	[5:4]	DGMA6[9:8]	Digital gamma V15 reference.	
		R/W	[3:2]	DGMA7[9:8]	Digital gamma V23 reference.	
		R/W	[1:0]	DGMA8[9:8]	Digital gamma V31 reference.	
0xCB	05h/05h/05h	R/W	[7:6]	DGMA9[9:8]	Digital gamma V47 reference.	
		R/W	[5:4]	DGMA10[9:8]	Digital gamma V63 reference.	
		R/W	[3:2]	DGMA11[9:8]	Digital gamma V95 reference.	
		R/W	[1:0]	DGMA12[9:8]	Digital gamma V127 reference.	
0xCC	6Bh/6Bh/6Bh	R/W	[7:6]	DGMA13[9:8]	Digital gamma V128 reference.	
		R/W	[5:4]	DGMA14[9:8]	Digital gamma V160 reference.	
		R/W	[3:2]	DGMA15[9:8]	Digital gamma V192 reference.	
		R/W	[1:0]	DGMA16[9:8]	Digital gamma V208 reference.	
0xCD	FFh/FFh/FFh	R/W	[7:6]	DGMA17[9:8]	Digital gamma V224 reference.	
		R/W	[5:4]	DGMA18[9:8]	Digital gamma V232 reference.	
		R/W	[3:2]	DGMA19[9:8]	Digital gamma V240 reference.	
		R/W	[1:0]	DGMA20[9:8]	Digital gamma V244 reference.	
0xCE	FFh/FFh/FFh	R/W	[7:6]	DGMA21[9:8]	Digital gamma V248 reference.	
		R/W	[5:4]	DGMA22[9:8]	Digital gamma V252 reference.	
		R/W	[3:2]	DGMA23[9:8]	Digital gamma V254 reference.	
		R/W	[1:0]	DGMA24[9:8]	Digital gamma V255 rReference.	

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9. Register Description

9.1 Registers of page 0

Address [7:0]	Default [7:0]	Data[7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xB1	00h	0	0	0	0	0	0	0	0	VENDER_ID	The register defines vender ID for customer.	V
0xB2	02h	-	-	-	0	-	-	-	-	LR_XOR	H-scan direction control. 0: Left to right. 1: Right to left.	V
		-	-	-	-	0	-	-	-	ZIGZAG_SEL	Panel type selection. 0: Stripe. 1: Zigzag.	V
		-	-	-	-	-	-	1	-	NBW_SEL	0: Normal White. 1: Normal Black.	V
		-	-	-	-	-	-	-	0	BIST	BIST mode control **(To pin BIST_EN do XOR operation)**.	V

Note: (1)

Combination logic	Truth table PIN	REG	To TCON	To TCON
BIST_EN (Pin), BIST (Reg) → XOR → To TCON BIST	0	0	0	1: BIST mode. 0: Normal mode.
	0	1	1	
	1	0	1	
	1	1	0	

0xB3	03h	0	0	-	-	-	-	-	-	ZTYPE_SEL	Zigzag type slection **(Panel type please refer to section 6.3)** 0: Zigzag type 0. 1: Zigzag type 1. 2: Zigzag type 2. 3: Zigzag type 3.	V
		-	-	0	0	-	-	-	-	PWRMD_XOR	POWR ON mode. **(To pin PWRMD[1:0] do XOR operation)**. PWRMD[1:0] to TCON: **PWRMD [1:0]** / **Driving mode** 00 – Support HX5186-C power mode. 01 – Suppot PFM circuit power mode. 10 – External VSP, VSN, VGH, VGL power mode. 11 – External VSP, VSN power mode.	V
		-	-	-	-	0	-	-	-	VRES_FIX	Display vertical Line decided by. 0: VRES[8:0]. 1: RES[2:0].	V
		-	-	-	-	-	0	1	1	RES	Resolution selection **(Please refer to section 6.1 for resolution)**.	V

Note: (2)

Combination logic	Truth table PIN/MIPI	REG	To TCON	To TCON
PWRMD1 (Pin), RWRMD1 (Reg) → XOR → To TCON PWRMD1	0	0	0	00b: Support HX5186-C power mode. 01b: Support PFM circuit power mode. 10b: External VSP, VSN, VGH, VGL power mode. 11b: External VSP, VSN power mode. **(Default)**
	0	1	1	
	1	0	1 **(Default)**	
	1	1	0	
PWRMD0 (Pin), RWRMD0 (Reg) → XOR → To TCON PWRMD0	0	0	0	
	0	1	1	
	1	0	1 **(Default)**	
	1	1	0	

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Address [7:0]	Default [7:0]	Data[7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xB4	F0h	1	1	1	1	0	0	0	0	VRES[7:0]	When VRES_FIX is set Low, the veritical display line decide by this register. The VRES range is 80 ~ 320 and step is 8H. The max veritical display line must be less than V-display. e.g. RES[3:0] is set 1280RGBx2048, VRES[8:0] < 0x100 If VRES[8:0]=0x0F0, resolution is 1280RGBx1920.	V
0xB5	A5h	1	0	1	0	0	1	0	1	ZDATA	ZigZag dummy data select.	V
0xB6	00h	0	-	-	-	-	-	-	-	VRES[8]	Vertical resolution selection **(With RB4h[7:0] at Page.0)**.	V
		-	-	0	-	-	-	-	-	ZZ_2PL	Zigzag 2H panel type selection. 0: 1H zigzag panel. 1: 2H zigzag panel.	V
		-	-	-	0	-	-	-	-	RB_SWAP	RB data swap enable. 0: Non-swap. 1: Swap.	V
		-	-	-	-	0	0	-	-	CABC_CTRL	CABC Mode selection. 00: Bypass Mode. 01: Still-Mode. 10: UI-Mode. 11: Moving-Mode.	V
		-	-	-	-	-	-	0	-	DITHER_EN	Dithering enable selection. 0: Disable. 1: Enable.	V
		-	-	-	-	-	-	-	0	D_GAM_EN	Digital gamma enable selection. 0: Disable. 1: Enable.	V
0xB7	02h	-	-	-	-	-	-	1	-	BC_CTRL	Back light control output at GPO pin.	V
		-	-	-	-	-	-	-	0	CASCADE_XOR(2)	CASCADE mode **(To pin CASCADE _ENB do XOR operation)**. CASCADE _ENB to TCON: **CASCADE _ENB** / **CASCADE mode** 0 / CASCADE 1 / Single.	V

Note: (3)

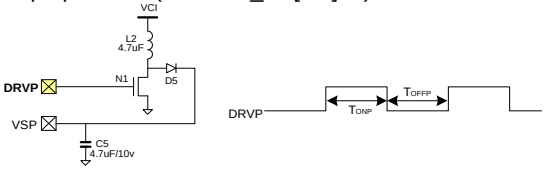
Combination Logic	Truth table PIN	REG	To TCON	To TCON
CASCADE_ENB (Pin), CASCADE_XOR (Reg) → XOR → To TCON CASCADE	0	0	0	1: Single. **(Default)** 0 : CASCADE
	0	1	1	
	1	0	1	
	1	1	0	

0xB8	C0h	1	-	-	-	-	-	-	-	GOA_EN	0: GOA disable. 1: GOA enable.	V
		-	1	-	-	-	-	-	-	VCOM_EN	0: VCOM disable. 1: VCOM enable.	V
		-	-	0	-	-	-	-	-	PNSW_XOR	MIPI P-N Swap **(To pin PNSW do XOR operation)**. PNSW to TCON: **PNSW** / **P-N swap** 0 / P-N swap. 1 / P-N normal. **(Default)**	V
		-	-	-	-	0	0	-	-	LNSW_XOR	MIPI Lane Swap. **(To pin LNSW do XOR operation)** LNSW to TCON **(Please refer to section 6.5 for Lane swap)**.	V
		-	-	-	-	-	-	0	0	LANSEL_XOR	MIPI lane selection **(To pin LANSEL do XOR operation)**. LANSEL to TCON: **LANSEL [1:0]** / **MIPI lane** 00 / Reserved. 01 / 2 Lane. 10 / Reserved. 11 / 4 Lane. **(Default)**	V

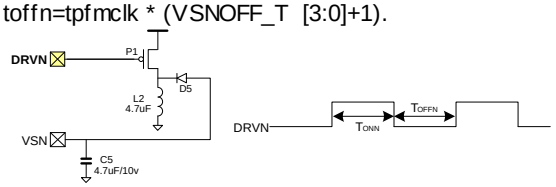
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<table>
<tr><th>Address [7:0]</th><th>Default [7:0]</th><th colspan="8">Data[7:0]
7 6 5 4 3 2 1 0</th><th>Name</th><th>Description</th><th>OTP</th></tr>
<tr><td rowspan="4">0xBA</td><td rowspan="4">43h</td><td>0</td><td>1</td><td>0</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>BLREV</td><td>Source output at V-blanking.
<table><tr><th>BLREV [2:0]</th><th>SD output at V-blanking</th></tr><tr><td>000</td><td>SD keep output the last line.</td></tr><tr><td>001</td><td>Hi-Z.</td></tr><tr><td>010</td><td>GND.</td></tr><tr><td>011</td><td>SD keep output and POL keep toggle.</td></tr><tr><td>100</td><td>SD keep output the last line but POL inversion.</td></tr></table></td><td>V</td></tr>
<tr><td>-</td><td>-</td><td>-</td><td>0</td><td>-</td><td>-</td><td>-</td><td>-</td><td>BLREVONOFF</td><td>Source output at power on & off.
0: GND.
1: Hi-Z.</td><td>V</td></tr>
<tr><td>-</td><td>-</td><td>-</td><td>-</td><td>0</td><td>-</td><td>-</td><td>-</td><td>VDD_OTP_SW</td><td>VDD_OTP power path switch selection.
0: Off.
1: On.</td><td>V</td></tr>
<tr><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>0</td><td>1</td><td>1</td><td>INV_SEL</td><td>POL inversion setting.
<table><tr><th>INV_SEL [2:0]</th><th>POL inversion</th></tr><tr><td>000</td><td>Line inversion.</td></tr><tr><td>001</td><td>1+2 Line inversion.</td></tr><tr><td>010</td><td>2 Line inversion.</td></tr><tr><td>011</td><td>Column inversion.</td></tr><tr><td>101</td><td>4 Line inversion.</td></tr></table></td><td>V</td></tr>
<tr><td>0xBB</td><td>67h</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>VCOMS</td><td>VCOM voltage setting (Step=10mV).
VCOM= -0.2 - 0.01*VCOMS[7:0].
<table><tr><th>VCOMS[7:0]</th><th>VCOM voltage</th></tr><tr><td>00000000</td><td>-0.2V</td></tr><tr><td>00000001</td><td>-0.21V</td></tr><tr><td>00000010</td><td>-0.22V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>01100110</td><td>-1.23V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>11111111</td><td>-2.75V</td></tr></table></td><td>V</td></tr>
<tr><td>0xBC</td><td>0Ah</td><td>-</td><td>-</td><td>-</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>VSPS</td><td>VSP voltage setting (Step=0.1V).
VSP=4.0 + 0.1*VSPS[4:0].
<table><tr><th>VSPS [4:0]</th><th>VSP voltage</th></tr><tr><td>00000</td><td>4.0V</td></tr><tr><td>00001</td><td>4.1V</td></tr><tr><td>00010</td><td>4.2V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>01010</td><td>5.0V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>10111</td><td>6.3V (Saturation)</td></tr></table></td><td>V</td></tr>
<tr><td>0xBD</td><td>4Ah</td><td>0</td><td>1</td><td>0</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>VDD_OTPOS</td><td>VDD_OTP voltage setting (Step=0.1V).
VDD_OTP = 8.3 + 0.1*VDD_OTPOS[2:0].
<table><tr><th>VDD_OTPOS [2:0]</th><th>VDD_OTP voltage</th></tr><tr><td>000</td><td>8.3V</td></tr><tr><td>001</td><td>8.4V</td></tr><tr><td>010</td><td>8.5V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111</td><td>9.0V</td></tr></table></td><td>V</td></tr>
</table>

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<table>
<tr><th>Address [7:0]</th><th>Default [7:0]</th><th colspan="8">Data[7:0]
7 6 5 4 3 2 1 0</th><th>Name</th><th>Description</th><th>OTP</th></tr>
<tr><td></td><td></td><td>- - - 0 1 0 1 0</td><td>VSNS</td><td>VSN voltage setting (Step=0.1V).
VSN=-4.0 - 0.1*VSNS[4:0].
<table><tr><th>VSNS [4:0]</th><th>VSN voltage</th></tr><tr><td>00000</td><td>-4.0V</td></tr><tr><td>00001</td><td>-4.1V</td></tr><tr><td>00010</td><td>-4.2V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>01010</td><td>-5.0V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>10111</td><td>-6.3V (Saturation)</td></tr></table></td><td>V</td></tr>
<tr><td>0xBE</td><td>23h</td><td>- - 1 0 0 0 1 1</td><td>VGHS</td><td>VGH voltage setting (Step=0.2V).
VGH=8.0 + 0.2*VGHS[5:0].
<table><tr><th>VGHS [5:0]</th><th>VGH voltage</th></tr><tr><td>000000</td><td>8.0V</td></tr><tr><td>000001</td><td>8.2V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>100011</td><td>15.0V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>110111</td><td>19.0V (Saturation)</td></tr></table></td><td>V</td></tr>
<tr><td>0xBF</td><td>14h</td><td>- - 0 1 0 1 0 0</td><td>VGLS</td><td>VGL voltage setting (Step=0.2V).
VGL=-6.0 - 0.2*VGLS[5:0].
<table><tr><th>VGLS [5:0]</th><th>VGL voltage</th></tr><tr><td>000000</td><td>-6.0V</td></tr><tr><td>000001</td><td>-6.2V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>010100</td><td>-10.0V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>110010</td><td>-16.0V (Saturation)</td></tr></table></td><td>V</td></tr>
<tr><td rowspan="3">0xC0</td><td rowspan="3">14h</td><td>0 - - - - - - -</td><td>VDD_OTP_EN</td><td>VDD_OTP power enable selection.
0: Disable.
1: Enable.</td><td>V</td></tr>
<tr><td>- 0 - - - - - -</td><td>VGLO_EN</td><td>VGLO2 enable selection.
0: Disable.
1: Enable.</td><td>V</td></tr>
<tr><td>- - 0 1 0 1 0 0</td><td>VGLOS</td><td>VGLO2 voltage setting (Step=0.2V).
VGLO2= -15.0 + (63 - VGLOS[5:0])*0.2.
<table><tr><th>VGLOS [5:0]</th><th>VGLO2 voltage</th></tr><tr><td>000000</td><td>-5.0V</td></tr><tr><td>….</td><td>-5.0V</td></tr><tr><td>001101</td><td>-5.0V (Max.)</td></tr><tr><td>001110</td><td>-5.2V</td></tr><tr><td>001111</td><td>-5.4V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>010100</td><td>-6.4V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111111</td><td>-15.0V (Min.)</td></tr></table></td><td>V</td></tr>
<tr><td rowspan="2">0xC1</td><td rowspan="2">66h</td><td>0 1 1 0 - - - -</td><td>VSPON_T</td><td>Set PFM TON duty for VSP voltage.
tonp=tpfmclk * (VSPON_T [3:0]+1).</td><td>V</td></tr>
<tr><td>- - - - 0 1 1 0</td><td>VSPOFF_T</td><td>Set PFM TOFF duty for VSP voltage.
toffp=tpfmclk * (VSPOFF_T [3:0]+1).
 
</td><td>V</td></tr>
</table>

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Address [7:0]	Default [7:0]	Data[7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xC2	66h	0	1	1	0	-	-	-	-	VSNON_T	Set PFM TON duty for VSN voltage. tonn=tpfmclk * (VSNON_T [3:0]+1).	V
		-	-	-	-	0	1	1	0	VSNOFF_T	Set PFM TOFF duty for VSN voltage. toffn=tpfmclk * (VSNOFF_T [3:0]+1). 	V
0xC3	61h	0	-	-	-	-	-	-	-	VCL_cpctrl[2]	VCL charge pump clock frequency.	V
		-	1	-	-	-	-	-	-	VGLXSP	VGL boost function selection. 0: VSN-VSP. 1: 2VSN-VSP **(Default)**.	V
		-	-	1	0	-	-	-	-	VGHXSP	VGH boost function selection. 00: 2VSP. 01: 2VSP+|VSN| **(Default)**. 1x: 2VSP+2|VSN|.	V
		-	-	-	-	0	-	-	-	PFMFREN	Enable PFM frequency randomizer of VSP and VSN.	V
		-	-	-	-	-	0	-	-	tontoff_offset	PFM ton/toff offset.	V
		-	-	-	-	-	-	0	1	dpfm_clk	DPFM clock selection **(1/tpfmclk)**. 00: 6.25MHz. 01: 12.5MHz **(Default)**. 1x: 25MHz.	V
0xC5	00h	0	0	-	-	-	-	-	-	VCL_cpctrl[1:0]	VCL/VGL/VGH charge pump clock frequency.	V
		-	-	0	0	0	-	-	-	VGL_cpctrl[2:0]		V
		-	-	-	-	-	0	0	0	VGH_cpctrl[2:0]		V
0xC6	ADh	1	0	-	-	-	-	-	-	cgpp_inv	Set DRVN/DRVP polarity when PWRMD[1:0]=00b. **(HX5186-C)** 0: Non-invert. 1: Invert. **(cgpp_inv=1: Set the DRVN signal. cgpp_inv=0: Set the DRVP signal.)**	V
		-	-	1	-	-	-	-	-	cgpp_soft_en	Enable charge pump soft start when PWRMD[1:0]=00b. **(HX5186-C)** 0: Disable. 1: Enable.	V
0xC6	ADh	-	-	-	0	1	1	-	-	cgpp_clksel	Charge Pump clock select when PWRMD[1:0]=00b. **(HX5186-C)**	V

cpctrl[2:0]	Charge pump clock frequency	Unit
000	X2 **(Default)**	Line
001	X4	Line
010	X8	Line
011	Reserve	Line
100	100	KHz
101	300	KHz
110	500	KHz
111	700	KHz

clksel[2:0]	Frequency
000	403 KHz
001	595 KHz
010	781 KHz
011	963 KHz **(Default)**
100	1136 KHz
101	1389 KHz
110	1786 KHz
111	2083 KHz

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<table>
<tr><th>Address [7:0]</th><th>Default [7:0]</th><th colspan="8">Data[7:0]
7 6 5 4 3 2 1 0</th><th>Name</th><th>Description</th><th>OTP</th></tr>
<tr><td></td><td></td><td>- - - - - - 0 1</td><td>cgpp_cmd_sel</td><td>Charge Pump command select when PWRMD[1:0]=00b (HX5186-C).
<table><tr><th>cmd_sel[1:0]</th><th>Pump command</th></tr><tr><td>00</td><td>X1.5</td></tr><tr><td>01</td><td>X2 (Default)</td></tr><tr><td>10</td><td>X3</td></tr><tr><td>11</td><td>X3</td></tr></table></td><td>V</td></tr>
<tr><td>0xCB</td><td>1Ch</td><td>- - 0 1 1 1 0 0</td><td>VGMHP</td><td>VGMHP voltage setting (Step=0.05V):
VGMHP=3.05 + 0.05* VGMHP [5:0]
<table><tr><th>VGMHP[5:0]</th><th>VGMHP voltage</th></tr><tr><td>000000</td><td>3.05V</td></tr><tr><td>000001</td><td>3.1V</td></tr><tr><td>000010</td><td>3.15V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>011100</td><td>4.45V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111111</td><td>6.2V</td></tr></table></td><td>V</td></tr>
<tr><td>CC</td><td>04h</td><td>- - 0 0 0 1 0 0</td><td>VGMLP</td><td>VGMLP voltage setting (Step=0.05V).
VGMLP=0 + 0.05* VGMLP [5:0].
<table><tr><th>VGMLP[5:0]</th><th>VGMLP voltage</th></tr><tr><td>000000</td><td>0V</td></tr><tr><td>000001</td><td>0.05V</td></tr><tr><td>000010</td><td>0.1V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>000100</td><td>0.2V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111111</td><td>3.15V</td></tr></table></td><td>V</td></tr>
<tr><td>CD</td><td>1Ch</td><td>- - 0 1 1 1 0 0</td><td>VGMHN</td><td>VGMHN voltage setting (Step=-0.05V):
VGMHN= -3.05 - 0.05* VGMHN [5:0]
<table><tr><th>VGMHN[5:0]</th><th>VGMHN voltage</th></tr><tr><td>000000</td><td>-3.05V</td></tr><tr><td>000001</td><td>-3.1V</td></tr><tr><td>000010</td><td>-3.15V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>010111</td><td>-4.45V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111111</td><td>-6.2V</td></tr></table></td><td>V</td></tr>
<tr><td>CE</td><td>04h</td><td>- - 0 0 0 1 0 0</td><td>VGMLN</td><td>VGMLN voltage setting (Step= -0.05V):
VGMLN = 0 - 0.05* VGMLN [5:0]
<table><tr><th>VGMLN[5:0]</th><th>VGMLN voltage</th></tr><tr><td>000000</td><td>0V</td></tr><tr><td>000001</td><td>-0.05V</td></tr><tr><td>000010</td><td>-0.1V</td></tr><tr><td>….</td><td>….</td></tr><tr><td>000100</td><td>-0.2V (Default)</td></tr><tr><td>….</td><td>….</td></tr><tr><td>111111</td><td>-3.15V</td></tr></table></td><td>V</td></tr>
<tr><td>0xCF</td><td>00h</td><td>- 0 0 - - - - -</td><td>POCSD_CTL</td><td>Source output offset cancel method selection.
POCSD_CTL=00b
<table><tr><th></th><th>L1</th><th>L2</th><th>L3</th><th>L4</th><th>L5</th><th>L6</th><th>L7</th><th>L8</th></tr><tr><td>Frame1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>Frame2</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Frame3</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>Frame4</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Frame5</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>Frame6</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>POCSD_CTL=01b
<table><tr><th></th><th>L1</th><th>L2</th><th>L3</th><th>L4</th><th>L5</th><th>L6</th><th>L7</th><th>L8</th></tr><tr><td>Frame1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Frame2</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Frame3</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Frame4</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Frame5</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>Frame6</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>POCSD_CTL=10b
<table><tr><th></th><th>L1</th><th>L2</th><th>L3</th><th>L4</th><th>L5</th><th>L6</th><th>L7</th><th>L8</th></tr><tr><td>Frame1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Frame2</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Frame3</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Frame4</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>Frame5</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Frame6</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>POCSD_CTL=11b
<table><tr><th></th><th>L1</th><th>L2</th><th>L3</th><th>L4</th><th>L5</th><th>L6</th><th>L7</th><th>L8</th></tr><tr><td>Frame1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Frame2</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Frame3</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Frame4</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Frame5</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Frame6</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table></td><td>V</td></tr>
</table>

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<table>
<tr><th>Address [7:0]</th><th>Default [7:0]</th><th colspan="8">Data[7:0]
7 6 5 4 3 2 1 0</th><th>Name</th><th>Description</th><th>OTP</th></tr>
<tr><td></td><td></td><td>- - - 0 0 0 0 0</td><td>SD_EQW</td><td>Source EQ time setting.
tEQ=SD_EQW[4:0]* 4x(1/OSC25M)
(SD_EQW [4:0] minimum is 0x03h)</td><td>V</td></tr>
<tr><td rowspan="2">0xD0</td><td rowspan="2">00h</td><td>- 0 0 0 - - - -</td><td>ISSEL_N_CAS</td><td rowspan="2">Source P/N Bias current when CASCADE mode
<table><tr><th>ISSEL [2:0]</th><th>SD Bias currecnt</th></tr><tr><td>000</td><td>200%(Default)</td></tr><tr><td>001</td><td>150%</td></tr><tr><td>010</td><td>120%</td></tr><tr><td>011</td><td>100%</td></tr><tr><td>100</td><td>86%</td></tr><tr><td>101</td><td>75%</td></tr><tr><td>110</td><td>50%</td></tr><tr><td>111</td><td>38%</td></tr></table></td><td>V</td></tr>
<tr><td>- - - - - 0 0 0</td><td>ISSEL_P_CAS</td><td>V</td></tr>
<tr><td rowspan="2">0xD1</td><td rowspan="2">44h</td><td>- 1 0 0 - - - -</td><td>ISSEL_N_NOTCAS</td><td rowspan="2">Source P/N Bias current when NON-CASCADE mode.
<table><tr><th>ISSEL [2:0]</th><th>SD Bias currecnt</th></tr><tr><td>000</td><td>200%</td></tr><tr><td>001</td><td>150%</td></tr><tr><td>010</td><td>120%</td></tr><tr><td>011</td><td>100%</td></tr><tr><td>100</td><td>86% (Default)</td></tr><tr><td>101</td><td>75%</td></tr><tr><td>110</td><td>50%</td></tr><tr><td>111</td><td>38%</td></tr></table></td><td>V</td></tr>
<tr><td>- - - - - 1 0 0</td><td>ISSEL_P_NOTCAS</td><td>V</td></tr>
<tr><td rowspan="2">0xD2</td><td rowspan="2">0Ah</td><td>- - - - 1 0 - -</td><td>T_VBP</td><td rowspan="2">GAMMA P/N Bias current selection.
<table><tr><th>T_VBP/N [1:0]</th><th>GAMMA Bias currecnt</th></tr><tr><td>00</td><td>80%</td></tr><tr><td>01</td><td>100%</td></tr><tr><td>10</td><td>120% (Default)</td></tr><tr><td>11</td><td>140%</td></tr></table></td><td>V</td></tr>
<tr><td>- - - - - - 1 0</td><td>T_VBN</td><td>V</td></tr>
<tr><td rowspan="2">0xF0</td><td rowspan="2">92h</td><td>1 0 0 1 - - - -</td><td>GPO1_SEL</td><td rowspan="10">GPO0~8 output signal selection.
<table><tr><th>GPO_SEL [3:0]</th><th>Output signal</th></tr><tr><td>0000</td><td>GND</td></tr><tr><td>0001</td><td>MIPI_TE</td></tr><tr><td>0010</td><td>BC_CTRL</td></tr><tr><td>0011</td><td>DIMO</td></tr><tr><td>1001</td><td>TP_SYNC</td></tr></table></td><td>V</td></tr>
<tr><td>- - - - 0 0 1 0</td><td>GPO0_SEL</td><td>V</td></tr>
<tr><td rowspan="2">0xF1</td><td rowspan="2">23h</td><td>0 0 1 0 - - - -</td><td>GPO3_SEL</td><td>V</td></tr>
<tr><td>- - - - 0 0 1 1</td><td>GPO2_SEL</td><td>V</td></tr>
<tr><td rowspan="2">0xF2</td><td rowspan="2">00h</td><td>0 0 0 0 - - - -</td><td>GPO5_SEL</td><td>V</td></tr>
<tr><td>- - - - 0 0 0 0</td><td>GPO4_SEL</td><td>V</td></tr>
<tr><td rowspan="2">0xF3</td><td rowspan="2">00h</td><td>0 0 0 0 - - - -</td><td>GPO7_SEL</td><td>V</td></tr>
<tr><td>- - - - 0 0 0 0</td><td>GPO6_SEL</td><td>V</td></tr>
<tr><td>0xF4</td><td>00h</td><td>- - - - 0 0 0 0</td><td>GPO8_SEL</td><td>V</td></tr>
<tr><td>0xF5</td><td>00h</td><td>- - - - - - - 0</td><td>cta_en</td><td>CTA function enable selection.
0: Disable.
1: Enable.</td><td>V</td></tr>
<tr><td>0xF6</td><td>FFh</td><td>1 1 1 1 1 1 1 1</td><td>cta_gain_r</td><td rowspan="3">CTA R/G/B gain setting.:
R/G/B output data=(R/G/B pixel data *(256+cta_gain[7:0])/512))</td><td>V</td></tr>
<tr><td>0xF7</td><td>FFh</td><td>1 1 1 1 1 1 1 1</td><td>cta_gain_g</td><td>V</td></tr>
<tr><td>0xF8</td><td>FFh</td><td>1 1 1 1 1 1 1 1</td><td>cta_gain_b</td><td>V</td></tr>
</table>

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9.2 Registers of page 1

Address [7:0]	Default [7:0]	Data [7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xB1	00h	-	-	0	0	0	0	0	0	OTP_Group	OTP trimming group selection. The group range is from Group.0 to Group.44.	-
0xB2	5Ah	0	1	0	1	1	0	1	0	OTP_pwd	Set to 0xA5 to enable 0xB3 command.	-
0xB3	00h	-	-	-	-	-	0	-	-	OTP_Re_Load	OTP auto re-load control.	-
		-	-	-	-	-	-	0	-	OTP_RD	OTP read control.	-
		-	-	-	-	-	-	-	0	OTP_WR	OTP write control.	-
0xB4	00h	-	-	-	-	-	-	0	0	OTP_ADDR[9:8]	Set OTP address [9:8].	-
0xB5	00h	0	0	0	0	0	0	0	0	OTP_ADDR[7:0]	Set OTP address [7:0].	-
0xB6	00h	0	0	0	0	0	0	0	0	OTP_PDOB	Read data from OTP function.	-
(Please reference "10.3 OTP function" for OTP register function.)												
0xC0 0xC1 0xC2 0xC3 0xC4	25h 25h 02h 02h 00h	0	0	-	-	-	-	-	-	GOUTL[22:1]_STBYB	Set GOUTL_[22:1] signal voltage at standby mode. 00: VGL. 01: VGH. 10: GND. 11: Reserved.	V
0xC5 0xC6 0xC7 0xC8 0xC9 0xCA 0xCB 0xCC 0xCD 0xCE 0xCF 0xD0 0xD1 0xD2 0xD3 0xD4 0xD5	00h 00h 25h 08h 08h 06h 06h 0Ch 0Ch 0Ah 0Ah 00h 00h 00h 00h 00h 00h	-	-	0	0	0	0	0	0	GOUTL_[22:1]_SEL	Mux GOA signal to GOUTL pin.	V
0xD6 0xD7 0xD8 0xD9 0xDA	25h 25h 01h 01h 00h	0	0	-	-	-	-	-	-	GOUTR[22:1]_STBYB	Set GOUTR_[22:1] signal voltage at standby mode. 00: VGL. 01: VGH. 10: GND. 11: Reserved.	V
0xDB 0xDC 0xDD 0xDE 0xDF 0xE0 0xE1 0xE2 0xE3 0xE4 0xE5 0xE6 0xE7 0XE8 0xE9 0xEA 0xEB	00h 00h 25h 07h 07h 05h 05h 0Bh 0Bh 09h 09h 00h 00h 00h 00h 00h 00h	-	-	0	0	0	0	0	0	GOUTR_[22:1]_SEL	Mux GOA signal to GOUTR pin.	V

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Address [7:0]	Default [7:0]	Data [7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP

SEL[5:0]	GOA output
0x00	VGL
0x01	STV1
0x02	STV2
0x03	STV3
0x04	STV4
0x05	CKV1
0x06	CKV2
0x07	CKV3
0x08	CKV4
0x09	CKV5

SEL[5:0]	GOA output
0x0A	CKV6
0x0B	CKV7
0x0C	CKV8
0x0D	CLR1
0x0E	CLR2
0x0F	CLR3
0x10	CLR4
0x11	FLCA
0x12	FLCB
0x13	GCKA

SEL[5:0]	GOA output
0x14	GCKB
0x15	DIR
0x16	DIRB
0x17	STV5
0x18	STV6
0x19	STV7
0x1A	STV8
0x1B	CKV9
0x1C	CKV10
0x1D	CKV11

SEL[5:0]	GOA output
0x1E	CKV12
0x1F	CKV13
0x20	CKV14
0x21	CKV15
0x22	CKV16
0x23	FLC1B
0x24	FLC2B
0X25	VGH
other	VGH

0xEC	C0h	-	-	0	-	-	-	-	-	GOA_HZ_EN_L	GOUTL/R[22:1] "Hi-Z" enable selection. 0: Disable. 1: Enable.	V
		-	-	-	0	-	-	-	-	GOA_HZ_EN_R		

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9.3 Registers of page 2

Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
0xC0	00h	-	-	-	0	0	0	0	0	T_PVP[0]	Control 0st Positive gamma op's input voltage.	
0xC1	00h	-	-	-	0	0	0	0	0	T_PVP[1]	Control 1st Positive gamma op's input voltage.	
0xC2	0Fh	-	-	0	0	0	0	0	0	T_PVP[2:19]	Control 2~19st Positive gamma op's input voltage.	V
0xC3	17h											
0xC4	29h											
0xC5	27h											
0xC6	28h											
0xC7	28h											
0xC8	26h											
0xC9	23h											
0xCA	28h											
0xCB	27h											
0xCC	28h											
0xCD	25h											
0xCE	2Fh											
0xCF	2Dh											
0xD0	32h											
0xD1	32h											
0xD2	34h											
0xD3	32h											
0xD4	3Ah	-	-	-	0	0	0	0	0	T_PVP[20]	Control 20st Positive gamma op's input voltage.	
0xD5	1Fh	-	-	-	0	0	0	0	0	T_PVP[21]	Control 21st Positive gamma op's input voltage.	
0xD6	00h	-	-	-	0	0	0	0	0	T_PVN[0]	Control 0st Negative gamma op's input voltage.	
0xD7	00h	-	-	-	0	0	0	0	0	T_PVN[1]	Control 1st Negative gamma op's input voltage.	
0xD8	0Fh	-	-	0	0	0	0	0	0	T_PVN[2:19]	Control 2~19st Negative gamma op's input voltage.	V
0xD9	17h											
0xDA	29h											
0xDB	23h											
0xDC	2Bh											
0xDD	2Ch											
0xDE	2Ah											
0xDF	25h											
0xE0	28h											
0xE1	27h											
0xE2	34h											
0xE3	33h											
0xE4	3Dh											
0xE5	39h											
0xE6	3Eh											
0xE7	36h											
0XE8	34h											
0xE9	32h											
0xEA	3Ah	-	-	-	0	0	0	0	0	T_PVN[20]	Control 20st Negative gamma op's input voltage.	
0xEB	1Fh	-	-	-	0	0	0	0	0	T_PVN[21]	Control 21st Negative gamma op's input voltage.	

Note: (1) Gamma structure please reference "11.3 Gamma reference voltage generator"

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9.4 Registers of page 3

Address [7:0]	Default [7:0]	Data [7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0XC0	01h	-	-	-	-	-	0	0	1	GOA_PHASE[2:0]	GOA phase selection. 000b: 4 phase. 001b: 8 phase. 010b: 6 phase. 011b: 12phase. 100b: 16 phase. Others: Reserve.	V
		-	-	-	-	0	-	-	-	CKV_BLANKON	CKV blanking on enable. 0: Disable.	V

Note: (1)

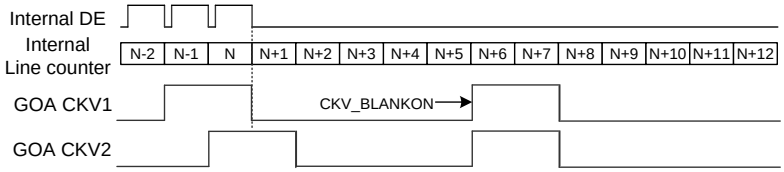


Figure 9.1: GOA CKV blanking on chart

0XC1	10h	-	-	0	1	-	-	-	-	GOA_Tn_FACTOR	GOA Timing **(T0~T3)** adjustment factor selection. 00: Tnx1. 01: Tnx2. 10: Tnx4. 11: Tnx8.	V
0xC2	00h	0	0	0	0	0	0	0	0	GOA_T0[7:0]	Odd STV falling position adjustment. T0=GOA_T0[7:0] x GOA_Tn_FACTOR.	V
0xC3	01h	0	0	0	0	0	0	0	1	GOA_T1[7:0]	Odd STV rising position adjustment. T1=GOA_T1[7:0] x GOA_Tn_FACTOR.	V
0xC4	00h	0	0	0	0	0	0	0	0	GOA_T2[7:0]	Odd CKV falling position adjustment. T2=GOA_T2[7:0] x GOA_Tn_FACTOR.	V
0xC5	23h	0	0	1	0	0	0	1	1	GOA_T3[7:0]	Odd CKV rising position adjustment. T3=GOA_T3[7:0] x GOA_Tn_FACTOR.	V

Note: (2) Unit is OSC25M.

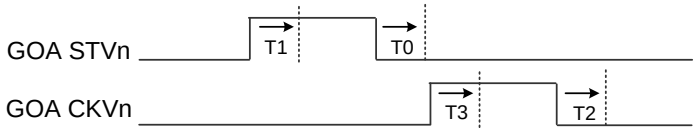


Figure 9.2: GOA timing tune

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Address [7:0]	Default [7:0]	Data [7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xC8	09h	-	-	-	0	1	0	0	1	STV_LEAD[4:0]	GOA STV leads time adjustment.	V
0xC9	07h	-	-	-	0	0	1	1	1	CKV_LEAD[4:0]	GOA CKV leads time adjustment.	V

Note: (3) a. Unit is H.
b. VT represents V-total.

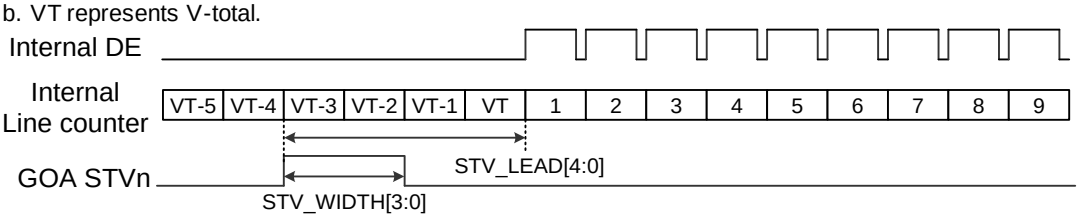


Figure 9.3: GOA STV time chart

(4) a. Unit is H.
b. VT represents V-total.

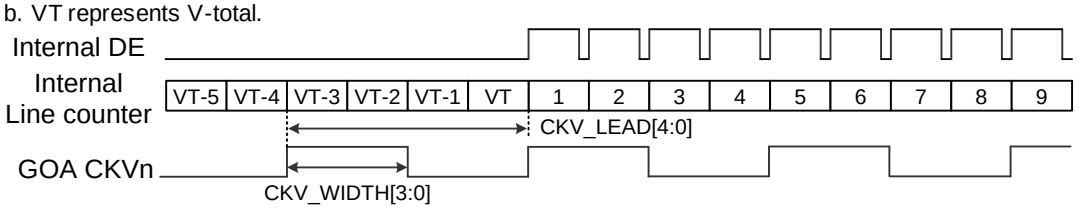


Figure 9.4: GOA CKV time chart

0xCA	41h	-	-	0	0	0	0	0	1	CKV_DUMMY[5:0]	GOA CKV dummy number in V-blanking. The total number is CKV_DUMMY[5:0]x GOA Phase.	V
		0	-	-	-	-	-	-	-	CKV_NONOVERLAP	CKV non-overlap control. 0: Non-overlap. 1: Overlap.	V

Note: (5) a. N represents V-Active lines.
b. DUMMY must be less than V-front proch.

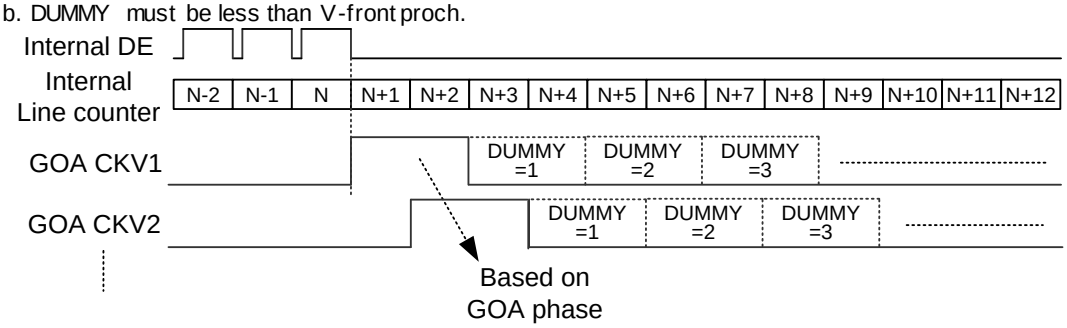


Figure 9.5: GOA CKV dummy chart

0xCB	34h	0	0	1	1	0	1	0	0	CKV_RISE_PREC[7:0]	CKV rising pre-charge width adjustment. 1 OSC25M/step	V
0xEB	00h	0	0	0	0	0	0	0	0	CKV_FALL_PREC[7:0]	CKV falling pre-charge width adjustment. 1 OSC25M/step	V

Note: (6)

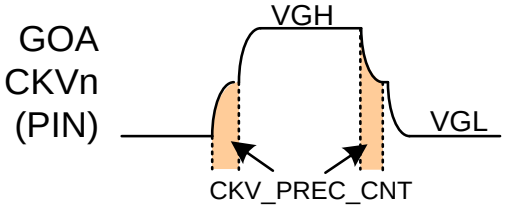


Figure 9.6: GOA precharge time chart

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Address [7:0]	Default [7:0]	Data [7:0] 7	6	5	4	3	2	1	0	Name	Description	OTP
0xCC	44h	-	-	-	-	0	1	0	0	CKV_WIDTH[3:0]	GOA CKV width adjustment.	V
		0	1	0	0	-	-	-	-	STV_WIDTH[3:0]	GOA STV width adjustment.	V
0xCD	07h	0	0	0	0	0	1	1	1	GOA_FLC[7:0]	GOA FLC toggle frame adjustment. The toggle frame range is 1 ~ 127 frames.	V

Note: (7)

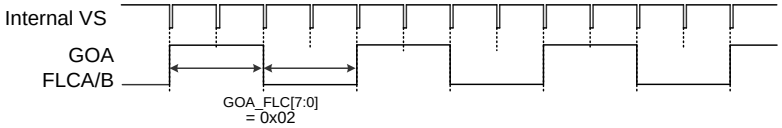


Figure 9.7: GOA FLC time chart

0xCE	03h	-	0	0	0	0	0	1	1	GOA_FLCA_LEAD[6:0]	FLCA lead last DE toggle number.	V
0xDB	00h	-	-	-	-	-	-	-	0-	CLR1_LEAD	GOA CLR1 offset direction setting.	V
0xCF	10h			0	-	-	-	-	-	CLR2_LEAD	GOA CLR2 offset direction setting.	V
		-	0	-	-	-	-	-	-	CLR3_LEAD	GOA CLR3 offset direction setting.	V
		0	-	-	-	-	-	-	-	CLR4_LEAD	GOA CLR4 offset direction setting.	V
		-	-	-	-	0	0	0	0	CLR1234_WD_MSB[3:0]	CLR1234 WD extend for clr1234_wd[11:8]	
0xD0	05h	0	0	0	0	0	1	0	1	CKV_BLANK_SHIFT[7:0]	Blank CKV shift from the last DE. H/step	V
0xD1	06h	0	0	0	0	0	1	1	0	CKV_BLANK_WIDTH[7:0]	Blank CKV width adjustment. H/step	V
0xD2	00h	0	0	0	0	0	0	0	0	CLR1_WD[7:0]	GOA CLR1 width adjustment CLR1_WIDTH[11:0]= CLR1234_WD_MSB[3:0]x256+ CLR1_WD[7:0]	V
0xD3	00h	0	0	0	0	0	0	0	0	CLR234_WD[7:0]	GOA CLR234 width adjustment. CLR234_WIDTH[11:0]= CLR1234_WD_MSB[3:0]x256+CLR234_WD[7:0]	V

Note: (8) a. Unit is H.
b. The total value must be less than V-front proch.

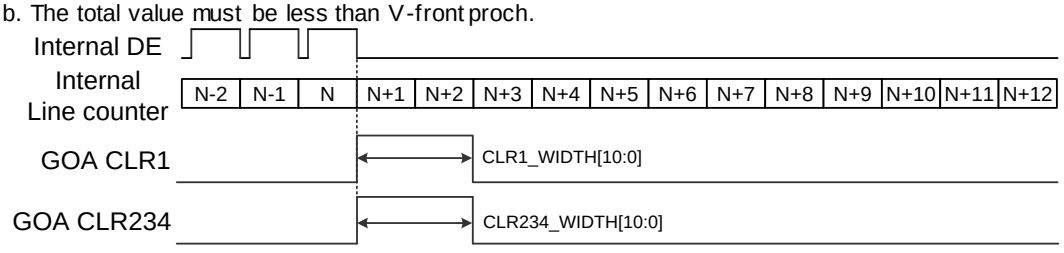


Figure 9.8: CLR width time chart

0xD4	88h	-	0	0	0	1	0	0	0	CLR1_START[6:0]	CLR1 start position. CLR1_ST[11:0]= CLR1_START_MSB[4:0]x128+CLR1_START[6:0]	V
		1	-	-	-	-	-	-	-	CLR1_POL	CLR1 polarity setting. 0: Non-inversion. 1: Inversion.	V
0xD5	09h	-	0	0	0	1	0	0	1	CLR2_START[6:0]	CLR2 start position. CLR2_ST[11:0]= CLR234_START_MSB[4:0]x128+CLR2_START[6:0]	V
		0	-	-	-	-	-	-	-	CLR2_POL	CLR2 polarity setting. 0: Non-inversion. 1: Inversion.	V
0xD6	03h	-	0	0	0	0	0	1	1	CLR3_START[6:0]	CLR3 start position. CLR3_ST[11:0]= CLR234_START_MSB[4:0]x128+CLR3_START[6:0]	V
		0	-	-	-	-	-	-	-	CLR3_POL	CLR3 polarity setting. 0: Non-inversion. 1: Inversion.	V

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0xD7	04h	-	0	0	0	0	1	0	0	CLR4_START[6:0]	CLR4 start position. CLR4_ST[11:0]= CLR234_START_MSB[4:0]x128+CLR4_START[6:0]	V
		0	-	-	-	-	-	-	-	CLR4_POL	CLR4 polarity setting. 0: Non-inversion. 1: Inversion.	V
0xD8	00h	-	-	-	0	0	0	0	0	CLR1_START_MSB[4:0]	MSB[11:7] of CLR1 start position.	V
0XD9	00h	-	-	-	0	0	0	0	0	CLR2/3/4_START_MSB[4:0]	MSB[11:7] of CLR2/3/4 start position.	V

Note: (9) a. Unit is H-total.
b. The total value must be less than V-total.

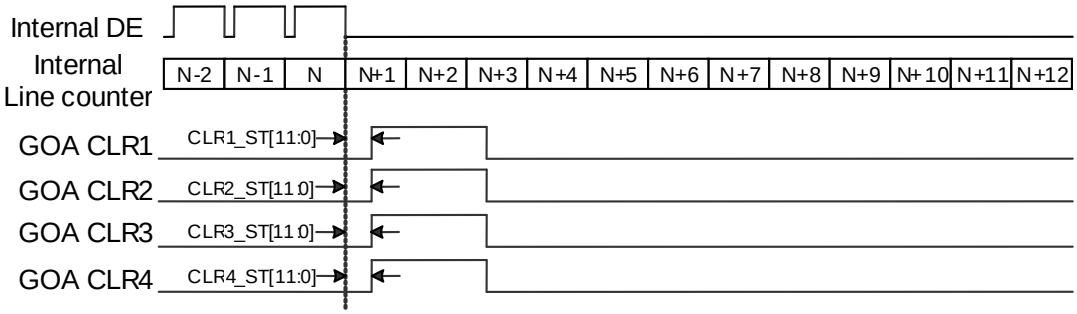


Figure 9.9: CLR start position time chart

0xDB	00h	-	-	-	-	0	0	0	-	FLC_NONOVLAP[1:0]	FLC nonoverlap setting. 00b: 0H x1b: 1H 10b: 2H	V
0xDD	01h	0	0	0	0	0	0	0	1	GOA_T2B[7:0]	Even CKV falling position adjustment. T0=GOA_T2B[7:0] x GOA_Tn_FACTOR.	V
0xDE	23h	0	0	1	0	0	0	1	1	GOA_T3B[7:0]	Even CKV rising position adjustment. T0=GOA_T3B[7:0] x GOA_Tn_FACTOR.	V
0xE6	00h	0	0	0	0	0	0	0	0	GOA_T0B[7:0]	Even STV falling position adjustment. T0=GOA_T0B[7:0] x GOA_Tn_FACTOR.	V
0xE7	01h	0	0	0	0	0	0	0	1	GOA_T1B[7:0]	Even STV rising position adjustment. T0=GOA_T1B[7:0] x GOA_Tn_FACTOR.	V
0XEA	00h	-	-	-	-	-	0	0	0	STV_PREC[3:0]	STV pre-charge width adjustment 1 OSC25M/step	V

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9.5 Registers of page 4

Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
0xB1	39h	-	0	-	-	-	-	-	-	DBG_UD_EN	DBG user define enable. 0: Disable. 1: Enable.	V
		-	-	1	-	-	-	-	-	DIM_EN	Back light dimming enable. 0: Disable. 1: Enable.	V
		-	-	-	1	-	-	-	-	BL_EN	CABC back light control enable. 0: Disable. 1: Enable.	V
		-	-	-	-	1	-	-	-	PWM_POL	Polarity of PWM control signal setting. 0: PWM output is inversion. 1: PWM output is non-inversion.	V
0xB2	21h	-	-	1	0	-	-	-	-	dim_step	Dimming frame scale A.	V
		-	-	-	-	-	0	0	1	dim_fm	Dimming frame scale B. **(dim_fm[2:0] minimum value is 1d)** Total dimming time = 2^(dim_step[1:0]+1) × dim_fm[2:0]. **(Unit: Frame)**	V
0xB4	10h	0	0	0	1	0	0	0	0	CABC_MB	Set PWM minimum duty.	V
0xB5	00h	0	0	0	0	0	0	0	0	PWM_PRD	PWM_FRQ_B for PWM period generation.	V
0xB6	02h	-	-	-	-	-	0	1	0	PWM_DIV	PWM_FRQ_A for PWM period generation. PWM output period = CLK period × 256 × 2^(PWM_PRD[7:0]) x (PWM_DIV[2:0] + 1).	V
0xB7	00h	0	0	0	0	0	0	0	0	DBG0	Back Light gain at frame average 0 when cabc in user define mode.	V
0xB8	20h	0	0	1	0	0	0	0	0	DBG1	Back Light gain at frame average 32 when cabc in user define mode.	V
0xB9	40h	0	1	0	0	0	0	0	0	DBG2	Back Light gain at frame average 64 when cabc in user define mode.	V
0xBA	60h	0	1	1	0	0	0	0	0	DBG3	Back Light gain at frame average 96 when cabc in user define mode.	V
0xBB	80h	1	0	0	0	0	0	0	0	DBG4	Back Light gain at frame average 128 when cabc in user define mode.	V
0xBC	A0h	1	0	1	0	0	0	0	0	DBG5	Back Light gain at frame average 160 when cabc in user define mode.	V
0xBD	C0h	1	1	0	0	0	0	0	0	DBG6	Back Light gain at frame average 192 when cabc in user define mode.	V
0xBE	E0h	1	1	1	1	0	0	0	0	DBG7	Back Light gain at frame average 224 when cabc in user define mode.	V

Note: (1) Please reference "10.2 CABC function" for CABC function.

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9.6 Registers of page 5

Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
0xB5	55h	-	1	0	1	-	-	-	-	RT3_M	Lane2/3 termination resistance control for Master.	V
		-	-	-	-	-	1	0	1	RT2_M		
0xB6	55h	-	1	0	1	-	-	-	-	RT1_M	Lane0/1 termination resistance control for Master.	V
		-	-	-	-	-	1	0	1	RT0_M		
0xB7	05h	-	-	-	-	-	1	0	1	RTC_M	Clock Lane termination resistance control for Master.	V

Note: (1) Termination resistance control:

RT[2:0]	Ω
111	81
110	90
101	102
100	118
011	128
010	153
001	192
000	255

9.7 Registers of page 6 (Engineer)

Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
0xB8	5Ah	0	1	0	1	1	0	1	0	ENGINEER_PWD	Engineer_pwd=8'hA5 enable engineer mode.	-
0xC0	5Ah	0	1	0	1	1	0	1	0	Function_EN	Function_EN=8'hA5 enable in-house function.	-
0xC2	23h	0	0	-	-	-	-	-	-	SD_OPM	Source slew rate adjust. 00: Default slew rate. 01: Hight slew rate. 10: Low slew rate. 11: Reserved.	V
0xC6	69h	-	-	-	-	-	-	0	1	IBSEL	Whole chip bias adjust. 00: 88%. 01: 100%. 10: 114%. 11: 133%.	V
0xD5	32h	0	0	1	1	0	0	1	0	GOE_WD	Source delay time adjustment **(CKV falling to Source off)**.	V

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9.8 Registers of page.7 ~ C

Page 7/A: Red positive/negative gamma
Page 8/B: Green positive/negative gamma
Page 9/C: Blue positive/negative gamma

Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
0xB1	00h/00h/00h	0	0	0	0	0	0	0	0	P/N-GMA1-R/G/B[7:0]	Positive/Negative gamma V1 Reference: Red/Green/Blue.	V
0xB2	04h/04h/03h	0	0	0	0	0	1	0	0	P/N-GMA2-R/G/B[7:0]	Positive/Negative gamma V2 Reference: Red/Green/Blue.	V
0xB3	0Bh/0Bh/09h	0	0	0	0	1	1	0	0	P/N-GMA3-R/G/B[7:0]	Positive/Negative gamma V3 Reference: Red/Green/Blue.	V
0xB4	19h/19h/16h	0	0	0	1	1	1	0	0	P/N-GMA4-R/G/B[7:0]	Positive/Negative gamma V4 Reference: Red/Green/Blue.	V
0xB5	28h/27h/23h	0	0	1	0	1	1	0	0	P/N-GMA5-R/G/B[7:0]	Positive/Negative gamma V5 Reference: Red/Green/Blue.	V
0xB6	37h/35h/2Fh	0	0	1	1	1	1	0	0	P/N-GMA6-R/G/B[7:0]	Positive/Negative gamma V6 Reference: Red/Green/Blue.	V
0xB7	55h/51h/48h	0	1	0	1	1	1	0	0	P/N-GMA7-R/G/B[7:0]	Positive/Negative gamma V7 Reference: Red/Green/Blue.	V
0xB8	71h/6Ah/67h	0	1	1	1	1	1	0	0	P/N-GMA8-R/G/B[7:0]	Positive/Negative gamma V8 Reference: Red/Green/Blue.	V
0xB9	AEh/A4h/A6h	1	0	1	1	1	1	0	0	P/N-GMA9-R/G/B[7:0]	Positive/Negative gamma V9 Reference: Red/Green/Blue.	V
0xBA	F1h/E5h/E9h	1	1	1	1	1	1	0	0	P/N-GMA10-R/G/B[7:0]	Positive/Negative gamma V10 Reference: Red/Green/Blue.	V
0xBB	7Ah/66h/74h	0	1	1	1	1	1	0	0	P/N-GMA11-R/G/B[7:0]	Positive/Negative gamma V11 Reference: Red/Green/Blue.	V
0xBC	F9h/E0h/F0h	1	1	1	1	1	1	0	0	P/N-GMA12-R/G/B[7:0]	Positive/Negative gamma V12 Reference: Red/Green/Blue.	V
0xBD	FDh/E4h/F4h	0	0	0	0	0	0	0	0	P/N-GMA13-R/G/B[7:0]	Positive/Negative gamma V13 Reference: Red/Green/Blue.	V
0xBE	7Ah/5Bh/70h	1	0	0	0	0	0	0	0	P/N-GMA14-R/G/B[7:0]	Positive/Negative gamma V14 Reference: Red/Green/Blue.	V
0xBF	FBh/D7h/ECh	0	0	0	0	0	0	0	0	P/N-GMA15-R/G/B[7:0]	Positive/Negative gamma V15 Reference: Red/Green/Blue.	V
0xC0	3Bh/14h/29h	0	1	0	0	0	0	0	0	P/N-GMA16-R/G/B[7:0]	Positive/Negative gamma V16 Reference: Red/Green/Blue.	V
0xC1	7Eh/52h/67h	1	0	0	0	0	0	0	0	P/N-GMA17-R/G/B[7:0]	Positive/Negative gamma V17 Reference: Red/Green/Blue.	V
0xC2	9Fh/71h/84h	1	0	1	0	0	0	0	0	P/N-GMA18-R/G/B[7:0]	Positive/Negative gamma V18 Reference: Red/Green/Blue.	V
0xC3	C0h/90h/A1h	1	1	0	0	0	0	0	0	P/N-GMA19-R/G/B[7:0]	Positive/Negative gamma V19 Reference: Red/Green/Blue.	V
0xC4	D1h/A0h/B1h	1	1	0	1	0	0	0	0	P/N-GMA20-R/G/B[7:0]	Positive/Negative gamma V20 Reference: Red/Green/Blue.	V
0xC5	E0h/AFh/C0h	1	1	1	0	0	0	0	0	P/N-GMA21-R/G/B[7:0]	Positive/Negative gamma V21 Reference: Red/Green/Blue.	V
0xC6	F1h/C0h/CCh	1	1	1	1	0	0	0	0	P/N-GMA22-R/G/B[7:0]	Positive/Negative gamma V22 Reference: Red/Green/Blue.	V
0xC7	F8h/C7h/D4h	1	1	1	1	1	0	0	0	P/N-GMA23-R/G/B[7:0]	Positive/Negative gamma V23 Reference: Red/Green/Blue.	V
0xC8	FCh/CCh/D8h	1	1	1	1	1	1	0	0	P/N-GMA24-R/G/B[7:0]	Positive/Negative gamma V24 Reference: Red/Green/Blue.	V
0xC9	00h/00h/00h	0	0	-	-	-	-	-	-	P/N-GMA1-R/G/B[9:8]	Positive/Negative gamma V1 Reference: Red/Green/Blue.	V
		-	-	0	0	-	-	-	-	P/N-GMA2-R/G/B[9:8]	Positive/Negative gamma V2 Reference: Red/Green/Blue.	V
		-	-	-	-	0	0	-	-	P/N-GMA3-R/G/B[9:8]	Positive/Negative gamma V3 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	0	0	P/N-GMA4-R/G/B[9:8]	Positive/Negative gamma V4 Reference:	V

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Address [7:0]	Default [7:0]	Data [7:0]								Name	Description	OTP
		7	6	5	4	3	2	1	0			
											Red/Green/Blue.	
0xCA	00h/00h/00h	0	0	-	-	-	-	-	-	P/N-GMA5-R/G/B[9:8]	Positive/Negative gamma V5 Reference: Red/Green/Blue.	V
		-	-	0	0	-	-	-	-	P/N-GMA6-R/G/B[9:8]	Positive/Negative gamma V6 Reference: Red/Green/Blue.	V
		-	-	-	-	0	0	-	-	P/N-GMA7-R/G/B[9:8]	Positive/Negative gamma V7 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	0	0	P/N-GMA8-R/G/B[9:8]	Positive/Negative gamma V8 Reference: Red/Green/Blue.	V
0xCB	05h/05h/05h	0	1	-	-	-	-	-	-	P/N-GMA9-R/G/B[9:8]	Positive/Negative gamma V9 Reference: Red/Green/Blue.	V
		-	-	0	1	-	-	-	-	P/N-GMA10-R/G/B[9:8]	Positive/Negative gamma V10 Reference: Red/Green/Blue.	V
		-	-	-	-	0	1	-	-	P/N-GMA11-R/G/B[9:8]	Positive/Negative gamma V11 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	0	1	P/N-GMA12-R/G/B[9:8]	Positive/Negative gamma V12 Reference: Red/Green/Blue.	V
0xCC	6Bh/6Bh/6Bh	0	1	-	-	-	-	-	-	P/N-GMA13-R/G/B[9:8]	Positive/Negative gamma V13 Reference: Red/Green/Blue.	V
		-	-	1	0	-	-	-	-	P/N-GMA14-R/G/B[9:8]	Positive/Negative gamma V14 Reference: Red/Green/Blue.	V
		-	-	-	-	1	0	-	-	P/N-GMA15-R/G/B[9:8]	Positive/Negative gamma V15 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	1	1	P/N-GMA16-R/G/B[9:8]	Positive/Negative gamma V16 Reference: Red/Green/Blue.	V
0xCD	FFh/FFh/FFh	1	1	-	-	-	-	-	-	P/N-GMA17-R/G/B[9:8]	Positive/Negative gamma V17 Reference: Red/Green/Blue.	V
		-	-	1	1	-	-	-	-	P/N-GMA18-R/G/B[9:8]	Positive/Negative gamma V18 Reference: Red/Green/Blue.	V
		-	-	-	-	1	1	-	-	P/N-GMA19-R/G/B[9:8]	Positive/Negative gamma V19 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	1	1	P/N-GMA20-R/G/B[9:8]	Positive/Negative gamma V20 Reference: Red/Green/Blue.	V
0xCE	FFh/FFh/FFh	1	1	-	-	-	-	-	-	P/N-GMA21-R/G/B[9:8]	Positive/Negative gamma V21 Reference: Red/Green/Blue.	V
		-	-	1	1	-	-	-	-	P/N-GMA22-R/G/B[9:8]	Positive/Negative gamma V22 Reference: Red/Green/Blue.	V
		-	-	-	-	1	1	-	-	P/N-GMA23-R/G/B[9:8]	Positive/Negative gamma V23 Reference: Red/Green/Blue.	V
		-	-	-	-	-	-	1	1	P/N-GMA24-R/G/B[9:8]	Positive/Negative gamma V24 Reference: Red/Green/Blue.	V

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10. Function Description

10.1 BIST pattern

When register BIST_EN is trigger to high, then HX8252-A will leave normal operation mode and starts to generate the BIST pattern to LCD panel without MIPI input signals.

The BIST pattern is illustrated as below figure.

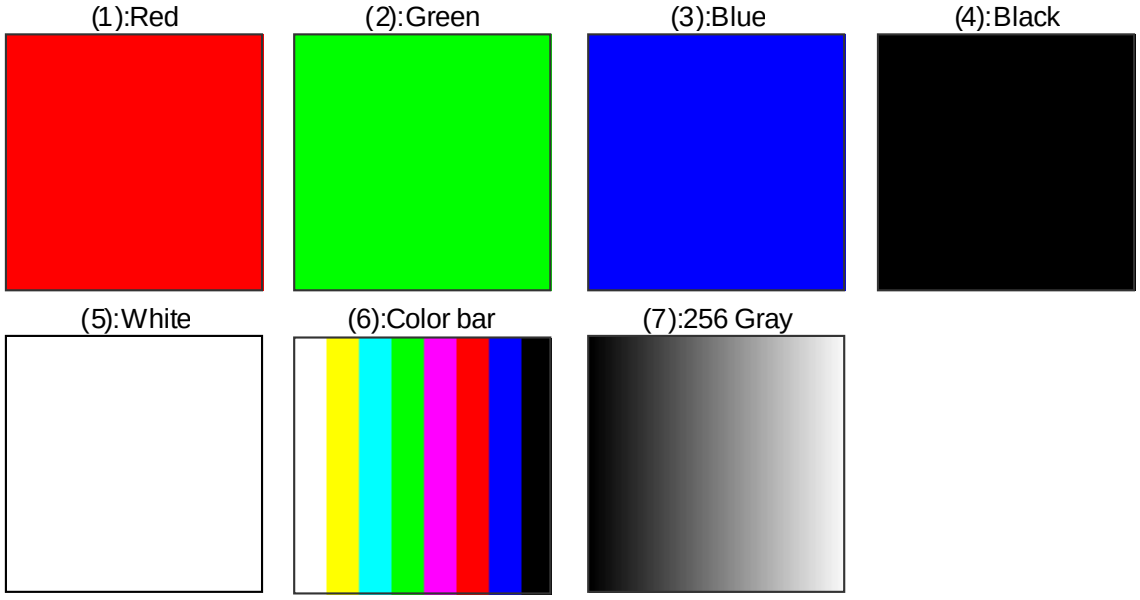


Figure 10.1: BIST pattern

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10.2 CABC function

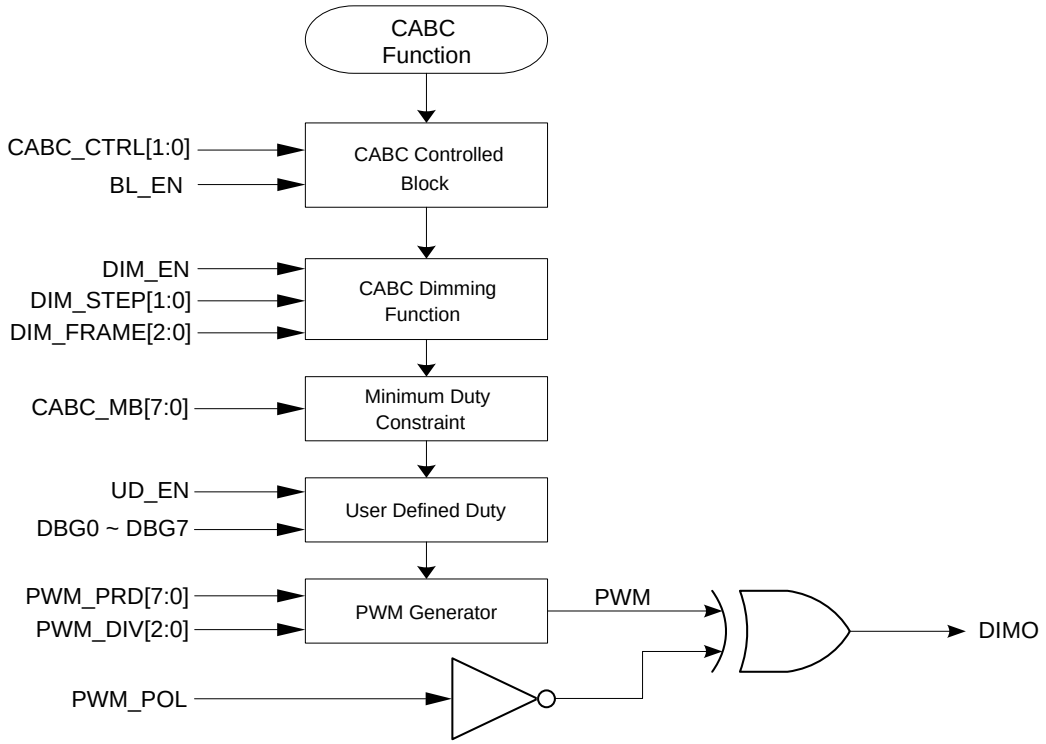


Figure 10.2: CABC flow chart

CABC controlled register:

- CABC_CTRL[1:0]: UI / Still / Moving / Bypass mode selection
- CABC_MB[7:0]: Minimum PWM duty constraint
- UD_EN & DBG0 ~ DBG7: Fixed PWM duty by user-defined when operating at Bypass mode
- PWM_POL: PWM duty polarity selection
- BL_EN: Back-light On/Off switch
- DIM_EN: Dimming function enable/ disable switch
- Dimming time selection: DIM_STEP[1:0] & DIM_FRME[2:0]
  - Total dimming time = 2^(DIM_STEP[1:0]+1) × DIM_FRME[2:0]
- PWM frequency selection: PWM_PRD[7:0] & PWM_DIV[2:0]
  - PWM output period = CLK period × 256 × 2^(PWM_PRD[7:0]) x (PWM_DIV[2:0] + 1)

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10.3 OTP function

10.3.1 OTP flow of programming and read

- OTP flow of programming by internal power. **(VDD_OTP)**
  Pleae add a 1.0μF CAP. between the VDD_OTP pin and GND.

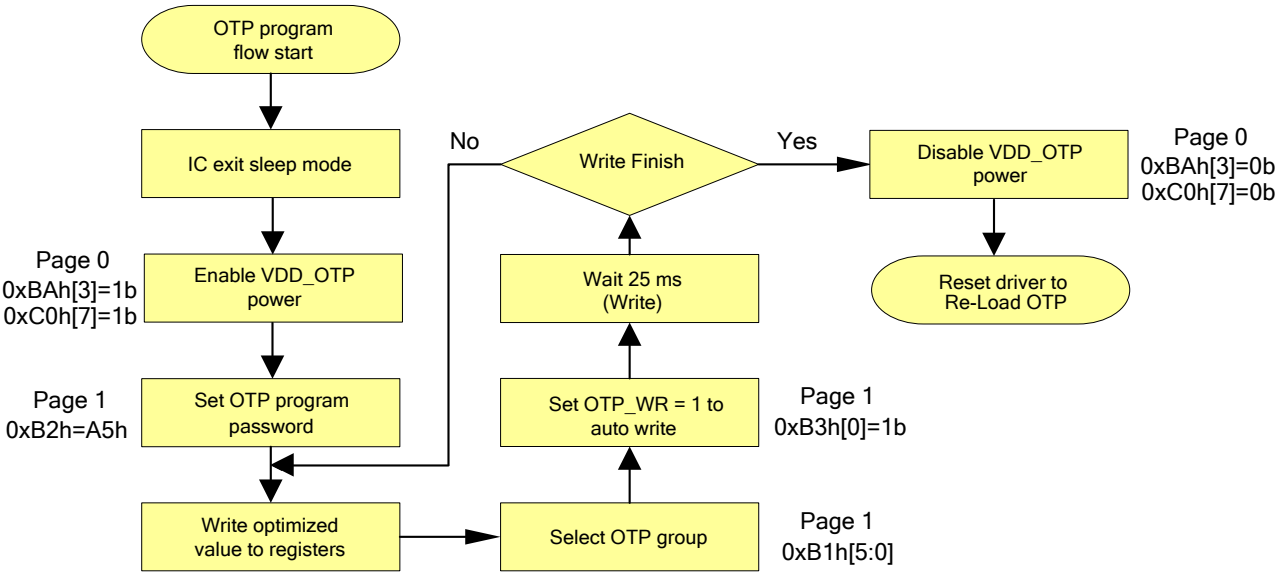


Figure 10.3: OTP program flow with internal power

- OTP flow of programming by external power **(VDD_OTP)**

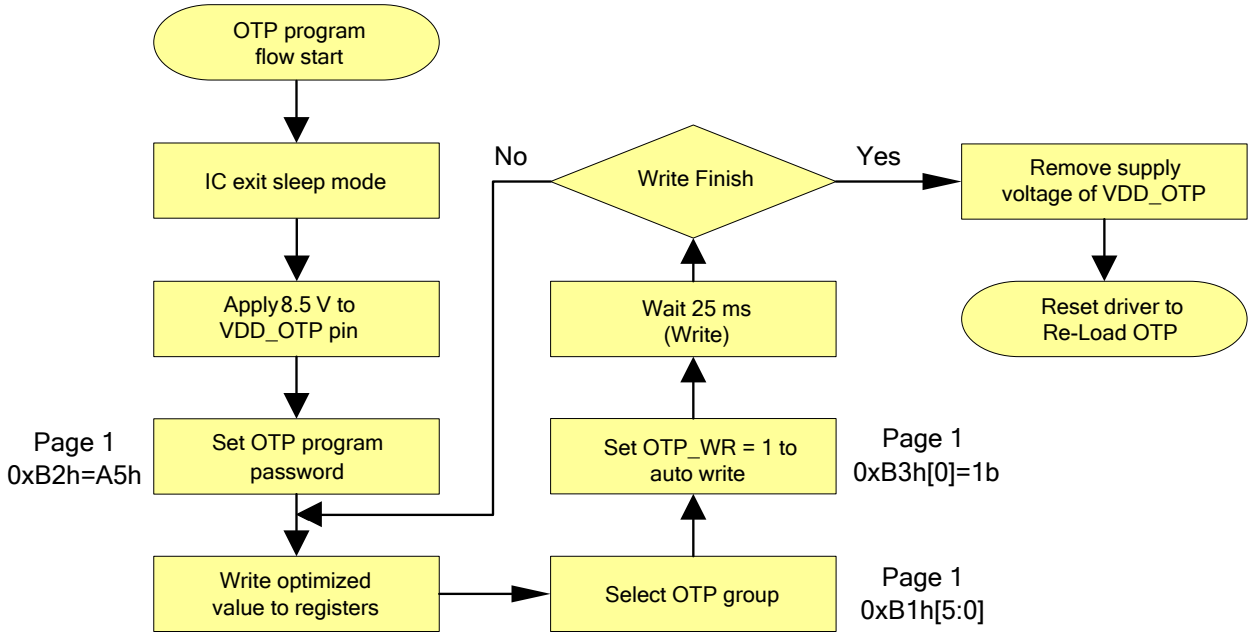


Figure 10.4: OTP program flow with external power

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- OTP flow of reading

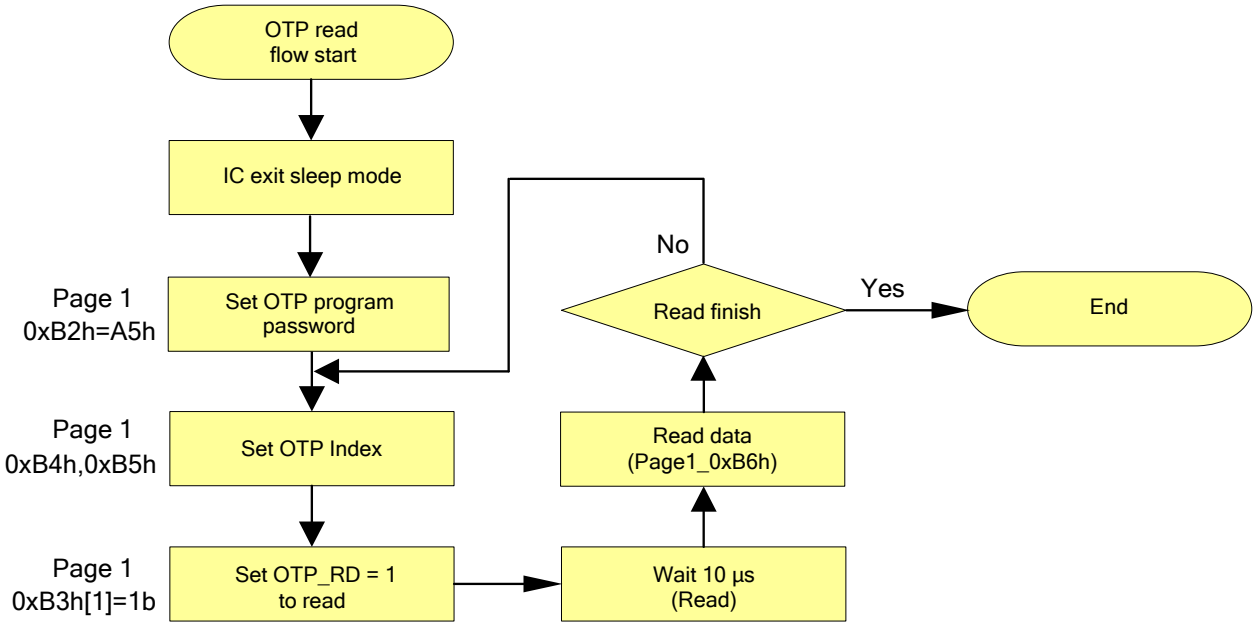


Figure 10.5: OTP read flow

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10.3.2 OTP table

OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
0	W1	-							GROUP0
1	T_VCOMS[7:0]								
2	W2	-							
3	T_VCOMS[7:0]								
4	W3	-							
5	T_VCOMS[7:0]								
6	W4	-							
7	T_VCOMS[7:0]								
8	W5	-							
9	T_VCOMS[7:0]								
10	W6	-							
11	T_VCOMS[7:0]								
12	W7	-							
13	T_VCOMS[7:0]								
14	W8	-							
15	T_VCOMS[7:0]								
16	W9	-							
17	T_VCOMS[7:0]								
18	W10	-							
19	T_VCOMS[7:0]								
22	W1	-	-	LR_XOR-	ZIGZAG_SEL	-	NBW_SEL	BIST_XOR	GROUP2
23	ZTYPE_SEL[1:0]		PWRMD_XOR[1:0]		VRES_FIX	RES[2:0]			
24	VRES[7:0]								
25	ZDATA[7:0]								
26	VRES[8]	-	ZZ_2PL	RB_SWAP	CABC_CTRL[1:0]		DITHER_EN	D_GAM_EN	
27	-		-	-	-	-	BC_CTRL	CASCADE_ENB_XOR	
28	GOA_EN	VCOM_EN	PNSW_XOR	-	LNSW_XOR[1:0]		LANSEL_XOR[1:0]		
29	RESERVE								
30	BLREV[2:0]			BLREVONOFF	VDD_OTP_SW	INV_SEL[2:0]			
31	W1	-	-	VSPS[4:0]					GROUP3
32	VDD_OTPOS[2:0]			VSNS[4:0]					
33	-	-	VGHS[5:0]						
34	-	-	VGLS[5:0]						
35	VDDOTP_EN	VGLO_EN	VGLO_S[5:0]						
36	VSPON_T[3:0]			VSPOFF_T[3:0]					
37	VSNON_T[3:0]			VSNOFF_T[3:0]					
38	VCL_cpctrl[2]	VGLXSP	VGHXSP[1:0]		PFMFREN	tontoff_offset	dpfm_clk		
39	PFM_SP_EN	PFMSPW[6:0]							
40	VCL_cpctrl[1:0]		VGL_cpctrl[2:0]			VGH_cpctrl[2:0]			
41	cgpp_inv		cgpp_soft_en	cgpp_clksel[2:0]			cgpp_cmd_sel[1:0]		
46	W1	-	VGMHP[5:0]						GROUP4
47	-	-	VGMLP [5:0]						
48	-	-	VGMHN [5:0]						
49	-	-	VGMLN [5:0]						
50	W1	POCSD_CTL[1:0]	SD_EQW[4:0]						GROUP5
51	-	ISSEL_N_CAS[2:0]			-	ISSEL_P_CAS[2:0]			GROUP6
52	-	ISSEL_N_NOTCAS[2:0]			-	ISSEL_P_NOTCAS[2:0]			
53	-	-	-	-	T_VBP[1:0]		T_VBN[1:0]		
60	W1	-	-						GROUP7
61	gpo1_sel[3:0]				gpo0_sel[3:0]				

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
62	gpo3_sel[3:0]				gpo2_sel[3:0]				GROUP7
63	gpo5_sel[3:0]				gpo4_sel[3:0]				
64	gpo7_sel[3:0]				gpo6_sel[3:0]				
65	-	-	-	-	gpo8_sel[3:0]				
66	W1	-						cta_en	GROUP8
67	cta_gain_r[7:0]								
68	cta_gain_g[7:0]								
69	cta_gain_b[7:0]								
70	RESERVED								
71	W1	-							GROUP9
72	GOUTL1_STBYB[1:0]	GOUTL_1_SEL[5:0]							
73	GOUTL2_STBYB[1:0]	GOUTL_2_SEL[5:0]							
74	GOUTL3_STBYB[1:0]	GOUTL_3_SEL[5:0]							
75	GOUTL4_STBYB[1:0]	GOUTL_4_SEL[5:0]							
76	GOUTL5_STBYB[1:0]	GOUTL_5_SEL[5:0]							
77	GOUTL6_STBYB[1:0]	GOUTL_6_SEL[5:0]							
78	GOUTL7_STBYB[1:0]	GOUTL_7_SEL[5:0]							
79	GOUTL8_STBYB[1:0]	GOUTL_8_SEL[5:0]							
80	GOUTL9_STBYB[1:0]	GOUTL_9_SEL[5:0]							
81	GOUTL10_STBYB[1:0]	GOUTL_10_SEL[5:0]							
82	GOUTL11_STBYB[1:0]	GOUTL_11_SEL[5:0]							
83	GOUTL12_STBYB[1:0]	GOUTL_12_SEL[5:0]							
84	GOUTL13_STBYB[1:0]	GOUTL_13_SEL[5:0]							
85	GOUTL14_STBYB[1:0]	GOUTL_14_SEL[5:0]							
86	GOUTL15_STBYB[1:0]	GOUTL_15_SEL[5:0]							
87	GOUTL16_STBYB[1:0]	GOUTL_16_SEL[5:0]							
88	GOUTL17_STBYB[1:0]	GOUTL_17_SEL[5:0]							
89	GOUTL18_STBYB[1:0]	GOUTL_18_SEL[5:0]							
90	GOUTL19_STBYB[1:0]	GOUTL_19_SEL[5:0]							
91	GOUTL20_STBYB[1:0]	GOUTL_20_SEL[5:0]							
92	GOUTL21_STBYB[1:0]	GOUTL_21_SEL[5:0]							
93	GOUTL22_STBYB[1:0]	GOUTL_22_SEL[5:0]							
94	GOUTR1_STBYB[1:0]	GOUTR_1_SEL[5:0]							
95	GOUTR2_STBYB[1:0]	GOUTR_2_SEL[5:0]							
96	GOUTR3_STBYB[1:0]	GOUTR_3_SEL[5:0]							
97	GOUTR4_STBYB[1:0]	GOUTR_4_SEL[5:0]							
98	GOUTR5_STBYB[1:0]	GOUTR_5_SEL[5:0]							
99	GOUTR6_STBYB[1:0]	GOUTR_6_SEL[5:0]							
100	GOUTR7_STBYB[1:0]	GOUTR_7_SEL[5:0]							
101	GOUTR8_STBYB[1:0]	GOUTR_8_SEL[5:0]							
102	GOUTR9_STBYB[1:0]	GOUTR_9_SEL[5:0]							
103	GOUTR10_STBYB[1:0]	GOUTR_10_SEL[5:0]							
104	GOUTR11_STBYB[1:0]	GOUTR_11_SEL[5:0]							
105	GOUTR12_STBYB[1:0]	GOUTR_12_SEL[5:0]							
106	GOUTR13_STBYB[1:0]	GOUTR_13_SEL[5:0]							
107	GOUTR14_STBYB[1:0]	GOUTR_14_SEL[5:0]							
108	GOUTR15_STBYB[1:0]	GOUTR_15_SEL[5:0]							
109	GOUTR16_STBYB[1:0]	GOUTR_16_SEL[5:0]							
110	GOUTR17_STBYB[1:0]	GOUTR_17_SEL[5:0]							
111	GOUTR18_STBYB[1:0]	GOUTR_18_SEL[5:0]							
112	GOUTR19_STBYB[1:0]	GOUTR_19_SEL[5:0]							
113	GOUTR20_STBYB[1:0]	GOUTR_20_SEL[5:0]							
114	GOUTR21_STBYB[1:0]	GOUTR_21_SEL[5:0]							
115	GOUTR22_STBYB[1:0]	GOUTR_22_SEL[5:0]							
116	-	-	GOA_HZ_ENL	GOA_HZ_ENR	-	-	-	-	
117	W2	-							
118	GOUTL1_STBYB[1:0]	GOUTL_1_SEL[5:0]							
119	GOUTL2_STBYB[1:0]	GOUTL_2_SEL[5:0]							
120	GOUTL3_STBYB[1:0]	GOUTL_3_SEL[5:0]							
121	GOUTL4_STBYB[1:0]	GOUTL_4_SEL[5:0]							
122	GOUTL5_STBYB[1:0]	GOUTL_5_SEL[5:0]							

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
123	GOUTL6_STBYB[1:0]		GOUTL_6_SEL[5:0]						
124	GOUTL7_STBYB[1:0]		GOUTL_7_SEL[5:0]						
125	GOUTL8_STBYB[1:0]		GOUTL_8_SEL[5:0]						
126	GOUTL9_STBYB[1:0]		GOUTL_9_SEL[5:0]						
127	GOUTL10_STBYB[1:0]		GOUTL_10_SEL[5:0]						
128	GOUTL11_STBYB[1:0]		GOUTL_11_SEL[5:0]						
129	GOUTL12_STBYB[1:0]		GOUTL_12_SEL[5:0]						
130	GOUTL13_STBYB[1:0]		GOUTL_13_SEL[5:0]						
131	GOUTL14_STBYB[1:0]		GOUTL_14_SEL[5:0]						
132	GOUTL15_STBYB[1:0]		GOUTL_15_SEL[5:0]						
133	GOUTL16_STBYB[1:0]		GOUTL_16_SEL[5:0]						
134	GOUTL17_STBYB[1:0]		GOUTL_17_SEL[5:0]						
135	GOUTL18_STBYB[1:0]		GOUTL_18_SEL[5:0]						
136	GOUTL19_STBYB[1:0]		GOUTL_19_SEL[5:0]						
137	GOUTL20_STBYB[1:0]		GOUTL_20_SEL[5:0]						
138	GOUTL21_STBYB[1:0]		GOUTL_21_SEL[5:0]						
139	GOUTL22_STBYB[1:0]		GOUTL_22_SEL[5:0]						
140	GOUTR1_STBYB[1:0]		GOUTR_1_SEL[5:0]						
141	GOUTR2_STBYB[1:0]		GOUTR_2_SEL[5:0]						
142	GOUTR3_STBYB[1:0]		GOUTR_3_SEL[5:0]						
143	GOUTR4_STBYB[1:0]		GOUTR_4_SEL[5:0]						
144	GOUTR5_STBYB[1:0]		GOUTR_5_SEL[5:0]						
145	GOUTR6_STBYB[1:0]		GOUTR_6_SEL[5:0]						
146	GOUTR7_STBYB[1:0]		GOUTR_7_SEL[5:0]						
147	GOUTR8_STBYB[1:0]		GOUTR_8_SEL[5:0]						
148	GOUTR9_STBYB[1:0]		GOUTR_9_SEL[5:0]						
149	GOUTR10_STBYB[1:0]		GOUTR_10_SEL[5:0]						
150	GOUTR11_STBYB[1:0]		GOUTR_11_SEL[5:0]						
151	GOUTR12_STBYB[1:0]		GOUTR_12_SEL[5:0]						
152	GOUTR13_STBYB[1:0]		GOUTR_13_SEL[5:0]						
153	GOUTR14_STBYB[1:0]		GOUTR_14_SEL[5:0]						
154	GOUTR15_STBYB[1:0]		GOUTR_15_SEL[5:0]						
155	GOUTR16_STBYB[1:0]		GOUTR_16_SEL[5:0]						
156	GOUTR17_STBYB[1:0]		GOUTR_17_SEL[5:0]						
157	GOUTR18_STBYB[1:0]		GOUTR_18_SEL[5:0]						
158	GOUTR19_STBYB[1:0]		GOUTR_19_SEL[5:0]						
159	GOUTR20_STBYB[1:0]		GOUTR_20_SEL[5:0]						
160	GOUTR21_STBYB[1:0]		GOUTR_21_SEL[5:0]						
161	GOUTR22_STBYB[1:0]		GOUTR_22_SEL[5:0]						
162	-	-	GOA_HZ_ENL	GOA_HZ_ENR	-	-	-	-	GROUP9
163	W1	-	-	T_PVP_0[4:0]					
164	-	-	-	T_PVP_1[4:0]					
165	-	-	T_PVP_2[5:0]						
166	-	-	T_PVP_3[5:0]						
167	-	-	T_PVP_4[5:0]						
168	-	-	T_PVP_5[5:0]						
169	-	-	T_PVP_6[5:0]						
170	-	-	T_PVP_7[5:0]						
171	-	-	T_PVP_8[5:0]						
172	-	-	T_PVP_9[5:0]						
173	-	-	T_PVP_10[5:0]						GROUP10
174	-	-	T_PVP_11[5:0]						
175	-	-	T_PVP_12[5:0]						
176	-	-	T_PVP_13[5:0]						
177	-	-	T_PVP_14[5:0]						
178	-	-	T_PVP_15[5:0]						
179	-	-	T_PVP_16[5:0]						
180	-	-	T_PVP_17[5:0]						
181	-	-	T_PVP_18[5:0]						
182	-	-	T_PVP_19[5:0]						

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
183	-	-	-	T_PVP_20[4:0]					
184	-	-	-	T_PVP_21[4:0]					
185	-	-	-	T_PVN_0[4:0]					
186	-	-	-	T_PVN_1[4:0]					
187	-	-	T_PVN_2[5:0]						
188	-	-	T_PVN_3[5:0]						
189	-	-	T_PVN_4[5:0]						
190	-	-	T_PVN_5[5:0]						
191	-	-	T_PVN_6[5:0]						
192	-	-	T_PVN_7[5:0]						
193	-	-	T_PVN_8[5:0]						
194	-	-	T_PVN_9[5:0]						
195	-	-	T_PVN_10[5:0]						
196	-	-	T_PVN_11[5:0]						
197	-	-	T_PVN_12[5:0]						
198	-	-	T_PVN_13[5:0]						
199	-	-	T_PVN_14[5:0]						
200	-	-	T_PVN_15[5:0]						
201	-	-	T_PVN_16[5:0]						
202	-	-	T_PVN_17[5:0]						
203	-	-	T_PVN_18[5:0]						
204	-	-	T_PVN_19[5:0]						
205	-	-	-	T_PVN_20[4:0]					
206	-	-	-	T_PVN_21[4:0]					
207	W2		-	T_PVP_0[4:0]					
208	-	-	-	T_PVP_1[4:0]					
209	-	-	T_PVP_2[5:0]						
210	-	-	T_PVP_3[5:0]						
211	-	-	T_PVP_4[5:0]						
212	-	-	T_PVP_5[5:0]						
213	-	-	T_PVP_6[5:0]						GROUP10
214	-	-	T_PVP_7[5:0]						
215	-	-	T_PVP_8[5:0]						
216	-	-	T_PVP_9[5:0]						
217	-	-	T_PVP_10[5:0]						
218	-	-	T_PVP_11[5:0]						
219	-	-	T_PVP_12[5:0]						
220	-	-	T_PVP_13[5:0]						
221	-	-	T_PVP_14[5:0]						
222	-	-	T_PVP_15[5:0]						
223	-	-	T_PVP_16[5:0]						
224	-	-	T_PVP_17[5:0]						
225	-	-	T_PVP_18[5:0]						
226	-	-	T_PVP_19[5:0]						
227	-	-	-	T_PVP_20[4:0]					
228	-	-	-	T_PVP_21[4:0]					
229	-	-	-	T_PVN_0[4:0]					
230	-	-	-	T_PVN_1[4:0]					
231	-	-	T_PVN_2[5:0]						
232	-	-	T_PVN_3[5:0]						
233	-	-	T_PVN_4[5:0]						
234	-	-	T_PVN_5[5:0]						
235	-	-	T_PVN_6[5:0]						
236	-	-	T_PVN_7[5:0]						
237	-	-	T_PVN_8[5:0]						
238	-	-	T_PVN_9[5:0]						
239	-	-	T_PVN_10[5:0]						
240	-	-	T_PVN_11[5:0]						
241	-	-	T_PVN_12[5:0]						
242	-	-	T_PVN_13[5:0]						
243	-	-	T_PVN_14[5:0]						

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
244	-	-	T_PVN_15[5:0]						
245	-	-	T_PVN_16[5:0]						
246	-	-	T_PVN_17[5:0]						
247	-	-	T_PVN_18[5:0]						
248	-	-	T_PVN_19[5:0]						
249	-	-	-	T_PVN_20[4:0]					
250	-	-	-	T_PVN_21[4:0]					
251	W3	-	-	T_PVP_0[4:0]					
252	-	-	-	T_PVP_1[4:0]					
253	-	-	T_PVP_2[5:0]						
254	-	-	T_PVP_3[5:0]						
255	-	-	T_PVP_4[5:0]						
256	-	-	T_PVP_5[5:0]						
257	-	-	T_PVP_6[5:0]						
258	-	-	T_PVP_7[5:0]						
259	-	-	T_PVP_8[5:0]						
260	-	-	T_PVP_9[5:0]						
261	-	-	T_PVP_10[5:0]						
262	-	-	T_PVP_11[5:0]						
263	-	-	T_PVP_12[5:0]						
264	-	-	T_PVP_13[5:0]						
265	-	-	T_PVP_14[5:0]						
266	-	-	T_PVP_15[5:0]						
267	-	-	T_PVP_16[5:0]						
268	-	-	T_PVP_17[5:0]						
269	-	-	T_PVP_18[5:0]						GROUP10
270	-	-	T_PVP_19[5:0]						
271	-	-	-	T_PVP_20[4:0]					
272	-	-	-	T_PVP_21[4:0]					
273	-	-	-	T_PVN_0[4:0]					
274	-	-	-	T_PVN_1[4:0]					
275	-	-	T_PVN_2[5:0]						
276	-	-	T_PVN_3[5:0]						
277	-	-	T_PVN_4[5:0]						
278	-	-	T_PVN_5[5:0]						
279	-	-	T_PVN_6[5:0]						
280	-	-	T_PVN_7[5:0]						
281	-	-	T_PVN_8[5:0]						
282	-	-	T_PVN_9[5:0]						
283	-	-	T_PVN_10[5:0]						
284	-	-	T_PVN_11[5:0]						
285	-	-	T_PVN_12[5:0]						
286	-	-	T_PVN_13[5:0]						
287	-	-	T_PVN_14[5:0]						
288	-	-	T_PVN_15[5:0]						
289	-	-	T_PVN_16[5:0]						
290	-	-	T_PVN_17[5:0]						
291	-	-	T_PVN_18[5:0]						
292	-	-	T_PVN_19[5:0]						
293	-	-	-	T_PVN_20[4:0]					
294	-	-	-	T_PVN_21[4:0]					
297	W1	-	-	-	CKV_BLANKON	GOA_PHASE[2:0]			
299	GOA_T0[7:0]								
300	GOA_T1[7:0]								
301	GOA_T2[7:0]								
302	GOA_T3[7:0]								GROUP12
305	-	-	STV_LEAD[4:0]						
306	-	-	CKV_LEAD[4:0]						
307	CKV_NONOVERLAP	-	CKV_DUMMY[5:0]						
308	CKV_RISE_PREC[7:0]								

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
309	STV_WIDTH[3:0]				CKV_WIDTH[3:0]				GROUP12
310	GOA_FLC[7:0]								
311	-	GOA_FLCA_LEAD[6:0]							
312	CLR4_LEAD	CLR3_LEAD	CLR2_LEAD	-					
313	CKV_BLANK_SHIFT[7:0]								
314	CKV_BLANK_WIDTH[7:0]								
315	CLR1_WD[7:0]								
316	CLR234_WD[7:0]								
317	CLR1_POL	CLR1_START[6:0]							
318	CLR2_POL	CLR2_START[6:0]							
319	CLR3_POL	CLR3_START[6:0]							
320	CLR4_POL	CLR4_START[6:0]							
321	-			CLR1_START_MSB[4:0]					
322	-			CLR2/3/4_START_MSB[4:0]					
324	-				FLC_NONOVLAP[1:0]			CLR1_LEAD	
326	GOA_T2B[7:0]								
327	GOA_T3B[7:0]								
335	GOA_T0B[7:0]								
336	GOA_T1B[7:0]								
337	W2	-	-	-	CKV_BLANK ON	GOA_PHASE[2:0]			
339	GOA_T0[7:0]								
340	GOA_T1[7:0]								
341	GOA_T2[7:0]								
342	GOA_T3[7:0]								
345	-	-	STV_LEAD[4:0]						
346	-	-	CKV_LEAD[4:0]						
347	CKV_NONOVERLAP	-	CKV_DUMMY[5:0]						
348	CKV_RISE_PREC[7:0]								
349	STV_WIDTH[3:0]				CKV_WIDTH[3:0]				
350	GOA_FLC[7:0]								
351	-	GOA_FLCA_LEAD[6:0]							
352	CLR4_LEAD	CLR3_LEAD	CLR2_LEAD	-					
353	CKV_BLANK_SHIFT[7:0]								
354	CKV_BLANK_WIDTH[7:0]								
355	CLR1_WD[7:0]								
356	CLR234_WD[7:0]								
357	CLR1_POL	CLR1_START[6:0]							
358	CLR2_POL	CLR2_START[6:0]							
359	CLR3_POL	CLR3_START[6:0]							
360	CLR4_POL	CLR4_START[6:0]							
361	-			CLR1_START_MSB[4:0]					
362	-			CLR2/3/4_START_MSB[4:0]					
364	-				FLC_NONOVLAP[1:0]			CLR1_LEAD	
366	GOA_T2B[7:0]								
367	GOA_T3B[7:0]								
375	GOA_T0B[7:0]								
376	GOA_T1B[7:0]								
377	W3	-	-	-	CKV_BLANKON	GOA_PHASE[2:0]			
379	GOA_T0[7:0]								
380	GOA_T1[7:0]								
381	GOA_T2[7:0]								
382	GOA_T3[7:0]								

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
385	-	-	STV_LEAD[4:0]						GROUP12
386	-	-	CKV_LEAD[4:0]						
387	CKV_NON OVERLAP	-	CKV_DUMMY[5:0]						
388	CKV_RISE_PREC[7:0]								
389	STV_WIDTH[3:0]				CKV_WIDTH[3:0]				
390	GOA_FLC[7:0]								
391	-	GOA_FLCA_LEAD[6:0]							
392	CLR4_LEAD	CLR3_LEAD	CLR2_LEAD	-					
393	CKV_BLANK_SHIFT[7:0]								
394	CKV_BLANK_WIDTH[7:0]								
395	CLR1_WD[7:0]								
396	CLR234_WD[7:0]								
397	CLR1_POL	CLR1_START[6:0]							
398	CLR2_POL	CLR2_START[6:0]							
399	CLR3_POL	CLR3_START[6:0]							
400	CLR4_POL	CLR4_START[6:0]							
401	-			CLR1_START_MSB[4:0]					
402	-			CLR2/3/4_START_MSB[4:0]					
404	-				FLC_NONOVLAP[1:0]			CLR1_LEAD	
406	GOA_T2B[7:0]								
407	GOA_T3B[7:0]								
415	GOA_T0B[7:0]								
416	GOA_T1B[7:0]								
419	W1	-				STV_PREC[2:0]			GROUP13
420	CKV_FALL_PREC[7:0]								
424	W2	-				STV_PREC[2:0]			
425	CKV_FALL_PREC[7:0]								
429	W3	-				STV_PREC[2:0]			
430	CKV_FALL_PREC[7:0]								
504	W1	-							GROUP30
507	GOE_WD								
512	W1	-							GROUP31
513	PGMA1R[7:0]								
514	PGMA2R[7:0]								
515	PGMA3R[7:0]								
516	PGMA4R[7:0]								
517	PGMA5R[7:0]								
518	PGMA6R[7:0]								
519	PGMA7R[7:0]								
520	PGMA8R[7:0]								
521	PGMA9R[7:0]								
522	PGMA10R[7:0]								
523	PGMA11R[7:0]								
524	PGMA12R[7:0]								
525	PGMA13R[7:0]								
526	PGMA14R[7:0]								
527	PGMA15R[7:0]								
528	PGMA16R[7:0]								
529	PGMA17R[7:0]								
530	PGMA18R[7:0]								
531	PGMA19R[7:0]								
532	PGMA20R[7:0]								
533	PGMA21R[7:0]								
534	PGMA22R[7:0]								
535	PGMA23R[7:0]								
536	PGMA24R[7:0]								

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537	PGMA1R[9:8]		PGMA2R[9:8]		PGMA3R[9:8]		PGMA4R[9:8]		GROUP31
538	PGMA5R[9:8]		PGMA6R[9:8]		PGMA7R[9:8]		PGMA8R[9:8]		
539	PGMA9R[9:8]		PGMA10R[9:8]		PGMA11R[9:8]		PGMA12R[9:8]		
540	PGMA13R[9:8]		PGMA14R[9:8]		PGMA15R[9:8]		PGMA16R[9:8]		
541	PGMA17R[9:8]		PGMA18R[9:8]		PGMA19R[9:8]		PGMA20R[9:8]		
542	PGMA21R[9:8]		PGMA22R[9:8]		PGMA23R[9:8]		PGMA24R[9:8]		
543	W2	-							
544	PGMA1R[7:0]								
545	PGMA2R[7:0]								
546	PGMA3R[7:0]								
547	PGMA4R[7:0]								
548	PGMA5R[7:0]								
549	PGMA6R[7:0]								
550	PGMA7R[7:0]								
551	PGMA8R[7:0]								
552	PGMA9R[7:0]								
553	PGMA10R[7:0]								
554	PGMA11R[7:0]								
555	PGMA12R[7:0]								
556	PGMA13R[7:0]								
557	PGMA14R[7:0]								
558	PGMA15R[7:0]								
559	PGMA16R[7:0]								
560	PGMA17R[7:0]								
561	PGMA18R[7:0]								
562	PGMA19R[7:0]								
563	PGMA20R[7:0]								
564	PGMA21R[7:0]								
565	PGMA22R[7:0]								
566	PGMA23R[7:0]								
567	PGMA24R[7:0]								
568	PGMA1R[9:8]		PGMA2R[9:8]		PGMA3R[9:8]		PGMA4R[9:8]		
569	PGMA5R[9:8]		PGMA6R[9:8]		PGMA7R[9:8]		PGMA8R[9:8]		
570	PGMA9R[9:8]		PGMA10R[9:8]		PGMA11R[9:8]		PGMA12R[9:8]		
571	PGMA13R[9:8]		PGMA14R[9:8]		PGMA15R[9:8]		PGMA16R[9:8]		
572	PGMA17R[9:8]		PGMA18R[9:8]		PGMA19R[9:8]		PGMA20R[9:8]		
573	PGMA21R[9:8]		PGMA22R[9:8]		PGMA23R[9:8]		PGMA24R[9:8]		
574	W1	-							GROUP32
575	PGMA1G[7:0]								
576	PGMA2G[7:0]								
577	PGMA3G[7:0]								
578	PGMA4G[7:0]								
579	PGMA5G[7:0]								
580	PGMA6G[7:0]								
581	PGMA7G[7:0]								
582	PGMA8G[7:0]								
583	PGMA9G[7:0]								
584	PGMA10G[7:0]								
585	PGMA11G[7:0]								
586	PGMA12G[7:0]								
587	PGMA13G[7:0]								
588	PGMA14G[7:0]								
589	PGMA15G[7:0]								
590	PGMA16G[7:0]								
591	PGMA17G[7:0]								
592	PGMA18G[7:0]								
593	PGMA19G[7:0]								
594	PGMA20G[7:0]								
595	PGMA21G[7:0]								
596	PGMA22G[7:0]								
597	PGMA23G[7:0]								

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
598	PGMA24G[7:0]								GROUP32
599	PGMA1G[9:8]		PGMA1G[9:8]		PGMA1G[9:8]		PGMA1G[9:8]		
600	PGMA5G[9:8]		PGMA5G[9:8]		PGMA5G[9:8]		PGMA5G[9:8]		
601	PGMA9G[9:8]		PGMA9G[9:8]		PGMA9G[9:8]		PGMA9G[9:8]		
602	PGMA13G[9:8]		PGMA13G[9:8]		PGMA13G[9:8]		PGMA13G[9:8]		
603	PGMA17G[9:8]		PGMA17G[9:8]		PGMA17G[9:8]		PGMA17G[9:8]		
604	PGMA21G[9:8]		PGMA21G[9:8]		PGMA21G[9:8]		PGMA21G[9:8]		
605	W2	-							
606	PGMA1G[7:0]								
607	PGMA2G[7:0]								
608	PGMA3G[7:0]								
609	PGMA4G[7:0]								
610	PGMA5G[7:0]								
611	PGMA6G[7:0]								
612	PGMA7G[7:0]								
613	PGMA8G[7:0]								
614	PGMA9G[7:0]								
615	PGMA10G[7:0]								
616	PGMA11G[7:0]								
617	PGMA12G[7:0]								
618	PGMA13G[7:0]								
619	PGMA14G[7:0]								
620	PGMA15G[7:0]								
621	PGMA16G[7:0]								
622	PGMA17G[7:0]								
623	PGMA18G[7:0]								
624	PGMA19G[7:0]								
625	PGMA20G[7:0]								
626	PGMA21G[7:0]								
627	PGMA22G[7:0]								
628	PGMA23G[7:0]								
629	PGMA24G[7:0]								
630	PGMA1G[9:8]		PGMA1G[9:8]		PGMA1G[9:8]		PGMA1G[9:8]		
631	PGMA5G[9:8]		PGMA5G[9:8]		PGMA5G[9:8]		PGMA5G[9:8]		
632	PGMA9G[9:8]		PGMA9G[9:8]		PGMA9G[9:8]		PGMA9G[9:8]		
633	PGMA13G[9:8]		PGMA13G[9:8]		PGMA13G[9:8]		PGMA13G[9:8]		
634	PGMA17G[9:8]		PGMA17G[9:8]		PGMA17G[9:8]		PGMA17G[9:8]		
635	PGMA21G[9:8]		PGMA21G[9:8]		PGMA21G[9:8]		PGMA21G[9:8]		
636	W1	-							GROUP33
637	PGMA1B[7:0]								
638	PGMA2B[7:0]								
639	PGMA3B[7:0]								
640	PGMA4B[7:0]								
641	PGMA5B[7:0]								
642	PGMA6B[7:0]								
643	PGMA7B[7:0]								
644	PGMA8B[7:0]								
645	PGMA9B[7:0]								
646	PGMA10B[7:0]								
647	PGMA11B[7:0]								
648	PGMA12B[7:0]								
649	PGMA13B[7:0]								
650	PGMA14B[7:0]								
651	PGMA15B[7:0]								
652	PGMA16B[7:0]								
653	PGMA17B[7:0]								
654	PGMA18B[7:0]								
655	PGMA19B[7:0]								
656	PGMA20B[7:0]								
657	PGMA21B[7:0]								
658	PGMA22B[7:0]								
659	PGMA23B[7:0]								

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
660	PGMA24B[7:0]								GROUP33
661	PGMA1B[9:8]		PGMA1B[9:8]		PGMA1B[9:8]		PGMA1B[9:8]		
662	PGMA5B[9:8]		PGMA5B[9:8]		PGMA5B[9:8]		PGMA5B[9:8]		
663	PGMA9B[9:8]		PGMA9B[9:8]		PGMA9B[9:8]		PGMA9B[9:8]		
664	PGMA13B[9:8]		PGMA13B[9:8]		PGMA13B[9:8]		PGMA13B[9:8]		
665	PGMA17B[9:8]		PGMA17B[9:8]		PGMA17B[9:8]		PGMA17B[9:8]		
666	PGMA21B[9:8]		PGMA21B[9:8]		PGMA21B[9:8]		PGMA21B[9:8]		
667	W2	-							
668	PGMA1B[7:0]								
669	PGMA2B[7:0]								
670	PGMA3B[7:0]								
671	PGMA4B[7:0]								
672	PGMA5B[7:0]								
673	PGMA6B[7:0]								
674	PGMA7B[7:0]								
675	PGMA8B[7:0]								
676	PGMA9B[7:0]								
677	PGMA10B[7:0]								
678	PGMA11B[7:0]								
679	PGMA12B[7:0]								
680	PGMA13B[7:0]								
681	PGMA14B[7:0]								
682	PGMA15B[7:0]								
683	PGMA16B[7:0]								
684	PGMA17B[7:0]								
685	PGMA18B[7:0]								
686	PGMA19B[7:0]								
687	PGMA20B[7:0]								
688	PGMA21B[7:0]								
689	PGMA22B[7:0]								
690	PGMA23B[7:0]								
691	PGMA24B[7:0]								
692	PGMA1B[9:8]		PGMA1B[9:8]		PGMA1B[9:8]		PGMA1B[9:8]		
693	PGMA5B[9:8]		PGMA5B[9:8]		PGMA5B[9:8]		PGMA5B[9:8]		
694	PGMA9B[9:8]		PGMA9B[9:8]		PGMA9B[9:8]		PGMA9B[9:8]		
695	PGMA13B[9:8]		PGMA13B[9:8]		PGMA13B[9:8]		PGMA13B[9:8]		
696	PGMA17B[9:8]		PGMA17B[9:8]		PGMA17B[9:8]		PGMA17B[9:8]		
697	PGMA21B[9:8]		PGMA21B[9:8]		PGMA21B[9:8]		PGMA21B[9:8]		
698	W1	-							GROUP34
699	NGMA1R[7:0]								
700	NGMA2R[7:0]								
701	NGMA3R[7:0]								
702	NGMA4R[7:0]								
703	NGMA5R[7:0]								
704	NGMA6R[7:0]								
705	NGMA7R[7:0]								
706	NGMA8R[7:0]								
707	NGMA9R[7:0]								
708	NGMA10R[7:0]								
709	NGMA11R[7:0]								
710	NGMA12R[7:0]								
711	NGMA13R[7:0]								
712	NGMA14R[7:0]								
713	NGMA15R[7:0]								
714	NGMA16R[7:0]								
715	NGMA17R[7:0]								
716	NGMA18R[7:0]								
717	NGMA19R[7:0]								
718	NGMA20R[7:0]								
719	NGMA21R[7:0]								
720	NGMA22R[7:0]								
721	NGMA23R[7:0]								

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722	NGMA24R[7:0]								GROUP34
723	NGMA1R[9:8]		NGMA2R[9:8]		NGMA3R[9:8]		NGMA4R[9:8]		
724	NGMA5R[9:8]		NGMA6R[9:8]		NGMA7R[9:8]		NGMA8R[9:8]		
725	NGMA9R[9:8]		NGMA10R[9:8]		NGMA11R[9:8]		NGMA12R[9:8]		
726	NGMA13R[9:8]		NGMA14R[9:8]		NGMA15R[9:8]		NGMA16R[9:8]		
727	NGMA17R[9:8]		NGMA18R[9:8]		NGMA19R[9:8]		NGMA20R[9:8]		
728	NGMA21R[9:8]		NGMA22R[9:8]		NGMA23R[9:8]		NGMA24R[9:8]		
729	W2	-							
730	NGMA1R[7:0]								
731	NGMA2R[7:0]								
732	NGMA3R[7:0]								
733	NGMA4R[7:0]								
734	NGMA5R[7:0]								
735	NGMA6R[7:0]								
736	NGMA7R[7:0]								
737	NGMA8R[7:0]								
738	NGMA9R[7:0]								
739	NGMA10R[7:0]								
740	NGMA11R[7:0]								
741	NGMA12R[7:0]								
742	NGMA13R[7:0]								
743	NGMA14R[7:0]								
744	NGMA15R[7:0]								
745	NGMA16R[7:0]								
746	NGMA17R[7:0]								
747	NGMA18R[7:0]								
748	NGMA19R[7:0]								
749	NGMA20R[7:0]								
750	NGMA21R[7:0]								
751	NGMA22R[7:0]								
752	NGMA23R[7:0]								
753	NGMA24R[7:0]								
754	NGMA1R[9:8]		NGMA2R[9:8]		NGMA3R[9:8]		NGMA4R[9:8]		
755	NGMA5R[9:8]		NGMA6R[9:8]		NGMA7R[9:8]		NGMA8R[9:8]		
756	NGMA9R[9:8]		NGMA10R[9:8]		NGMA11R[9:8]		NGMA12R[9:8]		
757	NGMA13R[9:8]		NGMA14R[9:8]		NGMA15R[9:8]		NGMA16R[9:8]		
758	NGMA17R[9:8]		NGMA18R[9:8]		NGMA19R[9:8]		NGMA20R[9:8]		
759	NGMA21R[9:8]		NGMA22R[9:8]		NGMA23R[9:8]		NGMA24R[9:8]		

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760	W1	-							
761	NGMA1G[7:0]								
762	NGMA2G[7:0]								
763	NGMA3G[7:0]								
764	NGMA4G[7:0]								
765	NGMA5G[7:0]								
766	NGMA6G[7:0]								
767	NGMA7G[7:0]								
768	NGMA8G[7:0]								
769	NGMA9G[7:0]								
770	NGMA10G[7:0]								
771	NGMA11G[7:0]								
772	NGMA12G[7:0]								
773	NGMA13G[7:0]								
774	NGMA14G[7:0]								
775	NGMA15G[7:0]								
776	NGMA16G[7:0]								
777	NGMA17G[7:0]								
778	NGMA18G[7:0]								
779	NGMA19G[7:0]								
780	NGMA20G[7:0]								
781	NGMA21G[7:0]								
782	NGMA22G[7:0]								
783	NGMA23G[7:0]								
784	NGMA24G[7:0]								
785	NGMA1G[9:8]		NGMA2G[9:8]		NGMA3G[9:8]		NGMA4G[9:8]		
786	NGMA5G[9:8]		NGMA6G[9:8]		NGMA7G[9:8]		NGMA8G[9:8]		
787	NGMA9G[9:8]		NGMA10G[9:8]		NGMA11G[9:8]		NGMA12G[9:8]		
788	NGMA13G[9:8]		NGMA14G[9:8]		NGMA15G[9:8]		NGMA16G[9:8]		
789	NGMA17G[9:8]		NGMA18G[9:8]		NGMA19G[9:8]		NGMA20G[9:8]		
790	NGMA21G[9:8]		NGMA22G[9:8]		NGMA23G[9:8]		NGMA24G[9:8]		
791	W2	-							GROUP35
792	NGMA1G[7:0]								
793	NGMA2G[7:0]								
794	NGMA3G[7:0]								
795	NGMA4G[7:0]								
796	NGMA5G[7:0]								
797	NGMA6G[7:0]								
798	NGMA7G[7:0]								
799	NGMA8G[7:0]								
800	NGMA9G[7:0]								
801	NGMA10G[7:0]								
802	NGMA11G[7:0]								
803	NGMA12G[7:0]								
804	NGMA13G[7:0]								
805	NGMA14G[7:0]								
806	NGMA15G[7:0]								
807	NGMA16G[7:0]								
808	NGMA17G[7:0]								
809	NGMA18G[7:0]								
810	NGMA19G[7:0]								
811	NGMA20G[7:0]								
812	NGMA21G[7:0]								
813	NGMA22G[7:0]								
814	NGMA23G[7:0]								
815	NGMA24G[7:0]								
816	NGMA1G[9:8]		NGMA2G[9:8]		NGMA3G[9:8]		NGMA4G[9:8]		
817	NGMA5G[9:8]		NGMA6G[9:8]		NGMA7G[9:8]		NGMA8G[9:8]		
818	NGMA9G[9:8]		NGMA10G[9:8]		NGMA11G[9:8]		NGMA12G[9:8]		
819	NGMA13G[9:8]		NGMA14G[9:8]		NGMA15G[9:8]		NGMA16G[9:8]		
820	NGMA17G[9:8]		NGMA18G[9:8]		NGMA19G[9:8]		NGMA20G[9:8]		
821	NGMA21G[9:8]		NGMA22G[9:8]		NGMA23G[9:8]		NGMA24G[9:8]		

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
822	W1	-							
823	NGMA1B[7:0]								
824	NGMA2B[7:0]								
825	NGMA3B[7:0]								
826	NGMA4B[7:0]								
827	NGMA5B[7:0]								
828	NGMA6B[7:0]								
829	NGMA7B[7:0]								
830	NGMA8B[7:0]								
831	NGMA9B[7:0]								
832	NGMA10B[7:0]								
833	NGMA11B[7:0]								
834	NGMA12B[7:0]								
835	NGMA13B[7:0]								
836	NGMA14B[7:0]								
837	NGMA15B[7:0]								
838	NGMA16B[7:0]								
839	NGMA17B[7:0]								
840	NGMA18B[7:0]								
841	NGMA19B[7:0]								
842	NGMA20B[7:0]								
843	NGMA21B[7:0]								
844	NGMA22B[7:0]								
845	NGMA23B[7:0]								
846	NGMA24B[7:0]								
847	NGMA1B[9:8]		NGMA2B[9:8]		NGMA3B[9:8]		NGMA4B[9:8]		
848	NGMA5B[9:8]		NGMA6B[9:8]		NGMA7B[9:8]		NGMA8B[9:8]		
849	NGMA9B[9:8]		NGMA10B[9:8]		NGMA11B[9:8]		NGMA12B[9:8]		
850	NGMA13B[9:8]		NGMA14B[9:8]		NGMA15B[9:8]		NGMA16B[9:8]		
851	NGMA17B[9:8]		NGMA18B[9:8]		NGMA19B[9:8]		NGMA20B[9:8]		
852	NGMA21B[9:8]		NGMA22B[9:8]		NGMA23B[9:8]		NGMA24B[9:8]		GROUP36
853	W2								
854	NGMA1B[7:0]								
855	NGMA2B[7:0]								
856	NGMA3B[7:0]								
857	NGMA4B[7:0]								
858	NGMA5B[7:0]								
859	NGMA6B[7:0]								
860	NGMA7B[7:0]								
861	NGMA8B[7:0]								
862	NGMA9B[7:0]								
863	NGMA10B[7:0]								
864	NGMA11B[7:0]								
865	NGMA12B[7:0]								
866	NGMA13B[7:0]								
867	NGMA14B[7:0]								
868	NGMA15B[7:0]								
869	NGMA16B[7:0]								
870	NGMA17B[7:0]								
871	NGMA18B[7:0]								
872	NGMA19B[7:0]								
873	NGMA20B[7:0]								
874	NGMA21B[7:0]								
875	NGMA22B[7:0]								
876	NGMA23B[7:0]								
877	NGMA24B[7:0]								
878	NGMA1B[9:8]		NGMA2B[9:8]		NGMA3B[9:8]		NGMA4B[9:8]		
879	NGMA5B[9:8]		NGMA6B[9:8]		NGMA7B[9:8]		NGMA8B[9:8]		
880	NGMA9B[9:8]		NGMA10B[9:8]		NGMA11B[9:8]		NGMA12B[9:8]		
881	NGMA13B[9:8]		NGMA14B[9:8]		NGMA15B[9:8]		NGMA16B[9:8]		
882	NGMA17B[9:8]		NGMA18B[9:8]		NGMA19B[9:8]		NGMA20B[9:8]		
883	NGMA21B[9:8]		NGMA22B[9:8]		NGMA23B[9:8]		NGMA24B[9:8]		

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OTP Index	D7	D6	D5	D4	D3	D2	D1	D0	Group
993	W1	-							GROUP44
994	VENDER_ID[7:0]								
995	W2	-							
996	VENDER_ID[7:0]								
997	W3	-							
998	VENDER_ID[7:0]								
999	W4	-							
1000	VENDER_ID[7:0]								
1001	W5	-							
1002	VENDER_ID[7:0]								
1003	W6	-							
1004	VENDER_ID[7:0]								
1005	W7	-							
1006	VENDER_ID[7:0]								
1007	W8	-							
1008	VENDER_ID[7:0]								

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10.4 No signal detect function

HX8252-A keeps detecting input signals for MIPI mode. The HS is only one signal by detected, and if the HS is missing, HX8252-A enters the no signal mode.

No signal mode:
A. All GOA signals will keep running
B. Keep PFM and charge pump running
C. Panel display will show black pattern

10.5 GAS function

When power is removed from an electronic device during display, the image still keeps on the LCD panel for a long time. GAS function can speed the process that image disappears.

The GAS function is a voltage detector. By GAS circuit, HX8252-A can detect low voltage of power and send control signal to discharge residual potential in LCD panel and remove image.

In any one of the following case, the chip will entry GAS function.
A. VCI is lower than 2V **(Case 1)**
B. VSP is lower than 3.5V **(Case 2)**
C. VSN is lower than -3.5V **(Case 3)**

GAS function:
A. Source output pull to VSSA
B. All GOA signals will be set to gas mode **(Setting by register)**
C. VCOM output pull to VSSA

The GAS function has debounce protection circuit. EX: If the duration of voltage drop on VCI is less than 20μs **(Ex: induced by ESD pulse)**, the GAS function will not be active even the VCI voltage level is less than 2V.

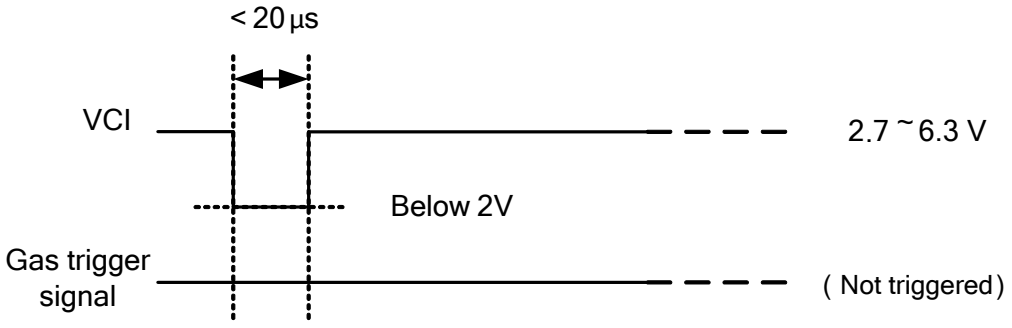


Figure 10.6: GAS function vs. VCI

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11. Gamma Adjustment Function

11.1 Digital gamma architecture

The HX8252-A incorporates digital gamma adjustment function. Gamma adjustment operation is implemented by deciding the 10-bit levels firstly in digital gamma adjustment control registers then dithering 8bit data to match the LCD panel. These registers are available for both polarities per RGB.

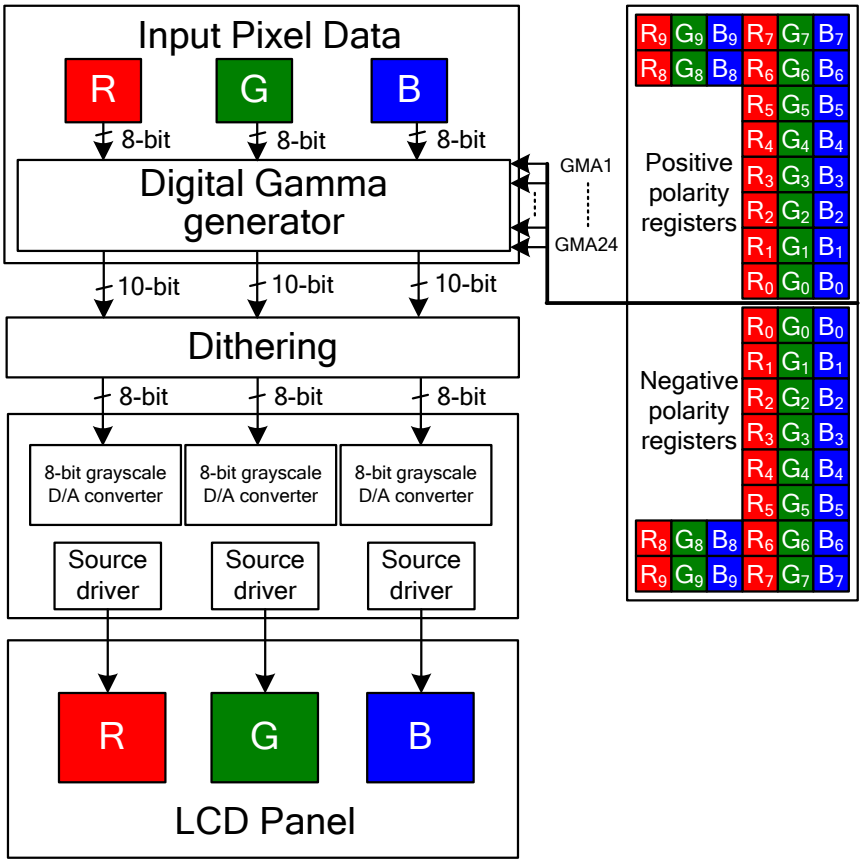
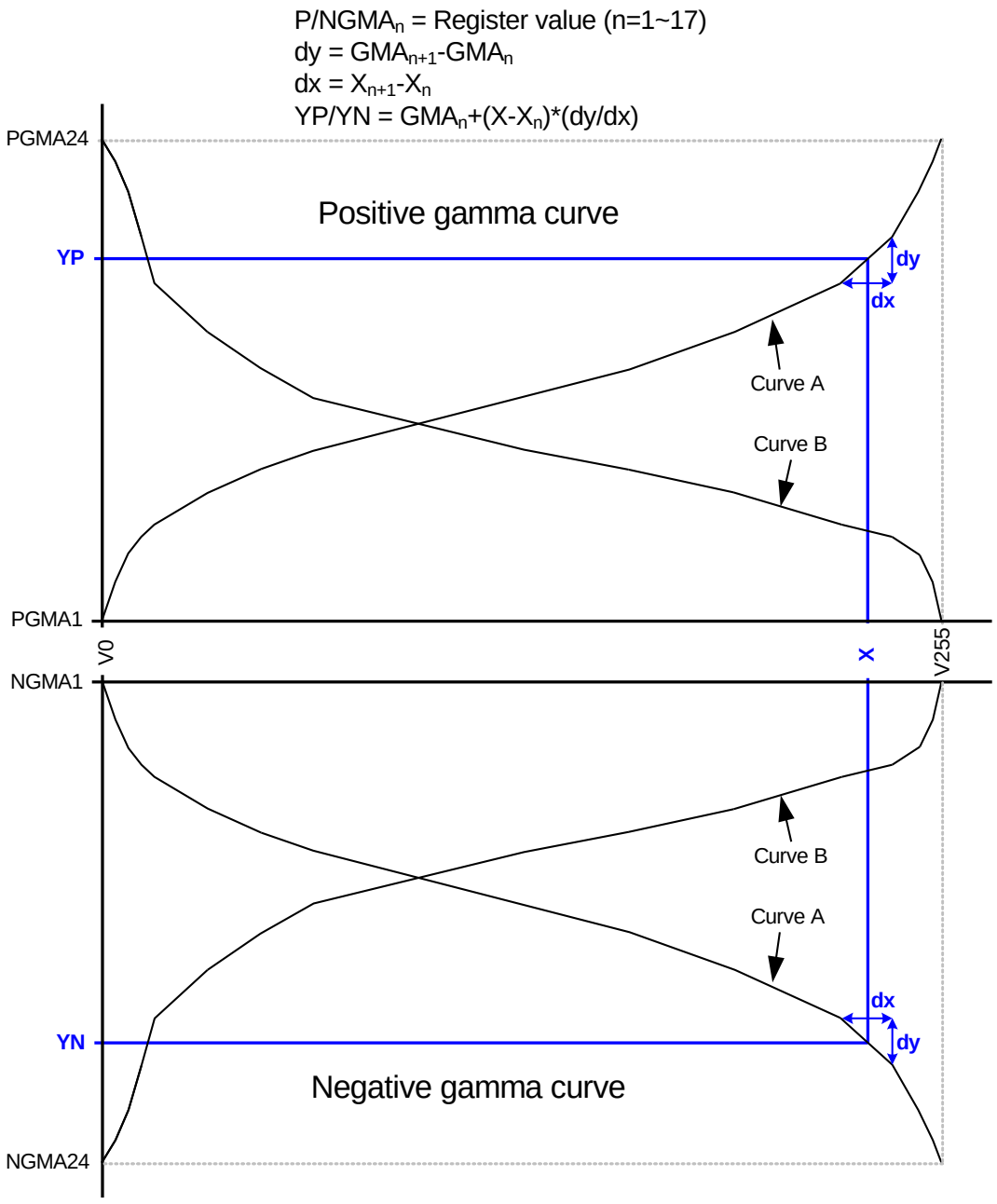


Figure 11.1: Gamma adjustment architecture

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11.2 Digital gamma curve adjustment

The gamma correction are done by 24-segment piecewise linear interpolation. The 24 segments are defined with 24 register values for level 0, 1, 3, 7, 11, 15, 23, 31, 47, 63, 95, 127, 128, 160, 192, 208, 224, 232, 240, 244, 248, 252, 254 and 255. That is, these 24 register values defined the positive and negative gamma curve and RGB data. The gamma correction output data is then fed to 10-bit data to TCON dithering IP then generates 8-bit DAC and OP to drive the source lines on the panel.



Note: (1) Curve A is for normal black panel. Curve B is for normal white panel.

Figure 11.2: Digital gamma curve adjustment

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11.3 Gamma reference voltage generator

The block consists of two gamma resister streams, one is for positive polarity and the other is for negative polarity. Gamma high/low voltage can be adjusted by VGMPH, VGMPL, VGMNH, and VGMNL.The gamma correction is done by 22 reference voltages. The 22 reference voltages are defined with 30 register values for level VP0/VP1/VP4/VP8/VP16/VP28/VP40/VP56/VP72/VP96/VP128/VP152/VP176/ VP192/VP204/VP216/VP228/VP240/VP248/VP252/VP254/VP255.

11.3.1 Positive gamma architecture

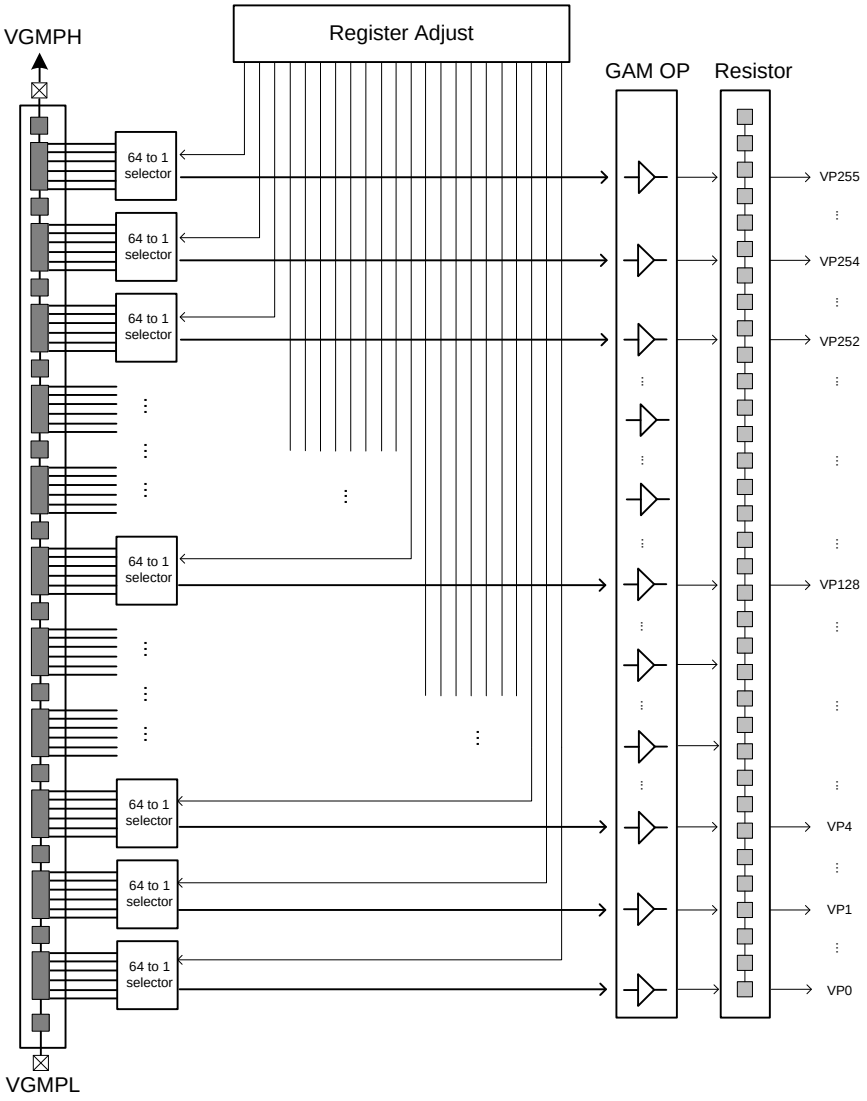


Figure 11.3: Positive gamma architecture

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11.3.2 Negative gamma architecture

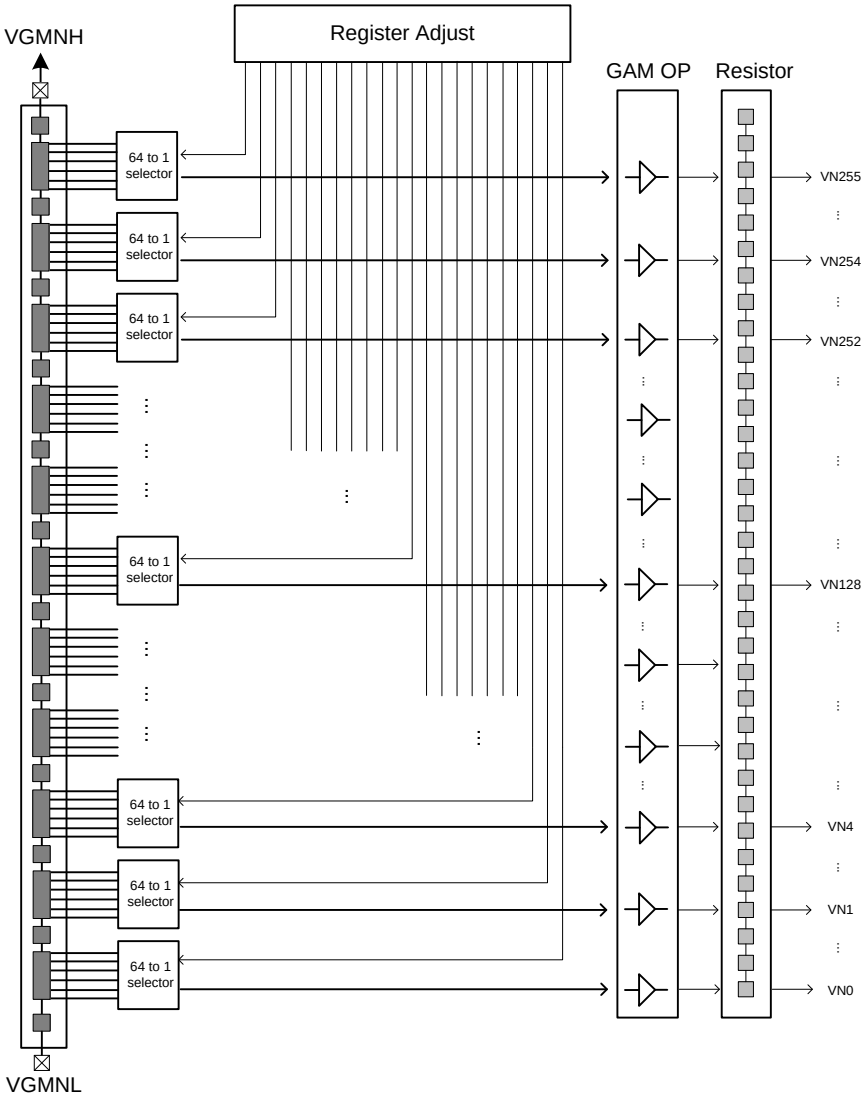


Figure 11.4: Negative gamma architecture

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12. DC Characteristics

12.1 Absolute maximum ratings

Parameter	Symbol	Spec.			Unit
		Min.	Typ.	Max.	
I/O voltage	VDDI	-0.3	-	3.96	V
Power input	VCI	-0.3	-	6.6	V
VSP voltage	VSP	-0.3	-	6.9	V
VSN voltage	VSN	-6.9	-	0.3	V
VDD_OTP **(OTP power)**	VDD_OTP	-0.3	-	8.9	V
Operating temperature(1)	T_{OPR}	-20	-	+85	℃
Storage temperature(1)	T_{STG}	-55	-	125	℃

Note: (1) Do not let condensation for low temperature.

12.2 Typical operating condition

Parameter	Symbol	Spec.			Unit
		Min.	Typ.	Max.	
VDDI voltage	VDDI	1.7	1.8	3.6	V
VCI voltage	VCI	2.7	3.3	6.5	V
VSP voltage	VSP	4.0	-	6.5	V
VSN voltage	VSN	-6.5	-	-4.0	V
VGH-VGL voltage	VGH-VGL	-	-	32	V
VDD_OTP **(OTP power)**	VDD_OTP	8.4	8.5	8.6	V

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12.3 DC electrical characteristics

(Test condition: VCI=2.7~3.6V, VSS=VSSA=0V, T_A=-20℃~+85℃)

Parameter		Symbol	Spec.			Unit
			Min.	Typ.	Max.	
VDDI Input high level voltage1		VIH1	0.8 x VDDI	-	VDDI	V
VDDI input low level voltage1		VIL1	VSS	-	0.2 x VDDI	V
Input leakage current		IL1	-1	-	+1	μA
VGLO2 output voltage		VGLO2	-15	-	-5	V
VGMAH output voltage		VGMAH	3.05	-	6.2	V
VGMAL output voltage		VGMAL	-6.2	-	-3.05	V
VCI_REG output voltage		VCI_REG	2.1	-	3.0	V
VGL output voltage		VGL	-16	-	-6	V
VGH output voltage		VGH	8	-	19	V
VCL output voltage		VCL	-2.1	-2.4	-3.0	V
VCOM output voltage		VCOM	-2.75	-1.48	-0.2	V
Input terminal pull-high resistance		RPU	-	300	-	KΩ
Input terminal pull-low resistance		RPD	-	300	-	KΩ
Source output level deviation	Graycode=0 ~ 14 Graycode=241 ~ 255	Sdev1	40	-	-	mV
	Graycode=15 ~ 31 Graycode=208 ~ 240	Sdev2	30	-	-	mV
	Graycode=32 ~ 207	Sdev3	20	-	-	mV
Source output offset deviation	Graycode=0 ~ 14 Graycode=241 ~ 255	Sdev4	50	-	-	mV
	Graycode=15 ~ 31 Graycode=208 ~ 240	Sdev5	40	-	-	mV
	Graycode=32 ~ 207	Sdev6	30	-	-	mV
Current consumption	Normal mode(1)	Ivdd	-	-	18.3	mA
		Ivci	-	-	2.5	
	STBYB mode	Ivdd	-	-	0.95	mA
		Ivci	-	-	0.55	
Rush current(2)		Ivddpeak	-	60	-	mA
VDD_OTP operation current		I_{VDD_OTP}	-	-	8	mA

Note: (1) Condition: VDDI=1.8V, VCI=3.3V, 640RGBx1920 resolution, MIPI frequence 740Mbps, frame rate=60Hz, PWRMD[1:0]=1/1, white pattern.
(2) IC internal only, excluded PFM circuit's peak current of VCI.

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12.4 MIPI DC electrical characteristics

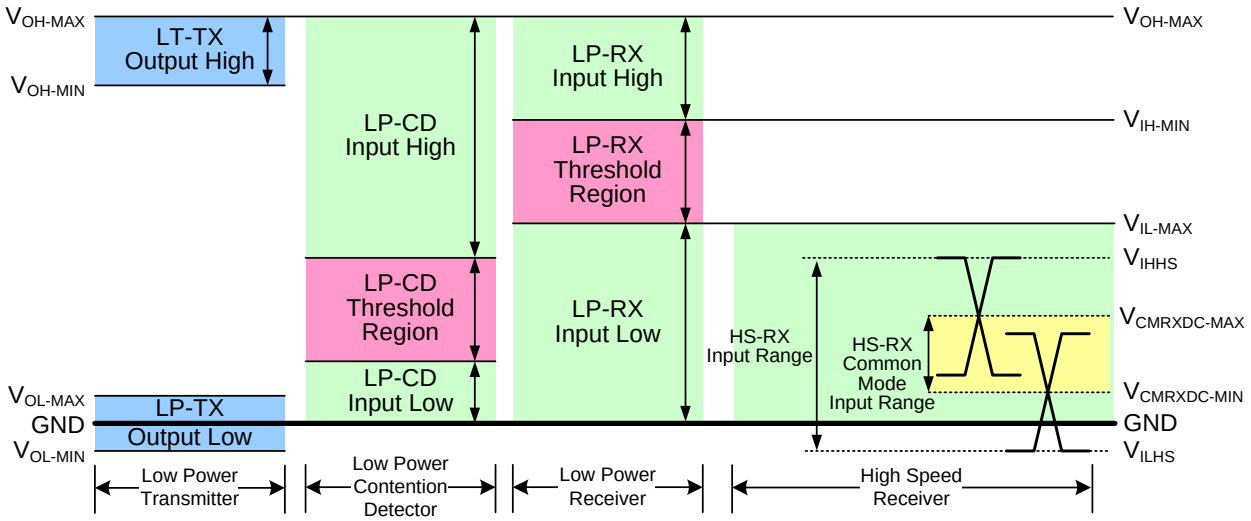


Figure 12.1: MIPI signaling and contention voltage levels

DC characteristics for MIPI LP mode

Parameter	Symbol	Spec.			Unit
		Min.	Typ.	Max.	
Logic 1 input voltage	V_{IH}	880	-	-	mV
Logic 0 input voltage	V_{IL}	0	-	550	mV
Logic 1 output voltage	V_{OH}	1.1	1.2	1.3	V
Logic 0 output voltage	V_{OL}	-50	-	50	mV

DC characteristics for MIPI HS mode

Parameter	Symbol	Spec.			Unit
		Min.	Typ.	Max.	
Common-mode voltage HS Receive mode	V_{CMRXDC}	70	-	330	mV
Differential input high threshold	V_{IDTH}	-	-	70	mV
Differential input low threshold	V_{IDTL}	70	-	-	mV
Single-ended input high voltage	V_{IHHS}	-	-	460	mV
Single-ended input low voltage	V_{ILHS}	-40	-	-	mV
Differential input impedance	Z_{ID}	80	100	125	Ω
HS transmit differential voltage **(VDP-VDN)**	|VOD|	140	200	270	mV

Note: (1) V_{IDTH} and V_{IDTL} only for reference, related to power and ground noise, this spec need to check on panel performance to fine tune

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13. AC Characteristics

13.1 MIPI AC characteristics

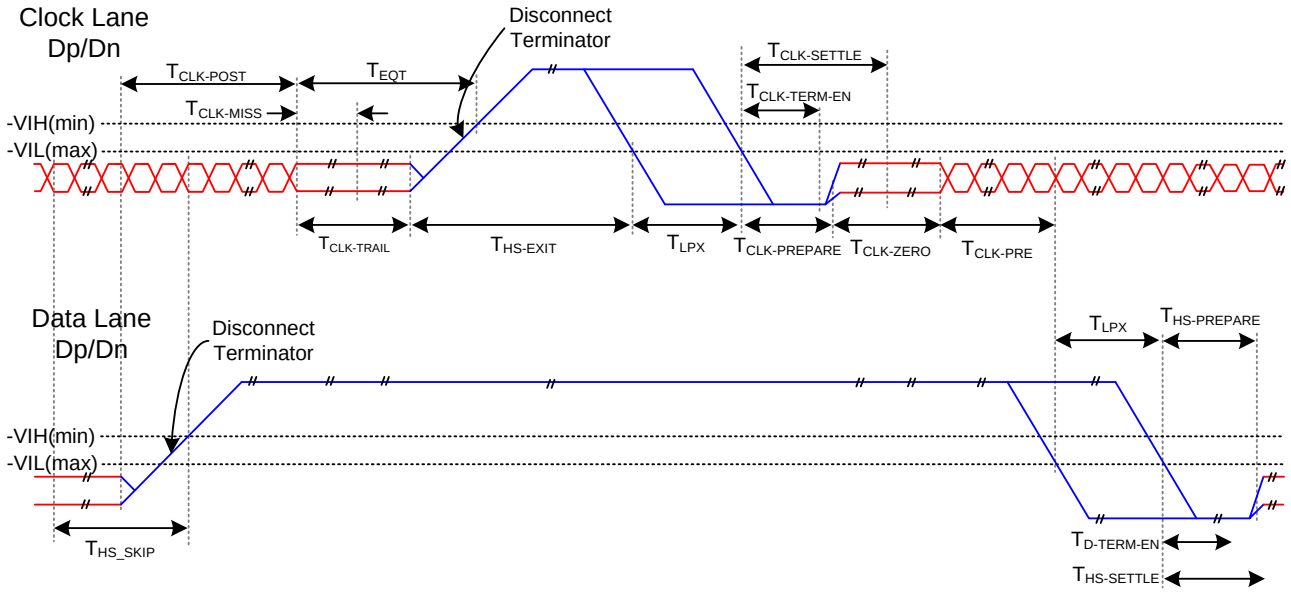


Figure 13.1: Switching the clock lane between clock transmission and low-power mode

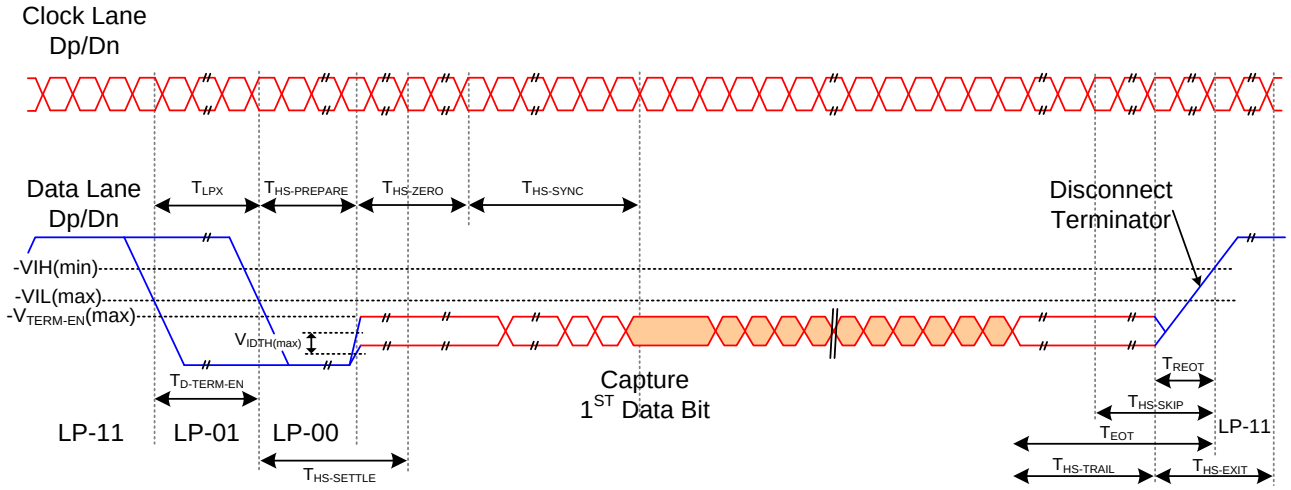


Figure 13.2: Timing of high-speed data transmission in bursts

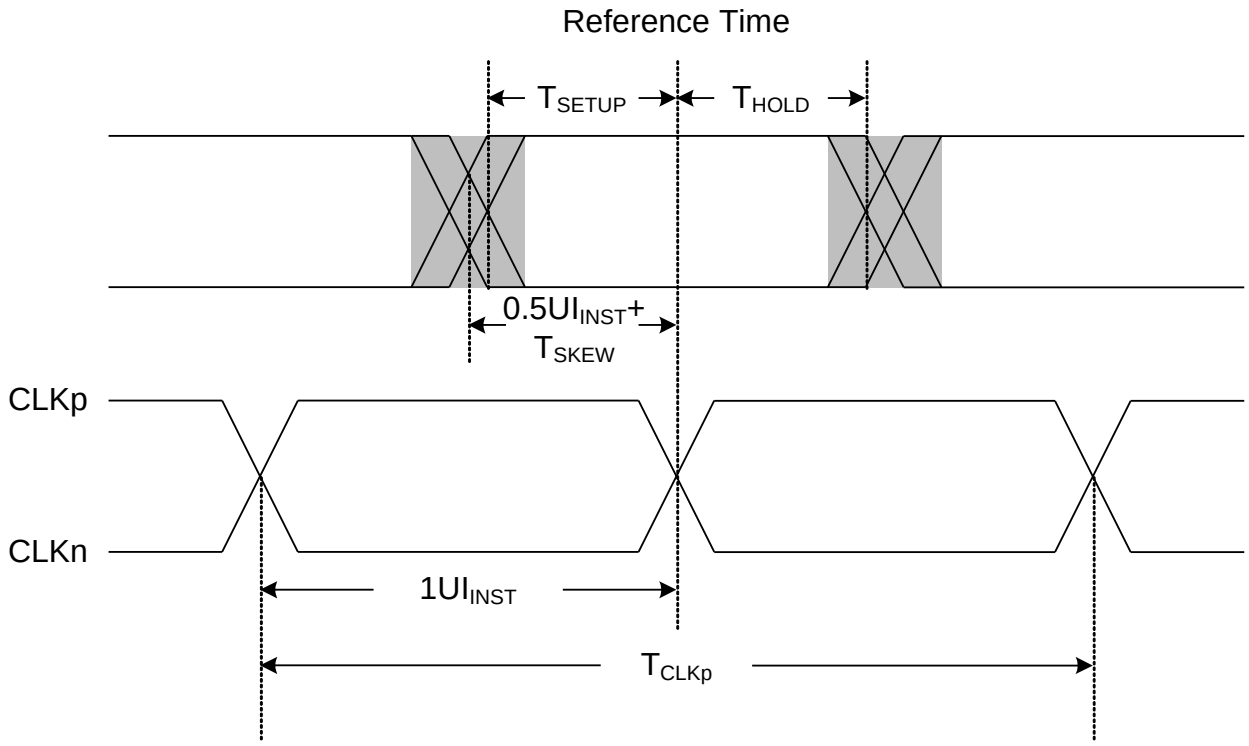
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Parameter	Description	Spec.			Unit
		Min	Typ	Max	
T_{REOT}	30%-85% rise time and fall time	-	-	35	ns
$T_{CLK-MISS}$	Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.	-	-	60	ns
$T_{CLK-POST}$*1	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of THS-TRAIL to the beginning of $T_{CLK-TRAIL}$.	60 ns + 52*UI **(For DCS)**	-	-	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8	-	-	ns
$T_{CLK-SETTLE}$	Time interval during which the HS receiver shall ignore any Clock Lane HS transitions, starting from the beginning of $T_{CLK-PRE}$.	95	-	300	ns
$T_{CLK-TERM-EN}$	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$	-	38	ns
$T_{HS-SETTLE}$	Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of $T_{HSPREPARE}$.	85 ns + 6*UI		145 ns + 10*UI	ns
T_{EOT}	Time from start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$ period to start of LP-11 state	-	-	105ns+48*UI	-
$T_{HS-EXIT}$(1)	time to drive LP-11 after HS burst	100	-	-	ns
$T_{HS-PREPARE}$	Time to drive LP-00 to prepare for HS transmission	40ns + 4*UI	-	85ns+6*UI	ns
$T_{HS-PREPARE}$ + $T_{HS-ZERO}$	$T_{HS-PREPARE}$ + Time to drive HS-0 before the Sync sequence	145ns + 10*UI	-	-	ns
$T_{HS-SKIP}$	Time-out at RX to ignore transition period of EoT	40	-	55ns+4*UI	ns
$T_{HS-TRAIL}$	Time to drive flipped differential state after last payload data bit of a HS transmission burst	60 + 4*UI	-	-	ns
T_{LPX}	Length of any Low-Power state period	50	-	-	ns
Ratio T_{LPX}	Ratio of $T_{LPX(MASTER)}/T_{LPS(SLAVE)}$ between Master and Slave side	2/3	-	3/2	-
T_{TA-GET}	Time to drive LP-00 by new TX	$5*T_{LPX}$			ns
T_{TA-GO}	Time to drive LP-00 after Turnaround Request	$4*T_{LPX}$			ns
$T_{TA-SURE}$	Time-out before new TX side starts driving	T_{LPX}	-	$2*T_{LPX}$	ns

Note: (1) For image transmission:
$T_{CLK-POST}$ min value =164 when MIPI max frequency per lane = 0.53Gbps.
$T_{CLK-POST}$ min value =112 when MIPI max frequency per lane = 1Gbps

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13.2 MIPI data-clock timing specification

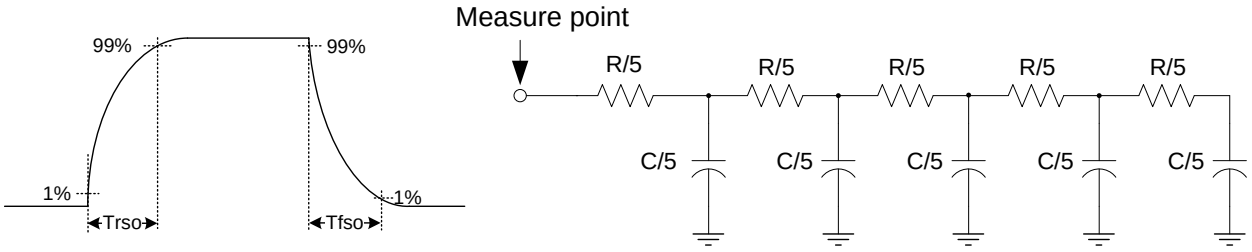


Parameter	Symbol	Spec.			Unit
		Min.	Typ.	Max.	
UI instantaneous	UI_{INST}	1.0	-	12.5(1)	ns
Data to clock setup time	T_{SETUP}	0.3	-	-	UI_{INST}
Data to clock hold time	T_{HOLD}	0.3	-	-	UI_{INST}

Note: (1) This value corresponds to a minimum 80 Mbps data rate.

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13.3 Source output timing (SOUT1 ~ SOUT2400, SL1, SR1)

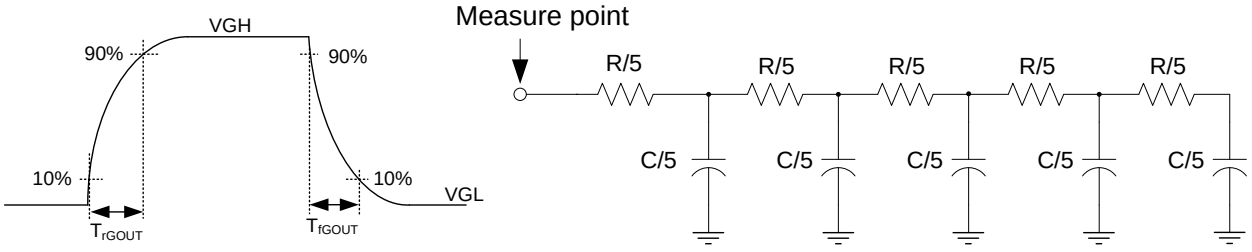


Parameter	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Source driver rising time	trSO	R=6.3KΩ, C=105.2pF	-	-	3.01	µs
Source driver falling time	tfSO		-	-	4.2	µs

Note: (1) Himax can support simulation for customer design.

Table 13.1: Source output timing

Panel control signal output 1**(GOUT1_L~GOUT22_L, GOUT1_R~GOUT22_R)**



Parameter	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Panel control signal rising time	T_{rGOUT}	R=4.42KΩ, C=156pF	-	-	1.74	µs
Panel control signal falling time	T_{fGOUT}		-	-	1.3	µs

Note: (1) Himax can support simulation for customer design.

Table 13.2: GOA output timing

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13.4 Timing requirements for RESETB

When RESETB of the reset pin equals to Low, it will be in the condition of reset. When it is in the condition of reset, it will make the device recover the initial set.

However, in order to avoid the reset noise cause reset, there is a mechanism to judge about whether the reset is needed or not.

The closed interval of Low can be shown as the following.

(VDDI=1.65V~3.6V, VSS=0V, T_A=-20℃~+85℃)

Parameter	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Reset low pulse width	T_{RST}	-	20	-	-	µs

Table 13.3: Reset timing

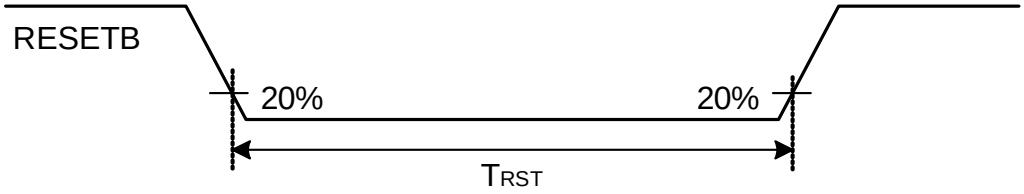


Figure 13.3: Reset timing

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14. Pin Assignment (IC Face View)

14.1 PAD sequence

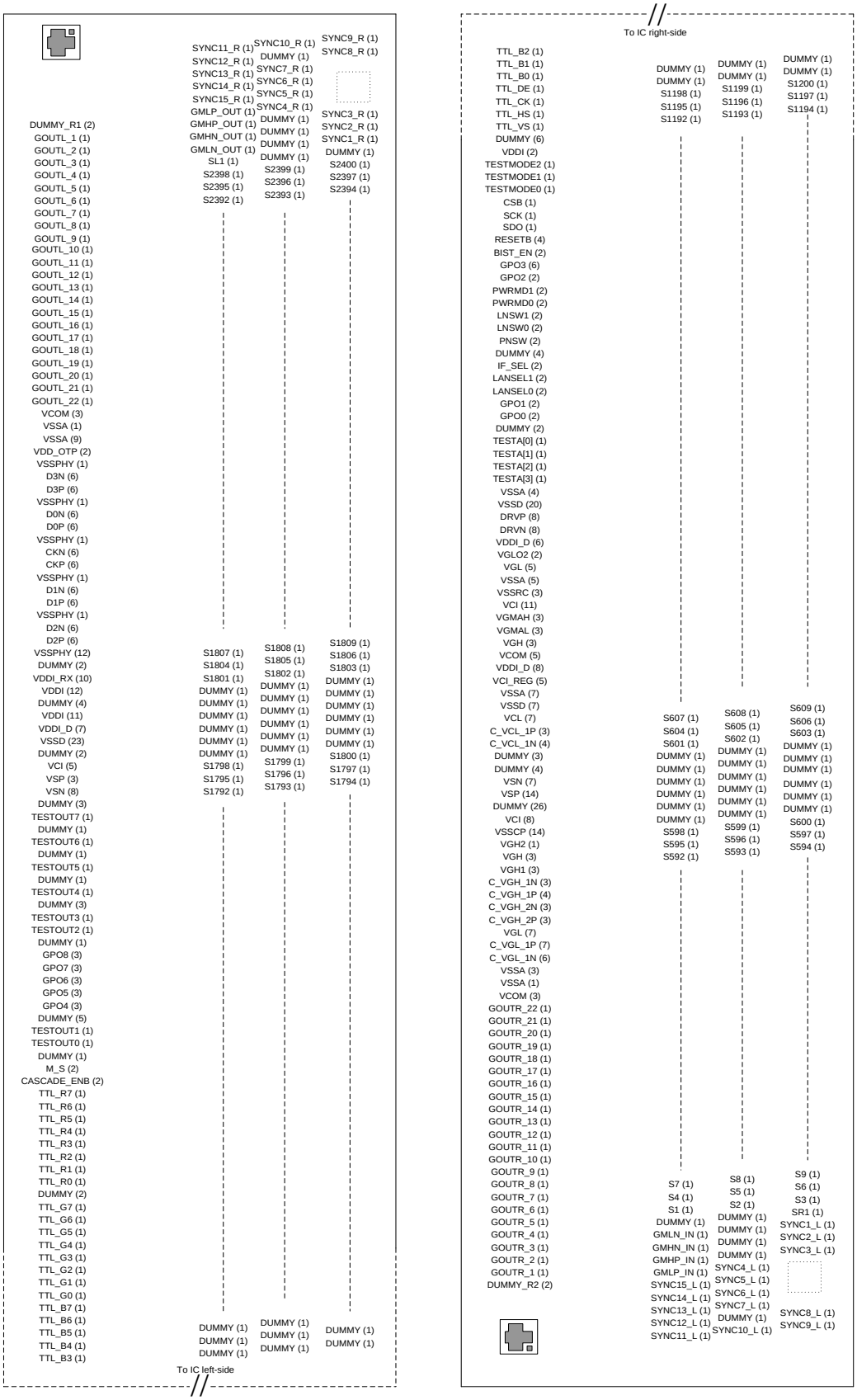
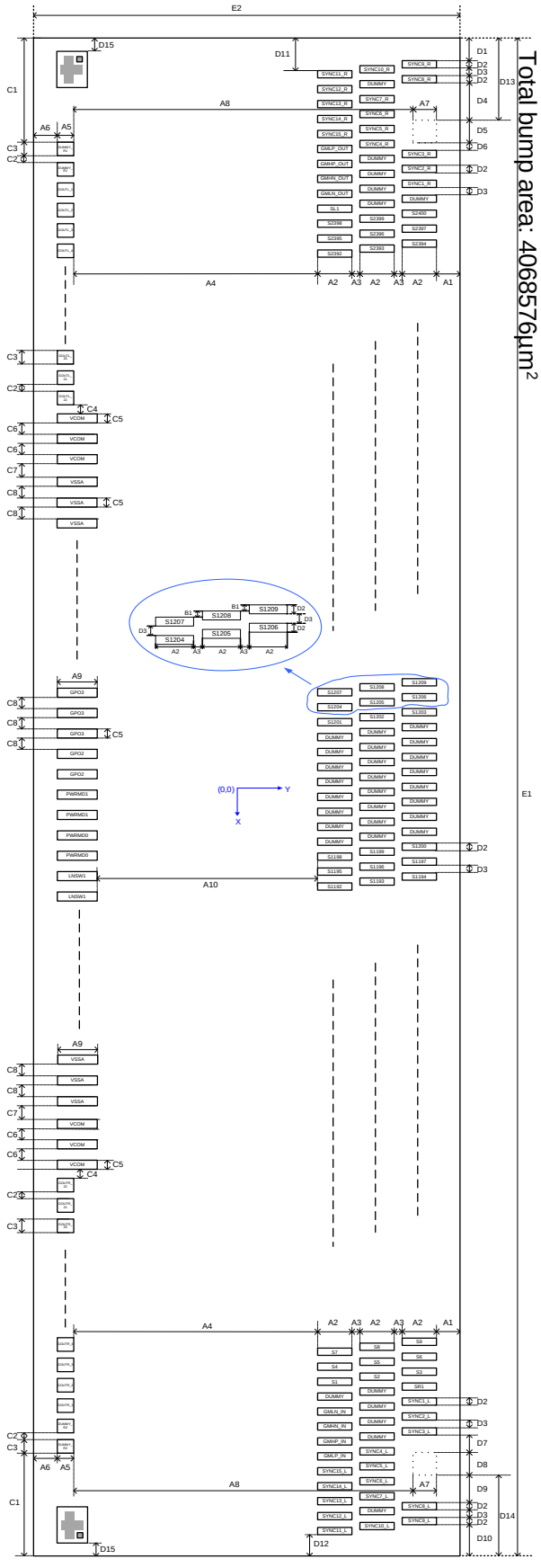


Figure 14.1: Pad sequence

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14.2 Bump information

14.2.1 Chip outline dimension



14.2.2 Pad information

Symbol	Dimension (μm)	Symbol	Dimension (μm)	Symbol	Dimension (μm)	Symbol	Dimension (μm)	Symbol	Dimension (μm)
A1	9	B1	11	C1	297.5	D1	117.5	E1	27960
A2	73	-	-	C2	15	D2	16	E2	825
A3	17	-	-	C3	30	D3	17	-	-
A4	519	-	-	C4	20	D4	82.5	-	-
A5	35	-	-	C5	20	D5	50	-	-
A6	9	-	-	C6	25	D6	16.5	-	-
A7	50	-	-	C7	35	D7	38.5	-	-
A8	722	-	-	C8	25	D8	50	-	-
A9	85	-	-	-	-	D9	60.5	-	-
A10	469	-	-	-	-	D10	139.5	-	-
-	-	-	-	-	-	D11	139.5	-	-
-	-	-	-	-	-	D12	117.5	-	-
-	-	-	-	-	-	D13	249	-	-
-	-	-	-	-	-	D14	249	-	-
-	-	-	-	-	-	D15	69	-	-

Table 14.1: Pad information

14.2.3 Alignment mark

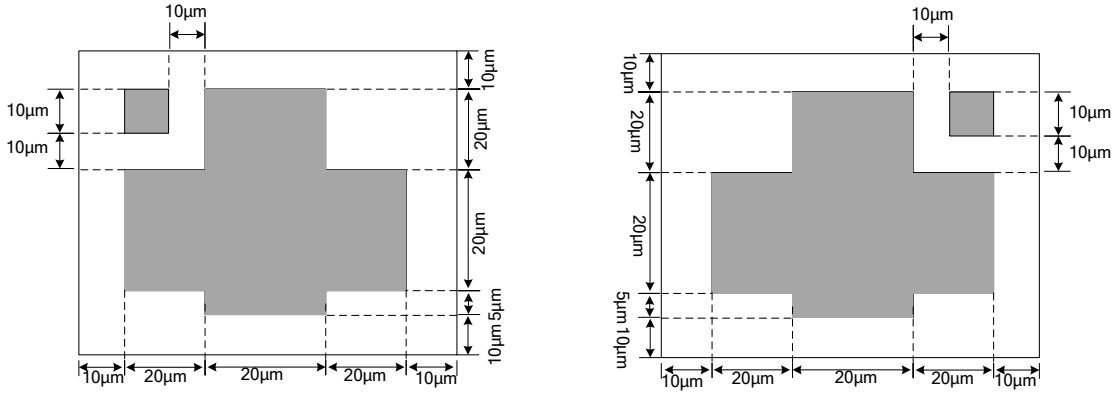
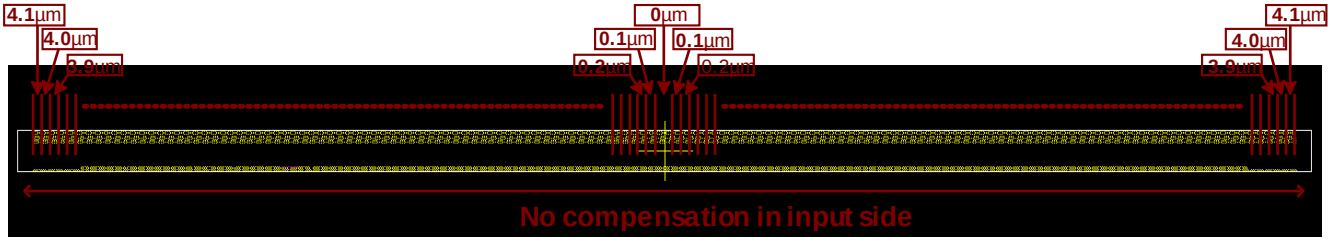


Figure 14.2: Alignment mark

14.3 Bump pad thermal compensation



IC Output Total Pitch: 27709μm
IC Shrinkage Size: 8.2μm
IC Shrinkage Rate: 0.03%

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14.4 Pad coordinates

X1: Pad coordinates before bump pad thermal compensation; X2 Pad coordinates after bump pad thermal compensation

No.	Name	X1	X2	Y	Bump size (μm)	No.	Name	X1	X2	Y	Bumpsize (μm)
1	DUMMY_R1	-13667.5	-13667.5	-386	30 x 35	51	D3P	-11407.5	-11407.5	-361	20 x 85
2	DUMMY_R1	-13622.5	-13622.5	-386	30 x 35	52	D3P	-11362.5	-11362.5	-361	20 x 85
3	GOUTL_1	-13577.5	-13577.5	-386	30 x 35	53	VSSPHY	-11317.5	-11317.5	-361	20 x 85
4	GOUTL_2	-13532.5	-13532.5	-386	30 x 35	54	D0N	-11272.5	-11272.5	-361	20 x 85
5	GOUTL_3	-13487.5	-13487.5	-386	30 x 35	55	D0N	-11227.5	-11227.5	-361	20 x 85
6	GOUTL_4	-13442.5	-13442.5	-386	30 x 35	56	D0N	-11182.5	-11182.5	-361	20 x 85
7	GOUTL_5	-13397.5	-13397.5	-386	30 x 35	57	D0N	-11137.5	-11137.5	-361	20 x 85
8	GOUTL_6	-13352.5	-13352.5	-386	30 x 35	58	D0N	-11092.5	-11092.5	-361	20 x 85
9	GOUTL_7	-13307.5	-13307.5	-386	30 x 35	59	D0N	-11047.5	-11047.5	-361	20 x 85
10	GOUTL_8	-13262.5	-13262.5	-386	30 x 35	60	D0P	-11002.5	-11002.5	-361	20 x 85
11	GOUTL_9	-13217.5	-13217.5	-386	30 x 35	61	D0P	-10957.5	-10957.5	-361	20 x 85
12	GOUTL_10	-13172.5	-13172.5	-386	30 x 35	62	D0P	-10912.5	-10912.5	-361	20 x 85
13	GOUTL_11	-13127.5	-13127.5	-386	30 x 35	63	D0P	-10867.5	-10867.5	-361	20 x 85
14	GOUTL_12	-13082.5	-13082.5	-386	30 x 35	64	D0P	-10822.5	-10822.5	-361	20 x 85
15	GOUTL_13	-13037.5	-13037.5	-386	30 x 35	65	D0P	-10777.5	-10777.5	-361	20 x 85
16	GOUTL_14	-12992.5	-12992.5	-386	30 x 35	66	VSSPHY	-10732.5	-10732.5	-361	20 x 85
17	GOUTL_15	-12947.5	-12947.5	-386	30 x 35	67	CKN	-10687.5	-10687.5	-361	20 x 85
18	GOUTL_16	-12902.5	-12902.5	-386	30 x 35	68	CKN	-10642.5	-10642.5	-361	20 x 85
19	GOUTL_17	-12857.5	-12857.5	-386	30 x 35	69	CKN	-10597.5	-10597.5	-361	20 x 85
20	GOUTL_18	-12812.5	-12812.5	-386	30 x 35	70	CKN	-10552.5	-10552.5	-361	20 x 85
21	GOUTL_19	-12767.5	-12767.5	-386	30 x 35	71	CKN	-10507.5	-10507.5	-361	20 x 85
22	GOUTL_20	-12722.5	-12722.5	-386	30 x 35	72	CKN	-10462.5	-10462.5	-361	20 x 85
23	GOUTL_21	-12677.5	-12677.5	-386	30 x 35	73	CKP	-10417.5	-10417.5	-361	20 x 85
24	GOUTL_22	-12632.5	-12632.5	-386	30 x 35	74	CKP	-10372.5	-10372.5	-361	20 x 85
25	VCOM	-12587.5	-12587.5	-361	20 x 85	75	CKP	-10327.5	-10327.5	-361	20 x 85
26	VCOM	-12542.5	-12542.5	-361	20 x 85	76	CKP	-10282.5	-10282.5	-361	20 x 85
27	VCOM	-12497.5	-12497.5	-361	20 x 85	77	CKP	-10237.5	-10237.5	-361	20 x 85
28	VSSA	-12442.5	-12442.5	-361	20 x 85	78	CKP	-10192.5	-10192.5	-361	20 x 85
29	VSSA	-12397.5	-12397.5	-361	20 x 85	79	VSSPHY	-10147.5	-10147.5	-361	20 x 85
30	VSSA	-12352.5	-12352.5	-361	20 x 85	80	D1N	-10102.5	-10102.5	-361	20 x 85
31	VSSA	-12307.5	-12307.5	-361	20 x 85	81	D1N	-10057.5	-10057.5	-361	20 x 85
32	VSSA	-12262.5	-12262.5	-361	20 x 85	82	D1N	-10012.5	-10012.5	-361	20 x 85
33	VSSA	-12217.5	-12217.5	-361	20 x 85	83	D1N	-9967.5	-9967.5	-361	20 x 85
34	VSSA	-12172.5	-12172.5	-361	20 x 85	84	D1N	-9922.5	-9922.5	-361	20 x 85
35	VSSA	-12127.5	-12127.5	-361	20 x 85	85	D1N	-9877.5	-9877.5	-361	20 x 85
36	VSSA	-12082.5	-12082.5	-361	20 x 85	86	D1P	-9832.5	-9832.5	-361	20 x 85
37	VSSA	-12037.5	-12037.5	-361	20 x 85	87	D1P	-9787.5	-9787.5	-361	20 x 85
38	VDD_OTP	-11992.5	-11992.5	-361	20 x 85	88	D1P	-9742.5	-9742.5	-361	20 x 85
39	VDD_OTP	-11947.5	-11947.5	-361	20 x 85	89	D1P	-9697.5	-9697.5	-361	20 x 85
40	VSSPHY	-11902.5	-11902.5	-361	20 x 85	90	D1P	-9652.5	-9652.5	-361	20 x 85
41	D3N	-11857.5	-11857.5	-361	20 x 85	91	D1P	-9607.5	-9607.5	-361	20 x 85
42	D3N	-11812.5	-11812.5	-361	20 x 85	92	VSSPHY	-9562.5	-9562.5	-361	20 x 85
43	D3N	-11767.5	-11767.5	-361	20 x 85	93	D2N	-9517.5	-9517.5	-361	20 x 85
44	D3N	-11722.5	-11722.5	-361	20 x 85	94	D2N	-9472.5	-9472.5	-361	20 x 85
45	D3N	-11677.5	-11677.5	-361	20 x 85	95	D2N	-9427.5	-9427.5	-361	20 x 85
46	D3N	-11632.5	-11632.5	-361	20 x 85	96	D2N	-9382.5	-9382.5	-361	20 x 85
47	D3P	-11587.5	-11587.5	-361	20 x 85	97	D2N	-9337.5	-9337.5	-361	20 x 85
48	D3P	-11542.5	-11542.5	-361	20 x 85	98	D2N	-9292.5	-9292.5	-361	20 x 85
49	D3P	-11497.5	-11497.5	-361	20 x 85	99	D2P	-9247.5	-9247.5	-361	20 x 85
50	D3P	-11452.5	-11452.5	-361	20 x 85	100	D2P	-9202.5	-9202.5	-361	20 x 85

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No.	Name	X1	X2	Y	Bump size (μm)
101	D2P	-9157.5	-9157.5	-361	20 x 85
102	D2P	-9112.5	-9112.5	-361	20 x 85
103	D2P	-9067.5	-9067.5	-361	20 x 85
104	D2P	-9022.5	-9022.5	-361	20 x 85
105	VSSPHY	-8977.5	-8977.5	-361	20 x 85
106	VSSPHY	-8932.5	-8932.5	-361	20 x 85
107	VSSPHY	-8887.5	-8887.5	-361	20 x 85
108	VSSPHY	-8842.5	-8842.5	-361	20 x 85
109	VSSPHY	-8797.5	-8797.5	-361	20 x 85
110	VSSPHY	-8752.5	-8752.5	-361	20 x 85
111	VSSPHY	-8707.5	-8707.5	-361	20 x 85
112	VSSPHY	-8662.5	-8662.5	-361	20 x 85
113	VSSPHY	-8617.5	-8617.5	-361	20 x 85
114	VSSPHY	-8572.5	-8572.5	-361	20 x 85
115	VSSPHY	-8527.5	-8527.5	-361	20 x 85
116	VSSPHY	-8482.5	-8482.5	-361	20 x 85
117	DUMMY	-8437.5	-8437.5	-361	20 x 85
118	DUMMY	-8392.5	-8392.5	-361	20 x 85
119	VDDI_RX	-8347.5	-8347.5	-361	20 x 85
120	VDDI_RX	-8302.5	-8302.5	-361	20 x 85
121	VDDI_RX	-8257.5	-8257.5	-361	20 x 85
122	VDDI_RX	-8212.5	-8212.5	-361	20 x 85
123	VDDI_RX	-8167.5	-8167.5	-361	20 x 85
124	VDDI_RX	-8122.5	-8122.5	-361	20 x 85
125	VDDI_RX	-8077.5	-8077.5	-361	20 x 85
126	VDDI_RX	-8032.5	-8032.5	-361	20 x 85
127	VDDI_RX	-7987.5	-7987.5	-361	20 x 85
128	VDDI_RX	-7942.5	-7942.5	-361	20 x 85
129	VDDI	-7897.5	-7897.5	-361	20 x 85
130	VDDI	-7852.5	-7852.5	-361	20 x 85
131	VDDI	-7807.5	-7807.5	-361	20 x 85
132	VDDI	-7762.5	-7762.5	-361	20 x 85
133	VDDI	-7717.5	-7717.5	-361	20 x 85
134	VDDI	-7672.5	-7672.5	-361	20 x 85
135	VDDI	-7627.5	-7627.5	-361	20 x 85
136	VDDI	-7582.5	-7582.5	-361	20 x 85
137	VDDI	-7537.5	-7537.5	-361	20 x 85
138	VDDI	-7492.5	-7492.5	-361	20 x 85
139	VDDI	-7447.5	-7447.5	-361	20 x 85
140	VDDI	-7402.5	-7402.5	-361	20 x 85
141	DUMMY	-7357.5	-7357.5	-361	20 x 85
142	DUMMY	-7312.5	-7312.5	-361	20 x 85
143	DUMMY	-7267.5	-7267.5	-361	20 x 85
144	DUMMY	-7222.5	-7222.5	-361	20 x 85
145	VDDI	-7177.5	-7177.5	-361	20 x 85
146	VDDI	-7132.5	-7132.5	-361	20 x 85
147	VDDI	-7087.5	-7087.5	-361	20 x 85
148	VDDI	-7042.5	-7042.5	-361	20 x 85
149	VDDI	-6997.5	-6997.5	-361	20 x 85
150	VDDI	-6952.5	-6952.5	-361	20 x 85

No.	Name	X1	X2	Y	Bumpsize (μm)
151	VDDI	-6907.5	-6907.5	-361	20 x 85
152	VDDI	-6862.5	-6862.5	-361	20 x 85
153	VDDI	-6817.5	-6817.5	-361	20 x 85
154	VDDI	-6772.5	-6772.5	-361	20 x 85
155	VDDI	-6727.5	-6727.5	-361	20 x 85
156	VDDI_D	-6682.5	-6682.5	-361	20 x 85
157	VDDI_D	-6637.5	-6637.5	-361	20 x 85
158	VDDI_D	-6592.5	-6592.5	-361	20 x 85
159	VDDI_D	-6547.5	-6547.5	-361	20 x 85
160	VDDI_D	-6502.5	-6502.5	-361	20 x 85
161	VDDI_D	-6457.5	-6457.5	-361	20 x 85
162	VDDI_D	-6412.5	-6412.5	-361	20 x 85
163	VSSD	-6367.5	-6367.5	-361	20 x 85
164	VSSD	-6322.5	-6322.5	-361	20 x 85
165	VSSD	-6277.5	-6277.5	-361	20 x 85
166	VSSD	-6232.5	-6232.5	-361	20 x 85
167	VSSD	-6187.5	-6187.5	-361	20 x 85
168	VSSD	-6142.5	-6142.5	-361	20 x 85
169	VSSD	-6097.5	-6097.5	-361	20 x 85
170	VSSD	-6052.5	-6052.5	-361	20 x 85
171	VSSD	-6007.5	-6007.5	-361	20 x 85
172	VSSD	-5962.5	-5962.5	-361	20 x 85
173	VSSD	-5917.5	-5917.5	-361	20 x 85
174	VSSD	-5872.5	-5872.5	-361	20 x 85
175	VSSD	-5827.5	-5827.5	-361	20 x 85
176	VSSD	-5782.5	-5782.5	-361	20 x 85
177	VSSD	-5737.5	-5737.5	-361	20 x 85
178	VSSD	-5692.5	-5692.5	-361	20 x 85
179	VSSD	-5647.5	-5647.5	-361	20 x 85
180	VSSD	-5602.5	-5602.5	-361	20 x 85
181	VSSD	-5557.5	-5557.5	-361	20 x 85
182	VSSD	-5512.5	-5512.5	-361	20 x 85
183	VSSD	-5467.5	-5467.5	-361	20 x 85
184	VSSD	-5422.5	-5422.5	-361	20 x 85
185	VSSD	-5377.5	-5377.5	-361	20 x 85
186	DUMMY	-5332.5	-5332.5	-361	20 x 85
187	DUMMY	-5287.5	-5287.5	-361	20 x 85
188	VCI	-5242.5	-5242.5	-361	20 x 85
189	VCI	-5197.5	-5197.5	-361	20 x 85
190	VCI	-5152.5	-5152.5	-361	20 x 85
191	VCI	-5107.5	-5107.5	-361	20 x 85
192	VCI	-5062.5	-5062.5	-361	20 x 85
193	VSP	-5017.5	-5017.5	-361	20 x 85
194	VSP	-4972.5	-4972.5	-361	20 x 85
195	VSP	-4927.5	-4927.5	-361	20 x 85
196	VSN	-4882.5	-4882.5	-361	20 x 85
197	VSN	-4837.5	-4837.5	-361	20 x 85
198	VSN	-4792.5	-4792.5	-361	20 x 85
199	VSN	-4747.5	-4747.5	-361	20 x 85
200	VSN	-4702.5	-4702.5	-361	20 x 85

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No.	Name	X1	X2	Y	Bump size (μm)
201	VSN	-4657.5	-4657.5	-361	20 x 85
202	VSN	-4612.5	-4612.5	-361	20 x 85
203	VSN	-4567.5	-4567.5	-361	20 x 85
204	DUMMY	-4522.5	-4522.5	-361	20 x 85
205	DUMMY	-4477.5	-4477.5	-361	20 x 85
206	DUMMY	-4432.5	-4432.5	-361	20 x 85
207	TESTOUT7	-4387.5	-4387.5	-361	20 x 85
208	DUMMY	-4342.5	-4342.5	-361	20 x 85
209	TESTOUT6	-4297.5	-4297.5	-361	20 x 85
210	DUMMY	-4252.5	-4252.5	-361	20 x 85
211	TESTOUT5	-4207.5	-4207.5	-361	20 x 85
212	DUMMY	-4162.5	-4162.5	-361	20 x 85
213	TESTOUT4	-4117.5	-4117.5	-361	20 x 85
214	DUMMY	-4072.5	-4072.5	-361	20 x 85
215	DUMMY	-4027.5	-4027.5	-361	20 x 85
216	DUMMY	-3982.5	-3982.5	-361	20 x 85
217	TESTOUT3	-3937.5	-3937.5	-361	20 x 85
218	TESTOUT2	-3892.5	-3892.5	-361	20 x 85
219	DUMMY	-3847.5	-3847.5	-361	20 x 85
220	GPO8	-3802.5	-3802.5	-361	20 x 85
221	GPO8	-3757.5	-3757.5	-361	20 x 85
222	GPO8	-3712.5	-3712.5	-361	20 x 85
223	GPO7	-3667.5	-3667.5	-361	20 x 85
224	GPO7	-3622.5	-3622.5	-361	20 x 85
225	GPO7	-3577.5	-3577.5	-361	20 x 85
226	GPO6	-3532.5	-3532.5	-361	20 x 85
227	GPO6	-3487.5	-3487.5	-361	20 x 85
228	GPO6	-3442.5	-3442.5	-361	20 x 85
229	GPO5	-3397.5	-3397.5	-361	20 x 85
230	GPO5	-3352.5	-3352.5	-361	20 x 85
231	GPO5	-3307.5	-3307.5	-361	20 x 85
232	GPO4	-3262.5	-3262.5	-361	20 x 85
233	GPO4	-3217.5	-3217.5	-361	20 x 85
234	GPO4	-3172.5	-3172.5	-361	20 x 85
235	DUMMY	-3127.5	-3127.5	-361	20 x 85
236	DUMMY	-3082.5	-3082.5	-361	20 x 85
237	DUMMY	-3037.5	-3037.5	-361	20 x 85
238	DUMMY	-2992.5	-2992.5	-361	20 x 85
239	DUMMY	-2947.5	-2947.5	-361	20 x 85
240	TESTOUT1	-2902.5	-2902.5	-361	20 x 85
241	TESTOUT0	-2857.5	-2857.5	-361	20 x 85
242	DUMMY	-2812.5	-2812.5	-361	20 x 85
243	M_S	-2767.5	-2767.5	-361	20 x 85
244	M_S	-2722.5	-2722.5	-361	20 x 85
245	CASCADE_ENB	-2677.5	-2677.5	-361	20 x 85
246	CASCADE_ENB	-2632.5	-2632.5	-361	20 x 85
247	TTL_R7	-2587.5	-2587.5	-361	20 x 85
248	TTL_R6	-2542.5	-2542.5	-361	20 x 85
249	TTL_R5	-2497.5	-2497.5	-361	20 x 85
250	TTL_R4	-2452.5	-2452.5	-361	20 x 85

No.	Name	X1	X2	Y	Bump size (μm)
251	TTL_R3	-2407.5	-2407.5	-361	20 x 85
252	TTL_R2	-2362.5	-2362.5	-361	20 x 85
253	TTL_R1	-2317.5	-2317.5	-361	20 x 85
254	TTL_R0	-2272.5	-2272.5	-361	20 x 85
255	DUMMY	-2227.5	-2227.5	-361	20 x 85
256	DUMMY	-2182.5	-2182.5	-361	20 x 85
257	TTL_G7	-2137.5	-2137.5	-361	20 x 85
258	TTL_G6	-2092.5	-2092.5	-361	20 x 85
259	TTL_G5	-2047.5	-2047.5	-361	20 x 85
260	TTL_G4	-2002.5	-2002.5	-361	20 x 85
261	TTL_G3	-1957.5	-1957.5	-361	20 x 85
262	TTL_G2	-1912.5	-1912.5	-361	20 x 85
263	TTL_G1	-1867.5	-1867.5	-361	20 x 85
264	TTL_G0	-1822.5	-1822.5	-361	20 x 85
265	TTL_B7	-1777.5	-1777.5	-361	20 x 85
266	TTL_B6	-1732.5	-1732.5	-361	20 x 85
267	TTL_B5	-1687.5	-1687.5	-361	20 x 85
268	TTL_B4	-1642.5	-1642.5	-361	20 x 85
269	TTL_B3	-1597.5	-1597.5	-361	20 x 85
270	TTL_B2	-1552.5	-1552.5	-361	20 x 85
271	TTL_B1	-1507.5	-1507.5	-361	20 x 85
272	TTL_B0	-1462.5	-1462.5	-361	20 x 85
273	TTL_DE	-1417.5	-1417.5	-361	20 x 85
274	TTL_CK	-1372.5	-1372.5	-361	20 x 85
275	TTL_HS	-1327.5	-1327.5	-361	20 x 85
276	TTL_VS	-1282.5	-1282.5	-361	20 x 85
277	DUMMY	-1237.5	-1237.5	-361	20 x 85
278	DUMMY	-1192.5	-1192.5	-361	20 x 85
279	DUMMY	-1147.5	-1147.5	-361	20 x 85
280	DUMMY	-1102.5	-1102.5	-361	20 x 85
281	DUMMY	-1057.5	-1057.5	-361	20 x 85
282	DUMMY	-1012.5	-1012.5	-361	20 x 85
283	VDDI	-967.5	-967.5	-361	20 x 85
284	VDDI	-922.5	-922.5	-361	20 x 85
285	TESTMODE2	-877.5	-877.5	-361	20 x 85
286	TESTMODE1	-832.5	-832.5	-361	20 x 85
287	TESTMODE0	-787.5	-787.5	-361	20 x 85
288	CSB	-742.5	-742.5	-361	20 x 85
289	SCK	-697.5	-697.5	-361	20 x 85
290	SDO	-652.5	-652.5	-361	20 x 85
291	RESETB	-607.5	-607.5	-361	20 x 85
292	RESETB	-562.5	-562.5	-361	20 x 85
293	RESETB	-517.5	-517.5	-361	20 x 85
294	RESETB	-472.5	-472.5	-361	20 x 85
295	BIST_EN	-427.5	-427.5	-361	20 x 85
296	BIST_EN	-382.5	-382.5	-361	20 x 85
297	GPO3	-337.5	-337.5	-361	20 x 85
298	GPO3	-292.5	-292.5	-361	20 x 85
299	GPO3	-247.5	-247.5	-361	20 x 85
300	GPO3	-202.5	-202.5	-361	20 x 85

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No.	Name	X1	X2	Y	Bump size (μm)
301	GPO3	-157.5	-157.5	-361	20 x 85
302	GPO3	-112.5	-112.5	-361	20 x 85
303	GPO2	-67.5	-67.5	-361	20 x 85
304	GPO2	-22.5	-22.5	-361	20 x 85
305	PWRMD1	22.5	22.5	-361	20 x 85
306	PWRMD1	67.5	67.5	-361	20 x 85
307	PWRMD0	112.5	112.5	-361	20 x 85
308	PWRMD0	157.5	157.5	-361	20 x 85
309	LNSW1	202.5	202.5	-361	20 x 85
310	LNSW1	247.5	247.5	-361	20 x 85
311	LNSW0	292.5	292.5	-361	20 x 85
312	LNSW0	337.5	337.5	-361	20 x 85
313	PNSW	382.5	382.5	-361	20 x 85
314	PNSW	427.5	427.5	-361	20 x 85
315	DUMMY	472.5	472.5	-361	20 x 85
316	DUMMY	517.5	517.5	-361	20 x 85
317	DUMMY	562.5	562.5	-361	20 x 85
318	DUMMY	607.5	607.5	-361	20 x 85
319	IF_SEL	652.5	652.5	-361	20 x 85
320	IF_SEL	697.5	697.5	-361	20 x 85
321	LANSEL1	742.5	742.5	-361	20 x 85
322	LANSEL1	787.5	787.5	-361	20 x 85
323	LANSEL0	832.5	832.5	-361	20 x 85
324	LANSEL0	877.5	877.5	-361	20 x 85
325	GPO1	922.5	922.5	-361	20 x 85
326	GPO1	967.5	967.5	-361	20 x 85
327	GPO0	1012.5	1012.5	-361	20 x 85
328	GPO0	1057.5	1057.5	-361	20 x 85
329	DUMMY	1102.5	1102.5	-361	20 x 85
330	DUMMY	1147.5	1147.5	-361	20 x 85
331	TESTA0	1192.5	1192.5	-361	20 x 85
332	TESTA1	1237.5	1237.5	-361	20 x 85
333	TESTA2	1282.5	1282.5	-361	20 x 85
334	TESTA3	1327.5	1327.5	-361	20 x 85
335	VSSA	1372.5	1372.5	-361	20 x 85
336	VSSA	1417.5	1417.5	-361	20 x 85
337	VSSA	1462.5	1462.5	-361	20 x 85
338	VSSA	1507.5	1507.5	-361	20 x 85
339	VSSD	1552.5	1552.5	-361	20 x 85
340	VSSD	1597.5	1597.5	-361	20 x 85
341	VSSD	1642.5	1642.5	-361	20 x 85
342	VSSD	1687.5	1687.5	-361	20 x 85
343	VSSD	1732.5	1732.5	-361	20 x 85
344	VSSD	1777.5	1777.5	-361	20 x 85
345	VSSD	1822.5	1822.5	-361	20 x 85
346	VSSD	1867.5	1867.5	-361	20 x 85
347	VSSD	1912.5	1912.5	-361	20 x 85
348	VSSD	1957.5	1957.5	-361	20 x 85
349	VSSD	2002.5	2002.5	-361	20 x 85
350	VSSD	2047.5	2047.5	-361	20 x 85

No.	Name	X1	X2	Y	Bumpsize (μm)
351	VSSD	2092.5	2092.5	-361	20 x 85
352	VSSD	2137.5	2137.5	-361	20 x 85
353	VSSD	2182.5	2182.5	-361	20 x 85
354	VSSD	2227.5	2227.5	-361	20 x 85
355	VSSD	2272.5	2272.5	-361	20 x 85
356	VSSD	2317.5	2317.5	-361	20 x 85
357	VSSD	2362.5	2362.5	-361	20 x 85
358	VSSD	2407.5	2407.5	-361	20 x 85
359	DRVP	2452.5	2452.5	-361	20 x 85
360	DRVP	2497.5	2497.5	-361	20 x 85
361	DRVP	2542.5	2542.5	-361	20 x 85
362	DRVP	2587.5	2587.5	-361	20 x 85
363	DRVP	2632.5	2632.5	-361	20 x 85
364	DRVP	2677.5	2677.5	-361	20 x 85
365	DRVP	2722.5	2722.5	-361	20 x 85
366	DRVP	2767.5	2767.5	-361	20 x 85
367	DRVN	2812.5	2812.5	-361	20 x 85
368	DRVN	2857.5	2857.5	-361	20 x 85
369	DRVN	2902.5	2902.5	-361	20 x 85
370	DRVN	2947.5	2947.5	-361	20 x 85
371	DRVN	2992.5	2992.5	-361	20 x 85
372	DRVN	3037.5	3037.5	-361	20 x 85
373	DRVN	3082.5	3082.5	-361	20 x 85
374	DRVN	3127.5	3127.5	-361	20 x 85
375	VDDI_D	3172.5	3172.5	-361	20 x 85
376	VDDI_D	3217.5	3217.5	-361	20 x 85
377	VDDI_D	3262.5	3262.5	-361	20 x 85
378	VDDI_D	3307.5	3307.5	-361	20 x 85
379	VDDI_D	3352.5	3352.5	-361	20 x 85
380	VDDI_D	3397.5	3397.5	-361	20 x 85
381	VGLO2	3442.5	3442.5	-361	20 x 85
382	VGLO2	3487.5	3487.5	-361	20 x 85
383	VGL	3532.5	3532.5	-361	20 x 85
384	VGL	3577.5	3577.5	-361	20 x 85
385	VGL	3622.5	3622.5	-361	20 x 85
386	VGL	3667.5	3667.5	-361	20 x 85
387	VGL	3712.5	3712.5	-361	20 x 85
388	VSSA	3757.5	3757.5	-361	20 x 85
389	VSSA	3802.5	3802.5	-361	20 x 85
390	VSSA	3847.5	3847.5	-361	20 x 85
391	VSSA	3892.5	3892.5	-361	20 x 85
392	VSSA	3937.5	3937.5	-361	20 x 85
393	VSSRC	3982.5	3982.5	-361	20 x 85
394	VSSRC	4027.5	4027.5	-361	20 x 85
395	VSSRC	4072.5	4072.5	-361	20 x 85
396	VCI	4117.5	4117.5	-361	20 x 85
397	VCI	4162.5	4162.5	-361	20 x 85
398	VCI	4207.5	4207.5	-361	20 x 85
399	VCI	4252.5	4252.5	-361	20 x 85
400	VCI	4297.5	4297.5	-361	20 x 85

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No.	Name	X1	X2	Y	Bump size (μm)
401	VCI	4342.5	4342.5	-361	20 x 85
402	VCI	4387.5	4387.5	-361	20 x 85
403	VCI	4432.5	4432.5	-361	20 x 85
404	VCI	4477.5	4477.5	-361	20 x 85
405	VCI	4522.5	4522.5	-361	20 x 85
406	VCI	4567.5	4567.5	-361	20 x 85
407	VGMAH	4612.5	4612.5	-361	20 x 85
408	VGMAH	4657.5	4657.5	-361	20 x 85
409	VGMAH	4702.5	4702.5	-361	20 x 85
410	VGMAL	4747.5	4747.5	-361	20 x 85
411	VGMAL	4792.5	4792.5	-361	20 x 85
412	VGMAL	4837.5	4837.5	-361	20 x 85
413	VGH	4882.5	4882.5	-361	20 x 85
414	VGH	4927.5	4927.5	-361	20 x 85
415	VGH	4972.5	4972.5	-361	20 x 85
416	VCOM	5017.5	5017.5	-361	20 x 85
417	VCOM	5062.5	5062.5	-361	20 x 85
418	VCOM	5107.5	5107.5	-361	20 x 85
419	VCOM	5152.5	5152.5	-361	20 x 85
420	VCOM	5197.5	5197.5	-361	20 x 85
421	VDDI_D	5242.5	5242.5	-361	20 x 85
422	VDDI_D	5287.5	5287.5	-361	20 x 85
423	VDDI_D	5332.5	5332.5	-361	20 x 85
424	VDDI_D	5377.5	5377.5	-361	20 x 85
425	VDDI_D	5422.5	5422.5	-361	20 x 85
426	VDDI_D	5467.5	5467.5	-361	20 x 85
427	VDDI_D	5512.5	5512.5	-361	20 x 85
428	VDDI_D	5557.5	5557.5	-361	20 x 85
429	VCI_REG	5602.5	5602.5	-361	20 x 85
430	VCI_REG	5647.5	5647.5	-361	20 x 85
431	VCI_REG	5692.5	5692.5	-361	20 x 85
432	VCI_REG	5737.5	5737.5	-361	20 x 85
433	VCI_REG	5782.5	5782.5	-361	20 x 85
434	VSSA	5827.5	5827.5	-361	20 x 85
435	VSSA	5872.5	5872.5	-361	20 x 85
436	VSSA	5917.5	5917.5	-361	20 x 85
437	VSSA	5962.5	5962.5	-361	20 x 85
438	VSSA	6007.5	6007.5	-361	20 x 85
439	VSSA	6052.5	6052.5	-361	20 x 85
440	VSSA	6097.5	6097.5	-361	20 x 85
441	VSSD	6142.5	6142.5	-361	20 x 85
442	VSSD	6187.5	6187.5	-361	20 x 85
443	VSSD	6232.5	6232.5	-361	20 x 85
444	VSSD	6277.5	6277.5	-361	20 x 85
445	VSSD	6322.5	6322.5	-361	20 x 85
446	VSSD	6367.5	6367.5	-361	20 x 85
447	VSSD	6412.5	6412.5	-361	20 x 85
448	VCL	6457.5	6457.5	-361	20 x 85
449	VCL	6502.5	6502.5	-361	20 x 85
450	VCL	6547.5	6547.5	-361	20 x 85

No.	Name	X1	X2	Y	Bump size (μm)
451	VCL	6592.5	6592.5	-361	20 x 85
452	VCL	6637.5	6637.5	-361	20 x 85
453	VCL	6682.5	6682.5	-361	20 x 85
454	VCL	6727.5	6727.5	-361	20 x 85
455	C_VCL_1P	6772.5	6772.5	-361	20 x 85
456	C_VCL_1P	6817.5	6817.5	-361	20 x 85
457	C_VCL_1P	6862.5	6862.5	-361	20 x 85
458	C_VCL_1N	6907.5	6907.5	-361	20 x 85
459	C_VCL_1N	6952.5	6952.5	-361	20 x 85
460	C_VCL_1N	6997.5	6997.5	-361	20 x 85
461	C_VCL_1N	7042.5	7042.5	-361	20 x 85
462	DUMMY	7087.5	7087.5	-361	20 x 85
463	DUMMY	7132.5	7132.5	-361	20 x 85
464	DUMMY	7177.5	7177.5	-361	20 x 85
465	DUMMY	7222.5	7222.5	-361	20 x 85
466	DUMMY	7267.5	7267.5	-361	20 x 85
467	DUMMY	7312.5	7312.5	-361	20 x 85
468	DUMMY	7357.5	7357.5	-361	20 x 85
469	VSN	7402.5	7402.5	-361	20 x 85
470	VSN	7447.5	7447.5	-361	20 x 85
471	VSN	7492.5	7492.5	-361	20 x 85
472	VSN	7537.5	7537.5	-361	20 x 85
473	VSN	7582.5	7582.5	-361	20 x 85
474	VSN	7627.5	7627.5	-361	20 x 85
475	VSN	7672.5	7672.5	-361	20 x 85
476	VSP	7717.5	7717.5	-361	20 x 85
477	VSP	7762.5	7762.5	-361	20 x 85
478	VSP	7807.5	7807.5	-361	20 x 85
479	VSP	7852.5	7852.5	-361	20 x 85
480	VSP	7897.5	7897.5	-361	20 x 85
481	VSP	7942.5	7942.5	-361	20 x 85
482	VSP	7987.5	7987.5	-361	20 x 85
483	VSP	8032.5	8032.5	-361	20 x 85
484	VSP	8077.5	8077.5	-361	20 x 85
485	VSP	8122.5	8122.5	-361	20 x 85
486	VSP	8167.5	8167.5	-361	20 x 85
487	VSP	8212.5	8212.5	-361	20 x 85
488	VSP	8257.5	8257.5	-361	20 x 85
489	VSP	8302.5	8302.5	-361	20 x 85
490	DUMMY	8347.5	8347.5	-361	20 x 85
491	DUMMY	8392.5	8392.5	-361	20 x 85
492	DUMMY	8437.5	8437.5	-361	20 x 85
493	DUMMY	8482.5	8482.5	-361	20 x 85
494	DUMMY	8527.5	8527.5	-361	20 x 85
495	DUMMY	8572.5	8572.5	-361	20 x 85
496	DUMMY	8617.5	8617.5	-361	20 x 85
497	DUMMY	8662.5	8662.5	-361	20 x 85
498	DUMMY	8707.5	8707.5	-361	20 x 85
499	DUMMY	8752.5	8752.5	-361	20 x 85
500	DUMMY	8797.5	8797.5	-361	20 x 85

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No.	Name	X1	X2	Y	Bump size (μm)
501	DUMMY	8842.5	8842.5	-361	20 x 85
502	DUMMY	8887.5	8887.5	-361	20 x 85
503	DUMMY	8932.5	8932.5	-361	20 x 85
504	DUMMY	8977.5	8977.5	-361	20 x 85
505	DUMMY	9022.5	9022.5	-361	20 x 85
506	DUMMY	9067.5	9067.5	-361	20 x 85
507	DUMMY	9112.5	9112.5	-361	20 x 85
508	DUMMY	9157.5	9157.5	-361	20 x 85
509	DUMMY	9202.5	9202.5	-361	20 x 85
510	DUMMY	9247.5	9247.5	-361	20 x 85
511	DUMMY	9292.5	9292.5	-361	20 x 85
512	DUMMY	9337.5	9337.5	-361	20 x 85
513	DUMMY	9382.5	9382.5	-361	20 x 85
514	DUMMY	9427.5	9427.5	-361	20 x 85
515	DUMMY	9472.5	9472.5	-361	20 x 85
516	VCI	9517.5	9517.5	-361	20 x 85
517	VCI	9562.5	9562.5	-361	20 x 85
518	VCI	9607.5	9607.5	-361	20 x 85
519	VCI	9652.5	9652.5	-361	20 x 85
520	VCI	9697.5	9697.5	-361	20 x 85
521	VCI	9742.5	9742.5	-361	20 x 85
522	VCI	9787.5	9787.5	-361	20 x 85
523	VCI	9832.5	9832.5	-361	20 x 85
524	VSSCP	9877.5	9877.5	-361	20 x 85
525	VSSCP	9922.5	9922.5	-361	20 x 85
526	VSSCP	9967.5	9967.5	-361	20 x 85
527	VSSCP	10012.5	10012.5	-361	20 x 85
528	VSSCP	10057.5	10057.5	-361	20 x 85
529	VSSCP	10102.5	10102.5	-361	20 x 85
530	VSSCP	10147.5	10147.5	-361	20 x 85
531	VSSCP	10192.5	10192.5	-361	20 x 85
532	VSSCP	10237.5	10237.5	-361	20 x 85
533	VSSCP	10282.5	10282.5	-361	20 x 85
534	VSSCP	10327.5	10327.5	-361	20 x 85
535	VSSCP	10372.5	10372.5	-361	20 x 85
536	VSSCP	10417.5	10417.5	-361	20 x 85
537	VSSCP	10462.5	10462.5	-361	20 x 85
538	VGH2	10507.5	10507.5	-361	20 x 85
539	VGH	10552.5	10552.5	-361	20 x 85
540	VGH	10597.5	10597.5	-361	20 x 85
541	VGH	10642.5	10642.5	-361	20 x 85
542	VGH1	10687.5	10687.5	-361	20 x 85
543	VGH1	10732.5	10732.5	-361	20 x 85
544	VGH1	10777.5	10777.5	-361	20 x 85
545	C_VGH_1N	10822.5	10822.5	-361	20 x 85
546	C_VGH_1N	10867.5	10867.5	-361	20 x 85
547	C_VGH_1N	10912.5	10912.5	-361	20 x 85
548	C_VGH_1P	10957.5	10957.5	-361	20 x 85
549	C_VGH_1P	11002.5	11002.5	-361	20 x 85
550	C_VGH_1P	11047.5	11047.5	-361	20 x 85

No.	Name	X1	X2	Y	Bumpsize (μm)
551	C_VGH_1P	11092.5	11092.5	-361	20 x 85
552	C_VGH_2N	11137.5	11137.5	-361	20 x 85
553	C_VGH_2N	11182.5	11182.5	-361	20 x 85
554	C_VGH_2N	11227.5	11227.5	-361	20 x 85
555	C_VGH_2P	11272.5	11272.5	-361	20 x 85
556	C_VGH_2P	11317.5	11317.5	-361	20 x 85
557	C_VGH_2P	11362.5	11362.5	-361	20 x 85
558	VGL	11407.5	11407.5	-361	20 x 85
559	VGL	11452.5	11452.5	-361	20 x 85
560	VGL	11497.5	11497.5	-361	20 x 85
561	VGL	11542.5	11542.5	-361	20 x 85
562	VGL	11587.5	11587.5	-361	20 x 85
563	VGL	11632.5	11632.5	-361	20 x 85
564	VGL	11677.5	11677.5	-361	20 x 85
565	C_VGL_1P	11722.5	11722.5	-361	20 x 85
566	C_VGL_1P	11767.5	11767.5	-361	20 x 85
567	C_VGL_1P	11812.5	11812.5	-361	20 x 85
568	C_VGL_1P	11857.5	11857.5	-361	20 x 85
569	C_VGL_1P	11902.5	11902.5	-361	20 x 85
570	C_VGL_1P	11947.5	11947.5	-361	20 x 85
571	C_VGL_1P	11992.5	11992.5	-361	20 x 85
572	C_VGL_1N	12037.5	12037.5	-361	20 x 85
573	C_VGL_1N	12082.5	12082.5	-361	20 x 85
574	C_VGL_1N	12127.5	12127.5	-361	20 x 85
575	C_VGL_1N	12172.5	12172.5	-361	20 x 85
576	C_VGL_1N	12217.5	12217.5	-361	20 x 85
577	C_VGL_1N	12262.5	12262.5	-361	20 x 85
578	VSSA	12307.5	12307.5	-361	20 x 85
579	VSSA	12352.5	12352.5	-361	20 x 85
580	VSSA	12397.5	12397.5	-361	20 x 85
581	VSSA	12442.5	12442.5	-361	20 x 85
582	VCOM	12497.5	12497.5	-361	20 x 85
583	VCOM	12542.5	12542.5	-361	20 x 85
584	VCOM	12587.5	12587.5	-361	20 x 85
585	GOUTR_22	12632.5	12632.5	-386	30 x 35
586	GOUTR_21	12677.5	12677.5	-386	30 x 35
587	GOUTR_20	12722.5	12722.5	-386	30 x 35
588	GOUTR_19	12767.5	12767.5	-386	30 x 35
589	GOUTR_18	12812.5	12812.5	-386	30 x 35
590	GOUTR_17	12857.5	12857.5	-386	30 x 35
591	GOUTR_16	12902.5	12902.5	-386	30 x 35
592	GOUTR_15	12947.5	12947.5	-386	30 x 35
593	GOUTR_14	12992.5	12992.5	-386	30 x 35
594	GOUTR_13	13037.5	13037.5	-386	30 x 35
595	GOUTR_12	13082.5	13082.5	-386	30 x 35
596	GOUTR_11	13127.5	13127.5	-386	30 x 35
597	GOUTR_10	13172.5	13172.5	-386	30 x 35
598	GOUTR_9	13217.5	13217.5	-386	30 x 35
599	GOUTR_8	13262.5	13262.5	-386	30 x 35
600	GOUTR_7	13307.5	13307.5	-386	30 x 35

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No.	Name	X1	X2	Y	Bump size (μm)	No.	Name	X1	X2	Y	Bumpsize (μm)
601	GOUTR_6	13352.5	13352.5	-386	30 x 35	651	S17	13348.5	13344.5	277	16 x 73
602	GOUTR_5	13397.5	13397.5	-386	30 x 35	652	S18	13337.5	13333.5	367	16 x 73
603	GOUTR_4	13442.5	13442.5	-386	30 x 35	653	S19	13326.5	13322.6	187	16 x 73
604	GOUTR_3	13487.5	13487.5	-386	30 x 35	654	S20	13315.5	13311.6	277	16 x 73
605	GOUTR_2	13532.5	13532.5	-386	30 x 35	655	S21	13304.5	13300.6	367	16 x 73
606	GOUTR_1	13577.5	13577.5	-386	30 x 35	656	S22	13293.5	13289.6	187	16 x 73
607	DUMMY_R2	13622.5	13622.5	-386	30 x 35	657	S23	13282.5	13278.6	277	16 x 73
608	DUMMY_R2	13667.5	13667.5	-386	30 x 35	658	S24	13271.5	13267.6	367	16 x 73
609	SYNC_L15	13722.5	13718.4	187	16 x 73	659	S25	13260.5	13256.6	187	16 x 73
610	SYNC_L14	13755.5	13751.4	187	16 x 73	660	S26	13249.5	13245.6	277	16 x 73
611	SYNC_L13	13788.5	13784.4	187	16 x 73	661	S27	13238.5	13234.6	367	16 x 73
612	SYNC_L12	13821.5	13817.4	187	16 x 73	662	S28	13227.5	13223.6	187	16 x 73
613	SYNC_L11	13854.5	13850.4	187	16 x 73	663	S29	13216.5	13212.6	277	16 x 73
614	DUMMY	13810.5	13806.4	277	16 x 73	664	S30	13205.5	13201.6	367	16 x 73
615	SYNC_L10	13843.5	13839.4	277	16 x 73	665	S31	13194.5	13190.6	187	16 x 73
616	SYNC_L9	13832.5	13828.4	367	16 x 73	666	S32	13183.5	13179.6	277	16 x 73
617	SYNC_L8	13799.5	13795.4	367	16 x 73	667	S33	13172.5	13168.6	367	16 x 73
618	SYNC_L7	13777.5	13773.4	277	16 x 73	668	S34	13161.5	13157.6	187	16 x 73
619	SYNC_L6	13744.5	13740.4	277	16 x 73	669	S35	13150.5	13146.6	277	16 x 73
620	SYNC_L5	13711.5	13707.4	277	16 x 73	670	S36	13139.5	13135.6	367	16 x 73
621	GMLP_IN	13689.5	13685.4	187	16 x 73	671	S37	13128.5	13124.6	187	16 x 73
622	SYNC_L4	13678.5	13674.4	277	16 x 73	672	S38	13117.5	13113.6	277	16 x 73
623	GMHP_IN	13656.5	13652.4	187	16 x 73	673	S39	13106.5	13102.6	367	16 x 73
624	DUMMY	13645.5	13641.4	277	16 x 73	674	S40	13095.5	13091.6	187	16 x 73
625	SYNC_L3	13634.5	13630.4	367	16 x 73	675	S41	13084.5	13080.6	277	16 x 73
626	GMHN_IN	13623.5	13619.4	187	16 x 73	676	S42	13073.5	13069.6	367	16 x 73
627	DUMMY	13612.5	13608.4	277	16 x 73	677	S43	13062.5	13058.6	187	16 x 73
628	SYNC_L2	13601.5	13597.4	367	16 x 73	678	S44	13051.5	13047.6	277	16 x 73
629	GMLN_IN	13590.5	13586.4	187	16 x 73	679	S45	13040.5	13036.6	367	16 x 73
630	DUMMY	13579.5	13575.4	277	16 x 73	680	S46	13029.5	13025.6	187	16 x 73
631	SYNC_L1	13568.5	13564.4	367	16 x 73	681	S47	13018.5	13014.6	277	16 x 73
632	DUMMY	13557.5	13553.4	187	16 x 73	682	S48	13007.5	13003.6	367	16 x 73
633	DUMMY	13546.5	13542.4	277	16 x 73	683	S49	12996.5	12992.7	187	16 x 73
634	SR1	13535.5	13531.4	367	16 x 73	684	S50	12985.5	12981.7	277	16 x 73
635	S1	13524.5	13520.5	187	16 x 73	685	S51	12974.5	12970.7	367	16 x 73
636	S2	13513.5	13509.5	277	16 x 73	686	S52	12963.5	12959.7	187	16 x 73
637	S3	13502.5	13498.5	367	16 x 73	687	S53	12952.5	12948.7	277	16 x 73
638	S4	13491.5	13487.5	187	16 x 73	688	S54	12941.5	12937.7	367	16 x 73
639	S5	13480.5	13476.5	277	16 x 73	689	S55	12930.5	12926.7	187	16 x 73
640	S6	13469.5	13465.5	367	16 x 73	690	S56	12919.5	12915.7	277	16 x 73
641	S7	13458.5	13454.5	187	16 x 73	691	S57	12908.5	12904.7	367	16 x 73
642	S8	13447.5	13443.5	277	16 x 73	692	S58	12897.5	12893.7	187	16 x 73
643	S9	13436.5	13432.5	367	16 x 73	693	S59	12886.5	12882.7	277	16 x 73
644	S10	13425.5	13421.5	187	16 x 73	694	S60	12875.5	12871.7	367	16 x 73
645	S11	13414.5	13410.5	277	16 x 73	695	S61	12864.5	12860.7	187	16 x 73
646	S12	13403.5	13399.5	367	16 x 73	696	S62	12853.5	12849.7	277	16 x 73
647	S13	13392.5	13388.5	187	16 x 73	697	S63	12842.5	12838.7	367	16 x 73
648	S14	13381.5	13377.5	277	16 x 73	698	S64	12831.5	12827.7	187	16 x 73
649	S15	13370.5	13366.5	367	16 x 73	699	S65	12820.5	12816.7	277	16 x 73
650	S16	13359.5	13355.5	187	16 x 73	700	S66	12809.5	12805.7	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
701	S67	12798.5	12794.7	187	16 x 73
702	S68	12787.5	12783.7	277	16 x 73
703	S69	12776.5	12772.7	367	16 x 73
704	S70	12765.5	12761.7	187	16 x 73
705	S71	12754.5	12750.7	277	16 x 73
706	S72	12743.5	12739.7	367	16 x 73
707	S73	12732.5	12728.7	187	16 x 73
708	S74	12721.5	12717.7	277	16 x 73
709	S75	12710.5	12706.7	367	16 x 73
710	S76	12699.5	12695.7	187	16 x 73
711	S77	12688.5	12684.7	277	16 x 73
712	S78	12677.5	12673.7	367	16 x 73
713	S79	12666.5	12662.8	187	16 x 73
714	S80	12655.5	12651.8	277	16 x 73
715	S81	12644.5	12640.8	367	16 x 73
716	S82	12633.5	12629.8	187	16 x 73
717	S83	12622.5	12618.8	277	16 x 73
718	S84	12611.5	12607.8	367	16 x 73
719	S85	12600.5	12596.8	187	16 x 73
720	S86	12589.5	12585.8	277	16 x 73
721	S87	12578.5	12574.8	367	16 x 73
722	S88	12567.5	12563.8	187	16 x 73
723	S89	12556.5	12552.8	277	16 x 73
724	S90	12545.5	12541.8	367	16 x 73
725	S91	12534.5	12530.8	187	16 x 73
726	S92	12523.5	12519.8	277	16 x 73
727	S93	12512.5	12508.8	367	16 x 73
728	S94	12501.5	12497.8	187	16 x 73
729	S95	12490.5	12486.8	277	16 x 73
730	S96	12479.5	12475.8	367	16 x 73
731	S97	12468.5	12464.8	187	16 x 73
732	S98	12457.5	12453.8	277	16 x 73
733	S99	12446.5	12442.8	367	16 x 73
734	S100	12435.5	12431.8	187	16 x 73
735	S101	12424.5	12420.8	277	16 x 73
736	S102	12413.5	12409.8	367	16 x 73
737	S103	12402.5	12398.8	187	16 x 73
738	S104	12391.5	12387.8	277	16 x 73
739	S105	12380.5	12376.8	367	16 x 73
740	S106	12369.5	12365.8	187	16 x 73
741	S107	12358.5	12354.8	277	16 x 73
742	S108	12347.5	12343.8	367	16 x 73
743	S109	12336.5	12332.9	187	16 x 73
744	S110	12325.5	12321.9	277	16 x 73
745	S111	12314.5	12310.9	367	16 x 73
746	S112	12303.5	12299.9	187	16 x 73
747	S113	12292.5	12288.9	277	16 x 73
748	S114	12281.5	12277.9	367	16 x 73
749	S115	12270.5	12266.9	187	16 x 73
750	S116	12259.5	12255.9	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
751	S117	12248.5	12244.9	367	16 x 73
752	S118	12237.5	12233.9	187	16 x 73
753	S119	12226.5	12222.9	277	16 x 73
754	S120	12215.5	12211.9	367	16 x 73
755	S121	12204.5	12200.9	187	16 x 73
756	S122	12193.5	12189.9	277	16 x 73
757	S123	12182.5	12178.9	367	16 x 73
758	S124	12171.5	12167.9	187	16 x 73
759	S125	12160.5	12156.9	277	16 x 73
760	S126	12149.5	12145.9	367	16 x 73
761	S127	12138.5	12134.9	187	16 x 73
762	S128	12127.5	12123.9	277	16 x 73
763	S129	12116.5	12112.9	367	16 x 73
764	S130	12105.5	12101.9	187	16 x 73
765	S131	12094.5	12090.9	277	16 x 73
766	S132	12083.5	12079.9	367	16 x 73
767	S133	12072.5	12068.9	187	16 x 73
768	S134	12061.5	12057.9	277	16 x 73
769	S135	12050.5	12046.9	367	16 x 73
770	S136	12039.5	12035.9	187	16 x 73
771	S137	12028.5	12024.9	277	16 x 73
772	S138	12017.5	12013.9	367	16 x 73
773	S139	12006.5	12003	187	16 x 73
774	S140	11995.5	11992	277	16 x 73
775	S141	11984.5	11981	367	16 x 73
776	S142	11973.5	11970	187	16 x 73
777	S143	11962.5	11959	277	16 x 73
778	S144	11951.5	11948	367	16 x 73
779	S145	11940.5	11937	187	16 x 73
780	S146	11929.5	11926	277	16 x 73
781	S147	11918.5	11915	367	16 x 73
782	S148	11907.5	11904	187	16 x 73
783	S149	11896.5	11893	277	16 x 73
784	S150	11885.5	11882	367	16 x 73
785	S151	11874.5	11871	187	16 x 73
786	S152	11863.5	11860	277	16 x 73
787	S153	11852.5	11849	367	16 x 73
788	S154	11841.5	11838	187	16 x 73
789	S155	11830.5	11827	277	16 x 73
790	S156	11819.5	11816	367	16 x 73
791	S157	11808.5	11805	187	16 x 73
792	S158	11797.5	11794	277	16 x 73
793	S159	11786.5	11783	367	16 x 73
794	S160	11775.5	11772	187	16 x 73
795	S161	11764.5	11761	277	16 x 73
796	S162	11753.5	11750	367	16 x 73
797	S163	11742.5	11739	187	16 x 73
798	S164	11731.5	11728	277	16 x 73
799	S165	11720.5	11717	367	16 x 73
800	S166	11709.5	11706	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
801	S167	11698.5	11695	277	16 x 73
802	S168	11687.5	11684	367	16 x 73
803	S169	11676.5	11673.1	187	16 x 73
804	S170	11665.5	11662.1	277	16 x 73
805	S171	11654.5	11651.1	367	16 x 73
806	S172	11643.5	11640.1	187	16 x 73
807	S173	11632.5	11629.1	277	16 x 73
808	S174	11621.5	11618.1	367	16 x 73
809	S75	11610.5	11607.1	187	16 x 73
810	S176	11599.5	11596.1	277	16 x 73
811	S177	11588.5	11585.1	367	16 x 73
812	S178	11577.5	11574.1	187	16 x 73
813	S179	11566.5	11563.1	277	16 x 73
814	S180	11555.5	11552.1	367	16 x 73
815	S181	11544.5	11541.1	187	16 x 73
816	S182	11533.5	11530.1	277	16 x 73
817	S183	11522.5	11519.1	367	16 x 73
818	S184	11511.5	11508.1	187	16 x 73
819	S185	11500.5	11497.1	277	16 x 73
820	S186	11489.5	11486.1	367	16 x 73
821	S187	11478.5	11475.1	187	16 x 73
822	S188	11467.5	11464.1	277	16 x 73
823	S189	11456.5	11453.1	367	16 x 73
824	S190	11445.5	11442.1	187	16 x 73
825	S191	11434.5	11431.1	277	16 x 73
826	S192	11423.5	11420.1	367	16 x 73
827	S193	11412.5	11409.1	187	16 x 73
828	S194	11401.5	11398.1	277	16 x 73
829	S195	11390.5	11387.1	367	16 x 73
830	S196	11379.5	11376.1	187	16 x 73
831	S197	11368.5	11365.1	277	16 x 73
832	S198	11357.5	11354.1	367	16 x 73
833	S199	11346.5	11343.2	187	16 x 73
834	S200	11335.5	11332.2	277	16 x 73
835	S201	11324.5	11321.2	367	16 x 73
836	S202	11313.5	11310.2	187	16 x 73
837	S203	11302.5	11299.2	277	16 x 73
838	S204	11291.5	11288.2	367	16 x 73
839	S205	11280.5	11277.2	187	16 x 73
840	S206	11269.5	11266.2	277	16 x 73
841	S207	11258.5	11255.2	367	16 x 73
842	S208	11247.5	11244.2	187	16 x 73
843	S209	11236.5	11233.2	277	16 x 73
844	S210	11225.5	11222.2	367	16 x 73
845	S211	11214.5	11211.2	187	16 x 73
846	S212	11203.5	11200.2	277	16 x 73
847	S213	11192.5	11189.2	367	16 x 73
848	S214	11181.5	11178.2	187	16 x 73
849	S215	11170.5	11167.2	277	16 x 73
850	S216	11159.5	11156.2	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
851	S217	11148.5	11145.2	187	16 x 73
852	S218	11137.5	11134.2	277	16 x 73
853	S219	11126.5	11123.2	367	16 x 73
854	S220	11115.5	11112.2	187	16 x 73
855	S221	11104.5	11101.2	277	16 x 73
856	S222	11093.5	11090.2	367	16 x 73
857	S223	11082.5	11079.2	187	16 x 73
858	S224	11071.5	11068.2	277	16 x 73
859	S225	11060.5	11057.2	367	16 x 73
860	S226	11049.5	11046.2	187	16 x 73
861	S227	11038.5	11035.2	277	16 x 73
862	S228	11027.5	11024.2	367	16 x 73
863	S229	11016.5	11013.3	187	16 x 73
864	S230	11005.5	11002.3	277	16 x 73
865	S231	10994.5	10991.3	367	16 x 73
866	S232	10983.5	10980.3	187	16 x 73
867	S233	10972.5	10969.3	277	16 x 73
868	S234	10961.5	10958.3	367	16 x 73
869	S235	10950.5	10947.3	187	16 x 73
870	S236	10939.5	10936.3	277	16 x 73
871	S237	10928.5	10925.3	367	16 x 73
872	S238	10917.5	10914.3	187	16 x 73
873	S239	10906.5	10903.3	277	16 x 73
874	S240	10895.5	10892.3	367	16 x 73
875	S241	10884.5	10881.3	187	16 x 73
876	S242	10873.5	10870.3	277	16 x 73
877	S243	10862.5	10859.3	367	16 x 73
878	S244	10851.5	10848.3	187	16 x 73
879	S245	10840.5	10837.3	277	16 x 73
880	S246	10829.5	10826.3	367	16 x 73
881	S247	10818.5	10815.3	187	16 x 73
882	S248	10807.5	10804.3	277	16 x 73
883	S249	10796.5	10793.3	367	16 x 73
884	S250	10785.5	10782.3	187	16 x 73
885	S251	10774.5	10771.3	277	16 x 73
886	S252	10763.5	10760.3	367	16 x 73
887	S253	10752.5	10749.3	187	16 x 73
888	S254	10741.5	10738.3	277	16 x 73
889	S255	10730.5	10727.3	367	16 x 73
890	S256	10719.5	10716.3	187	16 x 73
891	S257	10708.5	10705.3	277	16 x 73
892	S258	10697.5	10694.3	367	16 x 73
893	S259	10686.5	10683.4	187	16 x 73
894	S260	10675.5	10672.4	277	16 x 73
895	S261	10664.5	10661.4	367	16 x 73
896	S262	10653.5	10650.4	187	16 x 73
897	S263	10642.5	10639.4	277	16 x 73
898	S264	10631.5	10628.4	367	16 x 73
899	S265	10620.5	10617.4	187	16 x 73
900	S266	10609.5	10606.4	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
901	S267	10598.5	10595.4	367	16 x 73
902	S268	10587.5	10584.4	187	16 x 73
903	S269	10576.5	10573.4	277	16 x 73
904	S270	10565.5	10562.4	367	16 x 73
905	S271	10554.5	10551.4	187	16 x 73
906	S272	10543.5	10540.4	277	16 x 73
907	S273	10532.5	10529.4	367	16 x 73
908	S274	10521.5	10518.4	187	16 x 73
909	S275	10510.5	10507.4	277	16 x 73
910	S276	10499.5	10496.4	367	16 x 73
911	S277	10488.5	10485.4	187	16 x 73
912	S278	10477.5	10474.4	277	16 x 73
913	S279	10466.5	10463.4	367	16 x 73
914	S280	10455.5	10452.4	187	16 x 73
915	S281	10444.5	10441.4	277	16 x 73
916	S282	10433.5	10430.4	367	16 x 73
917	S283	10422.5	10419.4	187	16 x 73
918	S284	10411.5	10408.4	277	16 x 73
919	S285	10400.5	10397.4	367	16 x 73
920	S286	10389.5	10386.4	187	16 x 73
921	S287	10378.5	10375.4	277	16 x 73
922	S288	10367.5	10364.4	367	16 x 73
923	S289	10356.5	10353.5	187	16 x 73
924	S290	10345.5	10342.5	277	16 x 73
925	S291	10334.5	10331.5	367	16 x 73
926	S292	10323.5	10320.5	187	16 x 73
927	S293	10312.5	10309.5	277	16 x 73
928	S294	10301.5	10298.5	367	16 x 73
929	S295	10290.5	10287.5	187	16 x 73
930	S296	10279.5	10276.5	277	16 x 73
931	S297	10268.5	10265.5	367	16 x 73
932	S298	10257.5	10254.5	187	16 x 73
933	S299	10246.5	10243.5	277	16 x 73
934	S300	10235.5	10232.5	367	16 x 73
935	S301	10224.5	10221.5	187	16 x 73
936	S302	10213.5	10210.5	277	16 x 73
937	S303	10202.5	10199.5	367	16 x 73
938	S304	10191.5	10188.5	187	16 x 73
939	S305	10180.5	10177.5	277	16 x 73
940	S306	10169.5	10166.5	367	16 x 73
941	S307	10158.5	10155.5	187	16 x 73
942	S308	10147.5	10144.5	277	16 x 73
943	S309	10136.5	10133.5	367	16 x 73
944	S310	10125.5	10122.5	187	16 x 73
945	S311	10114.5	10111.5	277	16 x 73
946	S312	10103.5	10100.5	367	16 x 73
947	S313	10092.5	10089.5	187	16 x 73
948	S314	10081.5	10078.5	277	16 x 73
949	S315	10070.5	10067.5	367	16 x 73
950	S316	10059.5	10056.5	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
951	S317	10048.5	10045.5	277	16 x 73
952	S318	10037.5	10034.5	367	16 x 73
953	S319	10026.5	10023.6	187	16 x 73
954	S320	10015.5	10012.6	277	16 x 73
955	S321	10004.5	10001.6	367	16 x 73
956	S322	9993.5	9990.6	187	16 x 73
957	S323	9982.5	9979.6	277	16 x 73
958	S324	9971.5	9968.6	367	16 x 73
959	S325	9960.5	9957.6	187	16 x 73
960	S326	9949.5	9946.6	277	16 x 73
961	S327	9938.5	9935.6	367	16 x 73
962	S328	9927.5	9924.6	187	16 x 73
963	S329	9916.5	9913.6	277	16 x 73
964	S330	9905.5	9902.6	367	16 x 73
965	S331	9894.5	9891.6	187	16 x 73
966	S332	9883.5	9880.6	277	16 x 73
967	S333	9872.5	9869.6	367	16 x 73
968	S334	9861.5	9858.6	187	16 x 73
969	S335	9850.5	9847.6	277	16 x 73
970	S336	9839.5	9836.6	367	16 x 73
971	S337	9828.5	9825.6	187	16 x 73
972	S338	9817.5	9814.6	277	16 x 73
973	S339	9806.5	9803.6	367	16 x 73
974	S340	9795.5	9792.6	187	16 x 73
975	S341	9784.5	9781.6	277	16 x 73
976	S342	9773.5	9770.6	367	16 x 73
977	S343	9762.5	9759.6	187	16 x 73
978	S344	9751.5	9748.6	277	16 x 73
979	S345	9740.5	9737.6	367	16 x 73
980	S346	9729.5	9726.6	187	16 x 73
981	S347	9718.5	9715.6	277	16 x 73
982	S348	9707.5	9704.6	367	16 x 73
983	S349	9696.5	9693.7	187	16 x 73
984	S350	9685.5	9682.7	277	16 x 73
985	S351	9674.5	9671.7	367	16 x 73
986	S352	9663.5	9660.7	187	16 x 73
987	S353	9652.5	9649.7	277	16 x 73
988	S354	9641.5	9638.7	367	16 x 73
989	S355	9630.5	9627.7	187	16 x 73
990	S356	9619.5	9616.7	277	16 x 73
991	S357	9608.5	9605.7	367	16 x 73
992	S358	9597.5	9594.7	187	16 x 73
993	S359	9586.5	9583.7	277	16 x 73
994	S360	9575.5	9572.7	367	16 x 73
995	S361	9564.5	9561.7	187	16 x 73
996	S362	9553.5	9550.7	277	16 x 73
997	S363	9542.5	9539.7	367	16 x 73
998	S364	9531.5	9528.7	187	16 x 73
999	S365	9520.5	9517.7	277	16 x 73
1000	S366	9509.5	9506.7	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1001	S367	9498.5	9495.7	187	16 x 73
1002	S368	9487.5	9484.7	277	16 x 73
1003	S369	9476.5	9473.7	367	16 x 73
1004	S370	9465.5	9462.7	187	16 x 73
1005	S371	9454.5	9451.7	277	16 x 73
1006	S372	9443.5	9440.7	367	16 x 73
1007	S373	9432.5	9429.7	187	16 x 73
1008	S374	9421.5	9418.7	277	16 x 73
1009	S375	9410.5	9407.7	367	16 x 73
1010	S376	9399.5	9396.7	187	16 x 73
1011	S377	9388.5	9385.7	277	16 x 73
1012	S378	9377.5	9374.7	367	16 x 73
1013	S379	9366.5	9363.8	187	16 x 73
1014	S380	9355.5	9352.8	277	16 x 73
1015	S381	9344.5	9341.8	367	16 x 73
1016	S382	9333.5	9330.8	187	16 x 73
1017	S383	9322.5	9319.8	277	16 x 73
1018	S384	9311.5	9308.8	367	16 x 73
1019	S385	9300.5	9297.8	187	16 x 73
1020	S386	9289.5	9286.8	277	16 x 73
1021	S387	9278.5	9275.8	367	16 x 73
1022	S388	9267.5	9264.8	187	16 x 73
1023	S389	9256.5	9253.8	277	16 x 73
1024	S390	9245.5	9242.8	367	16 x 73
1025	S391	9234.5	9231.8	187	16 x 73
1026	S392	9223.5	9220.8	277	16 x 73
1027	S393	9212.5	9209.8	367	16 x 73
1028	S394	9201.5	9198.8	187	16 x 73
1029	S395	9190.5	9187.8	277	16 x 73
1030	S396	9179.5	9176.8	367	16 x 73
1031	S397	9168.5	9165.8	187	16 x 73
1032	S398	9157.5	9154.8	277	16 x 73
1033	S399	9146.5	9143.8	367	16 x 73
1034	S400	9135.5	9132.8	187	16 x 73
1035	S401	9124.5	9121.8	277	16 x 73
1036	S402	9113.5	9110.8	367	16 x 73
1037	S403	9102.5	9099.8	187	16 x 73
1038	S404	9091.5	9088.8	277	16 x 73
1039	S405	9080.5	9077.8	367	16 x 73
1040	S406	9069.5	9066.8	187	16 x 73
1041	S407	9058.5	9055.8	277	16 x 73
1042	S408	9047.5	9044.8	367	16 x 73
1043	S409	9036.5	9033.9	187	16 x 73
1044	S410	9025.5	9022.9	277	16 x 73
1045	S411	9014.5	9011.9	367	16 x 73
1046	S412	9003.5	9000.9	187	16 x 73
1047	S413	8992.5	8989.9	277	16 x 73
1048	S414	8981.5	8978.9	367	16 x 73
1049	S415	8970.5	8967.9	187	16 x 73
1050	S416	8959.5	8956.9	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1051	S417	8948.5	8945.9	367	16 x 73
1052	S418	8937.5	8934.9	187	16 x 73
1053	S419	8926.5	8923.9	277	16 x 73
1054	S420	8915.5	8912.9	367	16 x 73
1055	S421	8904.5	8901.9	187	16 x 73
1056	S422	8893.5	8890.9	277	16 x 73
1057	S423	8882.5	8879.9	367	16 x 73
1058	S424	8871.5	8868.9	187	16 x 73
1059	S425	8860.5	8857.9	277	16 x 73
1060	S426	8849.5	8846.9	367	16 x 73
1061	S427	8838.5	8835.9	187	16 x 73
1062	S428	8827.5	8824.9	277	16 x 73
1063	S429	8816.5	8813.9	367	16 x 73
1064	S430	8805.5	8802.9	187	16 x 73
1065	S431	8794.5	8791.9	277	16 x 73
1066	S432	8783.5	8780.9	367	16 x 73
1067	S433	8772.5	8769.9	187	16 x 73
1068	S434	8761.5	8758.9	277	16 x 73
1069	S435	8750.5	8747.9	367	16 x 73
1070	S436	8739.5	8736.9	187	16 x 73
1071	S437	8728.5	8725.9	277	16 x 73
1072	S438	8717.5	8714.9	367	16 x 73
1073	S439	8706.5	8704	187	16 x 73
1074	S440	8695.5	8693	277	16 x 73
1075	S441	8684.5	8682	367	16 x 73
1076	S442	8673.5	8671	187	16 x 73
1077	S443	8662.5	8660	277	16 x 73
1078	S444	8651.5	8649	367	16 x 73
1079	S445	8640.5	8638	187	16 x 73
1080	S446	8629.5	8627	277	16 x 73
1081	S447	8618.5	8616	367	16 x 73
1082	S448	8607.5	8605	187	16 x 73
1083	S449	8596.5	8594	277	16 x 73
1084	S450	8585.5	8583	367	16 x 73
1085	S451	8574.5	8572	187	16 x 73
1086	S452	8563.5	8561	277	16 x 73
1087	S453	8552.5	8550	367	16 x 73
1088	S454	8541.5	8539	187	16 x 73
1089	S455	8530.5	8528	277	16 x 73
1090	S456	8519.5	8517	367	16 x 73
1091	S457	8508.5	8506	187	16 x 73
1092	S458	8497.5	8495	277	16 x 73
1093	S459	8486.5	8484	367	16 x 73
1094	S460	8475.5	8473	187	16 x 73
1095	S461	8464.5	8462	277	16 x 73
1096	S462	8453.5	8451	367	16 x 73
1097	S463	8442.5	8440	187	16 x 73
1098	S464	8431.5	8429	277	16 x 73
1099	S465	8420.5	8418	367	16 x 73
1100	S466	8409.5	8407	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1101	S467	8398.5	8396	277	16 x 73
1102	S468	8387.5	8385	367	16 x 73
1103	S469	8376.5	8374.1	187	16 x 73
1104	S470	8365.5	8363.1	277	16 x 73
1105	S471	8354.5	8352.1	367	16 x 73
1106	S472	8343.5	8341.1	187	16 x 73
1107	S473	8332.5	8330.1	277	16 x 73
1108	S474	8321.5	8319.1	367	16 x 73
1109	S475	8310.5	8308.1	187	16 x 73
1110	S476	8299.5	8297.1	277	16 x 73
1111	S477	8288.5	8286.1	367	16 x 73
1112	S478	8277.5	8275.1	187	16 x 73
1113	S479	8266.5	8264.1	277	16 x 73
1114	S480	8255.5	8253.1	367	16 x 73
1115	S481	8244.5	8242.1	187	16 x 73
1116	S482	8233.5	8231.1	277	16 x 73
1117	S483	8222.5	8220.1	367	16 x 73
1118	S484	8211.5	8209.1	187	16 x 73
1119	S485	8200.5	8198.1	277	16 x 73
1120	S486	8189.5	8187.1	367	16 x 73
1121	S487	8178.5	8176.1	187	16 x 73
1122	S488	8167.5	8165.1	277	16 x 73
1123	S489	8156.5	8154.1	367	16 x 73
1124	S490	8145.5	8143.1	187	16 x 73
1125	S491	8134.5	8132.1	277	16 x 73
1126	S492	8123.5	8121.1	367	16 x 73
1127	S493	8112.5	8110.1	187	16 x 73
1128	S494	8101.5	8099.1	277	16 x 73
1129	S495	8090.5	8088.1	367	16 x 73
1130	S496	8079.5	8077.1	187	16 x 73
1131	S497	8068.5	8066.1	277	16 x 73
1132	S498	8057.5	8055.1	367	16 x 73
1133	S499	8046.5	8044.2	187	16 x 73
1134	S500	8035.5	8033.2	277	16 x 73
1135	S501	8024.5	8022.2	367	16 x 73
1136	S502	8013.5	8011.2	187	16 x 73
1137	S503	8002.5	8000.2	277	16 x 73
1138	S504	7991.5	7989.2	367	16 x 73
1139	S505	7980.5	7978.2	187	16 x 73
1140	S506	7969.5	7967.2	277	16 x 73
1141	S507	7958.5	7956.2	367	16 x 73
1142	S508	7947.5	7945.2	187	16 x 73
1143	S509	7936.5	7934.2	277	16 x 73
1144	S510	7925.5	7923.2	367	16 x 73
1145	S511	7914.5	7912.2	187	16 x 73
1146	S512	7903.5	7901.2	277	16 x 73
1147	S513	7892.5	7890.2	367	16 x 73
1148	S514	7881.5	7879.2	187	16 x 73
1149	S515	7870.5	7868.2	277	16 x 73
1150	S516	7859.5	7857.2	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1151	S517	7848.5	7846.2	187	16 x 73
1152	S518	7837.5	7835.2	277	16 x 73
1153	S519	7826.5	7824.2	367	16 x 73
1154	S520	7815.5	7813.2	187	16 x 73
1155	S521	7804.5	7802.2	277	16 x 73
1156	S522	7793.5	7791.2	367	16 x 73
1157	S523	7782.5	7780.2	187	16 x 73
1158	S524	7771.5	7769.2	277	16 x 73
1159	S525	7760.5	7758.2	367	16 x 73
1160	S526	7749.5	7747.2	187	16 x 73
1161	S527	7738.5	7736.2	277	16 x 73
1162	S528	7727.5	7725.2	367	16 x 73
1163	S529	7716.5	7714.3	187	16 x 73
1164	S530	7705.5	7703.3	277	16 x 73
1165	S531	7694.5	7692.3	367	16 x 73
1166	S532	7683.5	7681.3	187	16 x 73
1167	S533	7672.5	7670.3	277	16 x 73
1168	S534	7661.5	7659.3	367	16 x 73
1169	S535	7650.5	7648.3	187	16 x 73
1170	S536	7639.5	7637.3	277	16 x 73
1171	S537	7628.5	7626.3	367	16 x 73
1172	S538	7617.5	7615.3	187	16 x 73
1173	S539	7606.5	7604.3	277	16 x 73
1174	S540	7595.5	7593.3	367	16 x 73
1175	S541	7584.5	7582.3	187	16 x 73
1176	S542	7573.5	7571.3	277	16 x 73
1177	S543	7562.5	7560.3	367	16 x 73
1178	S544	7551.5	7549.3	187	16 x 73
1179	S545	7540.5	7538.3	277	16 x 73
1180	S546	7529.5	7527.3	367	16 x 73
1181	S547	7518.5	7516.3	187	16 x 73
1182	S548	7507.5	7505.3	277	16 x 73
1183	S549	7496.5	7494.3	367	16 x 73
1184	S550	7485.5	7483.3	187	16 x 73
1185	S551	7474.5	7472.3	277	16 x 73
1186	S552	7463.5	7461.3	367	16 x 73
1187	S553	7452.5	7450.3	187	16 x 73
1188	S554	7441.5	7439.3	277	16 x 73
1189	S555	7430.5	7428.3	367	16 x 73
1190	S556	7419.5	7417.3	187	16 x 73
1191	S557	7408.5	7406.3	277	16 x 73
1192	S558	7397.5	7395.3	367	16 x 73
1193	S559	7386.5	7384.4	187	16 x 73
1194	S560	7375.5	7373.4	277	16 x 73
1195	S561	7364.5	7362.4	367	16 x 73
1196	S562	7353.5	7351.4	187	16 x 73
1197	S563	7342.5	7340.4	277	16 x 73
1198	S564	7331.5	7329.4	367	16 x 73
1199	S565	7320.5	7318.4	187	16 x 73
1200	S566	7309.5	7307.4	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1201	S567	7298.5	7296.4	367	16 x 73
1202	S568	7287.5	7285.4	187	16 x 73
1203	S569	7276.5	7274.4	277	16 x 73
1204	S570	7265.5	7263.4	367	16 x 73
1205	S571	7254.5	7252.4	187	16 x 73
1206	S572	7243.5	7241.4	277	16 x 73
1207	S573	7232.5	7230.4	367	16 x 73
1208	S574	7221.5	7219.4	187	16 x 73
1209	S575	7210.5	7208.4	277	16 x 73
1210	S576	7199.5	7197.4	367	16 x 73
1211	S577	7188.5	7186.4	187	16 x 73
1212	S578	7177.5	7175.4	277	16 x 73
1213	S579	7166.5	7164.4	367	16 x 73
1214	S580	7155.5	7153.4	187	16 x 73
1215	S581	7144.5	7142.4	277	16 x 73
1216	S582	7133.5	7131.4	367	16 x 73
1217	S583	7122.5	7120.4	187	16 x 73
1218	S584	7111.5	7109.4	277	16 x 73
1219	S585	7100.5	7098.4	367	16 x 73
1220	S586	7089.5	7087.4	187	16 x 73
1221	S587	7078.5	7076.4	277	16 x 73
1222	S588	7067.5	7065.4	367	16 x 73
1223	S589	7056.5	7054.5	187	16 x 73
1224	S590	7045.5	7043.5	277	16 x 73
1225	S591	7034.5	7032.5	367	16 x 73
1226	S592	7023.5	7021.5	187	16 x 73
1227	S593	7012.5	7010.5	277	16 x 73
1228	S594	7001.5	6999.5	367	16 x 73
1229	S595	6990.5	6988.5	187	16 x 73
1230	S596	6979.5	6977.5	277	16 x 73
1231	S597	6968.5	6966.5	367	16 x 73
1232	S598	6957.5	6955.5	187	16 x 73
1233	S599	6946.5	6944.5	277	16 x 73
1234	S600	6935.5	6933.5	367	16 x 73
1235	DUMMY	6924.5	6922.5	187	16 x 73
1236	DUMMY	6913.5	6911.5	277	16 x 73
1237	DUMMY	6902.5	6900.5	367	16 x 73
1238	DUMMY	6891.5	6889.5	187	16 x 73
1239	DUMMY	6880.5	6878.5	277	16 x 73
1240	DUMMY	6869.5	6867.5	367	16 x 73
1241	DUMMY	6858.5	6856.5	187	16 x 73
1242	DUMMY	6847.5	6845.5	277	16 x 73
1243	DUMMY	6836.5	6834.5	367	16 x 73
1244	DUMMY	6825.5	6823.5	187	16 x 73
1245	DUMMY	6814.5	6812.5	277	16 x 73
1246	DUMMY	6803.5	6801.5	367	16 x 73
1247	DUMMY	6792.5	6790.5	187	16 x 73
1248	DUMMY	6781.5	6779.5	277	16 x 73
1249	DUMMY	6770.5	6768.5	367	16 x 73
1250	DUMMY	6759.5	6757.5	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1251	DUMMY	6748.5	6746.5	277	16 x 73
1252	DUMMY	6737.5	6735.5	367	16 x 73
1253	S601	6726.5	6724.6	187	16 x 73
1254	S602	6715.5	6713.6	277	16 x 73
1255	S603	6704.5	6702.6	367	16 x 73
1256	S604	6693.5	6691.6	187	16 x 73
1257	S605	6682.5	6680.6	277	16 x 73
1258	S606	6671.5	6669.6	367	16 x 73
1259	S607	6660.5	6658.6	187	16 x 73
1260	S608	6649.5	6647.6	277	16 x 73
1261	S609	6638.5	6636.6	367	16 x 73
1262	S610	6627.5	6625.6	187	16 x 73
1263	S611	6616.5	6614.6	277	16 x 73
1264	S612	6605.5	6603.6	367	16 x 73
1265	S613	6594.5	6592.6	187	16 x 73
1266	S614	6583.5	6581.6	277	16 x 73
1267	S615	6572.5	6570.6	367	16 x 73
1268	S616	6561.5	6559.6	187	16 x 73
1269	S617	6550.5	6548.6	277	16 x 73
1270	S618	6539.5	6537.6	367	16 x 73
1271	S619	6528.5	6526.6	187	16 x 73
1272	S620	6517.5	6515.6	277	16 x 73
1273	S621	6506.5	6504.6	367	16 x 73
1274	S622	6495.5	6493.6	187	16 x 73
1275	S623	6484.5	6482.6	277	16 x 73
1276	S624	6473.5	6471.6	367	16 x 73
1277	S625	6462.5	6460.6	187	16 x 73
1278	S626	6451.5	6449.6	277	16 x 73
1279	S627	6440.5	6438.6	367	16 x 73
1280	S628	6429.5	6427.6	187	16 x 73
1281	S629	6418.5	6416.6	277	16 x 73
1282	S630	6407.5	6405.6	367	16 x 73
1283	S631	6396.5	6394.7	187	16 x 73
1284	S632	6385.5	6383.7	277	16 x 73
1285	S633	6374.5	6372.7	367	16 x 73
1286	S634	6363.5	6361.7	187	16 x 73
1287	S635	6352.5	6350.7	277	16 x 73
1288	S636	6341.5	6339.7	367	16 x 73
1289	S637	6330.5	6328.7	187	16 x 73
1290	S638	6319.5	6317.7	277	16 x 73
1291	S639	6308.5	6306.7	367	16 x 73
1292	S640	6297.5	6295.7	187	16 x 73
1293	S641	6286.5	6284.7	277	16 x 73
1294	S642	6275.5	6273.7	367	16 x 73
1295	S643	6264.5	6262.7	187	16 x 73
1296	S644	6253.5	6251.7	277	16 x 73
1297	S645	6242.5	6240.7	367	16 x 73
1298	S646	6231.5	6229.7	187	16 x 73
1299	S647	6220.5	6218.7	277	16 x 73
1300	S648	6209.5	6207.7	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1301	S649	6198.5	6196.7	187	16 x 73
1302	S650	6187.5	6185.7	277	16 x 73
1303	S651	6176.5	6174.7	367	16 x 73
1304	S652	6165.5	6163.7	187	16 x 73
1305	S653	6154.5	6152.7	277	16 x 73
1306	S654	6143.5	6141.7	367	16 x 73
1307	S655	6132.5	6130.7	187	16 x 73
1308	S656	6121.5	6119.7	277	16 x 73
1309	S657	6110.5	6108.7	367	16 x 73
1310	S658	6099.5	6097.7	187	16 x 73
1311	S659	6088.5	6086.7	277	16 x 73
1312	S660	6077.5	6075.7	367	16 x 73
1313	S661	6066.5	6064.8	187	16 x 73
1314	S662	6055.5	6053.8	277	16 x 73
1315	S663	6044.5	6042.8	367	16 x 73
1316	S664	6033.5	6031.8	187	16 x 73
1317	S665	6022.5	6020.8	277	16 x 73
1318	S666	6011.5	6009.8	367	16 x 73
1319	S667	6000.5	5998.8	187	16 x 73
1320	S668	5989.5	5987.8	277	16 x 73
1321	S669	5978.5	5976.8	367	16 x 73
1322	S670	5967.5	5965.8	187	16 x 73
1323	S671	5956.5	5954.8	277	16 x 73
1324	S672	5945.5	5943.8	367	16 x 73
1325	S673	5934.5	5932.8	187	16 x 73
1326	S674	5923.5	5921.8	277	16 x 73
1327	S675	5912.5	5910.8	367	16 x 73
1328	S676	5901.5	5899.8	187	16 x 73
1329	S677	5890.5	5888.8	277	16 x 73
1330	S678	5879.5	5877.8	367	16 x 73
1331	S679	5868.5	5866.8	187	16 x 73
1332	S680	5857.5	5855.8	277	16 x 73
1333	S681	5846.5	5844.8	367	16 x 73
1334	S682	5835.5	5833.8	187	16 x 73
1335	S683	5824.5	5822.8	277	16 x 73
1336	S684	5813.5	5811.8	367	16 x 73
1337	S685	5802.5	5800.8	187	16 x 73
1338	S686	5791.5	5789.8	277	16 x 73
1339	S687	5780.5	5778.8	367	16 x 73
1340	S688	5769.5	5767.8	187	16 x 73
1341	S689	5758.5	5756.8	277	16 x 73
1342	S690	5747.5	5745.8	367	16 x 73
1343	S691	5736.5	5734.9	187	16 x 73
1344	S692	5725.5	5723.9	277	16 x 73
1345	S693	5714.5	5712.9	367	16 x 73
1346	S694	5703.5	5701.9	187	16 x 73
1347	S695	5692.5	5690.9	277	16 x 73
1348	S696	5681.5	5679.9	367	16 x 73
1349	S697	5670.5	5668.9	187	16 x 73
1350	S698	5659.5	5657.9	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1351	S699	5648.5	5646.9	367	16 x 73
1352	S700	5637.5	5635.9	187	16 x 73
1353	S701	5626.5	5624.9	277	16 x 73
1354	S702	5615.5	5613.9	367	16 x 73
1355	S703	5604.5	5602.9	187	16 x 73
1356	S704	5593.5	5591.9	277	16 x 73
1357	S705	5582.5	5580.9	367	16 x 73
1358	S706	5571.5	5569.9	187	16 x 73
1359	S707	5560.5	5558.9	277	16 x 73
1360	S708	5549.5	5547.9	367	16 x 73
1361	S709	5538.5	5536.9	187	16 x 73
1362	S710	5527.5	5525.9	277	16 x 73
1363	S711	5516.5	5514.9	367	16 x 73
1364	S712	5505.5	5503.9	187	16 x 73
1365	S713	5494.5	5492.9	277	16 x 73
1366	S714	5483.5	5481.9	367	16 x 73
1367	S715	5472.5	5470.9	187	16 x 73
1368	S716	5461.5	5459.9	277	16 x 73
1369	S717	5450.5	5448.9	367	16 x 73
1370	S718	5439.5	5437.9	187	16 x 73
1371	S719	5428.5	5426.9	277	16 x 73
1372	S720	5417.5	5415.9	367	16 x 73
1373	S721	5406.5	5405	187	16 x 73
1374	S722	5395.5	5394	277	16 x 73
1375	S723	5384.5	5383	367	16 x 73
1376	S724	5373.5	5372	187	16 x 73
1377	S725	5362.5	5361	277	16 x 73
1378	S726	5351.5	5350	367	16 x 73
1379	S727	5340.5	5339	187	16 x 73
1380	S728	5329.5	5328	277	16 x 73
1381	S729	5318.5	5317	367	16 x 73
1382	S730	5307.5	5306	187	16 x 73
1383	S731	5296.5	5295	277	16 x 73
1384	S732	5285.5	5284	367	16 x 73
1385	S733	5274.5	5273	187	16 x 73
1386	S734	5263.5	5262	277	16 x 73
1387	S735	5252.5	5251	367	16 x 73
1388	S736	5241.5	5240	187	16 x 73
1389	S737	5230.5	5229	277	16 x 73
1390	S738	5219.5	5218	367	16 x 73
1391	S739	5208.5	5207	187	16 x 73
1392	S740	5197.5	5196	277	16 x 73
1393	S741	5186.5	5185	367	16 x 73
1394	S742	5175.5	5174	187	16 x 73
1395	S743	5164.5	5163	277	16 x 73
1396	S744	5153.5	5152	367	16 x 73
1397	S745	5142.5	5141	187	16 x 73
1398	S746	5131.5	5130	277	16 x 73
1399	S747	5120.5	5119	367	16 x 73
1400	S748	5109.5	5108	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1401	S749	5098.5	5097	277	16 x 73
1402	S750	5087.5	5086	367	16 x 73
1403	S751	5076.5	5075.1	187	16 x 73
1404	S752	5065.5	5064.1	277	16 x 73
1405	S753	5054.5	5053.1	367	16 x 73
1406	S754	5043.5	5042.1	187	16 x 73
1407	S755	5032.5	5031.1	277	16 x 73
1408	S756	5021.5	5020.1	367	16 x 73
1409	S757	5010.5	5009.1	187	16 x 73
1410	S758	4999.5	4998.1	277	16 x 73
1411	S759	4988.5	4987.1	367	16 x 73
1412	S760	4977.5	4976.1	187	16 x 73
1413	S761	4966.5	4965.1	277	16 x 73
1414	S762	4955.5	4954.1	367	16 x 73
1415	S763	4944.5	4943.1	187	16 x 73
1416	S764	4933.5	4932.1	277	16 x 73
1417	S765	4922.5	4921.1	367	16 x 73
1418	S766	4911.5	4910.1	187	16 x 73
1419	S767	4900.5	4899.1	277	16 x 73
1420	S768	4889.5	4888.1	367	16 x 73
1421	S769	4878.5	4877.1	187	16 x 73
1422	S770	4867.5	4866.1	277	16 x 73
1423	S771	4856.5	4855.1	367	16 x 73
1424	S772	4845.5	4844.1	187	16 x 73
1425	S773	4834.5	4833.1	277	16 x 73
1426	S774	4823.5	4822.1	367	16 x 73
1427	S775	4812.5	4811.1	187	16 x 73
1428	S776	4801.5	4800.1	277	16 x 73
1429	S777	4790.5	4789.1	367	16 x 73
1430	S778	4779.5	4778.1	187	16 x 73
1431	S779	4768.5	4767.1	277	16 x 73
1432	S780	4757.5	4756.1	367	16 x 73
1433	S781	4746.5	4745.2	187	16 x 73
1434	S782	4735.5	4734.2	277	16 x 73
1435	S783	4724.5	4723.2	367	16 x 73
1436	S784	4713.5	4712.2	187	16 x 73
1437	S785	4702.5	4701.2	277	16 x 73
1438	S786	4691.5	4690.2	367	16 x 73
1439	S787	4680.5	4679.2	187	16 x 73
1440	S788	4669.5	4668.2	277	16 x 73
1441	S789	4658.5	4657.2	367	16 x 73
1442	S790	4647.5	4646.2	187	16 x 73
1443	S791	4636.5	4635.2	277	16 x 73
1444	S792	4625.5	4624.2	367	16 x 73
1445	S793	4614.5	4613.2	187	16 x 73
1446	S794	4603.5	4602.2	277	16 x 73
1447	S795	4592.5	4591.2	367	16 x 73
1448	S796	4581.5	4580.2	187	16 x 73
1449	S797	4570.5	4569.2	277	16 x 73
1450	S798	4559.5	4558.2	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1451	S799	4548.5	4547.2	187	16 x 73
1452	S800	4537.5	4536.2	277	16 x 73
1453	S801	4526.5	4525.2	367	16 x 73
1454	S802	4515.5	4514.2	187	16 x 73
1455	S803	4504.5	4503.2	277	16 x 73
1456	S804	4493.5	4492.2	367	16 x 73
1457	S805	4482.5	4481.2	187	16 x 73
1458	S806	4471.5	4470.2	277	16 x 73
1459	S807	4460.5	4459.2	367	16 x 73
1460	S808	4449.5	4448.2	187	16 x 73
1461	S809	4438.5	4437.2	277	16 x 73
1462	S810	4427.5	4426.2	367	16 x 73
1463	S811	4416.5	4415.3	187	16 x 73
1464	S812	4405.5	4404.3	277	16 x 73
1465	S813	4394.5	4393.3	367	16 x 73
1466	S814	4383.5	4382.3	187	16 x 73
1467	S815	4372.5	4371.3	277	16 x 73
1468	S816	4361.5	4360.3	367	16 x 73
1469	S817	4350.5	4349.3	187	16 x 73
1470	S818	4339.5	4338.3	277	16 x 73
1471	S819	4328.5	4327.3	367	16 x 73
1472	S820	4317.5	4316.3	187	16 x 73
1473	S821	4306.5	4305.3	277	16 x 73
1474	S822	4295.5	4294.3	367	16 x 73
1475	S823	4284.5	4283.3	187	16 x 73
1476	S824	4273.5	4272.3	277	16 x 73
1477	S825	4262.5	4261.3	367	16 x 73
1478	S826	4251.5	4250.3	187	16 x 73
1479	S827	4240.5	4239.3	277	16 x 73
1480	S828	4229.5	4228.3	367	16 x 73
1481	S829	4218.5	4217.3	187	16 x 73
1482	S830	4207.5	4206.3	277	16 x 73
1483	S831	4196.5	4195.3	367	16 x 73
1484	S832	4185.5	4184.3	187	16 x 73
1485	S833	4174.5	4173.3	277	16 x 73
1486	S834	4163.5	4162.3	367	16 x 73
1487	S835	4152.5	4151.3	187	16 x 73
1488	S836	4141.5	4140.3	277	16 x 73
1489	S837	4130.5	4129.3	367	16 x 73
1490	S838	4119.5	4118.3	187	16 x 73
1491	S839	4108.5	4107.3	277	16 x 73
1492	S840	4097.5	4096.3	367	16 x 73
1493	S841	4086.5	4085.4	187	16 x 73
1494	S842	4075.5	4074.4	277	16 x 73
1495	S843	4064.5	4063.4	367	16 x 73
1496	S844	4053.5	4052.4	187	16 x 73
1497	S845	4042.5	4041.4	277	16 x 73
1498	S846	4031.5	4030.4	367	16 x 73
1499	S847	4020.5	4019.4	187	16 x 73
1500	S848	4009.5	4008.4	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1501	S849	3998.5	3997.4	367	16 x 73
1502	S850	3987.5	3986.4	187	16 x 73
1503	S851	3976.5	3975.4	277	16 x 73
1504	S852	3965.5	3964.4	367	16 x 73
1505	S853	3954.5	3953.4	187	16 x 73
1506	S854	3943.5	3942.4	277	16 x 73
1507	S855	3932.5	3931.4	367	16 x 73
1508	S856	3921.5	3920.4	187	16 x 73
1509	S857	3910.5	3909.4	277	16 x 73
1510	S858	3899.5	3898.4	367	16 x 73
1511	S859	3888.5	3887.4	187	16 x 73
1512	S860	3877.5	3876.4	277	16 x 73
1513	S861	3866.5	3865.4	367	16 x 73
1514	S862	3855.5	3854.4	187	16 x 73
1515	S863	3844.5	3843.4	277	16 x 73
1516	S864	3833.5	3832.4	367	16 x 73
1517	S865	3822.5	3821.4	187	16 x 73
1518	S866	3811.5	3810.4	277	16 x 73
1519	S867	3800.5	3799.4	367	16 x 73
1520	S868	3789.5	3788.4	187	16 x 73
1521	S869	3778.5	3777.4	277	16 x 73
1522	S870	3767.5	3766.4	367	16 x 73
1523	S871	3756.5	3755.5	187	16 x 73
1524	S872	3745.5	3744.5	277	16 x 73
1525	S873	3734.5	3733.5	367	16 x 73
1526	S874	3723.5	3722.5	187	16 x 73
1527	S875	3712.5	3711.5	277	16 x 73
1528	S876	3701.5	3700.5	367	16 x 73
1529	S877	3690.5	3689.5	187	16 x 73
1530	S878	3679.5	3678.5	277	16 x 73
1531	S879	3668.5	3667.5	367	16 x 73
1532	S880	3657.5	3656.5	187	16 x 73
1533	S881	3646.5	3645.5	277	16 x 73
1534	S882	3635.5	3634.5	367	16 x 73
1535	S883	3624.5	3623.5	187	16 x 73
1536	S884	3613.5	3612.5	277	16 x 73
1537	S885	3602.5	3601.5	367	16 x 73
1538	S886	3591.5	3590.5	187	16 x 73
1539	S887	3580.5	3579.5	277	16 x 73
1540	S888	3569.5	3568.5	367	16 x 73
1541	S889	3558.5	3557.5	187	16 x 73
1542	S890	3547.5	3546.5	277	16 x 73
1543	S891	3536.5	3535.5	367	16 x 73
1544	S892	3525.5	3524.5	187	16 x 73
1545	S893	3514.5	3513.5	277	16 x 73
1546	S894	3503.5	3502.5	367	16 x 73
1547	S895	3492.5	3491.5	187	16 x 73
1548	S896	3481.5	3480.5	277	16 x 73
1549	S897	3470.5	3469.5	367	16 x 73
1550	S898	3459.5	3458.5	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1551	S899	3448.5	3447.5	277	16 x 73
1552	S900	3437.5	3436.5	367	16 x 73
1553	S901	3426.5	3425.6	187	16 x 73
1554	S902	3415.5	3414.6	277	16 x 73
1555	S903	3404.5	3403.6	367	16 x 73
1556	S904	3393.5	3392.6	187	16 x 73
1557	S905	3382.5	3381.6	277	16 x 73
1558	S906	3371.5	3370.6	367	16 x 73
1559	S907	3360.5	3359.6	187	16 x 73
1560	S908	3349.5	3348.6	277	16 x 73
1561	S909	3338.5	3337.6	367	16 x 73
1562	S910	3327.5	3326.6	187	16 x 73
1563	S911	3316.5	3315.6	277	16 x 73
1564	S912	3305.5	3304.6	367	16 x 73
1565	S913	3294.5	3293.6	187	16 x 73
1566	S914	3283.5	3282.6	277	16 x 73
1567	S915	3272.5	3271.6	367	16 x 73
1568	S916	3261.5	3260.6	187	16 x 73
1569	S917	3250.5	3249.6	277	16 x 73
1570	S918	3239.5	3238.6	367	16 x 73
1571	S919	3228.5	3227.6	187	16 x 73
1572	S920	3217.5	3216.6	277	16 x 73
1573	S921	3206.5	3205.6	367	16 x 73
1574	S922	3195.5	3194.6	187	16 x 73
1575	S923	3184.5	3183.6	277	16 x 73
1576	S924	3173.5	3172.6	367	16 x 73
1577	S925	3162.5	3161.6	187	16 x 73
1578	S926	3151.5	3150.6	277	16 x 73
1579	S927	3140.5	3139.6	367	16 x 73
1580	S928	3129.5	3128.6	187	16 x 73
1581	S929	3118.5	3117.6	277	16 x 73
1582	S930	3107.5	3106.6	367	16 x 73
1583	S931	3096.5	3095.7	187	16 x 73
1584	S932	3085.5	3084.7	277	16 x 73
1585	S933	3074.5	3073.7	367	16 x 73
1586	S934	3063.5	3062.7	187	16 x 73
1587	S935	3052.5	3051.7	277	16 x 73
1588	S936	3041.5	3040.7	367	16 x 73
1589	S937	3030.5	3029.7	187	16 x 73
1590	S938	3019.5	3018.7	277	16 x 73
1591	S939	3008.5	3007.7	367	16 x 73
1592	S940	2997.5	2996.7	187	16 x 73
1593	S941	2986.5	2985.7	277	16 x 73
1594	S942	2975.5	2974.7	367	16 x 73
1595	S943	2964.5	2963.7	187	16 x 73
1596	S944	2953.5	2952.7	277	16 x 73
1597	S945	2942.5	2941.7	367	16 x 73
1598	S946	2931.5	2930.7	187	16 x 73
1599	S947	2920.5	2919.7	277	16 x 73
1600	S948	2909.5	2908.7	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1601	S949	2898.5	2897.7	187	16 x 73
1602	S950	2887.5	2886.7	277	16 x 73
1603	S951	2876.5	2875.7	367	16 x 73
1604	S952	2865.5	2864.7	187	16 x 73
1605	S953	2854.5	2853.7	277	16 x 73
1606	S954	2843.5	2842.7	367	16 x 73
1607	S955	2832.5	2831.7	187	16 x 73
1608	S956	2821.5	2820.7	277	16 x 73
1609	S957	2810.5	2809.7	367	16 x 73
1610	S958	2799.5	2798.7	187	16 x 73
1611	S959	2788.5	2787.7	277	16 x 73
1612	S960	2777.5	2776.7	367	16 x 73
1613	S961	2766.5	2765.8	187	16 x 73
1614	S962	2755.5	2754.8	277	16 x 73
1615	S963	2744.5	2743.8	367	16 x 73
1616	S964	2733.5	2732.8	187	16 x 73
1617	S965	2722.5	2721.8	277	16 x 73
1618	S966	2711.5	2710.8	367	16 x 73
1619	S967	2700.5	2699.8	187	16 x 73
1620	S968	2689.5	2688.8	277	16 x 73
1621	S969	2678.5	2677.8	367	16 x 73
1622	S970	2667.5	2666.8	187	16 x 73
1623	S971	2656.5	2655.8	277	16 x 73
1624	S972	2645.5	2644.8	367	16 x 73
1625	S973	2634.5	2633.8	187	16 x 73
1626	S974	2623.5	2622.8	277	16 x 73
1627	S975	2612.5	2611.8	367	16 x 73
1628	S976	2601.5	2600.8	187	16 x 73
1629	S977	2590.5	2589.8	277	16 x 73
1630	S978	2579.5	2578.8	367	16 x 73
1631	S979	2568.5	2567.8	187	16 x 73
1632	S980	2557.5	2556.8	277	16 x 73
1633	S981	2546.5	2545.8	367	16 x 73
1634	S982	2535.5	2534.8	187	16 x 73
1635	S983	2524.5	2523.8	277	16 x 73
1636	S984	2513.5	2512.8	367	16 x 73
1637	S985	2502.5	2501.8	187	16 x 73
1638	S986	2491.5	2490.8	277	16 x 73
1639	S987	2480.5	2479.8	367	16 x 73
1640	S988	2469.5	2468.8	187	16 x 73
1641	S989	2458.5	2457.8	277	16 x 73
1642	S990	2447.5	2446.8	367	16 x 73
1643	S991	2436.5	2435.9	187	16 x 73
1644	S992	2425.5	2424.9	277	16 x 73
1645	S993	2414.5	2413.9	367	16 x 73
1646	S994	2403.5	2402.9	187	16 x 73
1647	S995	2392.5	2391.9	277	16 x 73
1648	S996	2381.5	2380.9	367	16 x 73
1649	S997	2370.5	2369.9	187	16 x 73
1650	S998	2359.5	2358.9	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1651	S999	2348.5	2347.9	367	16 x 73
1652	S1000	2337.5	2336.9	187	16 x 73
1653	S1001	2326.5	2325.9	277	16 x 73
1654	S1002	2315.5	2314.9	367	16 x 73
1655	S1003	2304.5	2303.9	187	16 x 73
1656	S1004	2293.5	2292.9	277	16 x 73
1657	S1005	2282.5	2281.9	367	16 x 73
1658	S1006	2271.5	2270.9	187	16 x 73
1659	S1007	2260.5	2259.9	277	16 x 73
1660	S1008	2249.5	2248.9	367	16 x 73
1661	S1009	2238.5	2237.9	187	16 x 73
1662	S1010	2227.5	2226.9	277	16 x 73
1663	S1011	2216.5	2215.9	367	16 x 73
1664	S1012	2205.5	2204.9	187	16 x 73
1665	S1013	2194.5	2193.9	277	16 x 73
1666	S1014	2183.5	2182.9	367	16 x 73
1667	S1015	2172.5	2171.9	187	16 x 73
1668	S1016	2161.5	2160.9	277	16 x 73
1669	S1017	2150.5	2149.9	367	16 x 73
1670	S1018	2139.5	2138.9	187	16 x 73
1671	S1019	2128.5	2127.9	277	16 x 73
1672	S1020	2117.5	2116.9	367	16 x 73
1673	S1021	2106.5	2106	187	16 x 73
1674	S1022	2095.5	2095	277	16 x 73
1675	S1023	2084.5	2084	367	16 x 73
1676	S1024	2073.5	2073	187	16 x 73
1677	S1025	2062.5	2062	277	16 x 73
1678	S1026	2051.5	2051	367	16 x 73
1679	S1027	2040.5	2040	187	16 x 73
1680	S1028	2029.5	2029	277	16 x 73
1681	S1029	2018.5	2018	367	16 x 73
1682	S1030	2007.5	2007	187	16 x 73
1683	S1031	1996.5	1996	277	16 x 73
1684	S1032	1985.5	1985	367	16 x 73
1685	S1033	1974.5	1974	187	16 x 73
1686	S1034	1963.5	1963	277	16 x 73
1687	S1035	1952.5	1952	367	16 x 73
1688	S1036	1941.5	1941	187	16 x 73
1689	S1037	1930.5	1930	277	16 x 73
1690	S1038	1919.5	1919	367	16 x 73
1691	S1039	1908.5	1908	187	16 x 73
1692	S1040	1897.5	1897	277	16 x 73
1693	S1041	1886.5	1886	367	16 x 73
1694	S1042	1875.5	1875	187	16 x 73
1695	S1043	1864.5	1864	277	16 x 73
1696	S1044	1853.5	1853	367	16 x 73
1697	S1045	1842.5	1842	187	16 x 73
1698	S1046	1831.5	1831	277	16 x 73
1699	S1047	1820.5	1820	367	16 x 73
1700	S1048	1809.5	1809	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1701	S1049	1798.5	1798	277	16 x 73
1702	S1050	1787.5	1787	367	16 x 73
1703	S1051	1776.5	1776.1	187	16 x 73
1704	S1052	1765.5	1765.1	277	16 x 73
1705	S1053	1754.5	1754.1	367	16 x 73
1706	S1054	1743.5	1743.1	187	16 x 73
1707	S1055	1732.5	1732.1	277	16 x 73
1708	S1056	1721.5	1721.1	367	16 x 73
1709	S1057	1710.5	1710.1	187	16 x 73
1710	S1058	1699.5	1699.1	277	16 x 73
1711	S1059	1688.5	1688.1	367	16 x 73
1712	S1060	1677.5	1677.1	187	16 x 73
1713	S1061	1666.5	1666.1	277	16 x 73
1714	S1062	1655.5	1655.1	367	16 x 73
1715	S1063	1644.5	1644.1	187	16 x 73
1716	S1064	1633.5	1633.1	277	16 x 73
1717	S1065	1622.5	1622.1	367	16 x 73
1718	S1066	1611.5	1611.1	187	16 x 73
1719	S1067	1600.5	1600.1	277	16 x 73
1720	S1068	1589.5	1589.1	367	16 x 73
1721	S1069	1578.5	1578.1	187	16 x 73
1722	S1070	1567.5	1567.1	277	16 x 73
1723	S1071	1556.5	1556.1	367	16 x 73
1724	S1072	1545.5	1545.1	187	16 x 73
1725	S1073	1534.5	1534.1	277	16 x 73
1726	S1074	1523.5	1523.1	367	16 x 73
1727	S1075	1512.5	1512.1	187	16 x 73
1728	S1076	1501.5	1501.1	277	16 x 73
1729	S1077	1490.5	1490.1	367	16 x 73
1730	S1078	1479.5	1479.1	187	16 x 73
1731	S1079	1468.5	1468.1	277	16 x 73
1732	S1080	1457.5	1457.1	367	16 x 73
1733	S1081	1446.5	1446.2	187	16 x 73
1734	S1082	1435.5	1435.2	277	16 x 73
1735	S1083	1424.5	1424.2	367	16 x 73
1736	S1084	1413.5	1413.2	187	16 x 73
1737	S1085	1402.5	1402.2	277	16 x 73
1738	S1086	1391.5	1391.2	367	16 x 73
1739	S1087	1380.5	1380.2	187	16 x 73
1740	S1088	1369.5	1369.2	277	16 x 73
1741	S1089	1358.5	1358.2	367	16 x 73
1742	S1090	1347.5	1347.2	187	16 x 73
1743	S1091	1336.5	1336.2	277	16 x 73
1744	S1092	1325.5	1325.2	367	16 x 73
1745	S1093	1314.5	1314.2	187	16 x 73
1746	S1094	1303.5	1303.2	277	16 x 73
1747	S1095	1292.5	1292.2	367	16 x 73
1748	S1096	1281.5	1281.2	187	16 x 73
1749	S1097	1270.5	1270.2	277	16 x 73
1750	S1098	1259.5	1259.2	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1751	S1099	1248.5	1248.2	187	16 x 73
1752	S1100	1237.5	1237.2	277	16 x 73
1753	S1101	1226.5	1226.2	367	16 x 73
1754	S1102	1215.5	1215.2	187	16 x 73
1755	S1103	1204.5	1204.2	277	16 x 73
1756	S1104	1193.5	1193.2	367	16 x 73
1757	S1105	1182.5	1182.2	187	16 x 73
1758	S1106	1171.5	1171.2	277	16 x 73
1759	S1107	1160.5	1160.2	367	16 x 73
1760	S1108	1149.5	1149.2	187	16 x 73
1761	S1109	1138.5	1138.2	277	16 x 73
1762	S1110	1127.5	1127.2	367	16 x 73
1763	S1111	1116.5	1116.3	187	16 x 73
1764	S1112	1105.5	1105.3	277	16 x 73
1765	S1113	1094.5	1094.3	367	16 x 73
1766	S1114	1083.5	1083.3	187	16 x 73
1767	S1115	1072.5	1072.3	277	16 x 73
1768	S1116	1061.5	1061.3	367	16 x 73
1769	S1117	1050.5	1050.3	187	16 x 73
1770	S1118	1039.5	1039.3	277	16 x 73
1771	S1119	1028.5	1028.3	367	16 x 73
1772	S1120	1017.5	1017.3	187	16 x 73
1773	S1121	1006.5	1006.3	277	16 x 73
1774	S1122	995.5	995.3	367	16 x 73
1775	S1123	984.5	984.3	187	16 x 73
1776	S1124	973.5	973.3	277	16 x 73
1777	S1125	962.5	962.3	367	16 x 73
1778	S1126	951.5	951.3	187	16 x 73
1779	S1127	940.5	940.3	277	16 x 73
1780	S1128	929.5	929.3	367	16 x 73
1781	S1129	918.5	918.3	187	16 x 73
1782	S1130	907.5	907.3	277	16 x 73
1783	S1131	896.5	896.3	367	16 x 73
1784	S1132	885.5	885.3	187	16 x 73
1785	S1133	874.5	874.3	277	16 x 73
1786	S1134	863.5	863.3	367	16 x 73
1787	S1135	852.5	852.3	187	16 x 73
1788	S1136	841.5	841.3	277	16 x 73
1789	S1137	830.5	830.3	367	16 x 73
1790	S1138	819.5	819.3	187	16 x 73
1791	S1139	808.5	808.3	277	16 x 73
1792	S1140	797.5	797.3	367	16 x 73
1793	S1141	786.5	786.4	187	16 x 73
1794	S1142	775.5	775.4	277	16 x 73
1795	S1143	764.5	764.4	367	16 x 73
1796	S1144	753.5	753.4	187	16 x 73
1797	S1145	742.5	742.4	277	16 x 73
1798	S1146	731.5	731.4	367	16 x 73
1799	S1147	720.5	720.4	187	16 x 73
1800	S1148	709.5	709.4	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1801	S1149	698.5	698.4	367	16 x 73
1802	S1150	687.5	687.4	187	16 x 73
1803	S1151	676.5	676.4	277	16 x 73
1804	S1152	665.5	665.4	367	16 x 73
1805	S1153	654.5	654.4	187	16 x 73
1806	S1154	643.5	643.4	277	16 x 73
1807	S1155	632.5	632.4	367	16 x 73
1808	S1156	621.5	621.4	187	16 x 73
1809	S1157	610.5	610.4	277	16 x 73
1810	S1158	599.5	599.4	367	16 x 73
1811	S1159	588.5	588.4	187	16 x 73
1812	S1160	577.5	577.4	277	16 x 73
1813	S1161	566.5	566.4	367	16 x 73
1814	S1162	555.5	555.4	187	16 x 73
1815	S1163	544.5	544.4	277	16 x 73
1816	S1164	533.5	533.4	367	16 x 73
1817	S1165	522.5	522.4	187	16 x 73
1818	S1166	511.5	511.4	277	16 x 73
1819	S1167	500.5	500.4	367	16 x 73
1820	S1168	489.5	489.4	187	16 x 73
1821	S1169	478.5	478.4	277	16 x 73
1822	S1170	467.5	467.4	367	16 x 73
1823	S1171	456.5	456.5	187	16 x 73
1824	S1172	445.5	445.5	277	16 x 73
1825	S1173	434.5	434.5	367	16 x 73
1826	S1174	423.5	423.5	187	16 x 73
1827	S1175	412.5	412.5	277	16 x 73
1828	S1176	401.5	401.5	367	16 x 73
1829	S1177	390.5	390.5	187	16 x 73
1830	S1178	379.5	379.5	277	16 x 73
1831	S1179	368.5	368.5	367	16 x 73
1832	S1180	357.5	357.5	187	16 x 73
1833	S1181	346.5	346.5	277	16 x 73
1834	S1182	335.5	335.5	367	16 x 73
1835	S1183	324.5	324.5	187	16 x 73
1836	S1184	313.5	313.5	277	16 x 73
1837	S1185	302.5	302.5	367	16 x 73
1838	S1186	291.5	291.5	187	16 x 73
1839	S1187	280.5	280.5	277	16 x 73
1840	S1188	269.5	269.5	367	16 x 73
1841	S1189	258.5	258.5	187	16 x 73
1842	S1190	247.5	247.5	277	16 x 73
1843	S1191	236.5	236.5	367	16 x 73
1844	S1192	225.5	225.5	187	16 x 73
1845	S1193	214.5	214.5	277	16 x 73
1846	S1194	203.5	203.5	367	16 x 73
1847	S1195	192.5	192.5	187	16 x 73
1848	S1196	181.5	181.5	277	16 x 73
1849	S1197	170.5	170.5	367	16 x 73
1850	S1198	159.5	159.5	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1851	S1199	148.5	148.5	277	16 x 73
1852	S1200	137.5	137.5	367	16 x 73
1853	DUMMY	126.5	126.5	187	16 x 73
1854	DUMMY	115.5	115.5	277	16 x 73
1855	DUMMY	104.5	104.5	367	16 x 73
1856	DUMMY	93.5	93.5	187	16 x 73
1857	DUMMY	82.5	82.5	277	16 x 73
1858	DUMMY	71.5	71.5	367	16 x 73
1859	DUMMY	60.5	60.5	187	16 x 73
1860	DUMMY	49.5	49.5	277	16 x 73
1861	DUMMY	38.5	38.5	367	16 x 73
1862	DUMMY	27.5	27.5	187	16 x 73
1863	DUMMY	16.5	16.5	277	16 x 73
1864	DUMMY	5.5	5.5	367	16 x 73
1865	DUMMY	-5.5	-5.5	187	16 x 73
1866	DUMMY	-16.5	-16.5	277	16 x 73
1867	DUMMY	-27.5	-27.5	367	16 x 73
1868	DUMMY	-38.5	-38.5	187	16 x 73
1869	DUMMY	-49.5	-49.5	277	16 x 73
1870	DUMMY	-60.5	-60.5	367	16 x 73
1871	DUMMY	-71.5	-71.5	187	16 x 73
1872	DUMMY	-82.5	-82.5	277	16 x 73
1873	DUMMY	-93.5	-93.5	367	16 x 73
1874	DUMMY	-104.5	-104.5	187	16 x 73
1875	DUMMY	-115.5	-115.5	277	16 x 73
1876	DUMMY	-126.5	-126.5	367	16 x 73
1877	S1201	-137.5	-137.5	187	16 x 73
1878	S1202	-148.5	-148.5	277	16 x 73
1879	S1203	-159.5	-159.5	367	16 x 73
1880	S1204	-170.5	-170.5	187	16 x 73
1881	S1205	-181.5	-181.5	277	16 x 73
1882	S1206	-192.5	-192.5	367	16 x 73
1883	S1207	-203.5	-203.5	187	16 x 73
1884	S1208	-214.5	-214.5	277	16 x 73
1885	S1209	-225.5	-225.5	367	16 x 73
1886	S1210	-236.5	-236.5	187	16 x 73
1887	S1211	-247.5	-247.5	277	16 x 73
1888	S1212	-258.5	-258.5	367	16 x 73
1889	S1213	-269.5	-269.5	187	16 x 73
1890	S1214	-280.5	-280.5	277	16 x 73
1891	S1215	-291.5	-291.5	367	16 x 73
1892	S1216	-302.5	-302.5	187	16 x 73
1893	S1217	-313.5	-313.5	277	16 x 73
1894	S1218	-324.5	-324.5	367	16 x 73
1895	S1219	-335.5	-335.4	187	16 x 73
1896	S1220	-346.5	-346.4	277	16 x 73
1897	S1221	-357.5	-357.4	367	16 x 73
1898	S1222	-368.5	-368.4	187	16 x 73
1899	S1223	-379.5	-379.4	277	16 x 73
1900	S1224	-390.5	-390.4	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
1901	S1225	-401.5	-401.4	187	16 x 73
1902	S1226	-412.5	-412.4	277	16 x 73
1903	S1227	-423.5	-423.4	367	16 x 73
1904	S1228	-434.5	-434.4	187	16 x 73
1905	S1229	-445.5	-445.4	277	16 x 73
1906	S1230	-456.5	-456.4	367	16 x 73
1907	S1231	-467.5	-467.4	187	16 x 73
1908	S1232	-478.5	-478.4	277	16 x 73
1909	S1233	-489.5	-489.4	367	16 x 73
1910	S1234	-500.5	-500.4	187	16 x 73
1911	S1235	-511.5	-511.4	277	16 x 73
1912	S1236	-522.5	-522.4	367	16 x 73
1913	S1237	-533.5	-533.4	187	16 x 73
1914	S1238	-544.5	-544.4	277	16 x 73
1915	S1239	-555.5	-555.4	367	16 x 73
1916	S1240	-566.5	-566.4	187	16 x 73
1917	S1241	-577.5	-577.4	277	16 x 73
1918	S1242	-588.5	-588.4	367	16 x 73
1919	S1243	-599.5	-599.4	187	16 x 73
1920	S1244	-610.5	-610.4	277	16 x 73
1921	S1245	-621.5	-621.4	367	16 x 73
1922	S1246	-632.5	-632.4	187	16 x 73
1923	S1247	-643.5	-643.4	277	16 x 73
1924	S1248	-654.5	-654.4	367	16 x 73
1925	S1249	-665.5	-665.3	187	16 x 73
1926	S1250	-676.5	-676.3	277	16 x 73
1927	S1251	-687.5	-687.3	367	16 x 73
1928	S1252	-698.5	-698.3	187	16 x 73
1929	S1253	-709.5	-709.3	277	16 x 73
1930	S1254	-720.5	-720.3	367	16 x 73
1931	S1255	-731.5	-731.3	187	16 x 73
1932	S1256	-742.5	-742.3	277	16 x 73
1933	S1257	-753.5	-753.3	367	16 x 73
1934	S1258	-764.5	-764.3	187	16 x 73
1935	S1259	-775.5	-775.3	277	16 x 73
1936	S1260	-786.5	-786.3	367	16 x 73
1937	S1261	-797.5	-797.3	187	16 x 73
1938	S1262	-808.5	-808.3	277	16 x 73
1939	S1263	-819.5	-819.3	367	16 x 73
1940	S1264	-830.5	-830.3	187	16 x 73
1941	S1265	-841.5	-841.3	277	16 x 73
1942	S1266	-852.5	-852.3	367	16 x 73
1943	S1267	-863.5	-863.3	187	16 x 73
1944	S1268	-874.5	-874.3	277	16 x 73
1945	S1269	-885.5	-885.3	367	16 x 73
1946	S1270	-896.5	-896.3	187	16 x 73
1947	S1271	-907.5	-907.3	277	16 x 73
1948	S1272	-918.5	-918.3	367	16 x 73
1949	S1273	-929.5	-929.3	187	16 x 73
1950	S1274	-940.5	-940.3	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
1951	S1275	-951.5	-951.3	367	16 x 73
1952	S1276	-962.5	-962.3	187	16 x 73
1953	S1277	-973.5	-973.3	277	16 x 73
1954	S1278	-984.5	-984.3	367	16 x 73
1955	S1279	-995.5	-995.2	187	16 x 73
1956	S1280	-1006.5	-1006.2	277	16 x 73
1957	S1281	-1017.5	-1017.2	367	16 x 73
1958	S1282	-1028.5	-1028.2	187	16 x 73
1959	S1283	-1039.5	-1039.2	277	16 x 73
1960	S1284	-1050.5	-1050.2	367	16 x 73
1961	S1285	-1061.5	-1061.2	187	16 x 73
1962	S1286	-1072.5	-1072.2	277	16 x 73
1963	S1287	-1083.5	-1083.2	367	16 x 73
1964	S1288	-1094.5	-1094.2	187	16 x 73
1965	S1289	-1105.5	-1105.2	277	16 x 73
1966	S1290	-1116.5	-1116.2	367	16 x 73
1967	S1291	-1127.5	-1127.2	187	16 x 73
1968	S1292	-1138.5	-1138.2	277	16 x 73
1969	S1293	-1149.5	-1149.2	367	16 x 73
1970	S1294	-1160.5	-1160.2	187	16 x 73
1971	S1295	-1171.5	-1171.2	277	16 x 73
1972	S1296	-1182.5	-1182.2	367	16 x 73
1973	S1297	-1193.5	-1193.2	187	16 x 73
1974	S1298	-1204.5	-1204.2	277	16 x 73
1975	S1299	-1215.5	-1215.2	367	16 x 73
1976	S1300	-1226.5	-1226.2	187	16 x 73
1977	S1301	-1237.5	-1237.2	277	16 x 73
1978	S1302	-1248.5	-1248.2	367	16 x 73
1979	S1303	-1259.5	-1259.2	187	16 x 73
1980	S1304	-1270.5	-1270.2	277	16 x 73
1981	S1305	-1281.5	-1281.2	367	16 x 73
1982	S1306	-1292.5	-1292.2	187	16 x 73
1983	S1307	-1303.5	-1303.2	277	16 x 73
1984	S1308	-1314.5	-1314.2	367	16 x 73
1985	S1309	-1325.5	-1325.1	187	16 x 73
1986	S1310	-1336.5	-1336.1	277	16 x 73
1987	S1311	-1347.5	-1347.1	367	16 x 73
1988	S1312	-1358.5	-1358.1	187	16 x 73
1989	S1313	-1369.5	-1369.1	277	16 x 73
1990	S1314	-1380.5	-1380.1	367	16 x 73
1991	S1315	-1391.5	-1391.1	187	16 x 73
1992	S1316	-1402.5	-1402.1	277	16 x 73
1993	S1317	-1413.5	-1413.1	367	16 x 73
1994	S1318	-1424.5	-1424.1	187	16 x 73
1995	S1319	-1435.5	-1435.1	277	16 x 73
1996	S1320	-1446.5	-1446.1	367	16 x 73
1997	S1321	-1457.5	-1457.1	187	16 x 73
1998	S1322	-1468.5	-1468.1	277	16 x 73
1999	S1323	-1479.5	-1479.1	367	16 x 73
2000	S1324	-1490.5	-1490.1	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2001	S1325	-1501.5	-1501.1	277	16 x 73
2002	S1326	-1512.5	-1512.1	367	16 x 73
2003	S1327	-1523.5	-1523.1	187	16 x 73
2004	S1328	-1534.5	-1534.1	277	16 x 73
2005	S1329	-1545.5	-1545.1	367	16 x 73
2006	S1330	-1556.5	-1556.1	187	16 x 73
2007	S1331	-1567.5	-1567.1	277	16 x 73
2008	S1332	-1578.5	-1578.1	367	16 x 73
2009	S1333	-1589.5	-1589.1	187	16 x 73
2010	S1334	-1600.5	-1600.1	277	16 x 73
2011	S1335	-1611.5	-1611.1	367	16 x 73
2012	S1336	-1622.5	-1622.1	187	16 x 73
2013	S1337	-1633.5	-1633.1	277	16 x 73
2014	S1338	-1644.5	-1644.1	367	16 x 73
2015	S1339	-1655.5	-1655	187	16 x 73
2016	S1340	-1666.5	-1666	277	16 x 73
2017	S1341	-1677.5	-1677	367	16 x 73
2018	S1342	-1688.5	-1688	187	16 x 73
2019	S1343	-1699.5	-1699	277	16 x 73
2020	S1344	-1710.5	-1710	367	16 x 73
2021	S1345	-1721.5	-1721	187	16 x 73
2022	S1346	-1732.5	-1732	277	16 x 73
2023	S1347	-1743.5	-1743	367	16 x 73
2024	S1348	-1754.5	-1754	187	16 x 73
2025	S1349	-1765.5	-1765	277	16 x 73
2026	S1350	-1776.5	-1776	367	16 x 73
2027	S1351	-1787.5	-1787	187	16 x 73
2028	S1352	-1798.5	-1798	277	16 x 73
2029	S1353	-1809.5	-1809	367	16 x 73
2030	S1354	-1820.5	-1820	187	16 x 73
2031	S1355	-1831.5	-1831	277	16 x 73
2032	S1356	-1842.5	-1842	367	16 x 73
2033	S1357	-1853.5	-1853	187	16 x 73
2034	S1358	-1864.5	-1864	277	16 x 73
2035	S1359	-1875.5	-1875	367	16 x 73
2036	S1360	-1886.5	-1886	187	16 x 73
2037	S1361	-1897.5	-1897	277	16 x 73
2038	S1362	-1908.5	-1908	367	16 x 73
2039	S1363	-1919.5	-1919	187	16 x 73
2040	S1364	-1930.5	-1930	277	16 x 73
2041	S1365	-1941.5	-1941	367	16 x 73
2042	S1366	-1952.5	-1952	187	16 x 73
2043	S1367	-1963.5	-1963	277	16 x 73
2044	S1368	-1974.5	-1974	367	16 x 73
2045	S1369	-1985.5	-1984.9	187	16 x 73
2046	S1370	-1996.5	-1995.9	277	16 x 73
2047	S1371	-2007.5	-2006.9	367	16 x 73
2048	S1372	-2018.5	-2017.9	187	16 x 73
2049	S1373	-2029.5	-2028.9	277	16 x 73
2050	S1374	-2040.5	-2039.9	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2051	S1375	-2051.5	-2050.9	187	16 x 73
2052	S1376	-2062.5	-2061.9	277	16 x 73
2053	S1377	-2073.5	-2072.9	367	16 x 73
2054	S1378	-2084.5	-2083.9	187	16 x 73
2055	S1379	-2095.5	-2094.9	277	16 x 73
2056	S1380	-2106.5	-2105.9	367	16 x 73
2057	S1381	-2117.5	-2116.9	187	16 x 73
2058	S1382	-2128.5	-2127.9	277	16 x 73
2059	S1383	-2139.5	-2138.9	367	16 x 73
2060	S1384	-2150.5	-2149.9	187	16 x 73
2061	S1385	-2161.5	-2160.9	277	16 x 73
2062	S1386	-2172.5	-2171.9	367	16 x 73
2063	S1387	-2183.5	-2182.9	187	16 x 73
2064	S1388	-2194.5	-2193.9	277	16 x 73
2065	S1389	-2205.5	-2204.9	367	16 x 73
2066	S1390	-2216.5	-2215.9	187	16 x 73
2067	S1391	-2227.5	-2226.9	277	16 x 73
2068	S1392	-2238.5	-2237.9	367	16 x 73
2069	S1393	-2249.5	-2248.9	187	16 x 73
2070	S1394	-2260.5	-2259.9	277	16 x 73
2071	S1395	-2271.5	-2270.9	367	16 x 73
2072	S1396	-2282.5	-2281.9	187	16 x 73
2073	S1397	-2293.5	-2292.9	277	16 x 73
2074	S1398	-2304.5	-2303.9	367	16 x 73
2075	S1399	-2315.5	-2314.8	187	16 x 73
2076	S1400	-2326.5	-2325.8	277	16 x 73
2077	S1401	-2337.5	-2336.8	367	16 x 73
2078	S1402	-2348.5	-2347.8	187	16 x 73
2079	S1403	-2359.5	-2358.8	277	16 x 73
2080	S1404	-2370.5	-2369.8	367	16 x 73
2081	S1405	-2381.5	-2380.8	187	16 x 73
2082	S1406	-2392.5	-2391.8	277	16 x 73
2083	S1407	-2403.5	-2402.8	367	16 x 73
2084	S1408	-2414.5	-2413.8	187	16 x 73
2085	S1409	-2425.5	-2424.8	277	16 x 73
2086	S1410	-2436.5	-2435.8	367	16 x 73
2087	S1411	-2447.5	-2446.8	187	16 x 73
2088	S1412	-2458.5	-2457.8	277	16 x 73
2089	S1413	-2469.5	-2468.8	367	16 x 73
2090	S1414	-2480.5	-2479.8	187	16 x 73
2091	S1415	-2491.5	-2490.8	277	16 x 73
2092	S1416	-2502.5	-2501.8	367	16 x 73
2093	S1417	-2513.5	-2512.8	187	16 x 73
2094	S1418	-2524.5	-2523.8	277	16 x 73
2095	S1419	-2535.5	-2534.8	367	16 x 73
2096	S1420	-2546.5	-2545.8	187	16 x 73
2097	S1421	-2557.5	-2556.8	277	16 x 73
2098	S1422	-2568.5	-2567.8	367	16 x 73
2099	S1423	-2579.5	-2578.8	187	16 x 73
2100	S1424	-2590.5	-2589.8	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2101	S1425	-2601.5	-2600.8	367	16 x 73
2102	S1426	-2612.5	-2611.8	187	16 x 73
2103	S1427	-2623.5	-2622.8	277	16 x 73
2104	S1428	-2634.5	-2633.8	367	16 x 73
2105	S1429	-2645.5	-2644.7	187	16 x 73
2106	S1430	-2656.5	-2655.7	277	16 x 73
2107	S1431	-2667.5	-2666.7	367	16 x 73
2108	S1432	-2678.5	-2677.7	187	16 x 73
2109	S1433	-2689.5	-2688.7	277	16 x 73
2110	S1434	-2700.5	-2699.7	367	16 x 73
2111	S1435	-2711.5	-2710.7	187	16 x 73
2112	S1436	-2722.5	-2721.7	277	16 x 73
2113	S1437	-2733.5	-2732.7	367	16 x 73
2114	S1438	-2744.5	-2743.7	187	16 x 73
2115	S1439	-2755.5	-2754.7	277	16 x 73
2116	S1440	-2766.5	-2765.7	367	16 x 73
2117	S1441	-2777.5	-2776.7	187	16 x 73
2118	S1442	-2788.5	-2787.7	277	16 x 73
2119	S1443	-2799.5	-2798.7	367	16 x 73
2120	S1444	-2810.5	-2809.7	187	16 x 73
2121	S1445	-2821.5	-2820.7	277	16 x 73
2122	S1446	-2832.5	-2831.7	367	16 x 73
2123	S1447	-2843.5	-2842.7	187	16 x 73
2124	S1448	-2854.5	-2853.7	277	16 x 73
2125	S1449	-2865.5	-2864.7	367	16 x 73
2126	S1450	-2876.5	-2875.7	187	16 x 73
2127	S1451	-2887.5	-2886.7	277	16 x 73
2128	S1452	-2898.5	-2897.7	367	16 x 73
2129	S1453	-2909.5	-2908.7	187	16 x 73
2130	S1454	-2920.5	-2919.7	277	16 x 73
2131	S1455	-2931.5	-2930.7	367	16 x 73
2132	S1456	-2942.5	-2941.7	187	16 x 73
2133	S1457	-2953.5	-2952.7	277	16 x 73
2134	S1458	-2964.5	-2963.7	367	16 x 73
2135	S1459	-2975.5	-2974.6	187	16 x 73
2136	S1460	-2986.5	-2985.6	277	16 x 73
2137	S1461	-2997.5	-2996.6	367	16 x 73
2138	S1462	-3008.5	-3007.6	187	16 x 73
2139	S1463	-3019.5	-3018.6	277	16 x 73
2140	S1464	-3030.5	-3029.6	367	16 x 73
2141	S1465	-3041.5	-3040.6	187	16 x 73
2142	S1466	-3052.5	-3051.6	277	16 x 73
2143	S1467	-3063.5	-3062.6	367	16 x 73
2144	S1468	-3074.5	-3073.6	187	16 x 73
2145	S1469	-3085.5	-3084.6	277	16 x 73
2146	S1470	-3096.5	-3095.6	367	16 x 73
2147	S1471	-3107.5	-3106.6	187	16 x 73
2148	S1472	-3118.5	-3117.6	277	16 x 73
2149	S1473	-3129.5	-3128.6	367	16 x 73
2150	S1474	-3140.5	-3139.6	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2151	S1475	-3151.5	-3150.6	277	16 x 73
2152	S1476	-3162.5	-3161.6	367	16 x 73
2153	S1477	-3173.5	-3172.6	187	16 x 73
2154	S1478	-3184.5	-3183.6	277	16 x 73
2155	S1479	-3195.5	-3194.6	367	16 x 73
2156	S1480	-3206.5	-3205.6	187	16 x 73
2157	S1481	-3217.5	-3216.6	277	16 x 73
2158	S1482	-3228.5	-3227.6	367	16 x 73
2159	S1483	-3239.5	-3238.6	187	16 x 73
2160	S1484	-3250.5	-3249.6	277	16 x 73
2161	S1485	-3261.5	-3260.6	367	16 x 73
2162	S1486	-3272.5	-3271.6	187	16 x 73
2163	S1487	-3283.5	-3282.6	277	16 x 73
2164	S1488	-3294.5	-3293.6	367	16 x 73
2165	S1489	-3305.5	-3304.5	187	16 x 73
2166	S1490	-3316.5	-3315.5	277	16 x 73
2167	S1491	-3327.5	-3326.5	367	16 x 73
2168	S1492	-3338.5	-3337.5	187	16 x 73
2169	S1493	-3349.5	-3348.5	277	16 x 73
2170	S1494	-3360.5	-3359.5	367	16 x 73
2171	S1495	-3371.5	-3370.5	187	16 x 73
2172	S1496	-3382.5	-3381.5	277	16 x 73
2173	S1497	-3393.5	-3392.5	367	16 x 73
2174	S1498	-3404.5	-3403.5	187	16 x 73
2175	S1499	-3415.5	-3414.5	277	16 x 73
2176	S1500	-3426.5	-3425.5	367	16 x 73
2177	S1501	-3437.5	-3436.5	187	16 x 73
2178	S1502	-3448.5	-3447.5	277	16 x 73
2179	S1503	-3459.5	-3458.5	367	16 x 73
2180	S1504	-3470.5	-3469.5	187	16 x 73
2181	S1505	-3481.5	-3480.5	277	16 x 73
2182	S1506	-3492.5	-3491.5	367	16 x 73
2183	S1507	-3503.5	-3502.5	187	16 x 73
2184	S1508	-3514.5	-3513.5	277	16 x 73
2185	S1509	-3525.5	-3524.5	367	16 x 73
2186	S1510	-3536.5	-3535.5	187	16 x 73
2187	S1511	-3547.5	-3546.5	277	16 x 73
2188	S1512	-3558.5	-3557.5	367	16 x 73
2189	S1513	-3569.5	-3568.5	187	16 x 73
2190	S1514	-3580.5	-3579.5	277	16 x 73
2191	S1515	-3591.5	-3590.5	367	16 x 73
2192	S1516	-3602.5	-3601.5	187	16 x 73
2193	S1517	-3613.5	-3612.5	277	16 x 73
2194	S1518	-3624.5	-3623.5	367	16 x 73
2195	S1519	-3635.5	-3634.4	187	16 x 73
2196	S1520	-3646.5	-3645.4	277	16 x 73
2197	S1521	-3657.5	-3656.4	367	16 x 73
2198	S1522	-3668.5	-3667.4	187	16 x 73
2199	S1523	-3679.5	-3678.4	277	16 x 73
2200	S1524	-3690.5	-3689.4	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2201	S1525	-3701.5	-3700.4	187	16 x 73
2202	S1526	-3712.5	-3711.4	277	16 x 73
2203	S1527	-3723.5	-3722.4	367	16 x 73
2204	S1528	-3734.5	-3733.4	187	16 x 73
2205	S1529	-3745.5	-3744.4	277	16 x 73
2206	S1530	-3756.5	-3755.4	367	16 x 73
2207	S1531	-3767.5	-3766.4	187	16 x 73
2208	S1532	-3778.5	-3777.4	277	16 x 73
2209	S1533	-3789.5	-3788.4	367	16 x 73
2210	S1534	-3800.5	-3799.4	187	16 x 73
2211	S1535	-3811.5	-3810.4	277	16 x 73
2212	S1536	-3822.5	-3821.4	367	16 x 73
2213	S1537	-3833.5	-3832.4	187	16 x 73
2214	S1538	-3844.5	-3843.4	277	16 x 73
2215	S1539	-3855.5	-3854.4	367	16 x 73
2216	S1540	-3866.5	-3865.4	187	16 x 73
2217	S1541	-3877.5	-3876.4	277	16 x 73
2218	S1542	-3888.5	-3887.4	367	16 x 73
2219	S1543	-3899.5	-3898.4	187	16 x 73
2220	S1544	-3910.5	-3909.4	277	16 x 73
2221	S1545	-3921.5	-3920.4	367	16 x 73
2222	S1546	-3932.5	-3931.4	187	16 x 73
2223	S1547	-3943.5	-3942.4	277	16 x 73
2224	S1548	-3954.5	-3953.4	367	16 x 73
2225	S1549	-3965.5	-3964.3	187	16 x 73
2226	S1550	-3976.5	-3975.3	277	16 x 73
2227	S1551	-3987.5	-3986.3	367	16 x 73
2228	S1552	-3998.5	-3997.3	187	16 x 73
2229	S1553	-4009.5	-4008.3	277	16 x 73
2230	S1554	-4020.5	-4019.3	367	16 x 73
2231	S1555	-4031.5	-4030.3	187	16 x 73
2232	S1556	-4042.5	-4041.3	277	16 x 73
2233	S1557	-4053.5	-4052.3	367	16 x 73
2234	S1558	-4064.5	-4063.3	187	16 x 73
2235	S1559	-4075.5	-4074.3	277	16 x 73
2236	S1560	-4086.5	-4085.3	367	16 x 73
2237	S1561	-4097.5	-4096.3	187	16 x 73
2238	S1562	-4108.5	-4107.3	277	16 x 73
2239	S1563	-4119.5	-4118.3	367	16 x 73
2240	S1564	-4130.5	-4129.3	187	16 x 73
2241	S1565	-4141.5	-4140.3	277	16 x 73
2242	S1566	-4152.5	-4151.3	367	16 x 73
2243	S1567	-4163.5	-4162.3	187	16 x 73
2244	S1568	-4174.5	-4173.3	277	16 x 73
2245	S1569	-4185.5	-4184.3	367	16 x 73
2246	S1570	-4196.5	-4195.3	187	16 x 73
2247	S1571	-4207.5	-4206.3	277	16 x 73
2248	S1572	-4218.5	-4217.3	367	16 x 73
2249	S1573	-4229.5	-4228.3	187	16 x 73
2250	S1574	-4240.5	-4239.3	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2251	S1575	-4251.5	-4250.3	367	16 x 73
2252	S1576	-4262.5	-4261.3	187	16 x 73
2253	S1577	-4273.5	-4272.3	277	16 x 73
2254	S1578	-4284.5	-4283.3	367	16 x 73
2255	S1579	-4295.5	-4294.2	187	16 x 73
2256	S1580	-4306.5	-4305.2	277	16 x 73
2257	S1581	-4317.5	-4316.2	367	16 x 73
2258	S1582	-4328.5	-4327.2	187	16 x 73
2259	S1583	-4339.5	-4338.2	277	16 x 73
2260	S1584	-4350.5	-4349.2	367	16 x 73
2261	S1585	-4361.5	-4360.2	187	16 x 73
2262	S1586	-4372.5	-4371.2	277	16 x 73
2263	S1587	-4383.5	-4382.2	367	16 x 73
2264	S1588	-4394.5	-4393.2	187	16 x 73
2265	S1589	-4405.5	-4404.2	277	16 x 73
2266	S1590	-4416.5	-4415.2	367	16 x 73
2267	S1591	-4427.5	-4426.2	187	16 x 73
2268	S1592	-4438.5	-4437.2	277	16 x 73
2269	S1593	-4449.5	-4448.2	367	16 x 73
2270	S1594	-4460.5	-4459.2	187	16 x 73
2271	S1595	-4471.5	-4470.2	277	16 x 73
2272	S1596	-4482.5	-4481.2	367	16 x 73
2273	S1597	-4493.5	-4492.2	187	16 x 73
2274	S1598	-4504.5	-4503.2	277	16 x 73
2275	S1599	-4515.5	-4514.2	367	16 x 73
2276	S1600	-4526.5	-4525.2	187	16 x 73
2277	S1601	-4537.5	-4536.2	277	16 x 73
2278	S1602	-4548.5	-4547.2	367	16 x 73
2279	S1603	-4559.5	-4558.2	187	16 x 73
2280	S1604	-4570.5	-4569.2	277	16 x 73
2281	S1605	-4581.5	-4580.2	367	16 x 73
2282	S1606	-4592.5	-4591.2	187	16 x 73
2283	S1607	-4603.5	-4602.2	277	16 x 73
2284	S1608	-4614.5	-4613.2	367	16 x 73
2285	S1609	-4625.5	-4624.1	187	16 x 73
2286	S1610	-4636.5	-4635.1	277	16 x 73
2287	S1611	-4647.5	-4646.1	367	16 x 73
2288	S1612	-4658.5	-4657.1	187	16 x 73
2289	S1613	-4669.5	-4668.1	277	16 x 73
2290	S1614	-4680.5	-4679.1	367	16 x 73
2291	S1615	-4691.5	-4690.1	187	16 x 73
2292	S1616	-4702.5	-4701.1	277	16 x 73
2293	S1617	-4713.5	-4712.1	367	16 x 73
2294	S1618	-4724.5	-4723.1	187	16 x 73
2295	S1619	-4735.5	-4734.1	277	16 x 73
2296	S1620	-4746.5	-4745.1	367	16 x 73
2297	S1621	-4757.5	-4756.1	187	16 x 73
2298	S1622	-4768.5	-4767.1	277	16 x 73
2299	S1623	-4779.5	-4778.1	367	16 x 73
2300	S1624	-4790.5	-4789.1	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2301	S1625	-4801.5	-4800.1	277	16 x 73
2302	S1626	-4812.5	-4811.1	367	16 x 73
2303	S1627	-4823.5	-4822.1	187	16 x 73
2304	S1628	-4834.5	-4833.1	277	16 x 73
2305	S1629	-4845.5	-4844.1	367	16 x 73
2306	S1630	-4856.5	-4855.1	187	16 x 73
2307	S1631	-4867.5	-4866.1	277	16 x 73
2308	S1632	-4878.5	-4877.1	367	16 x 73
2309	S1633	-4889.5	-4888.1	187	16 x 73
2310	S1634	-4900.5	-4899.1	277	16 x 73
2311	S1635	-4911.5	-4910.1	367	16 x 73
2312	S1636	-4922.5	-4921.1	187	16 x 73
2313	S1637	-4933.5	-4932.1	277	16 x 73
2314	S1638	-4944.5	-4943.1	367	16 x 73
2315	S1639	-4955.5	-4954	187	16 x 73
2316	S1640	-4966.5	-4965	277	16 x 73
2317	S1641	-4977.5	-4976	367	16 x 73
2318	S1642	-4988.5	-4987	187	16 x 73
2319	S1643	-4999.5	-4998	277	16 x 73
2320	S1644	-5010.5	-5009	367	16 x 73
2321	S1645	-5021.5	-5020	187	16 x 73
2322	S1646	-5032.5	-5031	277	16 x 73
2323	S1647	-5043.5	-5042	367	16 x 73
2324	S1648	-5054.5	-5053	187	16 x 73
2325	S1649	-5065.5	-5064	277	16 x 73
2326	S1650	-5076.5	-5075	367	16 x 73
2327	S1651	-5087.5	-5086	187	16 x 73
2328	S1652	-5098.5	-5097	277	16 x 73
2329	S1653	-5109.5	-5108	367	16 x 73
2330	S1654	-5120.5	-5119	187	16 x 73
2331	S1655	-5131.5	-5130	277	16 x 73
2332	S1656	-5142.5	-5141	367	16 x 73
2333	S1657	-5153.5	-5152	187	16 x 73
2334	S1658	-5164.5	-5163	277	16 x 73
2335	S1659	-5175.5	-5174	367	16 x 73
2336	S1660	-5186.5	-5185	187	16 x 73
2337	S1661	-5197.5	-5196	277	16 x 73
2338	S1662	-5208.5	-5207	367	16 x 73
2339	S1663	-5219.5	-5218	187	16 x 73
2340	S1664	-5230.5	-5229	277	16 x 73
2341	S1665	-5241.5	-5240	367	16 x 73
2342	S1666	-5252.5	-5251	187	16 x 73
2343	S1667	-5263.5	-5262	277	16 x 73
2344	S1668	-5274.5	-5273	367	16 x 73
2345	S1669	-5285.5	-5283.9	187	16 x 73
2346	S1670	-5296.5	-5294.9	277	16 x 73
2347	S1671	-5307.5	-5305.9	367	16 x 73
2348	S1672	-5318.5	-5316.9	187	16 x 73
2349	S1673	-5329.5	-5327.9	277	16 x 73
2350	S1674	-5340.5	-5338.9	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2351	S1675	-5351.5	-5349.9	187	16 x 73
2352	S1676	-5362.5	-5360.9	277	16 x 73
2353	S1677	-5373.5	-5371.9	367	16 x 73
2354	S1678	-5384.5	-5382.9	187	16 x 73
2355	S1679	-5395.5	-5393.9	277	16 x 73
2356	S1680	-5406.5	-5404.9	367	16 x 73
2357	S1681	-5417.5	-5415.9	187	16 x 73
2358	S1682	-5428.5	-5426.9	277	16 x 73
2359	S1683	-5439.5	-5437.9	367	16 x 73
2360	S1684	-5450.5	-5448.9	187	16 x 73
2361	S1685	-5461.5	-5459.9	277	16 x 73
2362	S1686	-5472.5	-5470.9	367	16 x 73
2363	S1687	-5483.5	-5481.9	187	16 x 73
2364	S1688	-5494.5	-5492.9	277	16 x 73
2365	S1689	-5505.5	-5503.9	367	16 x 73
2366	S1690	-5516.5	-5514.9	187	16 x 73
2367	S1691	-5527.5	-5525.9	277	16 x 73
2368	S1692	-5538.5	-5536.9	367	16 x 73
2369	S1693	-5549.5	-5547.9	187	16 x 73
2370	S1694	-5560.5	-5558.9	277	16 x 73
2371	S1695	-5571.5	-5569.9	367	16 x 73
2372	S1696	-5582.5	-5580.9	187	16 x 73
2373	S1697	-5593.5	-5591.9	277	16 x 73
2374	S1698	-5604.5	-5602.9	367	16 x 73
2375	S1699	-5615.5	-5613.8	187	16 x 73
2376	S1700	-5626.5	-5624.8	277	16 x 73
2377	S1701	-5637.5	-5635.8	367	16 x 73
2378	S1702	-5648.5	-5646.8	187	16 x 73
2379	S1703	-5659.5	-5657.8	277	16 x 73
2380	S1704	-5670.5	-5668.8	367	16 x 73
2381	S1705	-5681.5	-5679.8	187	16 x 73
2382	S1706	-5692.5	-5690.8	277	16 x 73
2383	S1707	-5703.5	-5701.8	367	16 x 73
2384	S1708	-5714.5	-5712.8	187	16 x 73
2385	S1709	-5725.5	-5723.8	277	16 x 73
2386	S1710	-5736.5	-5734.8	367	16 x 73
2387	S1711	-5747.5	-5745.8	187	16 x 73
2388	S1712	-5758.5	-5756.8	277	16 x 73
2389	S1713	-5769.5	-5767.8	367	16 x 73
2390	S1714	-5780.5	-5778.8	187	16 x 73
2391	S1715	-5791.5	-5789.8	277	16 x 73
2392	S1716	-5802.5	-5800.8	367	16 x 73
2393	S1717	-5813.5	-5811.8	187	16 x 73
2394	S1718	-5824.5	-5822.8	277	16 x 73
2395	S1719	-5835.5	-5833.8	367	16 x 73
2396	S1720	-5846.5	-5844.8	187	16 x 73
2397	S1721	-5857.5	-5855.8	277	16 x 73
2398	S1722	-5868.5	-5866.8	367	16 x 73
2399	S1723	-5879.5	-5877.8	187	16 x 73
2400	S1724	-5890.5	-5888.8	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2401	S1725	-5901.5	-5899.8	367	16 x 73
2402	S1726	-5912.5	-5910.8	187	16 x 73
2403	S1727	-5923.5	-5921.8	277	16 x 73
2404	S1728	-5934.5	-5932.8	367	16 x 73
2405	S1729	-5945.5	-5943.7	187	16 x 73
2406	S1730	-5956.5	-5954.7	277	16 x 73
2407	S1731	-5967.5	-5965.7	367	16 x 73
2408	S1732	-5978.5	-5976.7	187	16 x 73
2409	S1733	-5989.5	-5987.7	277	16 x 73
2410	S1734	-6000.5	-5998.7	367	16 x 73
2411	S1735	-6011.5	-6009.7	187	16 x 73
2412	S1736	-6022.5	-6020.7	277	16 x 73
2413	S1737	-6033.5	-6031.7	367	16 x 73
2414	S1738	-6044.5	-6042.7	187	16 x 73
2415	S1739	-6055.5	-6053.7	277	16 x 73
2416	S1740	-6066.5	-6064.7	367	16 x 73
2417	S1741	-6077.5	-6075.7	187	16 x 73
2418	S1742	-6088.5	-6086.7	277	16 x 73
2419	S1743	-6099.5	-6097.7	367	16 x 73
2420	S1744	-6110.5	-6108.7	187	16 x 73
2421	S1745	-6121.5	-6119.7	277	16 x 73
2422	S1746	-6132.5	-6130.7	367	16 x 73
2423	S1747	-6143.5	-6141.7	187	16 x 73
2424	S1748	-6154.5	-6152.7	277	16 x 73
2425	S1749	-6165.5	-6163.7	367	16 x 73
2426	S1750	-6176.5	-6174.7	187	16 x 73
2427	S1751	-6187.5	-6185.7	277	16 x 73
2428	S1752	-6198.5	-6196.7	367	16 x 73
2429	S1753	-6209.5	-6207.7	187	16 x 73
2430	S1754	-6220.5	-6218.7	277	16 x 73
2431	S1755	-6231.5	-6229.7	367	16 x 73
2432	S1756	-6242.5	-6240.7	187	16 x 73
2433	S1757	-6253.5	-6251.7	277	16 x 73
2434	S1758	-6264.5	-6262.7	367	16 x 73
2435	S1759	-6275.5	-6273.6	187	16 x 73
2436	S1760	-6286.5	-6284.6	277	16 x 73
2437	S1761	-6297.5	-6295.6	367	16 x 73
2438	S1762	-6308.5	-6306.6	187	16 x 73
2439	S1763	-6319.5	-6317.6	277	16 x 73
2440	S1764	-6330.5	-6328.6	367	16 x 73
2441	S1765	-6341.5	-6339.6	187	16 x 73
2442	S1766	-6352.5	-6350.6	277	16 x 73
2443	S1767	-6363.5	-6361.6	367	16 x 73
2444	S1768	-6374.5	-6372.6	187	16 x 73
2445	S1769	-6385.5	-6383.6	277	16 x 73
2446	S1770	-6396.5	-6394.6	367	16 x 73
2447	S1771	-6407.5	-6405.6	187	16 x 73
2448	S1772	-6418.5	-6416.6	277	16 x 73
2449	S1773	-6429.5	-6427.6	367	16 x 73
2450	S1774	-6440.5	-6438.6	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2451	S1775	-6451.5	-6449.6	277	16 x 73
2452	S1776	-6462.5	-6460.6	367	16 x 73
2453	S1777	-6473.5	-6471.6	187	16 x 73
2454	S1778	-6484.5	-6482.6	277	16 x 73
2455	S1779	-6495.5	-6493.6	367	16 x 73
2456	S1780	-6506.5	-6504.6	187	16 x 73
2457	S1781	-6517.5	-6515.6	277	16 x 73
2458	S1782	-6528.5	-6526.6	367	16 x 73
2459	S1783	-6539.5	-6537.6	187	16 x 73
2460	S1784	-6550.5	-6548.6	277	16 x 73
2461	S1785	-6561.5	-6559.6	367	16 x 73
2462	S1786	-6572.5	-6570.6	187	16 x 73
2463	S1787	-6583.5	-6581.6	277	16 x 73
2464	S1788	-6594.5	-6592.6	367	16 x 73
2465	S1789	-6605.5	-6603.5	187	16 x 73
2466	S1790	-6616.5	-6614.5	277	16 x 73
2467	S1791	-6627.5	-6625.5	367	16 x 73
2468	S1792	-6638.5	-6636.5	187	16 x 73
2469	S1793	-6649.5	-6647.5	277	16 x 73
2470	S1794	-6660.5	-6658.5	367	16 x 73
2471	S1795	-6671.5	-6669.5	187	16 x 73
2472	S1796	-6682.5	-6680.5	277	16 x 73
2473	S1797	-6693.5	-6691.5	367	16 x 73
2474	S1798	-6704.5	-6702.5	187	16 x 73
2475	S1799	-6715.5	-6713.5	277	16 x 73
2476	S1800	-6726.5	-6724.5	367	16 x 73
2477	DUMMY	-6737.5	-6735.5	187	16 x 73
2478	DUMMY	-6748.5	-6746.5	277	16 x 73
2479	DUMMY	-6759.5	-6757.5	367	16 x 73
2480	DUMMY	-6770.5	-6768.5	187	16 x 73
2481	DUMMY	-6781.5	-6779.5	277	16 x 73
2482	DUMMY	-6792.5	-6790.5	367	16 x 73
2483	DUMMY	-6803.5	-6801.5	187	16 x 73
2484	DUMMY	-6814.5	-6812.5	277	16 x 73
2485	DUMMY	-6825.5	-6823.5	367	16 x 73
2486	DUMMY	-6836.5	-6834.5	187	16 x 73
2487	DUMMY	-6847.5	-6845.5	277	16 x 73
2488	DUMMY	-6858.5	-6856.5	367	16 x 73
2489	DUMMY	-6869.5	-6867.5	187	16 x 73
2490	DUMMY	-6880.5	-6878.5	277	16 x 73
2491	DUMMY	-6891.5	-6889.5	367	16 x 73
2492	DUMMY	-6902.5	-6900.5	187	16 x 73
2493	DUMMY	-6913.5	-6911.5	277	16 x 73
2494	DUMMY	-6924.5	-6922.5	367	16 x 73
2495	S1801	-6935.5	-6933.4	187	16 x 73
2496	S1802	-6946.5	-6944.4	277	16 x 73
2497	S1803	-6957.5	-6955.4	367	16 x 73
2498	S1804	-6968.5	-6966.4	187	16 x 73
2499	S1805	-6979.5	-6977.4	277	16 x 73
2500	S1806	-6990.5	-6988.4	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2501	S1807	-7001.5	-6999.4	187	16 x 73
2502	S1808	-7012.5	-7010.4	277	16 x 73
2503	S1809	-7023.5	-7021.4	367	16 x 73
2504	S1810	-7034.5	-7032.4	187	16 x 73
2505	S1811	-7045.5	-7043.4	277	16 x 73
2506	S1812	-7056.5	-7054.4	367	16 x 73
2507	S1813	-7067.5	-7065.4	187	16 x 73
2508	S1814	-7078.5	-7076.4	277	16 x 73
2509	S1815	-7089.5	-7087.4	367	16 x 73
2510	S1816	-7100.5	-7098.4	187	16 x 73
2511	S1817	-7111.5	-7109.4	277	16 x 73
2512	S1818	-7122.5	-7120.4	367	16 x 73
2513	S1819	-7133.5	-7131.4	187	16 x 73
2514	S1820	-7144.5	-7142.4	277	16 x 73
2515	S1821	-7155.5	-7153.4	367	16 x 73
2516	S1822	-7166.5	-7164.4	187	16 x 73
2517	S1823	-7177.5	-7175.4	277	16 x 73
2518	S1824	-7188.5	-7186.4	367	16 x 73
2519	S1825	-7199.5	-7197.4	187	16 x 73
2520	S1826	-7210.5	-7208.4	277	16 x 73
2521	S1827	-7221.5	-7219.4	367	16 x 73
2522	S1828	-7232.5	-7230.4	187	16 x 73
2523	S1829	-7243.5	-7241.4	277	16 x 73
2524	S1830	-7254.5	-7252.4	367	16 x 73
2525	S1831	-7265.5	-7263.3	187	16 x 73
2526	S1832	-7276.5	-7274.3	277	16 x 73
2527	S1833	-7287.5	-7285.3	367	16 x 73
2528	S1834	-7298.5	-7296.3	187	16 x 73
2529	S1835	-7309.5	-7307.3	277	16 x 73
2530	S1836	-7320.5	-7318.3	367	16 x 73
2531	S1837	-7331.5	-7329.3	187	16 x 73
2532	S1838	-7342.5	-7340.3	277	16 x 73
2533	S1839	-7353.5	-7351.3	367	16 x 73
2534	S1840	-7364.5	-7362.3	187	16 x 73
2535	S1841	-7375.5	-7373.3	277	16 x 73
2536	S1842	-7386.5	-7384.3	367	16 x 73
2537	S1843	-7397.5	-7395.3	187	16 x 73
2538	S1844	-7408.5	-7406.3	277	16 x 73
2539	S1845	-7419.5	-7417.3	367	16 x 73
2540	S1846	-7430.5	-7428.3	187	16 x 73
2541	S1847	-7441.5	-7439.3	277	16 x 73
2542	S1848	-7452.5	-7450.3	367	16 x 73
2543	S1849	-7463.5	-7461.3	187	16 x 73
2544	S1850	-7474.5	-7472.3	277	16 x 73
2545	S1851	-7485.5	-7483.3	367	16 x 73
2546	S1852	-7496.5	-7494.3	187	16 x 73
2547	S1853	-7507.5	-7505.3	277	16 x 73
2548	S1854	-7518.5	-7516.3	367	16 x 73
2549	S1855	-7529.5	-7527.3	187	16 x 73
2550	S1856	-7540.5	-7538.3	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2551	S1857	-7551.5	-7549.3	367	16 x 73
2552	S1858	-7562.5	-7560.3	187	16 x 73
2553	S1859	-7573.5	-7571.3	277	16 x 73
2554	S1860	-7584.5	-7582.3	367	16 x 73
2555	S1861	-7595.5	-7593.2	187	16 x 73
2556	S1862	-7606.5	-7604.2	277	16 x 73
2557	S1863	-7617.5	-7615.2	367	16 x 73
2558	S1864	-7628.5	-7626.2	187	16 x 73
2559	S1865	-7639.5	-7637.2	277	16 x 73
2560	S1866	-7650.5	-7648.2	367	16 x 73
2561	S1867	-7661.5	-7659.2	187	16 x 73
2562	S1868	-7672.5	-7670.2	277	16 x 73
2563	S1869	-7683.5	-7681.2	367	16 x 73
2564	S1870	-7694.5	-7692.2	187	16 x 73
2565	S1871	-7705.5	-7703.2	277	16 x 73
2566	S1872	-7716.5	-7714.2	367	16 x 73
2567	S1873	-7727.5	-7725.2	187	16 x 73
2568	S1874	-7738.5	-7736.2	277	16 x 73
2569	S1875	-7749.5	-7747.2	367	16 x 73
2570	S1876	-7760.5	-7758.2	187	16 x 73
2571	S1877	-7771.5	-7769.2	277	16 x 73
2572	S1878	-7782.5	-7780.2	367	16 x 73
2573	S1879	-7793.5	-7791.2	187	16 x 73
2574	S1880	-7804.5	-7802.2	277	16 x 73
2575	S1881	-7815.5	-7813.2	367	16 x 73
2576	S1882	-7826.5	-7824.2	187	16 x 73
2577	S1883	-7837.5	-7835.2	277	16 x 73
2578	S1884	-7848.5	-7846.2	367	16 x 73
2579	S1885	-7859.5	-7857.2	187	16 x 73
2580	S1886	-7870.5	-7868.2	277	16 x 73
2581	S1887	-7881.5	-7879.2	367	16 x 73
2582	S1888	-7892.5	-7890.2	187	16 x 73
2583	S1889	-7903.5	-7901.2	277	16 x 73
2584	S1890	-7914.5	-7912.2	367	16 x 73
2585	S1891	-7925.5	-7923.1	187	16 x 73
2586	S1892	-7936.5	-7934.1	277	16 x 73
2587	S1893	-7947.5	-7945.1	367	16 x 73
2588	S1894	-7958.5	-7956.1	187	16 x 73
2589	S1895	-7969.5	-7967.1	277	16 x 73
2590	S1896	-7980.5	-7978.1	367	16 x 73
2591	S1897	-7991.5	-7989.1	187	16 x 73
2592	S1898	-8002.5	-8000.1	277	16 x 73
2593	S1899	-8013.5	-8011.1	367	16 x 73
2594	S1900	-8024.5	-8022.1	187	16 x 73
2595	S1901	-8035.5	-8033.1	277	16 x 73
2596	S1902	-8046.5	-8044.1	367	16 x 73
2597	S1903	-8057.5	-8055.1	187	16 x 73
2598	S1904	-8068.5	-8066.1	277	16 x 73
2599	S1905	-8079.5	-8077.1	367	16 x 73
2600	S1906	-8090.5	-8088.1	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2601	S1907	-8101.5	-8099.1	277	16 x 73
2602	S1908	-8112.5	-8110.1	367	16 x 73
2603	S1909	-8123.5	-8121.1	187	16 x 73
2604	S1910	-8134.5	-8132.1	277	16 x 73
2605	S1911	-8145.5	-8143.1	367	16 x 73
2606	S1912	-8156.5	-8154.1	187	16 x 73
2607	S1913	-8167.5	-8165.1	277	16 x 73
2608	S1914	-8178.5	-8176.1	367	16 x 73
2609	S1915	-8189.5	-8187.1	187	16 x 73
2610	S1916	-8200.5	-8198.1	277	16 x 73
2611	S1917	-8211.5	-8209.1	367	16 x 73
2612	S1918	-8222.5	-8220.1	187	16 x 73
2613	S1919	-8233.5	-8231.1	277	16 x 73
2614	S1920	-8244.5	-8242.1	367	16 x 73
2615	S1921	-8255.5	-8253	187	16 x 73
2616	S1922	-8266.5	-8264	277	16 x 73
2617	S1923	-8277.5	-8275	367	16 x 73
2618	S1924	-8288.5	-8286	187	16 x 73
2619	S1925	-8299.5	-8297	277	16 x 73
2620	S1926	-8310.5	-8308	367	16 x 73
2621	S1927	-8321.5	-8319	187	16 x 73
2622	S1928	-8332.5	-8330	277	16 x 73
2623	S1929	-8343.5	-8341	367	16 x 73
2624	S1930	-8354.5	-8352	187	16 x 73
2625	S1931	-8365.5	-8363	277	16 x 73
2626	S1932	-8376.5	-8374	367	16 x 73
2627	S1933	-8387.5	-8385	187	16 x 73
2628	S1934	-8398.5	-8396	277	16 x 73
2629	S1935	-8409.5	-8407	367	16 x 73
2630	S1936	-8420.5	-8418	187	16 x 73
2631	S1937	-8431.5	-8429	277	16 x 73
2632	S1938	-8442.5	-8440	367	16 x 73
2633	S1939	-8453.5	-8451	187	16 x 73
2634	S1940	-8464.5	-8462	277	16 x 73
2635	S1941	-8475.5	-8473	367	16 x 73
2636	S1942	-8486.5	-8484	187	16 x 73
2637	S1943	-8497.5	-8495	277	16 x 73
2638	S1944	-8508.5	-8506	367	16 x 73
2639	S1945	-8519.5	-8517	187	16 x 73
2640	S1946	-8530.5	-8528	277	16 x 73
2641	S1947	-8541.5	-8539	367	16 x 73
2642	S1948	-8552.5	-8550	187	16 x 73
2643	S1949	-8563.5	-8561	277	16 x 73
2644	S1950	-8574.5	-8572	367	16 x 73
2645	S1951	-8585.5	-8582.9	187	16 x 73
2646	S1952	-8596.5	-8593.9	277	16 x 73
2647	S1953	-8607.5	-8604.9	367	16 x 73
2648	S1954	-8618.5	-8615.9	187	16 x 73
2649	S1955	-8629.5	-8626.9	277	16 x 73
2650	S1956	-8640.5	-8637.9	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2651	S1957	-8651.5	-8648.9	187	16 x 73
2652	S1958	-8662.5	-8659.9	277	16 x 73
2653	S1959	-8673.5	-8670.9	367	16 x 73
2654	S1960	-8684.5	-8681.9	187	16 x 73
2655	S1961	-8695.5	-8692.9	277	16 x 73
2656	S1962	-8706.5	-8703.9	367	16 x 73
2657	S1963	-8717.5	-8714.9	187	16 x 73
2658	S1964	-8728.5	-8725.9	277	16 x 73
2659	S1965	-8739.5	-8736.9	367	16 x 73
2660	S1966	-8750.5	-8747.9	187	16 x 73
2661	S1967	-8761.5	-8758.9	277	16 x 73
2662	S1968	-8772.5	-8769.9	367	16 x 73
2663	S1969	-8783.5	-8780.9	187	16 x 73
2664	S1970	-8794.5	-8791.9	277	16 x 73
2665	S1971	-8805.5	-8802.9	367	16 x 73
2666	S1972	-8816.5	-8813.9	187	16 x 73
2667	S1973	-8827.5	-8824.9	277	16 x 73
2668	S1974	-8838.5	-8835.9	367	16 x 73
2669	S1975	-8849.5	-8846.9	187	16 x 73
2670	S1976	-8860.5	-8857.9	277	16 x 73
2671	S1977	-8871.5	-8868.9	367	16 x 73
2672	S1978	-8882.5	-8879.9	187	16 x 73
2673	S1979	-8893.5	-8890.9	277	16 x 73
2674	S1980	-8904.5	-8901.9	367	16 x 73
2675	S1981	-8915.5	-8912.8	187	16 x 73
2676	S1982	-8926.5	-8923.8	277	16 x 73
2677	S1983	-8937.5	-8934.8	367	16 x 73
2678	S1984	-8948.5	-8945.8	187	16 x 73
2679	S1985	-8959.5	-8956.8	277	16 x 73
2680	S1986	-8970.5	-8967.8	367	16 x 73
2681	S1987	-8981.5	-8978.8	187	16 x 73
2682	S1988	-8992.5	-8989.8	277	16 x 73
2683	S1989	-9003.5	-9000.8	367	16 x 73
2684	S1990	-9014.5	-9011.8	187	16 x 73
2685	S1991	-9025.5	-9022.8	277	16 x 73
2686	S1992	-9036.5	-9033.8	367	16 x 73
2687	S1993	-9047.5	-9044.8	187	16 x 73
2688	S1994	-9058.5	-9055.8	277	16 x 73
2689	S1995	-9069.5	-9066.8	367	16 x 73
2690	S1996	-9080.5	-9077.8	187	16 x 73
2691	S1997	-9091.5	-9088.8	277	16 x 73
2692	S1998	-9102.5	-9099.8	367	16 x 73
2693	S1999	-9113.5	-9110.8	187	16 x 73
2694	S2000	-9124.5	-9121.8	277	16 x 73
2695	S2001	-9135.5	-9132.8	367	16 x 73
2696	S2002	-9146.5	-9143.8	187	16 x 73
2697	S2003	-9157.5	-9154.8	277	16 x 73
2698	S2004	-9168.5	-9165.8	367	16 x 73
2699	S2005	-9179.5	-9176.8	187	16 x 73
2700	S2006	-9190.5	-9187.8	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2701	S2007	-9201.5	-9198.8	367	16 x 73
2702	S2008	-9212.5	-9209.8	187	16 x 73
2703	S2009	-9223.5	-9220.8	277	16 x 73
2704	S2010	-9234.5	-9231.8	367	16 x 73
2705	S2011	-9245.5	-9242.7	187	16 x 73
2706	S2012	-9256.5	-9253.7	277	16 x 73
2707	S2013	-9267.5	-9264.7	367	16 x 73
2708	S2014	-9278.5	-9275.7	187	16 x 73
2709	S2015	-9289.5	-9286.7	277	16 x 73
2710	S2016	-9300.5	-9297.7	367	16 x 73
2711	S2017	-9311.5	-9308.7	187	16 x 73
2712	S2018	-9322.5	-9319.7	277	16 x 73
2713	S2019	-9333.5	-9330.7	367	16 x 73
2714	S2020	-9344.5	-9341.7	187	16 x 73
2715	S2021	-9355.5	-9352.7	277	16 x 73
2716	S2022	-9366.5	-9363.7	367	16 x 73
2717	S2023	-9377.5	-9374.7	187	16 x 73
2718	S2024	-9388.5	-9385.7	277	16 x 73
2719	S2025	-9399.5	-9396.7	367	16 x 73
2720	S2026	-9410.5	-9407.7	187	16 x 73
2721	S2027	-9421.5	-9418.7	277	16 x 73
2722	S2028	-9432.5	-9429.7	367	16 x 73
2723	S2029	-9443.5	-9440.7	187	16 x 73
2724	S2030	-9454.5	-9451.7	277	16 x 73
2725	S2031	-9465.5	-9462.7	367	16 x 73
2726	S2032	-9476.5	-9473.7	187	16 x 73
2727	S2033	-9487.5	-9484.7	277	16 x 73
2728	S2034	-9498.5	-9495.7	367	16 x 73
2729	S2035	-9509.5	-9506.7	187	16 x 73
2730	S2036	-9520.5	-9517.7	277	16 x 73
2731	S2037	-9531.5	-9528.7	367	16 x 73
2732	S2038	-9542.5	-9539.7	187	16 x 73
2733	S2039	-9553.5	-9550.7	277	16 x 73
2734	S2040	-9564.5	-9561.7	367	16 x 73
2735	S2041	-9575.5	-9572.6	187	16 x 73
2736	S2042	-9586.5	-9583.6	277	16 x 73
2737	S2043	-9597.5	-9594.6	367	16 x 73
2738	S2044	-9608.5	-9605.6	187	16 x 73
2739	S2045	-9619.5	-9616.6	277	16 x 73
2740	S2046	-9630.5	-9627.6	367	16 x 73
2741	S2047	-9641.5	-9638.6	187	16 x 73
2742	S2048	-9652.5	-9649.6	277	16 x 73
2743	S2049	-9663.5	-9660.6	367	16 x 73
2744	S2050	-9674.5	-9671.6	187	16 x 73
2745	S2051	-9685.5	-9682.6	277	16 x 73
2746	S2052	-9696.5	-9693.6	367	16 x 73
2747	S2053	-9707.5	-9704.6	187	16 x 73
2748	S2054	-9718.5	-9715.6	277	16 x 73
2749	S2055	-9729.5	-9726.6	367	16 x 73
2750	S2056	-9740.5	-9737.6	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2751	S2057	-9751.5	-9748.6	277	16 x 73
2752	S2058	-9762.5	-9759.6	367	16 x 73
2753	S2059	-9773.5	-9770.6	187	16 x 73
2754	S2060	-9784.5	-9781.6	277	16 x 73
2755	S2061	-9795.5	-9792.6	367	16 x 73
2756	S2062	-9806.5	-9803.6	187	16 x 73
2757	S2063	-9817.5	-9814.6	277	16 x 73
2758	S2064	-9828.5	-9825.6	367	16 x 73
2759	S2065	-9839.5	-9836.6	187	16 x 73
2760	S2066	-9850.5	-9847.6	277	16 x 73
2761	S2067	-9861.5	-9858.6	367	16 x 73
2762	S2068	-9872.5	-9869.6	187	16 x 73
2763	S2069	-9883.5	-9880.6	277	16 x 73
2764	S2070	-9894.5	-9891.6	367	16 x 73
2765	S2071	-9905.5	-9902.5	187	16 x 73
2766	S2072	-9916.5	-9913.5	277	16 x 73
2767	S2073	-9927.5	-9924.5	367	16 x 73
2768	S2074	-9938.5	-9935.5	187	16 x 73
2769	S2075	-9949.5	-9946.5	277	16 x 73
2770	S2076	-9960.5	-9957.5	367	16 x 73
2771	S2077	-9971.5	-9968.5	187	16 x 73
2772	S2078	-9982.5	-9979.5	277	16 x 73
2773	S2079	-9993.5	-9990.5	367	16 x 73
2774	S2080	-10004.5	-10001.5	187	16 x 73
2775	S2081	-10015.5	-10012.5	277	16 x 73
2776	S2082	-10026.5	-10023.5	367	16 x 73
2777	S2083	-10037.5	-10034.5	187	16 x 73
2778	S2084	-10048.5	-10045.5	277	16 x 73
2779	S2085	-10059.5	-10056.5	367	16 x 73
2780	S2086	-10070.5	-10067.5	187	16 x 73
2781	S2087	-10081.5	-10078.5	277	16 x 73
2782	S2088	-10092.5	-10089.5	367	16 x 73
2783	S2089	-10103.5	-10100.5	187	16 x 73
2784	S2090	-10114.5	-10111.5	277	16 x 73
2785	S2091	-10125.5	-10122.5	367	16 x 73
2786	S2092	-10136.5	-10133.5	187	16 x 73
2787	S2093	-10147.5	-10144.5	277	16 x 73
2788	S2094	-10158.5	-10155.5	367	16 x 73
2789	S2095	-10169.5	-10166.5	187	16 x 73
2790	S2096	-10180.5	-10177.5	277	16 x 73
2791	S2097	-10191.5	-10188.5	367	16 x 73
2792	S2098	-10202.5	-10199.5	187	16 x 73
2793	S2099	-10213.5	-10210.5	277	16 x 73
2794	S2100	-10224.5	-10221.5	367	16 x 73
2795	S2101	-10235.5	-10232.4	187	16 x 73
2796	S2102	-10246.5	-10243.4	277	16 x 73
2797	S2103	-10257.5	-10254.4	367	16 x 73
2798	S2104	-10268.5	-10265.4	187	16 x 73
2799	S2105	-10279.5	-10276.4	277	16 x 73
2800	S2106	-10290.5	-10287.4	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2801	S2107	-10301.5	-10298.4	187	16 x 73
2802	S2108	-10312.5	-10309.4	277	16 x 73
2803	S2109	-10323.5	-10320.4	367	16 x 73
2804	S2110	-10334.5	-10331.4	187	16 x 73
2805	S2111	-10345.5	-10342.4	277	16 x 73
2806	S2112	-10356.5	-10353.4	367	16 x 73
2807	S2113	-10367.5	-10364.4	187	16 x 73
2808	S2114	-10378.5	-10375.4	277	16 x 73
2809	S2115	-10389.5	-10386.4	367	16 x 73
2810	S2116	-10400.5	-10397.4	187	16 x 73
2811	S2117	-10411.5	-10408.4	277	16 x 73
2812	S2118	-10422.5	-10419.4	367	16 x 73
2813	S2119	-10433.5	-10430.4	187	16 x 73
2814	S2120	-10444.5	-10441.4	277	16 x 73
2815	S2121	-10455.5	-10452.4	367	16 x 73
2816	S2122	-10466.5	-10463.4	187	16 x 73
2817	S2123	-10477.5	-10474.4	277	16 x 73
2818	S2124	-10488.5	-10485.4	367	16 x 73
2819	S2125	-10499.5	-10496.4	187	16 x 73
2820	S2126	-10510.5	-10507.4	277	16 x 73
2821	S2127	-10521.5	-10518.4	367	16 x 73
2822	S2128	-10532.5	-10529.4	187	16 x 73
2823	S2129	-10543.5	-10540.4	277	16 x 73
2824	S2130	-10554.5	-10551.4	367	16 x 73
2825	S2131	-10565.5	-10562.3	187	16 x 73
2826	S2132	-10576.5	-10573.3	277	16 x 73
2827	S2133	-10587.5	-10584.3	367	16 x 73
2828	S2134	-10598.5	-10595.3	187	16 x 73
2829	S2135	-10609.5	-10606.3	277	16 x 73
2830	S2136	-10620.5	-10617.3	367	16 x 73
2831	S2137	-10631.5	-10628.3	187	16 x 73
2832	S2138	-10642.5	-10639.3	277	16 x 73
2833	S2139	-10653.5	-10650.3	367	16 x 73
2834	S2140	-10664.5	-10661.3	187	16 x 73
2835	S2141	-10675.5	-10672.3	277	16 x 73
2836	S2142	-10686.5	-10683.3	367	16 x 73
2837	S2143	-10697.5	-10694.3	187	16 x 73
2838	S2144	-10708.5	-10705.3	277	16 x 73
2839	S2145	-10719.5	-10716.3	367	16 x 73
2840	S2146	-10730.5	-10727.3	187	16 x 73
2841	S2147	-10741.5	-10738.3	277	16 x 73
2842	S2148	-10752.5	-10749.3	367	16 x 73
2843	S2149	-10763.5	-10760.3	187	16 x 73
2844	S2150	-10774.5	-10771.3	277	16 x 73
2845	S2151	-10785.5	-10782.3	367	16 x 73
2846	S2152	-10796.5	-10793.3	187	16 x 73
2847	S2153	-10807.5	-10804.3	277	16 x 73
2848	S2154	-10818.5	-10815.3	367	16 x 73
2849	S2155	-10829.5	-10826.3	187	16 x 73
2850	S2156	-10840.5	-10837.3	277	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2851	S2157	-10851.5	-10848.3	367	16 x 73
2852	S2158	-10862.5	-10859.3	187	16 x 73
2853	S2159	-10873.5	-10870.3	277	16 x 73
2854	S2160	-10884.5	-10881.3	367	16 x 73
2855	S2161	-10895.5	-10892.2	187	16 x 73
2856	S2162	-10906.5	-10903.2	277	16 x 73
2857	S2163	-10917.5	-10914.2	367	16 x 73
2858	S2164	-10928.5	-10925.2	187	16 x 73
2859	S2165	-10939.5	-10936.2	277	16 x 73
2860	S2166	-10950.5	-10947.2	367	16 x 73
2861	S2167	-10961.5	-10958.2	187	16 x 73
2862	S2168	-10972.5	-10969.2	277	16 x 73
2863	S2169	-10983.5	-10980.2	367	16 x 73
2864	S2170	-10994.5	-10991.2	187	16 x 73
2865	S2171	-11005.5	-11002.2	277	16 x 73
2866	S2172	-11016.5	-11013.2	367	16 x 73
2867	S2173	-11027.5	-11024.2	187	16 x 73
2868	S2174	-11038.5	-11035.2	277	16 x 73
2869	S2175	-11049.5	-11046.2	367	16 x 73
2870	S2176	-11060.5	-11057.2	187	16 x 73
2871	S2177	-11071.5	-11068.2	277	16 x 73
2872	S2178	-11082.5	-11079.2	367	16 x 73
2873	S2179	-11093.5	-11090.2	187	16 x 73
2874	S2180	-11104.5	-11101.2	277	16 x 73
2875	S2181	-11115.5	-11112.2	367	16 x 73
2876	S2182	-11126.5	-11123.2	187	16 x 73
2877	S2183	-11137.5	-11134.2	277	16 x 73
2878	S2184	-11148.5	-11145.2	367	16 x 73
2879	S2185	-11159.5	-11156.2	187	16 x 73
2880	S2186	-11170.5	-11167.2	277	16 x 73
2881	S2187	-11181.5	-11178.2	367	16 x 73
2882	S2188	-11192.5	-11189.2	187	16 x 73
2883	S2189	-11203.5	-11200.2	277	16 x 73
2884	S2190	-11214.5	-11211.2	367	16 x 73
2885	S2191	-11225.5	-11222.1	187	16 x 73
2886	S2192	-11236.5	-11233.1	277	16 x 73
2887	S2193	-11247.5	-11244.1	367	16 x 73
2888	S2194	-11258.5	-11255.1	187	16 x 73
2889	S2195	-11269.5	-11266.1	277	16 x 73
2890	S2196	-11280.5	-11277.1	367	16 x 73
2891	S2197	-11291.5	-11288.1	187	16 x 73
2892	S2198	-11302.5	-11299.1	277	16 x 73
2893	S2199	-11313.5	-11310.1	367	16 x 73
2894	S2200	-11324.5	-11321.1	187	16 x 73
2895	S2201	-11335.5	-11332.1	277	16 x 73
2896	S2202	-11346.5	-11343.1	367	16 x 73
2897	S2203	-11357.5	-11354.1	187	16 x 73
2898	S2204	-11368.5	-11365.1	277	16 x 73
2899	S2205	-11379.5	-11376.1	367	16 x 73
2900	S2206	-11390.5	-11387.1	187	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
2901	S2207	-11401.5	-11398.1	277	16 x 73
2902	S2208	-11412.5	-11409.1	367	16 x 73
2903	S2209	-11423.5	-11420.1	187	16 x 73
2904	S2210	-11434.5	-11431.1	277	16 x 73
2905	S2211	-11445.5	-11442.1	367	16 x 73
2906	S2212	-11456.5	-11453.1	187	16 x 73
2907	S2213	-11467.5	-11464.1	277	16 x 73
2908	S2214	-11478.5	-11475.1	367	16 x 73
2909	S2215	-11489.5	-11486.1	187	16 x 73
2910	S2216	-11500.5	-11497.1	277	16 x 73
2911	S2217	-11511.5	-11508.1	367	16 x 73
2912	S2218	-11522.5	-11519.1	187	16 x 73
2913	S2219	-11533.5	-11530.1	277	16 x 73
2914	S2220	-11544.5	-11541.1	367	16 x 73
2915	S2221	-11555.5	-11552	187	16 x 73
2916	S2222	-11566.5	-11563	277	16 x 73
2917	S2223	-11577.5	-11574	367	16 x 73
2918	S2224	-11588.5	-11585	187	16 x 73
2919	S2225	-11599.5	-11596	277	16 x 73
2920	S2226	-11610.5	-11607	367	16 x 73
2921	S2227	-11621.5	-11618	187	16 x 73
2922	S2228	-11632.5	-11629	277	16 x 73
2923	S2229	-11643.5	-11640	367	16 x 73
2924	S2230	-11654.5	-11651	187	16 x 73
2925	S2231	-11665.5	-11662	277	16 x 73
2926	S2232	-11676.5	-11673	367	16 x 73
2927	S2233	-11687.5	-11684	187	16 x 73
2928	S2234	-11698.5	-11695	277	16 x 73
2929	S2235	-11709.5	-11706	367	16 x 73
2930	S2236	-11720.5	-11717	187	16 x 73
2931	S2237	-11731.5	-11728	277	16 x 73
2932	S2238	-11742.5	-11739	367	16 x 73
2933	S2239	-11753.5	-11750	187	16 x 73
2934	S2240	-11764.5	-11761	277	16 x 73
2935	S2241	-11775.5	-11772	367	16 x 73
2936	S2242	-11786.5	-11783	187	16 x 73
2937	S2243	-11797.5	-11794	277	16 x 73
2938	S2244	-11808.5	-11805	367	16 x 73
2939	S2245	-11819.5	-11816	187	16 x 73
2940	S2246	-11830.5	-11827	277	16 x 73
2941	S2247	-11841.5	-11838	367	16 x 73
2942	S2248	-11852.5	-11849	187	16 x 73
2943	S2249	-11863.5	-11860	277	16 x 73
2944	S2250	-11874.5	-11871	367	16 x 73
2945	S2251	-11885.5	-11881.9	187	16 x 73
2946	S2252	-11896.5	-11892.9	277	16 x 73
2947	S2253	-11907.5	-11903.9	367	16 x 73
2948	S2254	-11918.5	-11914.9	187	16 x 73
2949	S2255	-11929.5	-11925.9	277	16 x 73
2950	S2256	-11940.5	-11936.9	367	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
2951	S2257	-11951.5	-11947.9	187	16 x 73
2952	S2258	-11962.5	-11958.9	277	16 x 73
2953	S2259	-11973.5	-11969.9	367	16 x 73
2954	S2260	-11984.5	-11980.9	187	16 x 73
2955	S2261	-11995.5	-11991.9	277	16 x 73
2956	S2262	-12006.5	-12002.9	367	16 x 73
2957	S2263	-12017.5	-12013.9	187	16 x 73
2958	S2264	-12028.5	-12024.9	277	16 x 73
2959	S2265	-12039.5	-12035.9	367	16 x 73
2960	S2266	-12050.5	-12046.9	187	16 x 73
2961	S2267	-12061.5	-12057.9	277	16 x 73
2962	S2268	-12072.5	-12068.9	367	16 x 73
2963	S2269	-12083.5	-12079.9	187	16 x 73
2964	S2270	-12094.5	-12090.9	277	16 x 73
2965	S2271	-12105.5	-12101.9	367	16 x 73
2966	S2272	-12116.5	-12112.9	187	16 x 73
2967	S2273	-12127.5	-12123.9	277	16 x 73
2968	S2274	-12138.5	-12134.9	367	16 x 73
2969	S2275	-12149.5	-12145.9	187	16 x 73
2970	S2276	-12160.5	-12156.9	277	16 x 73
2971	S2277	-12171.5	-12167.9	367	16 x 73
2972	S2278	-12182.5	-12178.9	187	16 x 73
2973	S2279	-12193.5	-12189.9	277	16 x 73
2974	S2280	-12204.5	-12200.9	367	16 x 73
2975	S2281	-12215.5	-12211.8	187	16 x 73
2976	S2282	-12226.5	-12222.8	277	16 x 73
2977	S2283	-12237.5	-12233.8	367	16 x 73
2978	S2284	-12248.5	-12244.8	187	16 x 73
2979	S2285	-12259.5	-12255.8	277	16 x 73
2980	S2286	-12270.5	-12266.8	367	16 x 73
2981	S2287	-12281.5	-12277.8	187	16 x 73
2982	S2288	-12292.5	-12288.8	277	16 x 73
2983	S2289	-12303.5	-12299.8	367	16 x 73
2984	S2290	-12314.5	-12310.8	187	16 x 73
2985	S2291	-12325.5	-12321.8	277	16 x 73
2986	S2292	-12336.5	-12332.8	367	16 x 73
2987	S2293	-12347.5	-12343.8	187	16 x 73
2988	S2294	-12358.5	-12354.8	277	16 x 73
2989	S2295	-12369.5	-12365.8	367	16 x 73
2990	S2296	-12380.5	-12376.8	187	16 x 73
2991	S2297	-12391.5	-12387.8	277	16 x 73
2992	S2298	-12402.5	-12398.8	367	16 x 73
2993	S2299	-12413.5	-12409.8	187	16 x 73
2994	S2300	-12424.5	-12420.8	277	16 x 73
2995	S2301	-12435.5	-12431.8	367	16 x 73
2996	S2302	-12446.5	-12442.8	187	16 x 73
2997	S2303	-12457.5	-12453.8	277	16 x 73
2998	S2304	-12468.5	-12464.8	367	16 x 73
2999	S2305	-12479.5	-12475.8	187	16 x 73
3000	S2306	-12490.5	-12486.8	277	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
3001	S2307	-12501.5	-12497.8	367	16 x 73
3002	S2308	-12512.5	-12508.8	187	16 x 73
3003	S2309	-12523.5	-12519.8	277	16 x 73
3004	S2310	-12534.5	-12530.8	367	16 x 73
3005	S2311	-12545.5	-12541.7	187	16 x 73
3006	S2312	-12556.5	-12552.7	277	16 x 73
3007	S2313	-12567.5	-12563.7	367	16 x 73
3008	S2314	-12578.5	-12574.7	187	16 x 73
3009	S2315	-12589.5	-12585.7	277	16 x 73
3010	S2316	-12600.5	-12596.7	367	16 x 73
3011	S2317	-12611.5	-12607.7	187	16 x 73
3012	S2318	-12622.5	-12618.7	277	16 x 73
3013	S2319	-12633.5	-12629.7	367	16 x 73
3014	S2320	-12644.5	-12640.7	187	16 x 73
3015	S2321	-12655.5	-12651.7	277	16 x 73
3016	S2322	-12666.5	-12662.7	367	16 x 73
3017	S2323	-12677.5	-12673.7	187	16 x 73
3018	S2324	-12688.5	-12684.7	277	16 x 73
3019	S2325	-12699.5	-12695.7	367	16 x 73
3020	S2326	-12710.5	-12706.7	187	16 x 73
3021	S2327	-12721.5	-12717.7	277	16 x 73
3022	S2328	-12732.5	-12728.7	367	16 x 73
3023	S2329	-12743.5	-12739.7	187	16 x 73
3024	S2330	-12754.5	-12750.7	277	16 x 73
3025	S2331	-12765.5	-12761.7	367	16 x 73
3026	S2332	-12776.5	-12772.7	187	16 x 73
3027	S2333	-12787.5	-12783.7	277	16 x 73
3028	S2334	-12798.5	-12794.7	367	16 x 73
3029	S2335	-12809.5	-12805.7	187	16 x 73
3030	S2336	-12820.5	-12816.7	277	16 x 73
3031	S2337	-12831.5	-12827.7	367	16 x 73
3032	S2338	-12842.5	-12838.7	187	16 x 73
3033	S2339	-12853.5	-12849.7	277	16 x 73
3034	S2340	-12864.5	-12860.7	367	16 x 73
3035	S2341	-12875.5	-12871.6	187	16 x 73
3036	S2342	-12886.5	-12882.6	277	16 x 73
3037	S2343	-12897.5	-12893.6	367	16 x 73
3038	S2344	-12908.5	-12904.6	187	16 x 73
3039	S2345	-12919.5	-12915.6	277	16 x 73
3040	S2346	-12930.5	-12926.6	367	16 x 73
3041	S2347	-12941.5	-12937.6	187	16 x 73
3042	S2348	-12952.5	-12948.6	277	16 x 73
3043	S2349	-12963.5	-12959.6	367	16 x 73
3044	S2350	-12974.5	-12970.6	187	16 x 73
3045	S2351	-12985.5	-12981.6	277	16 x 73
3046	S2352	-12996.5	-12992.6	367	16 x 73
3047	S2353	-13007.5	-13003.6	187	16 x 73
3048	S2354	-13018.5	-13014.6	277	16 x 73
3049	S2355	-13029.5	-13025.6	367	16 x 73
3050	S2356	-13040.5	-13036.6	187	16 x 73

No.	Name	X1	X2	Y	Bumpsize (μm)
3051	S2357	-13051.5	-13047.6	277	16 x 73
3052	S2358	-13062.5	-13058.6	367	16 x 73
3053	S2359	-13073.5	-13069.6	187	16 x 73
3054	S2360	-13084.5	-13080.6	277	16 x 73
3055	S2361	-13095.5	-13091.6	367	16 x 73
3056	S2362	-13106.5	-13102.6	187	16 x 73
3057	S2363	-13117.5	-13113.6	277	16 x 73
3058	S2364	-13128.5	-13124.6	367	16 x 73
3059	S2365	-13139.5	-13135.6	187	16 x 73
3060	S2366	-13150.5	-13146.6	277	16 x 73
3061	S2367	-13161.5	-13157.6	367	16 x 73
3062	S2368	-13172.5	-13168.6	187	16 x 73
3063	S2369	-13183.5	-13179.6	277	16 x 73
3064	S2370	-13194.5	-13190.6	367	16 x 73
3065	S2371	-13205.5	-13201.5	187	16 x 73
3066	S2372	-13216.5	-13212.5	277	16 x 73
3067	S2373	-13227.5	-13223.5	367	16 x 73
3068	S2374	-13238.5	-13234.5	187	16 x 73
3069	S2375	-13249.5	-13245.5	277	16 x 73
3070	S2376	-13260.5	-13256.5	367	16 x 73
3071	S2377	-13271.5	-13267.5	187	16 x 73
3072	S2378	-13282.5	-13278.5	277	16 x 73
3073	S2379	-13293.5	-13289.5	367	16 x 73
3074	S2380	-13304.5	-13300.5	187	16 x 73
3075	S2381	-13315.5	-13311.5	277	16 x 73
3076	S2382	-13326.5	-13322.5	367	16 x 73
3077	S2383	-13337.5	-13333.5	187	16 x 73
3078	S2384	-13348.5	-13344.5	277	16 x 73
3079	S2385	-13359.5	-13355.5	367	16 x 73
3080	S2386	-13370.5	-13366.5	187	16 x 73
3081	S2387	-13381.5	-13377.5	277	16 x 73
3082	S2388	-13392.5	-13388.5	367	16 x 73
3083	S2389	-13403.5	-13399.5	187	16 x 73
3084	S2390	-13414.5	-13410.5	277	16 x 73
3085	S2391	-13425.5	-13421.5	367	16 x 73
3086	S2392	-13436.5	-13432.5	187	16 x 73
3087	S2393	-13447.5	-13443.5	277	16 x 73
3088	S2394	-13458.5	-13454.5	367	16 x 73
3089	S2395	-13469.5	-13465.5	187	16 x 73
3090	S2396	-13480.5	-13476.5	277	16 x 73
3091	S2397	-13491.5	-13487.5	367	16 x 73
3092	S2398	-13502.5	-13498.5	187	16 x 73
3093	S2399	-13513.5	-13509.5	277	16 x 73
3094	S2400	-13524.5	-13520.5	367	16 x 73
3095	SL1	-13535.5	-13531.4	187	16 x 73
3096	DUMMY	-13546.5	-13542.4	277	16 x 73
3097	DUMMY	-13557.5	-13553.4	367	16 x 73
3098	GMLN_OUT	-13568.5	-13564.4	187	16 x 73
3099	DUMMY	-13579.5	-13575.4	277	16 x 73
3100	SYNC_R1	-13590.5	-13586.4	367	16 x 73

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No.	Name	X1	X2	Y	Bump size (μm)
3101	GMHN_OUT	-13601.5	-13597.4	187	16 x 73
3102	DUMMY	-13612.5	-13608.4	277	16 x 73
3103	SYNC_R2	-13623.5	-13619.4	367	16 x 73
3104	GMHP_OUT	-13634.5	-13630.4	187	16 x 73
3105	DUMMY	-13645.5	-13641.4	277	16 x 73
3106	SYNC_R3	-13656.5	-13652.4	367	16 x 73
3107	GMLP_OUT	-13667.5	-13663.4	277	16 x 73
3108	SYNC_R4	-13678.5	-13674.4	277	16 x 73
3109	SYNC_R5	-13711.5	-13707.4	367	16 x 73
3110	SYNC_R6	-13744.5	-13740.4	367	16 x 73
3111	SYNC_R7	-13777.5	-13773.4	277	16 x 73
3112	SYNC_R8	-13821.5	-13817.4	277	16 x 73
3113	SYNC_R9	-13854.5	-13850.4	277	16 x 73
3114	SYNC_R10	-13843.5	-13839.4	277	16 x 73
3115	DUMMY	-13810.5	-13806.4	187	16 x 73
3116	SYNC_R11	-13832.5	-13828.4	187	16 x 73
3117	SYNC_R12	-13799.5	-13795.4	187	16 x 73
3118	SYNC_R13	-13766.5	-13762.4	187	16 x 73
3119	SYNC_R14	-13733.5	-13729.4	187	16 x 73
3120	SYNC_R15	-13700.5	-13696.4	187	16 x 73
WQ_ALMARK_R		13871	13871	378.5	50 x 50
WQ_ALMARK_L		-13871	-13871	378.5	50 x 50

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15. Ordering Information

Part no.	Package type
HX8252-A00XPDxxxx	X: mean fab code PD: mean COG xxx: mean chip thickness **(μm)** x: compensation value of bump pad

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